CM-EIP — Study Guide

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CM-EIP — Study Guide

Capital Markets – Excluded Investment Products – Securities, Collective Investment Schemes (EIP) and Foreign Exchange

A guide to the CM-EIP examination: the securities, collective investment scheme and foreign exchange products an adviser may recommend to retail clients, and the analysis used to value them. It condenses the official study guide into what the examination tests, section by section.

- **Exam:** 90 multiple-choice and multiple-response questions in 2 hours 30 minutes, including 3 case studies carrying 9 questions (3 per case study); **70% to pass**.
- **Currency of the material:** the official guide is version 1.1 (March 2025). Check the current MAS notices, SGX and SGX-DT rules, and each product's prospectus or Product Highlights Sheet before relying on a rule or specification.
- This is a study aid, not a replacement for the prospectus, Product Highlights Sheet, trust deed, insurance policy or current law. If you will conduct regulated activity, you must also satisfy the MAS representative-notification and competency requirements under Notice SFA 04-N22.

How to use this guide

1. Learn what the investor owns before learning how it is priced. Every product in this syllabus is one of four claims: a deposit, a debt claim, an equity claim, or a contract whose value derives from something else.
2. For every product, ask: what is the underlying, what does the investor own, what drives the return, what can be lost, and who is the counterparty?
3. Work the calculations in Chapter 2 and Chapter 7 by hand until the arithmetic is automatic. The examination supplies a formulae sheet but not the judgement about which formula applies.
4. Practise the case studies in Chapter 12 last. They test the same method as the chapters, applied to a named investor with a stated objective.
5. Use `CMEIP_QUESTIONS.md` only after reading the relevant section; its answers explain the reasoning rather than merely naming a letter.

Examination orientation

- The examination tests knowledge, understanding and application of Excluded Investment Products.
- Format: 90 multiple-choice and multiple-response questions in 2 hours 30 minutes, including 3 case studies of 9 questions; the passing mark is 70%.
- Unless a question says otherwise, use a **365-day** convention for Singapore Government Securities and a **360-day** convention for all other products.
- A non-programmable financial calculator may be used and a formulae sheet is provided.
- Scope: financial markets and instruments, risk and return measurement, the drivers of market movements, foreign exchange, company and financial-statement analysis, equities, deposits and fixed income, portfolio management, ETFs, unit trusts, REITs, insurance, warrants, and technical and quantitative analysis.

Master mental model

Every product in this syllabus is a claim on someone else's cash flow, or a contract that transfers an exposure. Classify the product before analysing it.

Claim type	Products in this syllabus	What the investor owns
Deposit	Savings and fixed deposits, currency deposits, structured deposits	A debt claim on a bank
Money market	Treasury bills, commercial paper, CDs, banker's acceptances, repos	A short-dated debt claim
Bond	Government, corporate, convertible, covered, high-yield, zero-coupon	A contractual debt claim
Equity	Ordinary and preference shares, depositary receipts	Residual ownership
Fund	Unit trusts, ETFs, REITs, business trusts, hedge and private equity funds	Units in a pooled vehicle
Insurance	Whole life, endowment, annuities	A contract with an insurer
Derivative-linked	Warrants, structured warrants	A contract, not ownership

Coverage map

Chapter	Topic	Examination weight
1	Introduction to investing, investments and financial markets	Foundational; defines the vocabulary every later
2	Risk, return and time value calculations	The most calculation-heavy chapter; underpins C
3	Key drivers of market movements and asset values	Economic, market and behavioural drivers
4	Foreign exchange	Quotation conventions, forwards and interest rat
5	Company analysis and understanding financial statements	The three statements and the ratio families
6	Equity securities	Share types, indices, DCF and relative valuation
7	Deposits and fixed-income securities	The largest chapter; pricing, duration, credit and
8	Portfolio management	EMH, portfolio theory, CAPM, IPS and perform
9	ETFs, unit trusts, REITs and insurance	Product mechanics and suitability
10	Warrants	Terminology, premium, gearing and valuation bo
11	Technical and quantitative analysis	Indicators, chart patterns and their limits
12	Case studies	3 case studies, 9 questions — the applied synthe

Conventions used throughout

- **Money** is written as plain text (`S$10,000`, `USD1,027,500`), never as maths, so it renders literally.
- **Formulas** are written as LaTeX and rendered by KaTeX. Every symbol is defined immediately after the formula.
- **Percentages** are quoted to the precision the source uses; do not round intermediate steps in a calculation question.
- **Dates and thresholds** are reproduced as printed. Where a rule has changed, the current figure is given and the change is noted once.

Chapter 1 — Introduction to Investing, Investments and Financial Markets

This chapter establishes the vocabulary the rest of the syllabus assumes: who the investor is, what a financial asset is, and how markets are organised. Every later chapter classifies a product against the four investor categories and the four asset classes introduced here.

Learning Objectives

The candidate should be able to:

- Understand the definition and needs of different types of investors.
- Describe what makes up the financial markets.
- Define the term ”financial assets”.
- Distinguish between financial assets and real assets.
- Describe the types of money market securities.
- Describe the types of capital market securities.
- Explain the differences between the primary market and the secondary market.
- Identify the characteristics of a good financial market.
- Distinguish the money market from the capital market.
- Discuss broadly the bond markets, the equity markets, and the options and futures markets.
- Understand the inter-market relationships and how the economy is sustained by the financial markets.

1.1 Introduction

The investment environment is changing quickly. Regulatory infrastructure has developed, the mix of promoted products has shifted, and financial technology has widened access to investments while admitting new players who redefine how money, wealth and value are transferred and stored. A representative of a financial institution must keep expanding the breadth of investment knowledge to stay relevant.

Representatives must also be able to identify suitable products and services for the varying needs, expectations, risk profiles, preferences and limitations of their clients, subject to regulatory requirements and industry

ethical considerations. Understanding client needs is the basis of every good recommendation, but client needs and return expectations must match what the markets can actually deliver.

1.1.1 Definition and Needs of Different Types of Investors

Whether the client is institutional or retail, the first step is to understand their needs before any advice or product recommendation is made. Section 4A of the Securities and Futures Act 2001 (SFA) defines Accredited Investors (AIs), Expert Investors (EIs) and Institutional Investors (IIs); anyone outside those definitions is a Retail Investor.

Category	Statutory basis
Institutional Investor (II)	SFA s 4A: government, statutory boards, central banks, banks, insurers, CMS licensees, a
Accredited Investor (AI)	SFA s 4A: an individual meeting the wealth or income tests, or a corporation, trust, partn
Expert Investor (EI)	SFA s 4A: a person whose business involves acquiring, disposing of or holding capital mar
Retail Investor	Everyone not in the three categories above

Accredited Investor thresholds. An individual qualifies as an AI on any one of three tests:

- **Net personal assets** exceeding **S**$2*million*∗∗.*Theprimaryresidenceiscountednetofanyoutstandingsecuredcreditfac* ∗*cappedatS***1 million**.
- **Financial assets** (bank deposits, investment products and other prescribed assets, net of liabilities) exceeding **S$1 million**.
- **Income** in the preceding 12 months of not less than **S$300,000**.

A corporation or other entity qualifies with net assets exceeding **S$10 million**, as determined by its most recent audited balance sheet, or by a balance sheet it certifies as giving a true and fair view, dated within the preceding 12 months. A partnership (other than an LLP) qualifies when each partner is an AI; a corporation qualifies when its entire share capital is owned by one or more AIs; and a person holding a joint account with an AI qualifies in respect of dealings through that joint account.

The AI opt-in regime. MAS raised the qualifying benchmark for AIs and introduced an opt-in regime: a capital markets intermediary (CMI) can no longer automatically treat anyone who qualifies as an AI unless the investor opts to be treated as one. An AI may also elect to benefit from the full range of retail safeguards.

Institutional Investors. IIs are "super-investors" by size and market impact. Their capital and resources give them timely, wide access to market information and professional research. They generally have clearly defined investment goals with prescribed directives and explicit requirements covering costs, execution efficiency, research and reporting support, volatility and return ranges, and asset-allocation, concentration and exposure limits. Because they are sophisticated and act within an institutional structure, they are less likely to make uneducated investments and are typically permitted wider product access than other investor types.

IIs are commonly classified by the purpose for which the money is invested:

Type of II	Purpose of the funds
Pension funds	Providing retirement income
Insurance funds	Meeting expected claims
Endowment funds	Satisfying philanthropic spending commitments
Sovereign wealth funds	Stabilising the national economy, or saving for future generations

Expert Investors. An EI trades capital markets products in the ordinary course of business. Like AIs, EIs receive less regulatory protection than retail investors, but for a different reason: not greater wealth, but occupational understanding of product risk. An EI is effectively a sub-set of retail investors meeting specific experience requirements in the SFA, so the same factors generally influence their decisions. Two differences matter: market rumours that move retail investors may not move EIs, and EIs tend to have a clearer idea of their objectives and constraints and may prefer greater involvement in the investment process.

Retail Investors. A retail investor buys securities for a personal account rather than for an organisation. Retail investors are the opposite of IIs: they trade in small amounts, trade less frequently, and have little influence over corporate decisions or the capital markets. Lacking access to boardrooms or timely market information, they are assumed to be less financially savvy and more risk-averse than AIs and IIs, and are therefore given more protective regulation.

Three regulatory developments shape how retail investors are treated in Singapore:

- More stringent regulation of non-conventional products such as precious metal buy-back arrangements, and the regulation of collectively managed investment schemes as Collective Investment Schemes (CIS).
- Refinement of investor classifications under the SFA and the Financial Advisers Act 2001 (FAA).
- An option for AIs to benefit from the full range of retail safeguards.

Most factors influencing a retail investor's decision coincide with those of an individual AI, because an AI is a sub-set of retail investors meeting specific wealth and income requirements.

Factors influencing investment decisions. The factors differ between individual and corporate investors.

Individual	Corporate
Demographics (age, gender, education level)	Management outlook (growth vs decl
Lifestyle (confidence level, number of dependants)	Size of firm
Risk tolerance (volatility range, risk-return spectrum)	Corporate governance — balancing v
Market trends and forecasts	Market trends and forecasts
Tax (higher brackets prefer tax-exempt returns)	Tax (fiscal incentives or barriers on a
Investment horizon	Investment horizon
Liquidity	Cash flow
Psychological influences (desires, prejudices, emotions)	Corporate culture (aggressive vs pass
Personal values (social or religious values shaping attitude to certain asset classes)	Corporate values

Client profiling. A representative maps the client against a risk-and-return grid. The standard return expectations quoted in the guide are:

Objective	Expected return
Preserve wealth and purchasing power	2–4%
Income with some growth	4–6%
Steady balanced growth	6–8%
Aggressive growth	8–12%

The profile also records investment knowledge (limited and low, average and occasional, experienced and regular, sophisticated and high), risk tolerance (low, medium, high), investment horizon (less than 1 year, 2–5 years, more than 5 years), the type of service sought (discretionary management, advisory, or transaction ideas and execution), product preferences (equities, fixed income, funds, derivatives), and the amount to invest (less than S2$million, S$2–10 million, S10–50$million, more than S$50 million).

1.1.2 Investment Products and Services

Products range from those actively traded and aimed mainly at IIs to those designed for AIs, high net worth individuals (HNWIs) or retail investors. Most products built from traditional asset classes — deposits, foreign currency, fixed income and equity — seek to provide yield and/or capital gains. Financial innovation also lets investors take a bearish position using put options, futures and inverse exchange-traded funds (ETFs) to generate potentially higher yields.

Clients now have access to large amounts of online information, which raises their interest in the products available. Investment professionals must therefore widen their product knowledge to give the right information and timely advice.

Product or service	Defined interest, yield or dividend	Potential for higher yield	Potential capital gains	Re
Traditional asset classes				
Local deposit				
Foreign deposit				
Money market instruments				
Bonds				
Equities				
Convertible bonds				

Product or service	Defined interest, yield or dividend	Potential for higher yield	Potential capital gains	Re
Derivatives				
Futures				
Swaps				
Options				
Warrants				
Structured products				
Funds				
Portfolio management				
Investment funds				
ETFs and ETNs				
Alternative investments				
Hedge funds				
Private equity				
REITs and property funds				
Commodity funds				

The wider range of securities and derivative structures lets investors:

- Diversify their portfolios.
- Optimise the risk-return profile of their portfolios.
- Protect investments, or take advantage of downward market movement.
- Selectively exploit opportunities through a leverage effect.

1.1.3 Financial Assets and Real Assets from the Individual Investor's Perspective

Beyond financial assets, individuals may own real physical assets and business assets whose values and cash flows can be significantly larger than their financial assets. For individuals the flow of funds between business, physical and financial assets is closely related.

Business owners have business financing needs and require corporate finance products and services as well as cash management. Business salaries and profits provide income that can be invested in financial assets or ploughed back into the business. Purchases of real physical assets such as real estate require financing and can be funded from financial assets. A representative must therefore be alert to business needs, changes and decisions about business assets and physical assets, because those can have a major impact on financial assets.

Financial assets
Traditional asset classes: cash deposits, foreign currencies, short-term money market instruments, bonds and other fixe
Derivatives: exchange-traded derivatives, financial futures, options, swaps, warrants, OTC structured products
Investment funds: ETFs, private funds sold OTC, pure asset class vs diversified funds, strategy funds, capital guarantee
Alternative investments: hedge funds, commodity funds, REITs and real estate funds, private equity funds

1.2 Financial Assets

Financial assets are paper or electronic claims on an issuer. The issuer may be from the public sector — a government such as the Singapore Government, or an agency such as the Housing and Development Board — a supranational such as the World Bank, or the private sector, such as a corporation. Financial assets are broadly classified into debt, equity and derivative instruments or securities.

1.2.1 Local and Foreign Cash Deposits

The most basic asset class is cash: Singapore dollar deposits in current, savings or fixed (time) deposit accounts, plus foreign currency deposit accounts. In a low interest rate environment, holding investible funds in cash carries a high opportunity cost, because inflation erodes purchasing power. Foreign currency deposits carry currency exchange risk and have been relatively volatile since the global financial crisis.

1.2.2 Fixed Income Securities

Fixed income securities are primarily interest-bearing instruments providing a fixed return. They include treasury bills, fixed-rate bonds, certificates of deposit, commercial papers and debentures.

Short-term money market securities. Money market instruments are relatively low-risk, highly liquid, short-term debt instruments issued by governments, financial institutions and corporations. Minimum denominations are relatively large and transaction sizes can be substantial. Maturities range from **1 day to 1 year**, and are often **less than 90 days**.

Instrument	What it is
Treasury bill	Issued by a central bank on behalf of the government to raise short-term funds
Repurchase agreement (repo)	Sale of a security at a specified price with a simultaneous commitment by the seller (
Reverse repo	The converse: purchase of a security by the lending party combined with a forward sa
Commercial paper	Short-term **unsecured** promissory note issued by a corporation
Banker's acceptance	A time draft drawn on a bank by a customer, which the bank agrees to pay on a spec
Certificate of deposit (CD)	Certificate issued by a bank in exchange for a deposit placed with it

Debt securities. Debt securities are long-term fixed borrowings by issuers. The investor lends a principal to the issuer, which agrees to make periodic interest payments and repay the principal at maturity. Debt securities are also called bonds. Singapore has one of the most developed bond markets in Asia, offering Singapore Government securities, statutory board securities, supranational bonds and corporate bonds.

Bond label	Definition
Eurobond	An international bond denominated in a currency **not native** to the country where it is issued
Eurodollar bond	Denominated in US dollars and sold **outside the United States** to non-US investors
Euroyen bond	Denominated in yen and sold outside Japan — for example, a Japanese company issuing Euroyen bo
Yankee bond	Sold **in the United States**, denominated in US dollars, but issued by a **foreign** corporation or gov
Domestic bond	Sold by an issuer **within its own country** in that country's currency

1.2.3 Equity Securities

Equity securities represent an ownership interest in a company. There are two forms:

- **Ordinary shares.** Ordinary shareholders elect the directors and vote on company issues, usually with votes equal to the number of shares owned.
- **Preference shares.** Technically equity, but also a **hybrid** security: voting rights are limited (they can vote only in limited circumstances) and the instrument resembles both equity and fixed income.

Priority of claim on the company's assets runs **creditors first, then preference shareholders, then ordinary shareholders**, who hold the residual claim.

1.2.4 Derivatives

Derivatives derive their value from the price of an underlying asset such as a stock, a bond or a market index. They include options, warrants, futures and convertible securities.

Instrument	Definition
Structured product	A note or deposit bundled with one or more derivatives
Option	Gives the buyer the right to buy (call) or sell (put) a specified amount of an underlying wi
Company-issued warrant	Right to buy a given number of the company's shares at a specified price over a given perio
Structured warrant	Issued usually by banks; lets investors profit from upward (call warrant) or downward (put
Futures contract	Agreement for future delivery of an asset between buyer and seller at a predetermined pric
Convertible security	Holder may convert into a predetermined number of ordinary shares within a specified peri

1.2.5 Portfolio Management and Investment Funds

Clients may invest directly in single securities or entrust funds to portfolio managers. Professional management benefits clients who lack the time, information, expertise, discipline or emotional temperament to invest actively.

Different asset classes and securities are pooled and managed by professional fund managers, either as pure asset funds or diversified under asset allocation strategies matched to client risk profiles and objectives. Hedge fund managers offer alternative approaches that seek returns irrespective of market direction. Funds are offered on stock exchanges worldwide as ETFs, and over-the-counter (OTC) by financial institutions.

1.3 Financial Markets

Financial markets are where assets are bought or sold, borrowed or lent, or otherwise transacted among issuers, investors, speculators, hedgers and intermediaries. Investments may be made in equity securities (shares, equity unit trusts) and debt securities (short-term money market paper, long-term fixed income bonds); sophisticated investors may use derivatives such as equity options and futures. The resulting markets include the equities market, money market, fixed income market, commodities market, options and futures market, and foreign exchange market.

These markets have seen extensive liberalisation and global consolidation, the impact of information technology enabling alternative platforms such as Electronic Communications Networks (for example Instinet), and rapid adoption of online technology letting professional and retail investors use brokerage services through computer networks and personal devices.

Technology also enables **algorithmic trading**, which uses a computer program following a defined set of instructions to execute a trade and can slice large trades to minimise market impact. A form of algorithmic trading is **high frequency trading**, which relies on computer speed to analyse large amounts of market information and self-execute trades in fractions of a second.

Structure of the financial markets. Four asset-class pillars sit under the markets:

Debt securities	Equity securities	Derivatives	Foreign
Money market, fixed income	Ordinary shares, preference shares	Options, futures, swaps, structured products	Spots,

Participants in financial markets:

- **Investors** — retail, private bank and institutional investors, including fund management companies.
- **Issuers** — public and private sector companies.
- **Intermediaries** — broker/dealer firms and market makers.
- **Regulators.**
- **Electronic exchanges** — for equities, debt, derivatives and commodities.
- **Infrastructure providers** — clearing houses, central depositories, communications networks, trade repositories.
- **Financial institutions** — distribution and underwriting of initial public offerings (IPOs), and participation in the secondary markets.

How markets sustain economic growth. Capital seekers (issuers of stocks and bonds) pay through arrangers and intermediaries (investment banks, underwriters, distributors, brokerages) to reach wholesale capital providers (banks, hedge and pension funds, insurance companies), who in turn serve retail capital providers (individuals, small firms).

Beyond regulated exchange-based capital markets there is growing activity in private equity and mezzanine financing, where capital seekers and providers negotiate funding for mergers, acquisitions or buy-outs. In these pre-primary market sectors, capital is raised from private investors rather than from the public. Some forms are referred to as "shadow banking".

1.3.1 Primary Market

The primary market is where new issues of securities are listed on the stock exchange and sold to the public. New sales of debt or equity take place through IPOs. The **issuer receives the cash** from subscribers, who receive financial claims that previously did not exist.

1.3.2 Secondary Market

Once new securities have been sold in the primary market they are listed and begin trading in the secondary market, which involves buying and selling securities initially sold in the primary market. Proceeds from a secondary-market sale go to the **selling investor, not the original issuer**. Secondary markets provide the liquidity essential to the proper functioning of capital markets.

1.3.3 Characteristics of a Good Financial Market

Characteristic	What it requires
Availability of information	Timely, accurate information on the volume and prices of past transactions and on all cu
Liquidity	Ability to buy or sell quickly at prices not substantially different from previous transactio
Transaction costs	Low across all aspects: the cost of reaching the market, brokerage, and transferring the s
Bid-offer spread	Narrow, reflecting wide and deep market-making where market makers readily quote two-
Information efficiency	Prices adjust quickly to new information on supply or demand, so prices reflect all availa
Price discovery	The most important function of any financial market: with the other characteristics in pl

1.3.4 Bond Markets

The largest domestic bond market in the world is the **US** bond market. The US Treasury issues notes and bonds with maturities greater than **1 year**, and is the largest single issuer of debt globally, making the treasury market the most active and liquid in the world. Treasury bills, notes and bonds are usually issued through an **auction** where dealers submit competitive bids.

The **Eurobond** market remains important for investors seeking foreign currency bonds, particularly while quantitative easing held interest rates very low and pushed investors into long-term issues. Because participation is worldwide, dealing and settlement procedures are governed by the rules of the **Association of International Bond Dealers (AIBD)**.

Although many corporate bonds are listed on major exchanges such as the NYSE, bond trading is mostly **OTC through bond dealers**, which are mostly banks and large investment banking firms.

1.3.5 Equity Markets

Ordinary shares, preference shares, warrants and ETFs trade actively on equity markets. Equity markets differ across countries in size, liquidity, degree of regulation, trading and settlement systems, and restrictions on foreign ownership. Internet brokerage has made equity investment more international.

Size and structure facts to remember:

- The **NYSE and NASDAQ combined** account for slightly more than **half** of world stock market capitalisation.
- Singapore is **not** in the top twenty global exchanges by market capitalisation, but **SGX** is the world's most liquid offshore market for the benchmark equity indices of **China, India, Japan and ASEAN**, and offers commodity and currency derivative products.
- **Japan has 8 stock exchanges**, with the Tokyo Stock Exchange (TSE) dominating.
- The US has regional exchanges alongside the NYSE and the American Stock Exchange (AMEX), listing small companies with geographic interest.

Settlement periods vary by market:

Market	Settlement
China	**T+0** — settlement on the trade date, on the philosophy that the investor must have funds on hand
United States	**T+2**
Singapore	**T+2** (SGX ST Rule 9.2.1: 2 exchange business days after the trade day)
Hong Kong	**T+2**

Securities not listed on an organised exchange may trade in the **OTC market**. In the US this is a negotiated market of dealers or market makers standing ready to buy and sell at market prices. Unlike brokers, **dealers have a vested interest in transactions and earn the spread** between bid and offer.

1.3.6 Options and Futures Markets

A futures exchange is a marketplace for financial instruments and commodities where buying and selling of underlying products is contracted for future delivery.

Exchange	Position
Chicago Mercantile Exchange (CME) Group	Largest futures exchange; processes **3 billion contracts** of diverse
Intercontinental Exchange (ICE)	Second largest, created through mergers and acquisitions among exc
SGX	Operates the **largest derivatives market in Asia**, benefiting fron
Hong Kong Exchanges and Clearing Limited	Owns the London Metal Exchange in addition to its stock and futur

Futures trade on electronic exchanges, with trades cleared through the clearing house each business day. The clearing house lets participants reverse a position before maturity because it tracks each participant's outstanding contracts.

US options trade on an exchange floor such as the **Chicago Board Options Exchange (CBOE)** using a system of market makers who quote bid and ask prices, with floor brokers trading either with the market maker or with other floor brokers. The **Options Clearing Corporation (OCC)** acts as intermediary between the brokers representing buyers and sellers (writers): it becomes **the buyer for every seller and the seller for every buyer**, guaranteeing that all contract obligations are met.

1.4 Digital Disruption and Fintech

The financial industry has undergone significant technological change and digital disruption. With internet access, the roles of financial institutions and capital markets are being challenged by new entrants, and banks work with fintech companies to design equivalent internet platforms and solutions.

Digital development	Impact on the financial sector
Internet banking	Consumer and branch banking redefined; tasks split between v
Rise in non-bank internet lending	New non-bank entrants provide direct financing over the intern
Internet business transactions reducing cash float	Transactions that took days to clear through the banking syste
Non-bank digital wallets bypassing banks	Excess cash moves from current and savings accounts into digi
Digital currencies for transactions in cyberspace	Cryptocurrencies are not backed by reserves or central banks,

1.5 Summary

- Representatives must distinguish institutional, accredited, expert and retail investors in the local context and identify their investment needs.
- Financial assets are broadly classified as debt securities, equity securities, collective investment schemes, funds or derivatives.
- Money market instruments are highly liquid, short-term debt instruments with maturities of less than one year.
- Fixed income debt securities are borrowings by issuers who agree to make periodic interest payments and return principal at maturity.
- Equity securities represent ownership interest in a corporation; the two types are ordinary shares and preference shares.
- Derivative securities derive their value from underlying assets; options, warrants, futures and convertible bonds are examples.
- Investors may invest directly in single securities or entrust funds to portfolio managers.
- The primary market is where new securities are distributed by the issuer to investors; the secondary market is where those securities then trade.
- In evaluating the quality of a financial market, consider availability of information, liquidity, transaction cost, the bid-offer spread, and informational and price-discovery efficiency.
- Bond trading is mostly over-the-counter through bond dealers rather than through organised exchanges.
- Equity markets differ across countries in size, liquidity, degree of regulation, trading and settlement systems, and restrictions on foreign ownership.
- The financial sector and capital markets are going through a digital revolution, upgrading traditional functions and admitting new entrants and joint ventures with financial institutions.

Exam cues

- **Four investor categories, one statutory source.** SFA **s 4A** defines II, AI and EI; everyone else is retail. Retail gets the most protection; II gets the widest product access.
- **AI thresholds — memorise all three individual tests:** net personal assets $>$ **S**$2m * *(primary residence net of secured credit, **capped at S**$\mathbf{1m}$), financial assets $>$ **S**$1m**net of liabilities, or income in the p * 12 months \geq S$**300k**. Corporate AI: net assets $>$ **S$10m** on the latest audited balance sheet, dated within the preceding 12 months.
- **AI opt-in.** A CMI cannot automatically treat a qualifying investor as an AI; the investor must opt in, and may instead elect full retail safeguards.
- **EI is about occupation, not wealth.** An EI trades capital markets products in the ordinary course of business — a sub-set of retail investors meeting experience requirements.
- **Money market vs capital market.** Money market: maturity **1 day to 1 year**, often **$<$ 90 days**, large denominations, low risk, highly liquid. Capital market: **$>$ 1 year** (bonds, equities).
- **Discount vs interest-bearing.** Treasury bills, commercial paper and banker's acceptances are **discounted** instruments; certificates of deposit are quoted on an **interest-bearing** basis.
- **Repo vs reverse repo.** A repo is **borrowing** by the seller of the security; a reverse repo is **lending** by the buyer. The effective interest is the difference between the purchase and sale prices.
- **Bond labels turn on two questions:** which currency, and where sold. Eurobond = currency not native to the place of issue; Eurodollar = USD sold outside the US; Yankee = USD sold **in** the US by a foreign issuer; domestic = issuer's own country and currency.
- **Claim priority:** creditors → preference shareholders → ordinary shareholders (residual).
- **Primary vs secondary.** Primary: issuer receives the cash, new claims are created. Secondary: the selling investor receives the cash, and the market supplies liquidity.
- **Six characteristics of a good market:** information availability, liquidity (marketability + price continuity + depth), low transaction costs (internal efficiency), narrow bid-offer spread, information efficiency, price discovery.
- **Settlement:** China **T+0**; US, Singapore and Hong Kong **T+2**.
- **Clearing house roles:** futures clearing house tracks outstanding contracts so positions can be reversed; the **OCC** becomes buyer to every seller and seller to every buyer.
- **Trap:** a **dealer** earns the bid-offer spread and has a vested interest in the transaction; a **broker** acts as agent for a commission. Do not use the terms interchangeably.
- **Trap:** preference shares are **equity** in law but behave like debt in priority and payout — the examination tests the hybrid character, not a simple label.
- **Trap:** the AI net-personal-assets test counts the primary residence **net of the secured loan and capped at S$1m**, not at market value.

Chapter 2 — Risk, Return and Time Value Calculations

This is the most calculation-heavy chapter in the syllabus, and the one every later valuation chapter depends on. It defines six measures of return, four measures of risk, the classification of investment risk, and the time-value formulas used in Chapters 6, 7, 8 and 12.

Learning Objectives

The candidate should be able to:

- Explain the concepts of risk and return in the management of investments.
- Explain the relationship between risk and return in the context of the risk-return trade-off.
- Identify the components of the total return and compute expected return on an investment.
- Explain the statistical measures of risk.
- Compute standard deviation and coefficient of variation as risk measures.
- Identify the various risks associated with investments.
- Calculate the present and future value of a single cash flow.

2.1 Risk and Return

The risk-return (or risk-reward) principle holds that rational investors demand higher potential returns when they take on more risk, and do not demand as high a rate of return when risk is lower.

The choice of assets follows the investor's risk profile. An investor with a low appetite for risk fears losing

any part of capital and considers relatively safe money market instruments offering a steady flow of interest income. An investor with a higher appetite is prepared to lose some — and in some cases all — of capital, and considers riskier securities such as equities in expectation of higher return.

Risk is a property of the instrument, not the asset class. Several examples make the point:

- **Cash** is often considered least risky because it is a medium of exchange and is liquid, yet cash is volatile against other currencies. Currency volatility rises with political change, currency-management policy change, and fiscal and monetary policy change.
- **Government bonds**, once unquestionably secure, now carry more counterparty risk, changing the traditional view of "riskless" sovereign debt (Portugal, Italy, Greece, Spain).
- A **CCC-rated** (junk or speculative grade) bond is much riskier than a **AAA-rated** blue-chip corporate bond. The blue-chip company's **stock** may be safer than the CCC bond if its business and financial profile is conservative, even though equities are generally the riskier asset class.
- **Derivatives** sit at the upper end of the risk-return continuum because most structured products embed long and/or short options. The range is diverse, however: some are capital guaranteed and therefore safer than some equities.

A representative must therefore examine the riskiness inherent in the security itself, review the product documents (Product Highlights Sheet and prospectus), and ask the relevant questions. Only after understanding the product structure, underlying strategy and payoff scenarios should a recommendation be made.

The VIX. The Chicago Board Options Exchange (CBOE) Volatility Index — the "VIX" — is the bellwether risk indicator for the US stock market and, indirectly, for global equities because of the weight of the US economy. It measures the market's expectation of **30-day S&P 500 volatility** and is called the "investor fear gauge".

VIX level	Interpretation
Above 30	High volatility; usually spikes upward during periods of great market uncertainty
Below 20	Relative stability
89.5	The high reached at the height of the Global Financial Crisis in **October 2008**
48	The high reached during the European Debt Crisis

A higher VIX means more risk and volatility are expected. Market maturity also matters: financial assets in mature markets (the United States) are perceived as less risky than those in emerging markets (China) or frontier markets (Sri Lanka).

2.2 Calculating Returns

Six measures of return are examinable. Each answers a different question, and the examination tests whether you pick the right one.

Measure	Question it answers	Section
Total return	What did I earn over the holding period?	2.2.1
Expected rate of return	What do I expect, given probabilities?	2.2.2
Annualised rate of return	What is the equivalent one-year rate?	2.2.3
Real rate of return	What did I earn after inflation?	2.2.4
Currency-adjusted return	What did I earn in my reference currency?	2.2.5
Internal rate of return	What rate equates my actual cash flows?	2.2.6

2.2.1 Total Return

Total return on a pre-tax basis has two components:

- **Current income** — income received during the holding period, such as dividends from shares and interest from fixed income securities or fixed deposits.
- **Capital gain or loss** — the change in price of the security.

$$\text{Total Return} = \frac{\text{Income} + \text{Capital Gain or Loss}}{\text{Purchase Price}}$$

Because the capital gain or loss is the ending price minus the beginning price, the same relationship is written:

$$\text{Total Return} = \frac{CF_t + (P_E - P_B)}{P_B}$$

where

$$CF_t$$

is the sum of cash flows during measurement period

$$t$$

,

$$P_E$$

is the price at the end of period

$$t$$

(the sale price), and

$$P_B$$

is the price at the beginning of period

$$t$$

(the purchase price).

Worked example. You buy 1,000 shares of XYZ at \$5 each and receive a dividend per share of \$0.20. You sell one year later at \$6.

$$\text{Total Return} = \frac{0.20 + (6 - 5)}{5} = 0.24 = 24\%$$

2.2.2 Expected Rate of Return

The historical rate of return is known; the expected rate of return is the future rate, which may or may not be realised. A single figure such as "15%" is a **point estimate**. Because the future is uncertain, several returns are possible under different economic outlooks, and probabilities can be assigned to each.

$$E(R) = \sum_{i=1}^{n} r_i p_i$$

where

$$E(R)$$

is the expected return on the security,

$$r_i$$

is the

$$i$$

th possible return,

$$p_i$$

is the probability of that return, and

$$n$$

is the number of possible returns. The expected return is therefore the **probability-weighted average** of possible returns.

Worked example.

Economic scenario	Probability (p_i)	Return (r_i)
Economic boom	0.20	25%
Recession	0.20	−30%
Stable	0.60	15%

Economic scenario	Probability (p_i)	Return (r_i)
	1.00	

$$E(R) = (0.20 \times 0.25) + (0.20 \times -0.30) + (0.60 \times 0.15) = 0.05 - 0.06 + 0.09 = 0.08 = 8\%$$

2.2.3 Annualised Rate of Return

Returns are often presented on an annualised basis. The annualised rate compounds the holding-period return to a one-year equivalent.

$$\text{Annualised Rate of Return} = (1 + r_{\text{period}})^{12/m} - 1$$

where

$$m$$

is the number of months in the holding period.

Worked example. A product is expected to generate a maximum return of **2.5% over 3 months**.

$$(1.025)^{12/3} - 1 = 1.10381 - 1 = 0.104 \approx 10.4\%$$

The compounding logic, step by step, on \$1:

Period end	Value
3 months	\$1 × 1.025 = \$1.025
6 months	\$1.025 × 1.025 = \$1.0506
9 months	\$1.0506 × 1.025 = \$1.0769
12 months	\$1.0769 × 1.025 = \$1.1038

The annual return is therefore 10.38%, which is

$$(1.025)^4 - 1$$

, the same as

$$(1.025)^{12/3} - 1$$

. The assumption is that the investment plus interest is reinvested — the return is **compounded**, earning interest on interest.

Working backwards. To find the return achievable over 4 months from a product quoting an annualised target of 12%:

$$(1 + r_4)^{12/4} - 1 = 0.12$$
$$(1 + r_4) = (1.12)^{4/12}$$
$$r_4 = (1.12)^{1/3} - 1 = 0.0385 = 3.85\%$$

2.2.4 Real Rate of Return (Inflation-Adjusted Return)

The real rate removes the effect of inflation, because asset prices rise with general inflation independently of the investment's performance.

$$\text{Real Return} = \frac{1 + r_{\text{nominal}}}{1 + \text{inflation}} - 1$$

Worked example. With an annualised return of 10.4% and inflation of 2%:

$$\frac{1.104}{1.02} - 1 = 0.08235 = 8.24\%$$

With **deflation** of 1% (asset values fell 1%), the same formula gives:

$$\frac{1.104}{1 - 0.01} - 1 = 11.52\%$$

Note the sign convention: inflation is added to 1 in the denominator, deflation is subtracted.

2.2.5 Currency-Adjusted Return

Assets and income generated in a foreign currency must be restated in the investor's **base (reference) currency**. A fund invested in foreign securities must have its performance restated in the desired reference currency, usually the local currency.

Worked example. A Singapore portfolio manager reviews a JPY portfolio pegged to the Nikkei 225 and a USD portfolio pegged to the S&P 500 over January 2013 to December 2017.

	Jan 2013	Dec 2017	Change
Nikkei 225	10,688.11	23,714.53	+121.88%
SGD/JPY	71.4943	85.0038	+18.89%
S&P 500	1,466.47	2,743.15	+87.06%
USD/SGD	1.2272	1.3268	+8.12%

JPY portfolio, restated in SGD. The reference currency (SGD) is the **first** listed currency of SGD/JPY, so divide.

- Beginning Nikkei in SGD:
$$10{,}688.11/71.4943 = 149.4960$$
- Ending Nikkei in SGD:
$$23{,}714.53/85.0038 = 278.9820$$
- Currency-adjusted return:
$$(278.9820/149.4960) - 1 = 0.8662 = 86.62\%$$

This is **lower** than the Nikkei's 121.88% because the SGD strengthened against the JPY, so the JPY gains are adjusted down.

USD portfolio, restated in SGD. The reference currency (SGD) is the **second** listed currency of USD/SGD, so multiply.

- Beginning S&P 500 in SGD:
$$1{,}466.47 \times 1.2272 = 1{,}799.6520$$
- Ending S&P 500 in SGD:
$$2{,}743.15 \times 1.3268 = 3{,}639.6114$$
- Currency-adjusted return:
$$(3{,}639.6114/1{,}799.6520) - 1 = 1.0224 = 102.24\%$$

This is **higher** than the S&P 500's 87.06% because the SGD weakened against the USD, so the USD gains are adjusted up.

The shortcut rule. The same answers come from combining the portfolio return with the currency return directly:

- JPY portfolio:
$$(1 + 1.2188)/(1 + 0.1889) - 1 = 0.8662 = 86.62\%$$
- USD portfolio:
$$(1 + 0.8706) \times (1 + 0.0812) - 1 = 1.0224 = 102.24\%$$

The rule of thumb: **divide**

$$(1 + \text{portfolio return})$$

by

$$(1 + \text{currency return})$$

when the reference currency is the **first** listed currency of the pair, and **multiply** when it is the **second** listed currency.

In SGD terms the US portfolio outperformed the Japanese portfolio, because the SGD strengthened against the JPY but weakened against the USD.

Some corporate and high net worth (HNW) investors use a major world currency such as USD or EUR as their reference currency because they take an international view of their invested assets. The US remains the largest capital market and many instruments, including commodities, are still denominated in USD, so a USD reference is convenient. Some HNW individuals in Asia keep offshore investments separate from domestic ones and use USD as the reference; in some countries local currencies have proven more volatile than global currencies, so investors prefer not to use them as the reference.

2.2.6 Internal Rate of Return

The internal rate of return (IRR) equates all cash outflows (investments) against all cash inflows (proceeds) evaluated at the same point in time. It is used for projects and investments with **uneven cash flows**, such as private equity.

Worked example (1). An investment requires $1 million now and a further $2 million at the start of the next year. The client receives $3.5 million only at the end of year 3.

Year 1	Year 2	Year 3
−$1m	−$2m	+$3.5m

Evaluated at the end of year 3, with IRR denoted

$$r$$

:

$$1(1 + r)^3 + 2(1 + r)^2 + 0(1 + r) = 3.5$$

The first $1m compounds for 3 years, the second $2m for 2 years, and the $0 for 1 year.

Choosing a starting guess. Total invested is $3m and total received is $3.5m, implying

$$(3.5/3) - 1 = 16.7\%$$

, which divided by 3 years gives **5.55%**.

Trial rate	Left-hand side	Conclusion
5.55%	$3.4041m	Below $3.5m, so the rate must be higher
7.00%	$3.5148m	Marginally above $3.5m, so the rate must be slightly lower
6.80%	$3.4999m	Close; 6.81% improves accuracy

The IRR is approximately **6.8%**. In a spreadsheet the calculation is `=IRR(cell range, [guess])`, solved iteratively.

IRR versus compounded annualised return. The IRR is the **money-weighted** return in a portfolio context: it is the actual return generated for the client given the movement of their cash flows, and it is sensitive to the size and timing of those flows. The compounded annualised rate is the **time-weighted** return and is unaffected by them.

Worked example (2). A client invests \$5m in a fund at the end of 2016. The fund returns 2% in 2017, so the client withdraws \$3m at the end of the year. The fund returns 10% in 2018, so the client adds \$8m at the start of 2019, and the fund loses 4% in 2019.

(a) Fund value at the end of 2019:

Step	Calculation	Result
Value at end of 2017	\$5m × 1.02	\$5.1m
Withdrawal	−\$3.0m	
Value at end of 2018	\$2.10m × 1.10	\$2.31m
Contribution	+\$8.00m	
Value at end of 2019	\$10.31m × 0.96	**\$9.898m**

(b) The client's actual (money-weighted) return. Total outlay is \$5m + \$8m = \$13m; total proceeds are \$3m + \$9.898m = \$12.898m. A first estimate of the loss is

$$12.898/13 - 1 = -0.7846\%$$

. No division by the number of years is needed because contributions and withdrawals are spread relatively evenly through the period.

Restating the cash flows:

$$5(1+r)^3 - 3(1+r)^2 + 8(1+r) = 9.898$$

Trial rate	Left-hand side
−0.786%	9.867 — rate must be less negative
−0.700%	9.881 — rate must be less negative
−0.600%	9.898 — close

The IRR, and therefore the money-weighted return, is **−0.6%**.

(c) The annualised compounded (time-weighted) return. The cumulative return is

$$(1.02 \times 1.10 \times 0.96) - 1 = 0.0771 = 7.71\%$$

. Annualising:

$$(1+R)^3 = 1.0771 \implies R = (1.0771)^{1/3} - 1 = 0.0251 = 2.51\%$$

The fund itself made 2.51% a year; the client lost 0.6% a year because of the timing of their own contributions and withdrawals. That gap is the whole point of the money-weighted versus time-weighted distinction.

2.3 Measures of Risk

Risk is the uncertainty of returns associated with an investment — in simple terms, the degree of fluctuation in the value of assets such as stocks. Return and risk are inextricably linked.

2.3.1 Variance and Standard Deviation

Variance and standard deviation are the most widely used statistical measures of risk. They measure the **dispersion of returns from the average return**: the smaller the dispersion, the lower the risk.

$$\sigma^2 = \sum p_i \left[r_i - E(R)\right]^2$$

$$\sigma = \sqrt{\sum p_i \left[r_i - E(R)\right]^2} = \sqrt{\sigma^2}$$

where

$$\sigma^2$$

is the variance of returns and

$$\sigma$$

is the standard deviation, the positive square root of the variance.

Worked example, using the expected-return example from 2.2.2:

r_i	p_i	$r_i p_i$	$r_i - E(R)$	$[r_i - E(R)]^2$	$p_i[r_i - E(R)]^2$
0.25	0.20	0.05	0.17	0.0289	0.00578
−0.30	0.20	−0.06	−0.38	0.1444	0.02888
0.15	0.60	0.09	0.07	0.0049	0.00294
	1.00	**E(R) = 0.08**			**0.03760**

Variance

$$\sigma^2 = 0.03760$$

. Standard deviation

$$\sigma = \sqrt{0.03760} = 0.1939 = 19.39\%$$

.

Volatility is the standard deviation of the security. A share with a standard deviation of 19.39% calculated on daily price changes has a **daily volatility** of 19.39%. This is **historical volatility**, which differs from **implied volatility** — the volatility implied by the price of a share option.

2.3.2 Coefficient of Variation

The coefficient of variation (CV) measures the **risk per unit of expected return**, which lets investments with different returns and different standard deviations be compared.

$$CV = \frac{\text{Standard Deviation of Returns}}{\text{Expected Rate of Return}}$$

Worked example.

	Investment A	Investment B
Expected return	0.05	0.15
Standard deviation	0.08	0.20

On standard deviation alone, B looks less attractive because its risk is higher. But B also has a higher expected return:

$$CV_A = \frac{0.08}{0.05} = 1.60 \qquad CV_B = \frac{0.20}{0.15} = 1.33$$

On a risk-per-unit-of-return basis, **B is more attractive** — it has the lower coefficient of variation.

2.3.3 Shortfall Risk — Probability of Losses Using Normal Distribution

Assuming returns are normally distributed, the probability of observing a range of returns can be computed. Investors care most about the probability of a **loss**.

Worked example. A balanced diversified portfolio of stocks and bonds has, over the last 20 years, an average return of **6%** with a standard deviation of **12%**.

The return that deviates by exactly ±1 standard deviation from the expected return:

$$X - E(r) = \pm 1\sigma \implies X = 0.06 \pm 0.12 = -0.06 \text{ or } 0.18$$

The **Z-score** expresses how many standard deviations a return deviates from the expected return:

$$Z = \frac{X - E(r)}{\sigma}$$

For -6%:

$$Z = (-0.06 - 0.06)/0.12 = -1$$

. For $+18\%$:

$$Z = (0.18 - 0.06)/0.12 = +1$$

.

The standard normal table gives the cumulative probability — the area to the **left** of the specified return. For

$$Z = 1.00$$

the probability is **0.8413**, so there is an 84.13% probability of an annual return lower than 18%. Because the total area is 100%, the probability of a return above 18% is

$$1 - 0.84 = 0.16$$

. Because the distribution is symmetrical, the probability of a return below -6% (

$$Z = -1.00$$

) is also **0.16**.

The table carries only cumulative probabilities for Z-scores at or above zero — the right tail. The left tail is inferred from symmetry: to find the probability of -6% or worse, look up

$$Z = 1.00$$

(0.8413) and take

$$1 - 0.8413 = 0.16$$

.

The probability of a return between -6% and $+18\%$ is therefore

$$0.84 - 0.16 = 0.68$$

.

Range	Z-scores	Probability inside
± 1 standard deviation	± 1.00	**68%**
± 2 standard deviations	± 2.00	**95.4%** (97.7% − 2.3%)

Probability of a loss. The question "what is the likelihood of making a loss?" means the likelihood that the return

$$X$$

is less than 0:

$$Z = \frac{0 - 6\%}{12\%} = -0.5$$

For

$$Z = -0.5$$

the table gives a **69% probability of a positive return** and therefore a **31% chance of a negative return**.

2.3.4 Value at Risk

Value at risk (VaR) extends shortfall risk and is actively used to estimate portfolio risk. It asks: "What is the smallest proportion of our portfolio value that is at risk 2.5% or 0.5% of the time?"

Balanced portfolio — expected return 6%, standard deviation 12%.

Confidence	Left-tail probability	Z-score	Calculation	VaR
97.5%	0.025	**−1.96**	$-1.96(0.12) + 0.06 = -0.1752$	**−18%**
99.5%	0.005	**−2.58**	$-2.58(0.12) + 0.06 = -0.2496$	**−25%**

For the 0.025 tail, look up a probability of 0.975 (that is, $1 - 0.025$). For the 0.005 tail, look up 0.995; the table shows 0.9949 and 0.9951 but not 0.995, so 0.9951 is used, corresponding to

$$Z = -2.58$$

.

So there is a **2.5% chance of losing at least 18%** and a **0.5% chance of losing at least 25%** in a given year.

Defensive portfolio — expected return 4%, standard deviation 4%.

Confidence	Z-score	Calculation	VaR
97.5%	−1.96	$-1.96(0.04) + 0.04 = -0.0384$	**−4%**
99.5%	−2.58	$-2.58(0.04) + 0.04 = -0.0632$	**−7%**

There is a 2.5% chance of losing at least 4% in a given year — about **1 year in 40** — and a 0.5% chance of losing at least 7%.

Probability of any loss. For a 0% return,

$$Z = (0\% - 4\%)/4\% = -1$$

, which implies a **16% probability of loss**; 84% of the time the portfolio should yield positive returns.

The limitation. Using historical data to predict the probability of future returns assumes the factors and conditions influencing the next few years resemble those influencing the past. They may not. The financial community therefore also uses the **Monte Carlo method**, which computes VaR from randomly generated returns under distributional assumptions, and computer-based models.

2.4 Types of Risk and Risk Management

Section 2.3 defined risk as volatility — deviation from the expected return in either direction. The average investor is mainly concerned with **negative** deviation, the risk of loss. Risks are grouped into six categories.

Category	What it covers
Issuer and counterparty	Business risk, credit risk, fraud risk
Product-specific	Leverage structure; term and maturity; product features (option triggers, strike levels, pr
Market (systematic)	Market shocks; country, political and currency risk; unexpected economic developments;
Execution and operational	Execution error; system failure; process errors and restrictions; computer viruses; data br
Management	Risk profile mismatch; concentration risk (lack of diversification)
Liquidity	Lack of market depth

2.4.1 Risks in Investments

a. Issuer and counterparty risks.

Issuer risk is the risk of loss when issuers of securities — shares, bonds, commercial paper, certificates of deposit, structured notes and deposits — default on their obligations.

The Lehman Brothers case is the standard illustration. In 2008 Lehman had close to **USD680 billion of assets supported by only USD22.5 billion of the firm's capital**. As the housing market weakened, mortgage loans began to default, and the value of Lehman's mortgage-backed securities plummeted once it became apparent the issuer could not support interest and principal payments. Investors in Singapore, Hong Kong and elsewhere who bought Lehman-issued minibonds (structured notes) suffered major losses.

Counterparty risk is the risk that one or more parties to a contract fails to fulfil its contractual obligation. Some instruments are not exchange-traded but are OTC contracts between counterparties, which creates this exposure.

The counterparty may also be the issuer. When a bank designs a structured product and sells it to its clients, the bank is both counterparty and issuer. An issuer may increase sales through other banks that distribute and market the product, paying a **retrocession fee** or commission for doing so. In the Lehman minibond case, affected investors tried to claim against the distributing banks, which saw themselves as distributors rather than counterparties — a distinction not fully evident to some clients. Some banks provided partial compensation to preserve reputation and client relationships. The lesson: "caveat emptor" and standard disclaimers may not always work, especially where it is unclear whether clients are novices or sophisticated investors.

Both issuer and counterparty risk can arise from **business risk** and, in some cases, outright **fraud**.

i. Business risk is uncertainty about a company's ability to meet its financial goals. For equity investors the downside risks are changes in earnings growth or potential:

- Earnings failing to meet market expectations.
- An unexpected slowdown in earnings growth, warranting a downward re-rating of the stock.
- An unexpected loss, which may trigger a sell-off.
- Events signalling deterioration in the potential growth of the business.

Bond traders and bond fund managers must also watch weakening business conditions for issuers — Singapore's offshore oil and gas engineering industry in 2015–2016 is the example. At a minimum, when bond prices fall, portfolios are marked to market and the fall must be reported. If conditions become dire, issuers may default and losses must be realised.

ii. Fraud is relatively uncommon because financial institutions conduct due diligence. But outsourcing and open-architecture concepts have led many institutions and HNW individuals to invest in boutique, hedge and private equity funds. Bernie Madoff ran a Ponzi scheme for several years affecting **more than 4,800 clients**, including institutions; it was exposed only during the 2008 financial crisis when redemptions could not be met.

b. Product-specific risk. Equity shareholders, as part owners, bear higher business risk; bondholders focus on the firm's ability to finance interest and principal. Beyond business and credit risk, the risk of a product depends on its specific features and contractual terms:

- Leverage multiple.
- Maturity period.
- Payoff structure (timing, size, sequence, limits).
- Sale of put or call options within the structure.
- Currency.

Leveraged products — futures, warrants, daily leveraged certificates, contracts for difference — have significantly higher volatility, moving multiple times their underlying. Some structured products with attractive yields effectively require the investor to **sell put options** to the counterparty (equity-linked notes). Others require **increased accumulation** of securities as they fall in value (accumulators), creating more risk. Private equity requires long gestation periods and shows drawdowns in the initial years before potentially achieving high returns later.

Currency denomination matters too: payment flows in a foreign currency can significantly affect returns in the domestic currency, and **bond yields can be wiped out by currency losses** when flows are converted back.

c. Market risk. Market risk affects most assets in the market and therefore cannot be fully diversified. All asset classes are affected, though the impact differs across them, and market shocks can be short- or long-lived.

Shock type	What happens
Financial shocks	Ripple effect across markets and asset classes. During the GI
Political shocks	Sudden change in government creates uncertainty about the
Unexpected contractionary policy or negative data	Unexpected rate hikes, tax increases, inflation and unemploy
Natural disasters, wars, terrorism	Affect domestic and global market stability, but the impact i

d. Execution and operational risks. These can arise from a counterparty, the exchange, or the systems and platforms handling internet trades initiated from an investor's own device.

Human error remains a live risk. In **2015 a typing error caused Mizuho Securities to lose at least 27 billion yen (about $225 million)** on a single stock trade: a trader tried to sell **610,000 shares at 1 yen** each when he intended to sell **1 share at 610,000 yen** (about $5,041).

i. Execution risk within financial institutions. Wrong trades must be reversed. Most institutions pay for losses caused by their own error, but the loss of trust and reputation may drive clients away, which is often the greater concern. Staff involved in execution undergo regular training, and new systems and processes reduce error rates. During system upgrades, temporary glitches, or compliance restrictions imposed at an inopportune time, trading may freeze and a client may miss an opportunity — a **cost of lost opportunity**. A system might flag a breach of trading or funding limits while ignoring available funds from completed transactions.

The main conflict between clients and representatives arises in **communicating and confirming trade orders**; representatives are trained in best practice for receiving, confirming, recording and executing orders to minimise disputes. Execution risk is increasingly pushed to clients because trading platforms are made directly available to them; institutions do set limits on client trades, which caps the risk to the institution and the financial system.

ii. False orders and flash crashes. Some trading risk arises from market manipulation. In **2010** a software program was used to place and cancel large orders within fractions of a second; computer-based strategies reacted to the false selling orders with real selling orders, and the **Dow Jones Industrial Average plunged 9% and recovered within 36 minutes** of intraday trading — a **flash crash**. In **February 2018** high frequency trading was blamed for a smaller flash crash when the DJIA fell almost **1,600 points (4%)** in a few minutes. Regulators have not fully solved the problem. The representative's response is to be aware of conditions that may cause greater volatility, remind long-term investors not to panic, and recognise that such events may present opportunities to acquire quality investments for clients willing to hold.

iii. Market disruption is the effect of a large and rapid change in market prices that stops the market functioning normally. It is characterised by price declines, widespread panic, and a massive increase in trade execution and volatility. Regulators and exchanges mitigate it with circuit breakers, shock absorbers or price limits.

- **Circuit breakers** vary across exchanges. On **SGX**, a circuit breaker triggers if a stock's price would move by more than **10% of the traded price 5 minutes prior** (the reference price). A large move can still proceed if it is gradual over a longer timeframe; the breaker halts trading only when volatility is excessive within a very short time. Once triggered, a **5-minute cooling-off period** follows during which trading can only take place **±10%** from the reference price. Trading then resumes with new reference prices established during the cooling-off period.
- **Shock absorbers** are similar and the terms are sometimes used interchangeably. Some use "shock absorbers" for halting index futures trading when the futures price falls substantially; others use it for halting computer-based HFT in specific stocks.

iv. Technical disruptions. Market halts have occurred from hardware or software problems, communication data feed errors and system updates. They usually last from **half an hour to half a day**. Most exchanges maintain back-up systems tracking execution and trade data.

v. Computer and data protection breaches. Viruses, hacking and data corruption could seriously affect financial markets. Institutions and governments have built defences, and representatives are expected to follow best practice in handling messages and communications.

e. Management risk. This is the risk that investments are mismanaged, resulting in excessive risk-taking.

i. Risk profile mismatch. Clients expect professional management. Each client has a risk profile and return expectation determining which investment suits them; conservative clients take less risky investments

and accept modest returns, aggressive clients pick higher expected returns with greater volatility. Allocating the right proportion of assets to each client type based on their risk-return expectation is **strategic asset allocation**. When profiling is done poorly, or only to fit the available sales offering, a **risk profile mismatch** can result, creating potential grounds for mis-selling, negligence or mismanagement.

ii. Concentration risk, or lack of diversification. Fund managers normally ensure proper diversification and minimal concentration. Clients managing their own money may end up highly concentrated in specific securities, industries or markets because there is no formal diversification strategy. Remaining concentrated can be a deliberate client choice, but representatives have a responsibility to advise diversification for the longer term, especially where clients lack the time, expertise or temperament to manage their own investments.

f. Liquidity risk. Liquidity risk has two aspects.

- **Funding liquidity** — the availability of funding for investments. In a crisis a liquidity crunch can occur, driving interest rates abnormally high.
- **Market liquidity** — the ability to liquidate a position quickly at a fair price, which depends on the availability of buyers and sellers. An instrument has more liquidity risk if its trading volume is low or its available float is small, with major blocks controlled by strategic investors. Poor market-price liquidity widens the bid-offer spread and makes trading more costly.

Major surprises can cause liquidity to drop. If negative news is released and sellers try to offload while buyers exist only at much lower prices — say **12% lower** — the price can gap downwards, several times a day, as selling pressure mounts. The mirror case is a **short squeeze**: a short position in a derivative may face a sudden rise in the underlying on good news, and with an absence of sellers the trader may be forced to cover at substantially higher prices.

2.4.2 Risk Management

Risk management involves due diligence, timely research and risk processes, client profiling and product suitability, diversification, and process and system risk management.

a. Due diligence checks. Security selection typically uses both fundamental and technical analysis. Selecting fund managers — traditional or hedge — requires regular due diligence and audit checks that go beyond tracking performance. Personnel changes, structural changes, cost pressures and market pressures can cause lapses in management, so external service providers must have proper checks and balances. This matters most in segments or jurisdictions with weak regulatory oversight.

b. Research and risk management processes. Not every institution performs primary research; cost and efficiency considerations usually produce a mix of primary and secondary research (gathering, evaluating and summarising research from other institutions and providers). Whatever the approach, keeping research current and having timely review processes shortens reaction time to deterioration in an investment or market, and allows underweighting of markets, sectors and companies before conditions worsen.

c. Client profile and product suitability. The products recommended depend on the client's objectives, investment purpose, risk profile, return expectation, experience and sophistication, desired manager engagement and servicing, net worth, the size of the investment relative to net worth, financial obligations and limitations.

A client may have a different purpose for money placed at one bank than at another. The risk profile selected for core family investments to be kept safe may differ sharply from that for excess cash held for opportunistic investments. Products are therefore recommended against the client's **desired risk profile for those particular investments**. Some corporate clients have a dedicated purpose with a specific timeline — money set aside for a building project starting in 12 months — so the investments must fit the limits and parameters the client sets.

d. Process and system risk management. Most banks have upgraded processes and systems to handle a changing environment. The recent focus has been compliance with regulatory requirements (documentation, declarations, alerts, checks), which slowed execution and required re-tooling and re-training. Systems are also upgraded to protect information, to make execution ”error-proof” with prompts and computer checks that limit human error, and to provide back-ups so transaction records can be retrieved after unexpected market disruption.

2.5 Time Value of Money

Money has time value: a dollar received in the future is worth **less** than a dollar received today, because a dollar today can be invested immediately. The time value of a sum depends on the date of receipt and prevailing interest rates. Discounted cash flow techniques are therefore necessary for correct investment analysis, because bond interest and dividends are paid at different points in time.

2.5.1 Simple and Compound Interest

A deposit of $10,000 for one year at 2.5% has principal $10,000, an interest period of 1 year, and a rate of 2.5%. For a Singapore dollar deposit the **day basis is 365 days**.

$$\text{Simple Interest} = \frac{\text{Principal} \times \text{Interest Rate (in \%)} \times \text{Interest Period (in days)}}{36{,}500}$$

The denominator depends on the currency's day basis: **Singapore dollar interest uses a 365-day basis; US dollar interest uses a 360-day basis.**

Worked example — simple interest. A SGD 1 million deposit earning 8% for 91 days:

$$\frac{1{,}000{,}000 \times 8 \times 91}{36{,}500} = \text{SGD } 19{,}945.21$$

Worked example — compound interest. The same deposit rolled over for another 91 days at 8%:

$$\frac{1{,}019{,}945.21 \times 8 \times 91}{36{,}500} = \text{SGD } 20{,}343.02$$

Total interest is

$$19{,}945.21 + 20{,}343.02 = \text{SGD } 40{,}288.23$$

. The annualised compounded rate is:

$$\frac{40{,}288.23}{1{,}000{,}000} \times \frac{36{,}500}{182} = 8.08\%$$

The second period earns interest on the first period's interest, which is why the annualised rate exceeds the nominal 8%.

2.5.2 Future Value

$$FV_n = PV(1+k)^n$$

where

$$FV_n$$

is the future value at the end of

$$n$$

years,

$$PV$$

is the present value or original amount invested at the beginning of the first year,

$$n$$

is the number of years of compounding, and

$$k$$

is the annual interest rate.

Worked example. $1,000 in an account paying 5% compounded annually for 10 years:

$$FV_{10} = 1{,}000(1.05)^{10} = 1{,}000(1.62889) = \$1{,}628.89$$

2.5.3 Time Value of Money in Negative Interest Rate Environment

Traditionally, interest rates represent the time value of money determined by the demand for and supply of funds: users of funds compensate suppliers for the loss of liquidity and opportunity cost, so rates are expected to be positive.

Recent years have seen negative rates imposed by central banks. In **2017, 10–20 European countries had negative yields on their 2-year sovereign debt**, meaning commercial banks were required to pay for holding excess reserves with the central bank. The intent was to push banks to minimise excess balances and increase productive lending. Sovereign debt issued at a premium above par and redeemed at par produces an implicit interest cost to the bank.

Country	Yield (%)	Country	Yield (%)	Country	Yield (%)	Country	Yield (%)
Switzerland	−0.92	Denmark	−0.69	Slovakia	−0.57	Italy	−0.17
Czech Rep	−0.81	Finland	−0.67	France	−0.53	Japan	−0.13
Germany	−0.74	Austria	−0.65	Slovenia	−0.38	Portugal	−0.09
Netherlands	−0.71	Ireland	−0.58	Spain	−0.37	Cyprus	−0.08
Sweden	−0.70	Belgium	−0.58	Bulgaria	−0.33	Lithuania	−0.08

Banks have generally not charged negative interest on depositor cash, because of the risk of political backlash and because depositors would move money abroad into foreign currency deposits, causing a capital outflow the central bank does not want. Even with positive nominal deposit rates, when inflation exceeded deposit rates, **real** interest rates were effectively negative.

Impact on the short end of the yield curve. Negative central bank rates drive short-term rates on other fixed income investments down. Investors initially tried to lock in yields at the short end, which drove short-term fixed income values up. In time-value terms: **negative interest rates raise the present value of fixed income investments at the short end of the yield curve.**

2.5.4 Compounded Annual Growth Rate

The compounded annual growth rate (CAGR) is the effective annual rate earned on the initial cash flow.

$$CAGR = \left[\frac{FV}{PV}\right]^{1/n} - 1$$

where

$$FV$$

is the future value,

$$PV$$

the present value, and

$$n$$

the number of years.

Worked example. With

$$FV = \$1{,}628.89$$

,

$$PV = \$1{,}000$$

and

$$n = 10$$

:

$$CAGR = \left[\frac{1{,}628.89}{1{,}000}\right]^{1/10} - 1 = 0.05 = 5\%$$

Compounding more than once a year. If \$1,000 is invested for 5 years at 5% compounded semi-annually, the money is invested for **10 six-month periods** at **2.5%** each. Quarterly compounding gives **20 three-month periods** at **1.25%** each. The general formula is:

$$FV_n = PV\left(1 + \frac{k}{m}\right)^{mn}$$

where

$$m$$

is the number of interest periods in the year.

Worked example — quarterly compounding. \$1,000 for 10 years at 5% compounded quarterly:

$$FV_{10} = 1{,}000\left(1 + \frac{0.05}{4}\right)^{40} = 1{,}000(1.0125)^{40} = 1{,}000(1.64362) = \$1{,}643.62$$

2.5.5 Present Value of a Single Cash Flow

Discounting is the opposite of compounding: it finds the present value of a future sum, and is used to judge the attractiveness of investment projects.

$$PV = \frac{FV_n}{(1+k)^n}$$

where

$$PV$$

is the present value,

$$FV_n$$

the future value at the end of

$$n$$

years,

$$n$$

the number of years to receipt, and

$$k$$

the annual discount (or interest) rate.

Worked example. At a 6% discount rate, the present value of \$1,000 receivable in 5 years:

$$PV = \frac{1{,}000}{(1.06)^5} = \frac{1{,}000}{1.3382} = \$747.27$$

The discount rate. The interest rate in present value calculations is the **discount rate**, interpreted as the return available on alternative similar investments. From the security holder's perspective it is an opportunity cost, and therefore also the return expected by holders of the securities. Because it is frequently used to adjust cash flows for risk, it is also called the **risk-adjusted discount rate**. The actual discount rate is at least the **risk-free government bond yield of the same maturity** as the intended holding period.

2.6 Summary

- There is a positive relationship between risk and return: investors who want higher returns must accept higher risk, and vice versa.
- There are six main measures of return: total return, expected rate of return, compounded annualised rate of return, real rate of return, currency-adjusted return, and internal rate of return.
- Risk can be analysed through the variance and standard deviation of asset returns; the standard deviation is also the return volatility.
- The coefficient of variation measures risk per unit of expected return, which allows investments with different returns and risks to be compared.
- Assuming normally distributed returns, the probability of a loss can be estimated from the Z-score; ± 1 standard deviation covers about 68% of outcomes and ± 2 about 95.4%.
- Value at risk extends this to a stated tail probability: the balanced portfolio loses at least 18% with 2.5% probability and at least 25% with 0.5% probability.

- Investment risk falls into six categories: issuer and counterparty, product-specific, market (systematic), execution and operational, management, and liquidity.
- Market risk cannot be diversified away, and correlations tend toward 1 during a crisis.
- Risk management combines due diligence, timely research, client profiling and suitability, diversification, and process and system controls.
- Money has time value: a dollar received in the future is worth less than a dollar received today.
- Future value compounds at
$$FV_n = PV(1+k)^n$$
; present value discounts at
$$PV = FV_n/(1+k)^n$$
; the compounded annual growth rate reverses the future-value formula.
- Negative interest rates raise the present value of fixed income investments at the short end of the yield curve.

Exam cues

- **Six measures of return:** total, expected, annualised, real, currency-adjusted, internal rate of return. Match the measure to the question asked.
- **Total return**
$$= [CF_t + (P_E - P_B)]/P_B$$
. Worked: dividend 0.20, buy 5, sell 6 → **24%**.
- **Expected return**
$$= \sum r_i p_i$$
. Worked: 0.20×25% + 0.20×(−30%) + 0.60×15% = **8%**.
- **Annualising:**
$$(1+r)^{12/m} - 1$$
. 2.5% over 3 months → **10.38%**. Reverse: 12% annual → **3.85%** over 4 months.
- **Real return**
$$= (1+r)/(1+\text{inflation}) - 1$$
. 10.4% with 2% inflation → **8.24%**; with 1% deflation → **11.52%**.
- **Currency adjustment rule:** reference currency is the **first** currency of the pair → **divide**; **second** → **multiply**. JPY portfolio 121.88% with SGD/JPY +18.89% → **86.62%**; USD portfolio 87.06% with USD/SGD +8.12% → **102.24%**.
- **IRR is money-weighted; CAGR is time-weighted.** IRR is sensitive to the size and timing of cash flows; CAGR is not. Worked: fund returns 2%, 10%, −4% → CAGR **2.51%**, but the client's IRR is **−0.6%**.
- **Variance and standard deviation:**
$$\sigma^2 = \sum p_i[r_i - E(R)]^2$$
. Worked: variance **0.03760**,
$$\sigma$$
= **19.39%**.
- **Coefficient of variation**
$$= \sigma/E(R)$$
. A: 0.08/0.05 = **1.60**; B: 0.20/0.15 = **1.33** — B is better on risk per unit of return.
- **Normal distribution landmarks:** ±1 = **68%**, ±2 = **95.4%**. Z = (X − E(r))/ . For E(r) 6%, 12%: Z = −0.5 gives a **31% chance of a loss**.
- **VaR Z-scores:** 2.5% tail → **−1.96**; 0.5% tail → **−2.58**. Balanced portfolio (6%, 12%) → **−18%** and **−25%**. Defensive (4%, 4%) → **−4%** and **−7%**.
- **VIX:** measures expected **30-day S&P 500** volatility. Above **30** = high volatility; below **20** = relative stability. Peaks: **89.5** (Oct 2008), **48** (European Debt Crisis).
- **Six risk categories:** issuer/counterparty, product-specific, market (systematic), execution/operational, management, liquidity. Market risk **cannot be diversified away**; it can be hedged.
- **Correlations tend toward 1 during a crisis** — diversification fails exactly when it is needed.
- **SGX circuit breaker:** triggers on a move of more than **10%** from the reference price (the traded price **5 minutes** prior), followed by a **5-minute cooling-off** period with trading limited to **±10%**.

- **Day basis: 365** for Singapore dollars, **360** for US dollars. Simple interest denominator is **36,500**.
- **TVM formulas:**

$$FV_n = PV(1+k)^n$$

;

$$FV_n = PV(1+k/m)^{mn}$$

;

$$PV = FV_n/(1+k)^n$$

;

$$CAGR = (FV/PV)^{1/n} - 1$$

.
- **Worked TVM values to memorise:** $1,000 at 5 1,628.89**; quarterly compounding = **$1,643.62**; $1,000 discounted 5 years at 6 747.27**; SGD 1m at 8% for 91 days = **SGD 19,945.21**.
- **Negative rates raise the present value** of short-end fixed income investments.
- **Trap:** the standard normal table gives the area to the **left** (cumulative). For a left-tail probability, look up

$$1-\alpha$$

and subtract from 1 — do not read the negative Z directly, because the table has no negative entries.
- **Trap:** do not divide the money-weighted return estimate by the number of years when contributions and withdrawals are spread evenly through the period.
- **Trap:** a higher standard deviation does not by itself make an investment worse — compare coefficient of variation before concluding.

Chapter 3 — Key Drivers of Market Movements and Asset Values

Fair value comes from fundamental analysis; market price comes from three interacting forces. This chapter separates those forces — fundamental, market-driven and behavioural — and gives the frameworks the examination tests: fiscal and monetary policy, the business cycle, the three classes of economic indicator, the industry life cycle, Porter's five forces, and the behavioural biases that move prices away from value.

Learning Objectives

The candidate should be able to:

- Identify and explain the key drivers of asset values and market movement.
- Describe and explain how key economic factors affect market movement and asset values.
- Follow the economic calendar and explain the three main classes of economic indicators — leading, coincident and lagging.
- Discuss how industry and company-specific factors affect market value.
- Describe how market factors affect asset prices.
- Describe how investment behaviour influences investment decisions, market movements and asset prices.

3.1 Introduction to Market Movements and Asset Values

An analyst assessing asset values typically calculates an **intrinsic or fair value** from fundamental analysis using a set of assumptions and parameters, then compares it with the market value. Fair-value assessments and recommendations often differ across analysts because the assumptions and projections differ, and perceptions of fair value vary among investors because they have different horizons and objectives.

Market price is influenced by many factors, ultimately resolving into the price agreed for the asset given the supply of the security and its demand at a point in time. Three groups of influence act on it:

- **Fundamental factors** — economic, industry and company-specific factors affecting prospects and value.
- **Technical factors** — investment behaviour and sentiment.
- **Market-driven factors** — market structure, trading rules and parameters, the type of participants, and the flow of funds and assets available at any one time.

The Efficient Market Hypothesis (EMH), postulated by Eugene Fama, contends that markets are efficient and asset prices adjust rapidly to reflect new publicly available information. On that view an asset's price **is** its fair value and no individual can consistently outperform the market. Yet periods of information inefficiency appear to exist, producing under- or over-valuation over extended periods. Investors have also been found to react irrationally because of cognitive biases such as overconfidence, emotional over-reaction, information bias and other predictable errors in reasoning and information processing. For astute and alert investors, those periods of under- and over-valuation are the opportunity.

3.1.1 Factors Driving Asset Values

Three primary factors influence asset values and market movement:

Factor group	What it covers	R
Fundamental factors	Economic factors, industry factors, company factors	T
Market-driven factors	Liquidity and funds flow, new securities and IPOs, derivatives influence, market structure	T
Investment behaviour	Herd instinct (greed and fear), loss aversion, cognitive bias	H

Market participants must be aware of all three and incorporate them when analysing and making sound investment decisions.

3.2 Fundamental Factors and Analysis

Fundamental analysis attempts to understand and estimate the fair value of an asset by examining:

- **Macroeconomic factors** — fiscal and monetary policy.
- **Industry-specific factors** — the industry life cycle and competitive structure.
- **Firm-specific factors** — business strategy and financial condition.

3.2.1 Economic Analysis

Expectations about the future health of an economy have direct and significant effects on its financial markets, because economic data reflects the composite performance of industries and companies and forms the basis for projections. When investors expect steady growth, stock markets are typically bullish; expectations of significant slowdown or the threat of recession usually trigger corrections and bearish conditions.

Investment instruments are affected by fiscal and monetary policy. Policy changes affect interest rates, currency, inflation, employment, consumption and corporate profits, and **expectations** of policy changes also move asset prices. The investment community therefore tracks leading economic indicators to gauge the significance of policy changes and their impact on different investments and industries, and looks to invest in the securities likely to benefit most.

Economic analysis also drives allocation. In **top-down asset allocation**, a larger proportion of funds goes to developed economies, growing sectors and industries, and the leading corporate performers within them — those with stronger financials and better earnings-growth potential.

3.2.2 Fiscal Policy

Fiscal policy is the use of **government expenditure and taxation** to influence aggregate demand and supply. Changes in government spending on infrastructure influence the general economy.

Money from fiscal expenditure passes through the economy and eventually reaches consumers, producing a **multiplier effect**. Greater construction and infrastructure spending raises demand for construction materials, equipment, services and labour; wages and profits rise in those industries, creating higher disposable incomes; that triggers further consumption of goods and services, benefiting suppliers in other industries; and with healthier consumption, private capital expenditure rises too, adding to the expansion. The initial round of spending ripples through different industries.

Governments can also cut taxes, creating higher profits and disposable income, which triggers greater consumption and stimulus. To fund the initial spending, governments may run a **budget deficit**. Excess spending is funded by drawing down surpluses from prior years or by issuing government bonds. Most governments ran higher debt to support fiscal stimulus immediately after the 2008 Global Financial Crisis; **US federal government debt as a percentage of GDP was at its highest in 2017**, back to post-World War II levels, when heavy stimulus was needed.

3.2.3 Impact of Fiscal Measures on Gross Domestic Product

Gross Domestic Product (GDP) is the most widely used measure of economic performance. It measures the market value of goods and services produced in the domestic economy during a specific period, usually one year. Four components are considered:

$$GDP = C + I + G + NE$$

where

$$C$$

is consumption expenditure by individuals on goods and services,

$$I$$

is private investment in capital goods (capital expenditure for replacing equipment and expanding the supply of capital assets),

$$G$$

is government spending (both investment and consumption, including education, defence, capital investment, infrastructure and health services), and

$$NE$$

is net exports — total exports less total imports, which may be positive or negative. When a country sells more to foreigners than it buys, net exports are positive.

To evaluate how fiscal policy affects the economy, consider how it affects each component. With the intent of sustaining a higher GDP growth target of **3–4%** compared with prior periods averaging **2.2% (1998–2017)**, the US government adopted a more aggressive expansionary stance:

Measure	GDP component affected
More infrastructure development projects	Raises G
Tax incentives for US companies to relocate factory operations back to the US	Raises I
Lowering the personal income tax rate	Raises C
Tariffs on selected foreign goods, making them less attractive relative to US-made goods	Intended to improve NE, tradi
Increasing job creation and retention to raise the employment rate	Increases C

Expectations that the US government would follow through led to a major rally after the November 2016 election, and in **2017 the S&P 500 returned 21%**. Successful near-term stimulus can create future problems, however — the threat of inflation and higher budget deficits.

3.2.4 Monetary Policy

Monetary policy tools stimulate or cool the economy by increasing or decreasing liquidity and/or by decreasing or increasing the cost of funding for short-term liquidity and bank reserves.

Stance	Mechanism	Effect on asset values
Expansionary	Expand money supply, lower interest rates, increase aggregate demand	Lower deposit rates push fun
Contractionary	Decrease money supply, raise interest rates, contain inflation	Higher borrowing costs disco

For the US economy, an important facet is the rate at which the Federal Reserve allows money supply to grow. Studies of US monetary history show that **declines in the growth rate of money supply have preceded business contractions**, while increases have preceded expansions. The Fed influences money supply through three channels:

- **Open Market Operations (OMOs)** — the central bank buys and sells government securities.
- **The Fed Funds Rate** — the rate at which banks lend excess reserves to one another so banks with shortfalls can meet reserve requirements.
- **The discount window** — a lending facility extended to eligible financial institutions to meet temporary shortfalls in available credit caused by short-term illiquidity.

Of the three, the Fed has relied heavily on **OMOs** as its key method for influencing market liquidity, using them extensively to pump liquidity into the market in 2008. This type of liquidity change **first affects bond prices, then common stocks, and finally the aggregate economy**.

Quantitative easing. The 2008 crisis produced the greatest recession since the US Great Depression of the 1930s. To stimulate consumption, employment and growth, the US, UK and European central banks embarked on quantitative easing (QE) programmes, producing a protracted period of very low interest rates (**2009–2017**). Asian central banks also kept rates low to increase domestic liquidity.

The Fed openly tied QE to unemployment. Rates were held at a low of **0.25% from December 2008 until December 2016**, then raised by **25 basis points** at a time as employment and job creation improved. With US unemployment reaching **3.8% in May 2018** — the lowest in 48 years — and core inflation reaching the Federal target of **2.0%**, the Fed signalled continued normalisation and balance-sheet reduction. The Fed Funds Rate was raised to **2.25% in September 2018**.

When the economy overheats, inflation control becomes the focus. Central banks contain inflationary pressure and excessive capacity build-up for fear that unsustainable growth and eventual correction cause a hard landing with major unemployment, destruction of asset values and social hardship. They then adopt a contractionary policy to moderate growth, triggering consolidation and engineering a **soft landing**.

a. Singapore's trade-weighted exchange rate index. Because its economy is open, Singapore's monetary regime centres on managing the exchange rate, which has a major influence on local inflation and international competitiveness. MAS conducts policy by managing a **trade-weighted exchange rate index**. Money supply is largely driven from within and has only limited impact on Singapore's economic activity. Independent policy targets cannot be set for both exchange rates and interest rates, because one affects the other, so Singapore has **no independent policy targets for interest rates and money supply**.

b. Singapore's exchange-rate-based monetary policy. Since **1981**, monetary policy in Singapore has centred on exchange-rate management, with price stability as the basis for sustainable growth. The exchange rate is an ideal intermediate target for a small, open economy: it is relatively controllable through direct intervention in the foreign exchange market and bears a stable, predictable relationship with price stability over the medium term.

Feature	Detail
Basket	The Singapore dollar is managed against a basket of currencies of major trading partners and competit
Managed float	The trade-weighted exchange rate may fluctuate within a **policy band**, whose level and direction are a
Review cycle	The policy band is reviewed periodically — typically every **six months** — to stay consistent with func
Interest rates	Choosing the exchange rate as the intermediate target means MAS gives up control of domestic interes

The exchange rate has been an effective anti-inflation tool: domestic inflation averaged **1.9% per annum from 1981 to 2010**. The system's success owes much to strong fundamentals — prudent fiscal policy, flexible product and factor markets, a sound financial system and a robust corporate sector.

c. Money supply, credit creation and required reserves. Money supply also increases through credit creation when reserve requirements are lowered. A bank takes a new deposit and lends out 90%, keeping 10% as statutory reserves; the process repeats through other banks, producing a **10-fold increase** in money supply.

Round	New deposit	New investment (loan)	New reserves
1st	10,000	9,000	1,000
2nd	9,000	8,100	900
3rd	8,100	7,290	810
4th	7,290	6,561	729
…	…	…	…
After all rounds	**100,000**	**90,000**	**10,000**

When the economy overheats, the central bank raises required reserves, indirectly reducing money supply and liquidity. Lowering required reserves expands money supply and multiplies lending. China applies this actively: with US–China tariff trade wars looming and a slowdown expected, the Chinese central bank

lowered the Reserve Requirement Ratio three times in the first half of 2018 to increase liquidity and lending.

d. Quantity theory of money. When money supply increases, central banks fear that excess liquidity causes asset inflation. The quantity theory of money states a direct proportional relationship between the amount of money in the economy and the price of goods and services — too much money in the system causes inflation. Fisher's **equation of exchange**:

$$M \times V = P \times Q$$

where

$$M$$

is the stock of money (currency outside banks plus demand deposits),

$$V$$

is the velocity of money circulation (the number of times per period an average unit of money changes hands),

$$P$$

is the general price level or a price index, and

$$Q$$

is the number of transactions in an economy during a year.

When the equation is fully operative, the central bank cannot increase output by increasing money supply: as

$$M$$

rises,

$$P$$

rises (more money chasing goods, causing inflation), and velocity

$$V$$

— which is socially determined, decided by consumers — then determines output

$$Q$$

.

e. Inflation, interest rates and asset values.

Interest rates and bond prices. Lower interest rates mean lower discount rates, giving higher fair-value targets. When rates rise, higher discount rates applied to bond cash flows mean lower intrinsic bond values. Because interest-rate inflection points tend to coincide with business-cycle peaks, **the highest bond yields tend to occur close to the peak of the business cycle**, at the tail end of a series of rate hikes; the lowest yields tend to be offered at the bottom of a cycle before recovery.

Interest rates and stock prices. The relationship is more complex. The stock market can rise through periods of rising rates when global economies are strong, even with rising inflation. Expectations and announcements of hikes can bring short-term corrections, but markets can continue upward through several hikes if the economy keeps growing. At some point the market recognises that further hikes to control inflation will no longer be accompanied by strong earnings and economic performance; any further hike is then read as a negative trigger for an overvalued market, and a deeper correction follows.

Scenario	What happens
Positive	Rates rise with inflation and strong corporate earnings; firms pass cost increases on to consumers be
Slightly negative	Rates rise but expected earnings barely change because firms cannot raise prices to match higher co
Very negative	Rates rise while cash flows decline; sales revenue and earnings fall, firms cut wages and labour, lay-o

Rising rates are not negative for the whole market. Some industries react positively to inflation and rate hikes. During the initial expansion, **banks** benefit from widening the net interest spread — raising lending

rates faster than deposit rates — while still enjoying loan growth, so net interest income expands. Banks typically benefit from a gradual rise in rates.

f. Effectiveness of fiscal and monetary policy. Five factors limit efficacy:

- **Opposing policies cancel out.** In the US, fiscal policy is under the government and monetary policy under the Fed; politically motivated fiscal priorities can run contrary to monetary policy. In most countries, however, central bank policy is generally aligned with the ruling government.
- **High savings rates blunt fiscal stimulus.** In Asian markets with a culturally high propensity to save (Japan), higher disposable income from tax breaks produces a smaller-than-expected consumption response.
- **Open economies leak stimulus.** Where there is heavy dependence on major foreign companies and construction labour, infrastructure stimulus leaks abroad, lowering the domestic impact.
- **Policies take time.** The deeper the recession, the longer the stimulus takes to flow through.
- **Government strength and stability matter.** Fiscal policy relies on the ability to keep funding it; monetary policy relies on belief in the government's ability to control money supply. With political corruption, social unrest, lack of taxable income, or hyperinflation, neither works.

g. Negative interest rates as a monetary policy tool. After the Global Financial Crisis, a few central banks — **Sweden, Denmark, Switzerland and Japan** — set negative rates on commercial banks holding excess funds as reserve deposits with the central bank, so banks pay to keep excess funds on deposit. The aim for the Eurozone-related central banks was to stimulate growth and escape deflation, pushing banks to put money to productive use by lending rather than paying to hold cash.

h. Money supply from investment flows. Money supply also comes from external sources. External flows can come from **flights to quality**, as investors reallocate to markets with stronger currency, better economic outlook, healthier earnings growth and potentially higher returns. For example, there was strong demand for the US dollar for most of 2018 as money flowed out of emerging markets into the US on expectations of an emerging-market slowdown caused by US trade tariffs.

3.2.5 Business Cycles and Market Movement

Business cycles reflect the rise and fall in the general level of economic activity. No two cycles are identical, but they share common traits: a **business peak**, a **slowdown or contraction**, a **recessionary trough**, then **recovery and expansion**.

The stock market is predictive of economic conditions. Stock market peaks and troughs have typically preceded turning points in general business activity — stock prices turn ahead of the economy. Before GDP figures showing two quarters of contraction (a **technical recession**) are announced, equity markets have typically corrected. Equity investors read the trends from leading indicators and adjust; while conditions are still weak, the market waits for leading strategists to declare the worst is over, so as to participate in a rally ahead of reported improvement.

Phase	Typical observations and policy
1. Early bear to mid bear	Negative GDP; central bank cuts rates and raises liquidity; equity markets fall; asset defla
2. Late bear to early bull	Negative GDP growth persists; rate cuts continue; fiscal and monetary stimulus; provision
3. Early bull to mid bull	GDP turns positive; rates start to rise; corporate profits improve sharply; policy stimulus
4. Late bull to early bear	Inflationary pressures rise; central bank targets inflation; factor costs rise and productivit

The equity market reflects the broad range of companies and industries in the economy, so aggregated earnings forecasts represent the growth and performance of the economy. Given the strong correlation between business cycles, the stock market and the overall economy, insight into the phases of the cycle improves the ability to forecast major turning points in stock prices.

The **bond market** is predictive of expected interest-rate trends: with expectations of hikes, yields start rising ahead of actual central bank increases. The bond market also reflects the market demand for credit. When the economy is doing well, credit should expand, so expected long-term credit demand is higher and, on top of the liquidity premium, yields should be upward sloping. When longer-term credit demand appears weak, long-maturity rates may not rise as much even with Fed Funds hikes, **flattening the yield curve — one of the early predictors of an upcoming recession**.

3.2.6 Economic Calendars

Economic calendars document the release dates of key economic data and are published by various financial information providers. Investors monitor releases closely to evaluate their impact on financial markets. Price volatility of affected assets tends to increase in the period immediately before and after a release.

3.2.7 Economic Indicators

Economic indicators are key statistics giving cues and signals about the direction of the economy. Three broad categories are most widely used.

a. Leading indicators. Leading indicators have peaks and troughs that **precede** the economy and are useful short-term predictors. In the US, the **Composite Leading Economic Index (CLEI)** published by The Conference Board comprises **10 indicators**:

#	Leading indicator	Source and description
1	Average weekly hours (manufacturing)	Monthly average hourly earnings ar
2	Unemployment claims	Weekly first-time claims for state u
3	Manufacturers' shipments, inventories and orders	Monthly preliminary report from th
4	Manufacturers' shipments, inventories and orders for non-defence capital goods	Monthly preliminary report from th
5	Housing starts and building permits	Monthly report, US Census Bureau
6	10-year T-bond spread over the Fed rate	Spread between the 10-year US Tre
7	Adjusted M2 money supply	The Fed's inflation-adjusted measu
8	ISM Index of Manufacturing (new production orders)	Supplier deliveries, imports, produc
9	The S&P 500 Index	Free-float capitalisation-weighted in
10	Consumer sentiment	University of Michigan Consumer S

A downward spiral in the CLEI components typically signals a forthcoming recession; an upturn indicates an upcoming expansion. Composite leading indicators (CLIs) are also compiled by bodies such as the OECD. Such composites are useful as short-term indicators of turning points, but **they say nothing about the magnitude** of the shifts.

Singapore's CLI, published by the Singapore Department of Statistics, comprises:

- Total new companies formed.
- Money supply (M2).
- Stock Exchange of Singapore indices.
- Business expectations for wholesale trade.
- Business expectations for stock of finished goods (manufacturing).
- Purchasing Managers' Index (manufacturing).
- Total non-oil seaborne cargo handled.
- Domestic liquidity indicator.
- Total non-oil retained imports.

The Purchasing Managers' Index (PMI) is a widely followed leading indicator measuring the activity level of purchasing managers in the manufacturing, construction and services sectors. In the US it is compiled monthly by the Institute of Supply Management (ISM). **A reading of 50 or higher suggests expansion; below 50 indicates contraction.** The PMI is followed closely for China and the US, the two largest economies, because China is the factory of the world and expansion or contraction in either manufacturing sector ripples through other economies. In Singapore the PMI is compiled and published by the Singapore Institute of Purchasing and Material Management and is a key barometer of manufacturing health.

b. Coincident indicators. Coincident indicators have peaks and troughs that **align with** business cycles and move in tandem with the overall economy, reflecting the current state. Sales and personal income are examples: they peak or bottom as the economy does. In the US, the **Coincident Economic Indicators (CEI)** comprise **4 series**:

#	Coincident indicator	Source and description
1	Nonfarm payroll employment	Excludes farm workers, private household employees and non-profit
2	Industrial production index	Inflation-adjusted production output from manufacturing, mining ar
3	Manufacturing and trade sales index	Monthly inflation-adjusted sales index, Federal Reserve Bank of St.

#	Coincident indicator	Source and description
4	Personal income and consumption expenditure	Change in personal income reported by the Bureau of Economic An

Singapore's equivalent is the **Composite Coincident Index (CCI)**, published by the Singapore Department of Statistics, comprising:

- GDP at constant prices.
- Index of industrial production.
- Non-oil domestic exports at constant prices.
- Total employment.
- Retail sales index at constant prices, excluding motor vehicles.

c. Lagging indicators. Lagging indicators peak and trough **after** the aggregate economy. They change only after the economy has begun to form a pattern and are useful for **confirming** turning points, typically lagging by a few quarters. The Conference Board's index of lagging indicators comprises **7 components**:

- Average duration of employment.
- The manufacturing and trade inventory-to-sales ratio.
- The change in labour cost per unit of manufacturing output.
- The average prime rate charged by banks.
- The value of outstanding commercial and industrial loans.
- The consumer credit outstanding to personal income ratio.
- The change in the consumer price index for services.

3.2.8 Other Economic Factors Affecting Asset Values and Market Movements

a. Foreign currency reserves and balance of payments deficit. A country with a positive balance of payments (BoP) year after year generates a BoP surplus. Whether it comes from the trade account (exports exceed imports) or the capital account (inflows exceed outflows), foreign exchange reserves increase. A country with a widening trade deficit can become a target of speculative attack because the deficit makes it a "debtor nation". Such weaknesses are generally believed to have led to the **Asian Currency Crisis in 1997**, when weaker Asian currencies came under speculative attack and plummeted, in some cases by **more than 50%**.

Singapore and Hong Kong defended their currencies. Singapore has one of the largest foreign reserves in the world — **USD337 billion as at July 2023**, ranking 11th globally. Countries with no natural resources should maintain high reserves to weather financial storms and global downturns.

A country's exchange rate also affects asset values significantly. If the market expects a sharp fall in a currency, all securities denominated in it are affected, producing an outflow from investments in that currency.

b. Inflation and the Consumer Price Index (CPI). The CPI measures the price level of goods and services purchased by households as an indication of inflation. Most central banks are wary of inflation because it is difficult to control once it gathers momentum. The US Fed uses **Core Personal Consumption Expenditure (Core PCE)** as its primary measure and has traditionally set **2%** as the long-run inflation rate, raising rates when there are signs of inflationary pressure. 2017 was different: the Fed raised rates even though Core PCE remained below 2%.

Inflation is a major concern because it:

- Erodes purchasing power.
- Pressures consumers to buy ahead for fear of escalating prices, especially big-ticket items such as housing.
- Encourages businesses to build in buffer pricing, aggravating the expected increase.
- Forces central banks to raise borrowing costs to control it.
- Raises costs that affect quality of life for most of the population and can create social unrest.

In Singapore, the Department of Statistics rebased the CPI to **2019 in February 2020**. For 2020, both MAS Core Inflation and CPI-All Items came in at **−0.2%**, lower than their 2019 outturns of **1.0%** and **0.6%**.

Despite low reported inflation, the Economist Intelligence Unit (EIU) ranked Singapore the **costliest city in the world from 2014 to 2018**; the 2020 ranking placed it fourth. Two reasons explain the inconsistency:

the strong Singapore dollar kept costs lower for the local population than for expatriates, and the basket of goods differs — the EIU includes the cost of buying a car, which is not part of Singapore's CPI for a consumer using public transport.

c. Persistent deflation. Deflation is a fall in the real value of assets, which differs from a reduction in inflation (disinflation). Falling prices shift spending into the future, which can trigger a fall in consumption and investment and create an unwarranted economic contraction and negative growth.

Persistent deflation followed the implosion of the Japanese stock and real estate markets in **1989**. Japan's central bank deferred part of the huge corporate losses by allowing outstanding debts to be restructured into low-interest, longer-term government bonds — effectively a bailout. Those bonds were in turn purchased by local corporates, banks, public entities and the Japanese people. In deferring part of the pain from the crash, the Japanese economy endured **12 years of deflation**, low and negative GDP growth over **1989–2014**, and anaemic consumption. The efforts of the European and US central banks after 2008 were aimed at avoiding a similar fate of long-term deflation.

d. Factor costs. Crude oil is a key factor cost that has made a dramatic impact on global economic growth over the last century. Before fracking technology and the development of energy alternatives, the supply of crude oil was easily affected by output from the Organization of the Petroleum Exporting Countries (OPEC). Five of the last six US recessions since the 1970s — **1973, 1975, 1979, 1980/81, 1990 and 2008/09** — were preceded by spikes in crude oil prices.

From **2015 to 2017**, excess capacity and supply from new oil producers reversed the situation and crude oil prices plunged to **USD28 per barrel** at one point. The lower crude price reduced factor costs and inflation globally. While lower crude prices hurt oil-producing countries and oil companies, the rest of the world benefited from cost savings that supported the global economic recovery. US oil and refining companies emerged as a major producer and dominant player. Amid agreement from OPEC member nations and Russia to cap output in 2017, the excess supply was eventually absorbed and prices began to stabilise in 2017–2018.

Other major factor costs include rent and labour. High land prices, rental and labour costs raise the cost of doing business, stifle new business creation and skew the composition of viable industries, all of which has a negative long-run impact on growth.

3.2.9 Political Factors Affecting Markets and Asset Values

Country and political risk. To assess country risk, markets look beyond monetary and fiscal policy to the legal framework, foreign exchange reserves, political stability and credibility. Dedicated research organisations — BMI Group, Economist Intelligence Unit, Country Risk Solutions — have in-house assessment criteria, and Euromoney publishes survey reports gauging the market's view of country risk. Countries with stable governments, a good track record and consistency in driving their economic agenda, maintaining social peace and abiding by proper legal and accounting frameworks are ranked highly and attract investors because return uncertainty is lower. **Asset valuations are lower for higher-risk countries**, because higher yields are required to compensate investors for the country risk borne. For sovereign debt credit ratings, the renowned agencies are **Moody's, Standard & Poor's and Fitch Ratings**.

Political shocks and uncertainty. Brexit was a major surprise in 2016. The **51.9%** vote for Britain to leave the European Union produced panic selling of the GBP in June 2016: it **fell 10.5% in 4 days** and drifted down to a low of **GBP/USD 1.21714 (down 18%)** by October, because the investment and business community regarded exit as negative for Britain. During the night of the US election, when it became clear that Republican President Trump's unexpected win was imminent, the **DJIA plunged 800 points**; markets recovered soon after when the President highlighted fiscal stimulus plans including tax cuts and infrastructure spending.

Continuity and political will. Whatever the type of government — democratic, socialist or communist — investors want strong, capable leaders who uphold the legal framework and drive the economy for the good of people, businesses and industries. Uncertainty about succession and continuity, or a lack of political will to follow through, raises risk and tends to produce investment outflows.

Sectors and companies that fall in and out of favour. New governments' political and social agendas benefit some companies and threaten others. A socially conscious government may seek healthcare cost reductions and curb drug prices; a pro-defence government raises interest in the defence industry; governments supported by oil-rich or coal-producing states may be less inclined towards renewable energy than those seeking energy self-sufficiency; some governments welcome the gaming industry for taxable income while others curb it for social ills. Changes in government can therefore change the fortunes of businesses.

3.2.10 Industry Analysis

Some factors affect certain industries more favourably than others. At times industry-related factors outweigh economic factors and make certain industries more resilient to economic or financial crisis, which is usually reflected in share prices. Other factors — structural change and disruptive technology — can cause rapid decline.

Three approaches are considered: the **industry life cycle**, **Porter's five competitive forces**, and **business cycle analysis**.

a. Importance of industry analysis. Two reasons:

- Risks and rates of return vary widely across industries in a given period, because of different product and service mixes and different business-cycle sensitivities.
- Industries display different rates of return over time, so past industry performance cannot simply be extrapolated.

b. Industry life cycle. A framework for evaluating the general health and current stage of an industry. Development divides into five stages: **early development, rapid expansion, mature growth, stabilisation and market maturity, and deceleration of growth and decline**. Besides sales, profit margins and earnings growth can be analysed with the same framework.

Stage	Characteristics
i. Early development (pioneering or start-up)	Introduction of a new technology and a product that did not previousl
ii. Rapid growth	Awareness grows, a market develops, demand picks up significantly; ca
iii. Mature growth	Demand largely satisfied; sales growth stays above average but no long
iv. Stabilisation	Probably the longest phase. Industry growth tracks overall economic g
v. Decline	Sales growth declines from shifts in demand or the emergence of substi

An understanding of the growth pattern helps assess the growth potential of companies in the industry. The average investor may prefer industries in rapid or mature growth; aggressive investors may consider early development, which offers the highest potential returns and the greatest risks; industries in decline should generally be avoided. Comparing an industry's sales with those of firms in the overall economy helps identify its stage.

c. Competitive forces and industry structure. Michael Porter holds that an industry's competitive environment and the intensity of competition determine a firm's ability to sustain above-average returns on invested capital. Five forces determine industry profitability:

- Rivalry among existing competitors.
- Threat of new entrants.
- Threat of substitute products.
- Bargaining power of buyers.
- Bargaining power of suppliers.

Where the five forces are favourable, many companies earn attractive returns. Where pressure from one or more forces is intense, it is harder to command high returns despite management effort.

Force	What makes it intense
i. Rivalry among existing competitors	Many firms of relatively equal size, including foreign competitors; slow grow
ii. Threat of new entrants	Low current prices relative to costs, small capital requirements, few econom
iii. Threat of substitute products	Substitutes cap the price firms can charge; the closer the substitute in price
iv. Bargaining power of buyers	Buyers purchasing large volumes relative to supplier sales; the most vulnera
v. Bargaining power of suppliers	Few suppliers; critical inputs with few or no substitutes, creating a sellers' r

Because an industry's competitive structure is not static, regular review is necessary.

d. Business cycle analysis. Another way to analyse industries is by performance relative to the overall economy. Earnings of many companies move with business cycles, and some industries cope with recession reasonably well while others perform badly.

Industry type	Behaviour
Cyclical	Sales and earnings greatly affected by the business environment; do exceptionally well in expansions
Defensive	Least affected by recession and adversity
Growth	Earnings expected to be significantly above the average of all industries, and growth may occur not
Interest-sensitive	Sensitive to changes in interest-rate expectations

To predict industry performance, analyse the stage of the business cycle and its likely responses to macroeconomic changes — the balance of payments, the trade account, inflation and unemployment. If the economy heads into recession, cyclical industries are affected more than others and defensive industries least. The outcome of industry analysis lets the investor select the industry that better fits their risk and return requirements.

3.2.11 Firm-Specific Factors and Analysis

Just as industry factors affect firms across industries, unique characteristics of firms within an industry affect their attractiveness. Applying a **bottom-up** approach to stock-picking, firms with specific competitive advantages distinguish themselves — for example, what makes Alibaba and Tencent dominant among Chinese internet sales platforms, or why Nvidia traded at a significantly higher valuation in 2017 than other US semiconductor makers.

#	Firm-specific factor
1	Dominance and control of technology/research
2	Control over resource supply
3	Superior brand differentiation
4	Dominance in sales distribution/marketing
5	Cost efficiency, lowest-cost producer
6	Successful focus on target niche clients
7	Established diversified core business
8	Superior leadership, management and talent retention

Firm-specific factors are critical for making precise sales and earnings forecasts, which increases the accuracy of share-price valuation.

3.3 Market-Driven Factors and Asset Values

Market-driven factors are physical market conditions that affect short-term demand and supply and, consequently, the pricing of investment products.

3.3.1 Liquidity and Money Flow

Excessive liquidity. During periods of uncertainty investors sell off investments and hold more cash, and interest rates are usually low as the central bank tries to stimulate the economy. At the onset of a rally, aggressive investors and money managers begin investing, and the market keeps rising as long as there is sufficient liquidity and money flow to drive it upward.

Liquidity shortage. At some point the bulk of funds is almost fully invested. With insufficient funds to sustain the upward drive, the market plateaus and funds rotate to lagging sectors and assets. This is typically when parts of the market become overvalued. Lacking fresh funds to sustain a bull market, the market is poised for a trigger to cause correction and profit-taking.

Analysts track the flow of funds, the excess cash balances of mutual funds and bank deposit levels to predict the funds available for investing. When liquidity builds up and deposit rates are low, money eventually flows back into the stock market as investors seek higher returns, so there is more upside. When available or excess funds dwindle and investors are increasingly invested, there is less liquidity to push the market up — a signal to sell.

3.3.2 New Securities Issues

Excessive equities issued. A significant supply of securities can drain funds for a period. If several major IPOs come to the same market at once, funds are absorbed by them — in some cases deliberately priced at a discount to ensure demand — at the expense of similar stocks already trading. Some investors take profit on existing stocks to participate in the IPOs. The number of IPOs per year therefore affects demand for other similar securities, and when there are too many IPOs in a year, demand for some may suffer because funds are already committed to earlier issues.

Excessive Treasury bills and bonds issued. The same happens in fixed income. Excessive issuance of government T-bills and bonds of a particular maturity creates an oversupply in that maturity tranche, so bond prices may dip briefly to offer better yields for that tranche. The same occurs if there is a sudden reduction of interest by regular buyers of US Treasuries — if China or Japan decided to buy fewer US Treasury bonds, that would suddenly create excess supply.

3.3.3 Trading Influence from Derivatives and Other Asset Classes

a. Influence from structured products. Markets are generally efficient and difficult for any one investor or speculator to influence. But with structured products there may be a **collective incentive** for proprietary traders at financial institutions to move the price of certain stocks to reach strike-price targets for stocks or indices, because those institutions are usually the counterparty to the structured products and the profitability of their positions depends on prices moving favourably.

On thin-trading days or during specific periods of the day, certain parties may try to push prices in a specific direction. When more than one institution sells similar structured products with similar target price ranges, the collective ability to move the underlying price of securities and indices can be significant. Proving collusion and illegality is difficult, but such actions are unethical because the institution profits by working against investors to whom it sold a product. Some institutions argue that proprietary desks have no knowledge of the structured-product positions held by other investors and receive no instruction to act either way, but institutions and professionals should not tolerate practices that profit at clients' expense.

b. Influence from futures and options. In bearish markets, the sale of futures contracts and put options can add significant pressure on underlying assets. Futures and options are leveraged instruments traded on margin: an investor puts up only a small fraction of capital to own the full notional amount, so returns are magnified relative to the underlying. As underlying assets fall, futures and options prices fall further with magnified losses; derivative investors may be forced to reduce holdings to meet margin requirements, driving the market down further. This creates the risk of the derivative and underlying markets **feeding off each other in a downward spiral**. The reverse can occur but is less worrying, at least initially, because markets are rising.

c. Major common expiration dates. Many instruments mature and must be rolled over toward the end of quarterly periods, so volume and volatility can pick up in the **last two weeks of March, June, September and December**. In the US, on the **third Friday** of each of those months, the last hour of the stock market session (**3.00–4.00pm New York time**) is the **triple witching hour**: three kinds of securities mature that day — stock market index futures, stock market index options and stock options. During that hour, expirations generally increase trading volume in options, futures and the underlying stocks, and occasionally price volatility as well.

3.3.4 Other Trade Flows

Hot money. Global money from hedge and other large funds seeking temporary inefficiencies and arbitrage opportunities may flow into a market, sector or set of stocks and temporarily push up volume, interest and prices. Such hot money can be orchestrated with the dissemination of additional positive information to generate interest, only to exit once return targets are achieved. This is essentially **pump-and-dump** activity, but with more finesse and sophistication and at a global level.

Computer-driven momentum-based trading. Computer-driven trading follows a defined set of instructions to make high-speed decisions and transactions across many markets — commodity futures, stocks, currency, money market — generating profits at a frequency human traders cannot match. Such algorithms are predominantly influenced by technical market indicators that dictate continued buying while upticks and specific price and volume trends are in place, regardless of valuation, and continued selling on downward momentum. As a result of widespread use, prices of securities and assets have become more volatile.

3.3.5 Market Microstructure Influence on Price and Asset Values

In the short term, prices and values are also influenced by market structure — the trading and execution processes, rules, liquidity and strength of participants.

a. Differences in the price discovery process.

Market type	How price is discovered
Issuer-priced	The buyer simply looks at the price quote provided by the issuer, w
OTC negotiated	Prices are negotiated over the phone with OTC dealers or market r
Continuous stock market	A rule-based process with bid and ask quotes provided continuously
Call market (some European bourses, e.g. France)	All bids are submitted and batched for simultaneous execution once
IPO	Similar to a call market, with bids submitted, but with the added c
Auction of a pool of stocks	Bids are submitted and the auctioneer takes the highest bids to ma

The types of players in each market structure and the specific systems and rules applied therefore affect the pricing process.

b. Impact of computer-driven, momentum-based trading on the pricing process. Algorithmic trading is one of the most commonly applied strategies globally. **High frequency trading (HFT)**, the primary form, is the rapid execution of extremely short-duration trades, profiting by arbitraging small price differences across exchanges. Its success comes from processing information and executing trades simultaneously without the time and emotional burden of human decisions. HFT has improved market liquidity and produced higher price efficiency through synchronised statistical arbitrage, but it also poses risks of increased volatility and potential **predatory trading** at the expense of less sophisticated participants, destabilising the market.

c. Inventory costs and liquidity. Research indicates the quoted bid-ask spread can be influenced by **inventory costs** and by the amount of liquidity certain traders or investors possess. A dealer or market maker holding an inventory of stocks, and incurring costs, may price differently from one that does not. Other studies suggest parties with greater cash liquidity can have greater influence over pricing.

d. Dark pools and price efficiency. Dark pools are networks — private financial forums or exchanges — for trading securities, derivatives and other instruments. The bulk of dark pool trades are large trades by financial institutions conducted outside public exchanges, keeping trade details confidential. Dark pools are not new, since block trades and private transactions have long existed, but the pools of entities facilitating and financing private trading have grown, away from the exchange's view, until transactions are completed and brought into the open. Although the standard price discovery process is generally assumed to keep publicly traded prices relatively fair, in dark pools the internally agreed trade prices, together with the size and lack of transparency of participant identities, put less sophisticated participants at a disadvantage.

3.4 Effect of Investment Behaviour and Emotions on Asset Values

3.4.1 Herd Instinct — Greed and Fear

A herd instinct is an emotional phenomenon characterised by a lack of independent decision-making in a crowd acting without centralised direction. It is attributed to **greed and fear** and can drive asset values well above or below the levels justified by fundamentals. Behavioural studies show the herd instinct kicks in especially when **information is scarce and threats seem imminent**.

As the market declines to a trough or turning point, participants begin accumulating when they believe the worst is over. Once the trough is confirmed, confidence gradually returns as more participants and funds re-enter. Retail investors pile back in and **greed takes over** as participants lose sight of valuations and push the market to new heights. An **investment bubble** occurs when the market is driven by investors overlooking fundamentals and overcome with optimism about asset values, perpetuated by exuberant behaviour.

The **1997–2000 US Internet Dot.com Bubble** is the example of mass euphoria with fundamentals discarded: investors bought overpriced companies, some generating no revenue at all, let alone profit. **NASDAQ rose sharply from 1,000 to 5,000**, with a market price-earnings (P/E) ratio of **200x**. Then Fed Chairman Alan Greenspan warned of overvaluation by coining the term "**irrational exuberance**" in **December 1996**, but the bubble continued until 2001.

3.4.2 Loss Aversion in Portfolios

Loss aversion is associated with **prospect theory**. An implication is that individuals prefer **less** uncertainty for gains and **more** uncertainty for losses. When the market is falling, there may be higher fund redemptions and securities sold. Investors and managers may be unconsciously influenced by loss aversion to sell profitable holdings to lock in positive returns (lowering uncertainty) while holding on to book losses on poorer performers in the hope of a rebound (raising uncertainty).

The profitable securities sold may actually have more upside, but because of loss aversion investors sell those first and retain the losers, which may fall deeper into the red because they are trying to avoid realising losses. Loss aversion therefore has a potential **double negative impact**.

3.4.3 Cognitive Bias

Cognitive bias is the tendency to make conclusions based on one's past experiences, preferences or predetermined beliefs. **Confirmation bias** is the tendency to look for views and recommendations aligned with one's own. Investors who have committed funds to an investment are more prone to listen to and agree with research calling for appreciation than research calling for a sale, and then reinforce the view by committing even more funds. An investor less influenced by this bias actively looks for research opinions different from their own and checks whether the alternative views have merit.

3.5 Summary

- Three primary factors affect market movements and asset values: fundamental factors, market-driven factors and investment behavioural factors.
- Fundamental analysis revolves around economic, political, industry and firm-specific factors.
- Fiscal and monetary policy, business cycles and interest rates are the main areas of economic analysis.
- Expansionary fiscal and monetary policy stimulates growth; contractionary policy controls inflation and cools the economy. Expectations and announcements of policy actions can dramatically move markets.
- Business cycles reflect the general level of economic activity. Stock market peaks and troughs have typically preceded turning points in business activity, so insight into cycle phases improves forecasting of turning points in stock prices.
- The investment community monitors key information releases to see whether actual data match estimates and whether forecasts need revision. Price volatility of affected assets tends to rise immediately before and after a release.
- Three categories of economic indicator give early warning of changes in activity: leading, coincident and lagging.
- Other economic factors include foreign currency reserve levels, the balance of payments, inflation and deflation, and changes in factor costs.
- Market participants favour countries with stable, capable and honest governments. Investors should watch elections, changes in leadership, party dissent and stated party policies to mitigate return uncertainty.
- Industry analysis matters because risks and returns vary widely across industries, and industry profitability does not remain constant. Industry-specific factors can dominate economic factors.
- The industry life cycle divides development into five stages: early development, rapid expansion, mature growth, stabilisation and decline.
- Porter's five competitive forces determine the intensity of competition and therefore industry profitability.
- Industry business cycles should be examined by comparing company earnings performance with that of the overall economy.
- Firm-specific factors uniquely distinguish one firm from another and support more accurate earnings and cash flow forecasts.
- Market-driven factors are physical market conditions affecting short-term demand and supply and therefore pricing.
- Major market-driven factors include liquidity flows, new securities issues, trading influences from derivatives and other asset classes, hot-money flows, and computer-driven momentum trading. These must not be underestimated, as they are increasingly dominant over fundamentals.
- Market microstructure influences include varying pricing processes, inventory costs and dark pool liquidity.
- Investor behaviour matters: fear, greed, loss aversion and cognitive biases significantly affect asset values and market movements.

Exam cues

- **Three drivers of asset values:** fundamental, market-driven, investment behaviour. Fundamental and market-driven are the rational explanation; behaviour is the emotional one.
- **EMH (Fama):** prices adjust rapidly to new public information, so price equals fair value and no one consistently outperforms. The examination tests the *counter-argument* — persistent inefficiency and cognitive bias create opportunity.
- **Fiscal policy** = government expenditure and taxation. **Monetary policy** = money supply and the cost of funding. Do not mix them.
- **GDP identity:**
$$GDP = C + I + G + NE$$
. Match each fiscal measure to the component it raises: infrastructure →
$$G$$
; relocation incentives →
$$I$$
; income tax cuts →
$$C$$
; tariffs →
$$NE$$
.
- **Multiplier effect:** fiscal spending ripples through industries, raising wages, profits, disposable income and further consumption, then private capital expenditure.
- **Fed's three tools:** OMOs, the Fed Funds Rate, the discount window. **OMOs are the primary tool.** Liquidity changes hit **bond prices first, then stocks, then the economy**.
- **QE timeline:** rates held at **0.25% from Dec 2008 to Dec 2016**, then raised **25bp** at a time; unemployment **3.8% in May 2018** (48-year low); core inflation target **2.0%**; Fed Funds **2.25% in September 2018**.
- **Singapore's monetary policy is exchange-rate-based, since 1981.** Managed float against a **trade-weighted basket**, within a **policy band** announced **semi-annually**, reviewed about every **six months**, with an **MPS** and **Macroeconomic Review**. Singapore therefore has **no independent interest-rate or money-supply target**, and domestic rates have typically been **below US rates**.
- **Credit creation:** 10% reserve requirement → **10-fold** money supply expansion (10,000 deposit → 100,000 total).
- **Fisher's equation of exchange:**
$$M \times V = P \times Q$$
. Increasing
$$M$$
raises
$$P$$
, not
$$Q$$
.
- **Interest rates and bonds:** lower rates → lower discount rate → higher bond value. **Highest yields near the business-cycle peak**; lowest yields at the bottom before recovery.
- **Interest rates and stocks — three scenarios:** positive (
$$k$$
and
$$g$$
both rise, price stable), slightly negative (
$$k$$
rises,
$$g$$

constant, price falls), very negative (k rises, g falls, price falls sharply). Banks benefit from a **gradual** rise through a widening net interest spread.

- **Business cycle phases and strategy:** early/mid bear → cash, investment-grade bonds, precious metals; late bear/early bull → cyclicals and high-beta, reduce cash; early/mid bull → growth stocks, high-yield bonds, sector rotation; late bull/early bear → defensive high-dividend stocks, investment-grade bonds.
- **A flattening yield curve is an early predictor of recession.**
- **Three indicator classes: leading** precede (CLEI, 10 components; PMI), **coincident** align (CEI, 4 components), **lagging** confirm (7 components). Leading indicators signal turning points but **not magnitude**.
- **PMI: 50 is the dividing line** — 50 or above = expansion, below 50 = contraction.
- **Singapore's CLI** includes total new companies formed, M2, SGX indices, business expectations, PMI, non-oil seaborne cargo, domestic liquidity indicator and non-oil retained imports. **Singapore's CCI** includes GDP at constant prices, industrial production, non-oil domestic exports, total employment and retail sales excluding motor vehicles.
- **1997 Asian Currency Crisis:** widening trade deficits made countries "debtor nations" and targets of speculative attack; currencies fell **more than 50%** in some cases. Singapore's reserves: **USD337 billion as at July 2023**, 11th globally.
- **Inflation:** US Fed targets **Core PCE at 2%**. Singapore rebased CPI to **2019 in February 2020**; 2020 Core and All-Items both **−0.2%** (2019: 1.0% and 0.6%). EIU ranked Singapore the **costliest city 2014–2018**.
- **Brexit: 51.9%** leave vote; GBP **fell 10.5% in 4 days**, down **18%** to GBP/USD 1.21714 by October 2016. US election night: DJIA **plunged 800 points**, then recovered on fiscal-stimulus plans.
- **Industry life cycle — five stages:** early development, rapid expansion, mature growth, stabilisation, decline. Mature growth sales run **12–20%**; rapid-growth earnings compound at **30–50%**.
- **Porter's five forces:** rivalry, new entrants, substitutes, buyer power, supplier power. Kodak is the substitute example; the US government and Apple are buyer-power examples.
- **Industry types:** cyclical (durable goods), defensive (food, utilities), growth, interest-sensitive (financials, real estate).
- **Firm-specific factors:** eight listed — technology dominance, resource supply, brand, distribution, cost efficiency, niche focus, diversified core, leadership.
- **Liquidity cycle:** excess liquidity drives the rally; when funds are fully invested the market plateaus and rotates; dwindling liquidity signals a sell.
- **Triple witching:** third Friday of March, June, September and December, **3.00–4.00pm New York time**; index futures, index options and stock options all expire.
- **Behavioural biases:** herd instinct (greed and fear, strongest when information is scarce), loss aversion (prospect theory — sell winners, hold losers), cognitive bias (confirmation bias).
- **Dot.com bubble:** NASDAQ **1,000 → 5,000**, P/E **200x**, "irrational exuberance" coined **December 1996**, bubble ran to 2001.
- **Trap:** a **leading** indicator predicts, a **coincident** indicator confirms the present, a **lagging** indicator confirms the past. PMI is leading; nonfarm payrolls are coincident; the average prime rate is lagging.
- **Trap:** expansionary monetary policy lowers rates and *raises* asset values; expansionary fiscal policy raises GDP components but may raise inflation and deficits.
- **Trap:** rising interest rates are not uniformly bad for equities — banks can benefit during the initial expansion.

Chapter 4 — Foreign Exchange

Foreign exchange is the one asset class every other investment is exposed to, whether or not the investor intends it. This chapter covers what moves exchange rates, how rates are quoted and read, how forward rates are derived from interest differentials, and the four risks an FX position carries.

Learning Objectives

The candidate should be able to:

- Describe the features of the foreign exchange market.
- Understand how foreign exchange rates are quoted.
- Understand the difference between direct and indirect quotations and cross quotations.
- Calculate cross rates from indirect quotes in USD against other currencies.
- Understand the bid-offer spread.
- Distinguish between a spot and a forward transaction.
- Explain how forward rates are determined and calculate forward points.
- Explain interest rate parity and covered interest arbitrage.
- Describe non-deliverable forwards and their use.
- Identify the risks in foreign exchange and the methods of managing them.

4.1 Introduction

Investments in foreign currencies carry **foreign exchange risk**. Currency values fluctuate daily and can affect international investments significantly: fluctuations may enhance returns through exchange gains or depress them through exchange losses. A representative must therefore understand how exchange rates are determined, how the FX market operates, and how to use it in financial transactions.

Investing in a foreign currency also forces a second decision: **which currency funds the investment?** The investor may borrow in a foreign currency or convert funds from the base currency, and the two choices produce different FX and interest-rate risk exposures.

4.1.1 Definition and Terminology

A foreign exchange transaction is the purchase of one currency against the sale of another for settlement or delivery on a specified date. The **exchange rate** is the price per unit of one currency expressed in units of another.

Most FX transactions are **spot** deals, made for delivery in **two working days** (value spot). There is also a market for same-day delivery (**value today**) and next-day delivery (**value tomorrow**, or value tom).

Base and counter currency. Currencies are quoted in pairs, for example USD/SGD 1.2520. The first listed currency is the **base currency**; the second is the **counter currency** or **term currency**. In that example USD is the base and SGD the counter, and the quotation means USD 1 = SGD 1.2520.

For an investor, the base currency is the **operating or reference currency**: all profits and losses are measured in it. Holding investments in any other currency is a source of FX risk that must be managed.

Long and short positions. FX risk positions are described as long or short.

- **Long** means buying with the expectation that the purchase will rise in value — betting the base currency will strengthen against the term currency.
- **Short** means selling the base currency in the hope its price falls.

If an investor borrowed USD to fund a SGD investment, they sold USD spot for SGD, so the position is **short USD and long SGD**. To close it, they enter the opposite transaction — sell spot SGD and buy USD — and are said to have **squared off** the position, leaving no FX exposure.

4.2 Factors Influencing Exchange Rates

Besides interest rates and inflation, the exchange rate is one of the most important determinants of a country's overall economic health. Six leading factors explain exchange rate fluctuations:

- Interest rates.
- Inflation rates.
- Political stability.
- Foreign currency reserves.
- Trade account balances and terms of trade.
- Government or public debt.

4.2.1 Interest Rates

Interest rates, inflation and exchange rates are highly correlated. Governments and central banks vary interest rates to manage inflation and currency values. **Higher interest rates offer lenders a higher return relative to other countries, attracting foreign capital inflow, which causes the exchange rate**

to rise. The impact may be moderated by higher inflation in that country relative to others, or by other economic and political factors.

4.2.2 Inflation Rates

A country with a consistently **lower** inflation rate than others will typically see its currency **appreciate**. High inflation tends to produce **depreciation**.

Inflation can have a severe impact on currency value. IMF data for 2017–2018 showed **Venezuela, South Sudan and the Democratic Republic of the Congo** with the three highest inflation rates, all experiencing hyperinflation in 2018. The IMF indicated Venezuela's inflation would have hit **1,000,000% in 2018**. Venezuelans had already converted to USD, gold and even bitcoin as better stores of value, and the Venezuelan Bolivar was considered virtually worthless. Despite launching the New Venezuelan Bolivar in late August 2018, the new currency **fell 95% in value on the day of launch**.

In developed economies, inflation was negligible for most of the decade after the Global Financial Crisis of 2007/2008; several major economies were more concerned about deflation in the years immediately after. With a decade of monetary and fiscal stimulus in the US, Europe and elsewhere, and the recovery of oil prices, inflation returned to the fore.

Country	Exchange rate, September 2018	12-month LIBOR interbank	June 2018 CPI
United States	—	+2.88713	2.55%
Europe	EUR/USD 1.1756	−0.16700	2.00%
United Kingdom	GBP/USD 1.3238	+1.06000	2.37%
Switzerland	USD/CHF 0.9614	−0.51860	0.94%
Japan	USD/JPY 112.5000	+0.13900	0.96%
Canada	USD/CAD 1.2914	+1.78650	3.00%
Australia	AUD/USD 0.7278	+3.33800	2.10%

4.2.3 Political Stability

Investors seek stable countries with strong economies. Countries with stable governments executing sound financial and trade policies attract investment, and currencies tend to appreciate when there are strong capital inflows and demand for domestic investments. The opposite happens with corruption, social unrest, frequent changes in government and inconsistent policies, which lead to lost investor confidence and lower currency demand. Political uncertainty and social unrest also drive outflows from residents who feel it safer to hold wealth in more stable foreign currencies.

4.2.4 Foreign Currency Reserves

A larger foreign currency reserve does not necessarily lead to appreciation, because some governments do not want their currency to strengthen. But countries with strong reserves are generally better positioned to defend and manage their currency levels than those with weak reserves. During the Asian Currency Crisis, hedge funds and other speculators exploited the vulnerability of Asian currencies with weaker reserves, while countries with strong reserves — **Hong Kong, Singapore and Taiwan** — successfully defended their currencies.

Rank	Country	USD billions	As at	Rank	Country	USD billions	As at
1	China	3,480	June 2022	6	Taiwan	548	Dec 2021
2	Japan	1,376	June 2022	7	Hong Kong	504	June 2022
3	Switzerland	1,033	June 2022	8	Saudi Arabia	451	June 2022
4	Russia	630	June 2022	9	South Korea	449	June 2022
5	India	599	June 2022	10	Singapore	365	June 2022

4.2.5 Trade Account Balance and Terms of Trade

A **trade deficit** means the country spends more on imports than it receives from exports, so it must acquire more foreign currency to pay for imports, raising demand for foreign currency. A **trade surplus** means export receipts exceed import expenditure, producing a net increase in demand for the domestic currency. All else equal, a trade deficit should weaken demand for the local currency and a trade surplus should strengthen it.

Terms of trade is the ratio of export prices to import prices. Terms of trade **improve** when export prices rise faster than import prices, implying higher demand for the currency and appreciation. If export prices rise proportionally less than import prices, demand for the domestic currency falls and the currency depreciates.

Export-oriented countries may find upward pressure on the local currency difficult to manage, and a massive inflow from export-driven surpluses can encourage domestic inflation. To offset this, a government can take money out of the system by investing abroad, moderating the impact on its currency. **China** has invested in US Treasuries and invested abroad so the CNY could strengthen more gradually. **India**, which has run a trade deficit for 20 years, saw its deficit widen further in 2018, and the INR fell about **14% in the first nine months of 2018**.

US–China trade and foreign exchange. In 2018 the US government accused major trading partners, particularly China, of enjoying hundreds of billions of dollars of surpluses over the previous decade. The argument was that by keeping the CNY low and not erecting tariff barriers, China artificially made its exports more competitive while limiting imports. To protect US jobs, the US began imposing larger tariff barriers on a wider range of Chinese goods.

4.2.6 Government Debt

Government debt affects both interest rates and currency value. Nations with excessive public debt relative to their income-generating capacity and GDP are usually less attractive to foreign investors. Large debt is often needed to increase liquidity and stimulate the economy when tax and other income is insufficient.

When government debt is denominated in **USD**, a fall in the local currency makes the debt harder to repay. If public debt is repaid by **printing money**, the country can experience massive inflation that further devalues the currency, making further local-currency financing even harder.

Sometimes printing money is not even an option. The **Greek public debt crisis** began after the Prime Minister admitted in **2009** that budget deficit figures had been understated for years and could be double those originally reported, triggering rating downgrades. Because Greece is a Eurozone member and had no monetary policy independence, it could not inflate its way out of debt. With default appearing imminent, many foreign investors sold Greek bonds. Foreign investors are less willing to acquire securities in a country whose government is perceived to have high default risk. Greece has needed **three bail-out programmes totalling 300 billion euros**.

4.3 Foreign Exchange Markets

In the FX market, financial institutions, commercial banks, business firms, governments and central banks, tourists, speculators and investors buy and sell currencies. Unlike the stock or futures market, the FX market is **not confined to a specific geographical location**. Transactions are made through a complex communication network of computer terminals, telephones, faxes, telexes, cable and communication satellites, and personal devices.

The market works **round the clock**: although markets open and close through a 24-hour trading day, dealing runs continuously, starting in **Sydney, then Tokyo, Singapore/Hong Kong, London, New York and San Francisco**.

Market size and composition (BIS Triennial Central Bank Survey, April 2019):

Measure	Value
Average daily FX turnover	**USD6.6 trillion** (up from USD5.1 trillion in April 2016)
Spot trading	**USD2 trillion** per day (up from USD1.7 trillion), but a declining share
FX swaps	**USD3.2 trillion** per day — almost **half** of global FX trading
Outright forwards	**USD999 billion** per day, up **43%**
USD share	On one side of **88%** of all trades — the dominant vehicle currency
EUR share	**32%** (up from 31%)
JPY share	**17%**, down 5 percentage points
Emerging market currencies	**25%** of total turnover
CNY	**USD284 billion** turnover; the world's **eighth** most traded currency

USD-denominated assets also dominate reserves: about **USD6.8 trillion, or 62% of allocated reserves**, in Q2 2019 (IMF COFER). Currency pairs within the **G7** (Canada, France, Germany, Italy, Japan, UK, US)

continue to dominate the global market.

Currency characteristics.

Currency	Character	Behaviour
CHF (and occasionally **JPY**)	Low-volatility, **risk-off** currencies	The market gravitates to them when volatili
SGD	The regional safe haven in Southeast Asia	Sought when risk rises in the region
AUD and **ZAR**	**Commodity currencies**	Particularly strong during higher global grov

4.4 Foreign Exchange Rate Quotations

FX rates can be quoted on a **direct**, **indirect** or **cross** basis, and are usually quoted to the **fourth decimal place**. The last digit is a **pip** (= 0.0001), the smallest unit of price movement or quotation for FX, just as a basis point is for interest rates and fixed income.

4.4.1 Bid-Offer Spread

A bank or authorised FX market maker always provides **two** prices: the **bid**, at which it will buy a currency, and the **offer** (or ask), at which it will sell.

For **USD/SGD 1.2520-25**, the dealer pays SGD 1.2520 to buy 1 USD and sells 1 USD for SGD 1.2525. The difference, **0.0005 or 5 basis points**, is the spread or profit earned by the bank. The spread varies with market conditions and transaction size: **the greater the uncertainty and volatility, the wider the spread**. The party asking for a quote always **buys at the offer and sells at the bid** — in market parlance, the price is always **against the price taker**.

4.4.2 Direct Quotes / European Convention

A **direct quote** expresses the price of one unit of **foreign** currency in units of **domestic** currency: foreign/domestic. Because the USD is the most widely accepted currency, most direct quotes use it as the base:

- USD/CNY 6.6186 — 1 USD = CNY 6.6186
- USD/JPY 101.86 — 1 USD = JPY 101.86
- USD/SGD 1.2520 — 1 USD = SGD 1.2520

4.4.3 Indirect Quotes / American Convention

Currencies in the old **sterling bloc** — **GBP, AUD and NZD** — are the base currency when quoted against the USD, which becomes the counter currency. The **EUR** is quoted the same way:

- GBP/USD 1.6816 — 1 GBP = USD 1.6816
- AUD/USD 0.9235 — 1 AUD = USD 0.9235
- NZD/USD 0.8630 — 1 NZD = USD 0.8630
- EUR/USD 1.3622 — 1 EUR = USD 1.3622

An **indirect quote** expresses the price of one unit of **domestic** currency in units of **foreign** currency: domestic/foreign. So GBP/USD, as typically quoted, is an indirect quote **in the UK**.

Whether a quote is direct or indirect therefore depends on context — specifically, which currency is domestic and which is foreign. The indirect quote is simply the **reciprocal** of the direct quote: in Singapore, if the direct quote is USD/SGD 1.2520, the indirect quote is SGD/USD **0.7987** (1/1.2520).

4.4.4 Cross Quotes

Most FX transactions involve the USD, but the growing importance of other major currencies has produced direct quotation of two non-USD currencies. A **cross-currency rate**, or **cross rate**, refers to currency pairs that do not involve the USD. The idea is that two exchange rates sharing a common currency allow the rate between the remaining two to be calculated.

The general rule: divide the term currency on the same side of the spread by the base currency on the opposite side of the spread. To get the SGD/JPY **bid**, divide the USD/JPY **bid** by the USD/SGD **ask**. To get the SGD/JPY **ask**, divide the USD/JPY **ask** by the USD/SGD **bid**.

Currency pair	Bid	Ask
USD/SGD	1.3362	1.3382
USD/JPY	112.6530	112.7330
SGD/JPY	**84.1825**	**84.3684**

- SGD/JPY bid = 112.6530 / 1.3382 = **84.1825**
- SGD/JPY ask = 112.7330 / 1.3362 = **84.3684**

When the common currency is the term currency in both quotes, multiply on the same side of the spread. To get the GBP/SGD bid, multiply the GBP/USD bid by the USD/SGD bid; to get the GBP/SGD ask, multiply the GBP/USD ask by the USD/SGD ask.

Currency pair	Bid	Ask	
GBP/USD	1.3501	1.3510	GBP is base, USD is term
USD/SGD	1.3362	1.3382	USD is base, SGD is term
GBP/SGD	**1.8040**	**1.8079**	USD cancels out

- GBP/SGD bid = 1.3501 × 1.3362 = **1.8040** — the rate at which the market buys GBP and sells SGD.
- GBP/SGD ask = 1.3510 × 1.3382 = **1.8079** — the rate at which the market sells GBP and buys SGD.

The two rules are the same rule seen from different sides: whichever currency is common must cancel, so it is divided out when it is the base of the second quote and multiplied through when it is the term of both.

4.4.5 Settlement Dates

A **spot** transaction settles in **two working days**. A **forward** transaction is a contract to exchange one currency for another at a future date later than spot, at a rate determined on the trade date. The purpose of a forward is to establish and fix the cost of exchange for a known FX commitment at a future date.

4.5 Forward Foreign Exchange Contracts

An investor enters a forward exchange contract for one of two reasons:

- To **hedge** a currency exposure.
- To **speculate** by taking a position in the currency.

An investor who has bought or sold foreign securities may cover the FX exposure by fixing a forward rate. By locking in a known forward rate, the investor also **forfeits potential gains** from favourable market movements. A single forward transaction of this kind is an **outright**.

Some investors or traders have no initial exposure but hope to profit from a view on the currency's direction: they buy the forward position if the currency is expected to appreciate, and sell if it is expected to depreciate.

4.5.1 Forward Rates

A forward transaction has two components:

- The **spot exchange rate**.
- The **interest differential**, expressed in FX rate terms.

The forward rate therefore depends on the movement of the spot rate of the pair and the respective interest rates of the two currencies. The 3-month USD/SGD forward rate depends on the USD/SGD spot rate, the 3-month USD rate and the 3-month SGD rate.

The interest differential is converted into **forward (or swap) points**, which are added to or subtracted from the spot rate:

$$\text{Forward exchange rate} = \text{Spot rate} \pm \text{Forward points}$$

Forward points	Term	Meaning
Added to spot	Forward **premium**	Forward rate more expensive than spot
Subtracted from spot	Forward **discount**	Forward rate cheaper than spot

Generally, the currency of the country with the **higher** interest rate trades at a **forward discount**, and the currency with the **lower** interest rate trades at a **forward premium**.

4.5.2 Calculating Forward Points

$$\text{Forward Points} = \frac{S \times (R_c - R_b) \times n/360}{1 + (R_b \times n/360)}$$

where

$$S$$

is the spot rate,

$$R_c$$

is the annualised interest rate of the **counter** currency,

$$R_b$$

is the annualised interest rate of the **base** currency, and

$$n$$

is the number of days from the spot date to the forward date.

Sometimes supply or demand for outright forwards is so great that the forward rate does not reflect the interest differential. When that happens there is room for **arbitrage between the money market and the FX market**. Banks also run proprietary trading of the forward FX market, taking positions based on expectations of swap-point movements — the relative movement of the two currencies' interest rates — known as **forex gap trading**.

Forward points reflect **only** relative interest rates. They are **not** an indication of what the future spot rate will actually be.

Worked example. Spot USD/SGD = 1.2520,

$$R_b = R_{USD} = 1.75\%$$

p.a.,

$$R_c = R_{SGD} = 0.75\%$$

p.a.,

$$n = 30$$

days.

$$\text{Forward points} = \frac{1.2520 \times (0.0075 - 0.0175) \times 30/360}{1 + (0.0175 \times 30/360)} = \frac{-0.0010}{1.0015} = -\text{SGD } 0.0010$$

That is **−10 points**, so the 30-day USD/SGD forward is quoted at a **discount** of **USD/SGD 1.2510**. The USD has the higher interest rate, so it trades at a forward discount — consistent with the rule above.

4.5.3 Interest Rate Parity

The **interest rate parity** theorem states that, except for small transaction costs, the forward premium or discount should equal the difference in national interest rates for securities of the same maturity. If the relationship is violated, arbitrageurs can make risk-free profits, and their actions force the forward and spot rates back into alignment.

$$F = S\left[\frac{1 + R_c\left(\frac{n}{360}\right)}{1 + R_b\left(\frac{n}{360}\right)}\right]$$

where

$$F$$

is the forward rate,

$$S$$

the spot rate,

$$R_c$$

the annualised counter-currency rate,

$$R_b$$

the annualised base-currency rate, and

$$n$$

the number of days.

Worked example. With

$$R_{USD} = 0.06$$

,

$$R_{GBP} = 0.05$$

and GBP spot at USD 1.50, the 1-year forward is:

$$F = 1.50 \times \frac{1.06}{1.05} = \text{USD } 1.5143 \text{ per GBP}$$

Covered interest arbitrage. Suppose the forward price is quoted at **USD 1.5100** instead of 1.5143 — parity is violated. The arbitrage:

Step	Action	Amount
1	Borrow GBP 1,000,000 for 1 year at 5%	Repayment due: **GBP 1,050,000**
2	Sell the borrowed GBP at spot USD 1.50	Receive **USD 1,500,000**
3	Invest the USD for 1 year at 6%	Receive **USD 1,590,000**
4	Buy GBP 1,050,000 forward at USD 1.5100	Pay **USD 1,585,500**
5	Close out: receive USD 1,590,000, pay USD 1,585,500, repay GBP 1,050,000	**Profit USD 4,500**

The net position is zero because all transactions close out, but the arbitrageur keeps **USD 4,500**. Effectively they sold GBP at 1.5143 (1,590,000 / 1,050,000) and bought it forward at the lower 1.5100. The whole process — borrowing in one currency, investing in another, and covering the exchange position forward — is **covered interest arbitrage**.

4.5.4 Forward Rate Quotes

Most market information services show forward rates not as outright rates but as **margins in points** above or below spot.

	Spot	1-month	2-month
USD/JPY	119.22/25	52/51	106/104
AUD/USD	0.6475/80	17/21	36/43

Forward points often carry no sign. Like spot rates, market makers quote forward points on a bid-offer spread. For a currency at a **forward discount**, the bid (left) is **higher** than the offer (right): the market maker buys at a deeper discount and sells at a lesser discount. For a currency at a **forward premium**, the bid is **lower** than the offer: the maker buys at a lesser premium and sells at a higher premium.

To derive outright rates:

- **Subtract** the points from the corresponding spot rates if the points are **descending** (left higher than right, e.g. 52/51).
- **Add** the points if they are **ascending** (left lower than right, e.g. 17/21).

	Spot	1-month	Outright
USD/JPY	119.22/25	−52/−51	**119.2148 / 119.2449**
AUD/USD	0.6475/80	+17/+21	**0.6492 / 0.6501**

Intuitively, to maintain a bid-offer spread the market maker always quotes the **lower-valued** forward point first, followed by the higher. That can only be true if 52/51 are negative, that is −52/−51.

4.5.5 Non-Deliverable Forwards

Some countries impose tight capital and exchange controls to reduce currency outflow and prevent exchange-rate volatility. The currency market in those countries often has low liquidity and is inaccessible because of the restrictions. A **non-deliverable forward (NDF)** is a financial instrument commonly used to hedge exposure to currencies offshore that are not internationally traded, or whose trade is limited or legally restricted to the domestic market. The NDF market is an **OTC** market that developed over the last two decades with the growth of emerging economies such as **China, Taiwan, Vietnam and Indonesia**.

An NDF is similar to a regular FX forward, with one key difference: there is **no physical delivery or exchange of the domestic currency** in the offshore market. An NDF is a **cash-settled, short-term forward contract** settled at an agreed forward rate at maturity in a freely traded currency. NDFs are typically quoted with **USD as the reference currency**. The only exchange of cash flows, or calculation of profit and loss, is based on the difference between the agreed (contracted NDF) rate and the spot rate prevailing at settlement, with **no exchange of the notional amount**. The normal tenor ranges from **1 month to a year**.

Besides hedging currency risk in countries with rigid capital controls, NDFs provide arbitrage opportunities and allow speculative positions.

Worked example. A Singapore client imports goods from China. The supplier quotes in CNY but requires payment in USD at the prevailing rate. The invoice shows the USD equivalent of **CNY 10 million**, payable in 3 months. Current USD/CNY spot is **6.5845**.

	Current rate 6.5845	If USD/CNY rises 5% to 6.9137	If USD/CNY falls 5% to 6.2553
CNY 10 million in USD	USD 1,518,718	USD 1,446,404	USD 1,598,644
FX gain (loss)	—	FX saving: **USD 72,314**	FX loss: **USD 79,926**

If the client does nothing, the risk is that the **USD depreciates against the CNY**: at 6.2553 the CNY 10 million would require an additional **USD 79,926** in three months.

To remove the uncertainty, the client enters a **3-month USD/CNY NDF to sell USD (buy CNY)**, fixing USD/CNY at 6.5845 for the CNY 10 million notional. If USD/CNY falls about 5% to 6.2553, the bank pays the client about **USD 79,926**, offsetting the extra amount payable to the supplier. If instead the CNY falls and USD/CNY rises to 6.9137, the client pays the bank **USD 72,314** on the notional loss, offset by the USD 72,314 less needed for the supplier. The NDF fixes the rate at a comfortable level and removes the uncertainty.

4.6 Risks in Foreign Exchange

Most FX trading is very short-term, with proprietary trading making profits and losses over periods as brief as minutes. Four risks must be managed:

Risk	Description
Market risk	Movement and volatility of FX rates
Price liquidity	In less-traded currencies, and even in major currencies during an extreme market event such
Credit risk	The counterparty fails to honour the deal. In FX, credit risk is only a **fraction** of the total
Settlement/delivery risk	The counterparty fails to pay on time. This is a specific counterparty credit risk occurring i

4.7 Managing Foreign Exchange Risks

Several strategies reduce exchange-rate risk. Some — currency intervention and controls — can only be administered by governments or central banks. Besides money market instruments and exchange-traded FX derivatives such as forwards and futures, firms and individuals use options and swaps to hedge risk and exploit opportunities. Futures, options and swaps are covered in the CM-SIP syllabus; this chapter covers currency forwards and the alternative strategies below.

4.7.1 Exchange Rate Regimes

An **exchange rate regime** is the policy framework a central bank adopts to manage its currency against other currencies and the FX market. Each country is free to choose a regime aligned with its monetary or fiscal policy. Most choose a fixed rate pegged to a dominant world currency, a floating rate, or a variant in between such as a managed trade-weighted float.

Regime	Description
Fixed or pegged	The central bank or government manages its currency against a m
Managed peg (soft peg)	No official fixing, but the government uses another currency as a r
Trade-weighted band (trade-weighted peg or float)	Managed within a trading band based on a trade-weighted basket
Floating (or managed free float)	The currency floats freely and adjusts to demand and supply with

Other arrangements.

- **Currency substitution or dollarisation** — a country adopts another nation's currency as legal tender. The adopting country inherits the **credibility** of that currency, but **not** that country's creditworthiness.
- **Monetary (currency) union** — several countries share a common currency (EUR). Such unions are difficult because some countries and governments may be more extravagant than others.
- **Currency board** — an institutional arrangement based on full convertibility of a local currency into a reserve currency, backed by foreign currency reserves (HKD).

Singapore. The SGD is managed as a **trade-weighted currency band** based on Singapore's major trading partners and competitors. MAS manages the SGD on a **managed float within an undisclosed currency band**, which provides flexibility for short-term market fluctuations, and the band is reviewed from time to time against economic fundamentals. Under a **Currency Interchangeability Agreement in 1967**, the Brunei Dollar is interchangeable with the Singapore Dollar **at par**, and each is legal tender in the other's country.

The cost of defending a rate. It can be costly for a central bank to attain a desired exchange rate. The level of foreign currency reserves, capital outflows, economic fundamentals and market conditions all make it challenging. When the Chinese government moved towards a managed trade-weighted approach in **late 2015**, major currency outflows and pressure on the CNY followed. From **July 2014 to December 2016**, USD/CNY rose only about **11%**, from CNY 6.20 to CNY 6.90, but China ran down its foreign currency reserves by almost **USD 1 trillion (25%)** to defend the CNY.

4.7.2 Hedging Currency Risk by Borrowing in Similar Currency

Investors and entrepreneurs often borrow in the currency of the investment to mitigate a fall in that currency. A Singapore investor in UK property would establish borrowings in **GBP**: if GBP weakens, the GBP borrowing is a **natural hedge** against the GBP exposure, and because the investment was not funded in SGD, the impact of the fall is less damaging.

Worked example. A Singaporean firm wants to buy a **GBP 1 million** UK property and must borrow either SGD or GBP to fund **80%** of the value. GBP/SGD is **1.8060** now and the GBP depreciates **5% to 1.7157** in a year. Ignoring interest and transaction costs, and assuming the property is sold after a year at the same price:

If the firm borrows in SGD:

Item	Calculation	Amount
Down payment	$0.2 \times 1{,}000{,}000 \times 1.8060$	SGD 361,200

Item	Calculation	Amount
SGD loan	$0.8 \times 1{,}000{,}000 \times 1.8060$	SGD 1,444,800
Sale proceeds after 1 year	$1{,}000{,}000 \times 1.7157$	SGD 1,715,700
Total profit/(loss)	$1{,}715{,}700 - 361{,}200 - 1{,}444{,}800$	**−SGD 90,300**

The firm had to borrow SGD 1,444,800 and find another SGD 361,200 for the down payment, a total outlay of **SGD 1,806,000**, which it converted into GBP 1 million at 1.8060 to buy the house. After a year the house is sold for GBP 1 million, converted into SGD 1,715,700 at 1.7157, and the SGD 1,444,800 loan is repaid.

If the firm borrows in GBP:

Item	Calculation	Amount
Down payment	$0.2 \times 1{,}000{,}000 \times 1.8060$	SGD 361,200
Net proceeds after repaying the GBP 800,000 loan	$(1{,}000{,}000 - 800{,}000) \times 1.7157$	SGD 343,140
Total profit/(loss)	$343{,}140 - 361{,}200$	**−SGD 18,060**

The firm is better off borrowing in **GBP**, because the loss is smaller — **SGD 18,060** against SGD 90,300. The GBP loan hedged the currency exposure.

4.7.3 Carry Trades

A **carry trade** borrows at a low interest rate to invest in an asset likely to provide a higher return. A currency carry trade sells a currency with a relatively low interest rate and uses the funds to buy a different currency with higher yields. While paying low interest on the currency borrowed or sold, the investor captures the **interest rate differential**.

The strategy relies on **relative stability in currency prices**, because an adverse exchange-rate movement can wipe out the returns from the interest differential. Where possible, carry trades should be structured on stable currency pairs, or when one is confident of weakness in the borrowing currency. If the invested currency weakens, the investor loses on both the additional interest incurred and the FX loss. Investments become highly speculative when using currencies facing high inflation and weak economies with a history of weakness and devaluation.

Worked example — borrowing in JPY and investing in AUD, 2008–2016. A Singapore investor borrows **JPY 100 million** each year and invests in Australian corporate bonds, creating a **350–400+ basis point** spread annually over the Japanese loan rate. The strategy worked in most years but failed in **2008, 2011 and 2015**, when the AUD (the invested currency) fell sharply against the JPY (the borrowed currency).

	2008	2009	2010	2011	2012	2013	2014	2015	2
Yield on Australian bond	7.00	5.50	6.00	6.60	6.00	5.50	5.15	4.75	4
Average JPY loan rate	2.25	2.00	1.65	1.50	1.25	1.25	1.20	1.00	0
AUD/SGD start of year	1.2611	1.0178	1.2636	1.3142	1.3286	1.2831	1.1284	1.0523	1
AUD/SGD end of year	1.0178	1.2636	1.3142	1.3286	1.2831	1.1284	1.0523	1.0233	1
SGD/JPY start of year	77.7445	63.3266	66.1700	63.1530	59.2623	71.3046	82.9152	90.4936	8
SGD/JPY end of year	63.3266	66.1700	63.1530	59.2623	71.3046	82.9152	90.4936	83.7575	8
Net gain (loss), SGD	**−(503,946)**	526,960	56,409	**−(6,240)**	307,507	80,035	64,272	**−(80,223)**	7

The mechanics of the table, for 2008 as the worked year:

Step	2008	2009	2010	2011	2012	2013	20
Borrowing in JPY	100,000,000	100,000,000	100,000,000	100,000,000	100,000,000	100,000,000	10
Interest cost (JPY)	2,250,000	2,000,000	1,650,000	1,500,000	1,250,000	1,250,000	1,
Total repayment due (JPY)	102,250,000	102,000,000	101,650,000	101,500,000	101,250,000	101,250,000	10
Initial borrowed (SGD)	1,286,265	1,579,115	1,511,260	1,583,457	1,687,415	1,402,434	1,

Step	2008	2009	2010	2011	2012	2013	20
Japanese loan + interest (SGD)	1,614,645	1,541,485	1,609,584	1,712,726	1,419,964	1,221,128	1,
Initial AUD investment (AUD)	1,019,923	1,551,559	1,195,958	1,204,910	1,270,079	1,092,970	1,
Total proceeds in AUD	1,091,317	1,636,895	1,267,715	1,284,434	1,346,284	1,153,084	1,
AUD proceeds year end (SGD)	1,110,699	2,068,445	1,665,993	1,706,486	1,727,471	1,301,163	1,

In **2008**, JPY 100 million was borrowed at **2.25%**, converted into **AUD 1.020 million (SGD 1.286 million)** and invested at **7%**. Over the year the AUD fell **34.2% against the JPY** and **19% against the SGD**. Total Australian proceeds, converted back to SGD, were **SGD 1.111 million** — lower than the **SGD 1.615 million** equivalent of the JPY principal and interest. The carry trade lost money because the invested currency weakened and the borrowed currency strengthened.

The higher the differential between the AUD investment and the JPY borrowing cost, the better for the investor. As long as the borrowed currency stays flat or depreciates against the invested currency, the carry trade works.

4.8 Summary

- The leading factors explaining exchange-rate fluctuations are interest rates, inflation rates, political stability, foreign currency reserves, trade account balances and terms of trade, and government or public debt.
- Higher interest rates offer lenders a higher return relative to other countries, attracting foreign capital inflow and causing the exchange rate to rise.
- A country with consistently lower inflation than others typically sees its currency appreciate; high inflation tends to produce depreciation.
- Countries with stable governments executing sound financial and trade policies attract investment, and currencies tend to appreciate with strong capital inflows and demand for domestic investments. The opposite happens with corruption, social unrest, frequent government change and inconsistent policy.
- Countries with strong foreign currency reserves are better positioned to defend and manage their currency levels than those with weak reserves.
- If public debt is repaid by printing money, the country can experience massive inflation that further devalues the currency and makes local-currency financing even harder.
- An FX transaction is the purchase of one currency with the simultaneous sale of another for settlement or delivery on a specified date. Rates can be quoted on a direct, indirect or cross basis.
- A spot transaction exchanges one currency for another for settlement usually in two working days. A forward transaction exchanges currencies at a future date at a rate determined on the trade date.
- An investor may enter a forward exchange contract to hedge a currency exposure or to speculate by taking a currency position.
- The major factor determining the forward rate is the difference in interest rates in the two countries over the forward period. The interest differential converts into forward or swap points, added to or subtracted from the spot rate.

Exam cues

- **Base currency first, counter (term) currency second.** USD/SGD 1.2520 means USD 1 = SGD 1.2520. The base currency is the investor's reference currency, and all profit and loss is measured in it.
- **Pip = 0.0001**, the fourth decimal place — the FX equivalent of a basis point.
- **Bid is where the dealer buys; offer is where the dealer sells.** The price taker always **buys at the offer and sells at the bid**. USD/SGD 1.2520-25 → spread **5 basis points**. Wider spread when volatility and uncertainty rise.
- **Direct quote = foreign/domestic** (USD/SGD in Singapore). **Indirect quote = domestic/foreign** (GBP/USD in the UK). The indirect quote is the **reciprocal** of the direct: 1/1.2520 = **0.7987**.
- **Sterling-bloc currencies and the EUR are quoted as base against the USD:** GBP/USD, AUD/USD, NZD/USD, EUR/USD.
- **Cross-rate rule:** divide the term currency on the same side by the base currency on the **opposite** side. SGD/JPY bid = USD/JPY bid ÷ USD/SGD ask = 112.6530/1.3382 = **84.1825**; ask = 112.7330/1.3362 = **84.3684**.
- **Spot settles T+2.** Value today and value tomorrow also exist.

- **Forward = spot ± forward points.** Points **added** = forward **premium**; **subtracted** = forward **discount**. The **higher-interest-rate currency trades at a forward discount**; the lower-interest-rate currency at a premium.
- **Forward points formula:**
$$S(R_c - R_b)(n/360)/[1 + R_b(n/360)]$$
. Worked: spot 1.2520, USD 1.75%, SGD 0.75%, 30 days → **−10 points**, forward **1.2510**.
- **Forward points reflect relative interest rates only** — they are not a forecast of the future spot rate.
- **Interest rate parity:**
$$F = S[(1 + R_c n/360)/(1 + R_b n/360)]$$
. Worked: spot 1.50, USD 6%, GBP 5% → **1.5143**.
- **Covered interest arbitrage:** borrow GBP 1m at 5%, sell spot at 1.50, invest USD at 6%, buy GBP forward at 1.5100 → **profit USD 4,500**.
- **Descending points (52/51) are negative — subtract.** Ascending points (17/21) are positive — add. USD/JPY 119.22/25 with −52/−51 → **119.2148/119.2449**; AUD/USD 0.6475/80 with +17/+21 → **0.6492/0.6501**.
- **NDF:** cash-settled, short-term forward, **no delivery of the domestic currency**, no exchange of notional, USD reference currency, tenor **1 month to 1 year**. Used for currencies with capital controls (China, Taiwan, Vietnam, Indonesia).
- **NDF worked example:** CNY 10m at 6.5845 = USD 1,518,718. At 6.2553 → USD 1,598,644 (**loss USD 79,926**); at 6.9137 → USD 1,446,404 (**saving USD 72,314**).
- **Four FX risks:** market, price liquidity, credit, settlement/delivery. **Credit risk is only a fraction of the transaction amount** because FX is an exchange, not a loan. **Settlement risk arises 2 days before payment**; **CLS** reduces it.
- **Four exchange-rate regimes:** fixed peg, managed soft peg, trade-weighted band, free float. Plus dollarisation, monetary union and currency board.
- **Singapore:** trade-weighted band, managed float within an **undisclosed** band, reviewed periodically. **Brunei–Singapore Currency Interchangeability Agreement 1967 — at par.**
- **China's defence cost:** July 2014–December 2016, USD/CNY rose only about **11%** (6.20 → 6.90) while reserves fell almost **USD 1 trillion (25%)**.
- **Borrowing in the investment currency is a natural hedge.** Worked: GBP 1m property, 80% borrowed, GBP falls 5% from 1.8060 to 1.7157 → loss **SGD 90,300** if borrowing SGD, **SGD 18,060** if borrowing GBP.
- **Carry trade:** borrow low-yield, invest high-yield, capture the differential — but an adverse FX move wipes it out. JPY/AUD 2008–2016: failed in **2008, 2011, 2015**. In 2008 the AUD fell **34.2% vs JPY** and **19% vs SGD**, producing a **SGD 503,946 loss**.
- **Reserve rankings (June 2022):** China 3,480; Japan 1,376; Switzerland 1,033; Russia 630; India 599; Taiwan 548; Hong Kong 504; Saudi Arabia 451; South Korea 449; Singapore 365 (USD billions).
- **BIS 2019:** daily FX turnover **USD6.6 trillion**; FX swaps **USD3.2 trillion** (almost half); spot **USD2 trillion**; USD on one side of **88%** of trades; EUR **32%**; JPY **17%**; EM currencies **25%**; CNY **USD284 billion**, eighth most traded.
- **Trap:** a **direct** quote in one country is an **indirect** quote in another. The label depends on which currency is domestic — never answer without establishing the home currency.
- **Trap:** forward points are **not** a prediction of the future spot rate. They are the interest differential expressed in FX terms.
- **Trap:** a **higher** interest rate currency is at a forward **discount**, not a premium — the higher rate is offset by the cheaper forward rate.

Chapter 5 — Company Analysis and Understanding Financial Statements

Company analysis serves both equity and credit work: any security whose payments depend on an issuer's continuing viability is analysed through the same three statements. This chapter covers the qualitative advantages that precede the numbers, the structure of the balance sheet, income statement and cash flow statement, the per-share calculations, and the four families of ratio the examination tests.

Learning Objectives

The candidate should be able to:

- Explain company analysis in the context of fundamental analysis.
- Read and interpret the three major financial statements — the balance sheet, the income statement and the statement of changes in cash flow.
- Analyse a company's financial performance using financial ratios.
- Explain the liquidity of a company using the current, quick and cash ratios.
- Explain financial risk and leverage using the debt-to-equity ratio, total debt to total assets ratio, interest coverage and total fixed charge coverage.
- Analyse operating efficiency using total asset turnover, inventory turnover, days to sell inventory, receivables turnover and the collection period for accounts receivables.
- Analyse profitability in relation to sales or capital using gross profit margin, operating profit margin, net profit margin, return on assets, return on equity and earnings per share.
- Understand the use of valuation ratios in estimating the value of a share.
- Understand what good quality of financial statements and good quality of earnings mean.
- Explain the importance of comparative and trend analysis.
- Identify the limitations in the use of financial ratios.

5.1 Introduction

Company analysis does not relate only to equity analysis. It is equally relevant to **credit analysis** of bond investments and of any other security whose cash flow payments depend on the ongoing viability of the issuing company.

Company analysis divides into two parts:

- **Qualitative analysis.**
- **Quantitative analysis** (financial statement analysis).

Three principles govern the work:

- **Asset prices move on future prospects.** Historical statements show recent profitability and how the company has been managed. Historical numbers, ratios and relationships are used to make projections given the outlook. Prices move when expectations about future performance change.
- **Qualitative factors drive performance.** Price movements are first driven by news and information bearing on prospects. It matters more to focus on the qualitative conditions that allow success than to rely on ratios alone, because those conditions drive and precede the quantitative outcomes.
- **Excellent companies can look expensive and still be good investments.** Companies with strong business models and excellent earnings growth may see price run ahead of earnings in the near term. If the positive qualitative factors persist and steady, sustainable earnings growth is delivered, earnings catch up with valuations and prices rise. Buying weaker or loss-making companies that have been heavily sold down brings, at best, short-term trading opportunities; at worst it traps investors as prospects keep deteriorating.

5.2 Qualitative Analysis

Qualitative analysis identifies the characteristics that let a company outperform its peers. Eight competitive advantages are examinable.

#	Competitive advantage	Companies exhibiting it
1	Dominance and control of key technology, research and patents	Apple, Nvidia, Pfizer, Johnson & Johnson, Novartis
2	Control over key resource supply	De Beers, Bayer-Monsanto
3	Superior brand differentiation	Apple, Coca-Cola, Microsoft, BMW, Mercedes, Sam
4	Control of distribution channels	Alibaba, Amazon, Tencent
5	Cost efficiency, lowest-cost producer	Xiaomi, TSMC, McDonald's, IKEA
6	Successful focus on niche clients	Harley Davidson, Swiss private banks (UBS, Credit
7	Superior management, talent, connections	Apple, Alibaba, Microsoft, Berkshire Hathaway, Ch
8	Diversified core businesses with shared costs	General Electric, 3M, Honeywell

Technology, research and patents. Control over technology lets a company maintain higher prices and margins while growing revenue. Patented technology can give an almost monopolistic influence as the sole supplier in the high-end range, establishing significant price premiums. Apple continues to command a premium over other smartphone producers; Nvidia, the top-end chipmaker designing GPUs for gaming, mobile computing and automotive, enjoys higher margins than competitors because it focuses on top-end chips.

Dominance over supply. Substantial control over raw materials and resources is an advantage. Despite giving up almost monopolistic control of diamond supply decades ago, **De Beers still controls about 30–35% of world diamond supply**. Monsanto, a leading producer of genetically modified seeds, was acquired by Bayer, another leading producer; the combined group should hold more than half the GMO seed crop market.

Superior brand differentiation. A strong brand lets products command higher pricing, better sales and better margins. Luxury companies are especially conscious of brand image. Some brands are known for precision, quality and dependability (Mercedes, BMW, Honda, Toyota); others for superior service standards (Singapore Airlines).

Controlling the distribution channel. Dominating the sales channel is a critical advantage. The need for costly physical infrastructure is now less of a barrier because of the internet, which has redefined how companies market, communicate, collect payments and deliver. Traditional brick-and-mortar retailers have been forced into a blended approach. Companies with a first-mover advantage on internet platforms (Alibaba, Amazon, Tencent) take a larger share of sales commissions and advertising revenue on top of selling their own products.

Cost efficiency, lowest-cost producer. Some companies compete by being the most efficient producers, driving volume on very low margins through efficiency and economies of scale. Some have become outsourcing partners for manufacturers that no longer want to bear manufacturing cost and retooling risk. These partners have lower cost and scalability, operate on thin margins, and offer economies by aggregating orders from different companies into high volumes. The infrastructure cost needed for scale becomes a natural barrier to entry. Examples: **Taiwan Semiconductor Manufacturing, Foxconn and Samsung**, major subcontractors for other technology companies.

Successful focus on niche clients or products. Specialising in a particular client group often commands premium fees and pricing. Harley Davidson caters to a high-end niche of motorcycle enthusiasts; Swiss private banks and luxury goods manufacturers have long focused on high net worth clients.

Superior management, talent and relationships. Strong management and founders with foresight, expertise or relationships are hard to emulate. Entrepreneurial acumen and creative genius cannot easily be transferred, letting some companies dominate their fields. Others benefit from management's relationships with markets and governments. Examples: Steve Jobs and Apple, Bill Gates and Microsoft, Warren Buffett and Berkshire Hathaway, Jack Ma and Alibaba, Li Ka-shing and Cheung Kong Hutchinson Holdings.

Diversified core businesses with shared costs. Some companies regularly identify new core businesses, typically holding **3–4 core businesses** where they have a competitive advantage and lead market share. Resources focus on the cores, with a smaller portion for new business development. Sharing common costs (administrative, marketing, human resources) lets businesses compete more aggressively on pricing and leverage shared capabilities. When one business hits a cyclical downturn, the others support earnings until it recovers. Typically one or two cores become **cash cows** funding new growth. As margins narrow with industry maturity and product life cycles reach their tail end, promising peripheral businesses are converted into new cores. Examples: General Electric, Honeywell, 3M.

5.3 Quantitative Analysis (Financial Statements Analysis)

Financial statements show the resources available to management, how those resources are financed, and what the company has accomplished with them. They are prepared for many parties: creditors, shareholders and investors, tax authorities, industry regulators, senior managers, the board and employees.

Three statements are analysed:

- The **balance sheet**.
- The **income statement**.
- The **cash flow statement**.

5.3.1 Historical Financial Statements

A company's current value depends on potential earnings and future cash flow, not historical data. Why then examine history?

For most companies the most recent historical performance is still the best reference point for near-term forecasts. Although "history is not necessarily a good predictor of the future", the odds tilt in the investor's favour when investing in companies with a track record of solid earnings and financials. Companies that have faced difficulties recently are generally more likely to keep encountering problems unless conditions change significantly.

Analysing past growth and ratios helps identify leading companies with the right strategies and competitive advantages. Analysts combine that with research on future industry demand and supply to project earnings better.

5.3.2 Presenting Financial Statements

Financial statements must be **audited**. The investment community expects auditors to make qualifying comments where statements are not well prepared. **Qualified reports** are issued when auditors are not completely satisfied with how the statements are prepared.

Market	Reporting frequency	Accounting sta
United States	Quarterly statements to shareholders and the investment community is the norm	**US GAAP**, a
Europe	Semi-annual and annual statements are the minimum in some markets	**IFRS** for all li
Singapore	Most companies currently provide **quarterly** reporting in the interest of transparency	**IFRS**

Because US GAAP and IFRS differ, **adjustments and restatements** are needed to compare companies across markets properly.

5.3.3 Reviewing Financial Statements

Four steps:

- Examine key ratios over time and explain the reasons behind the company's performance and financial health.
- Forecast likely future performance from the research outlook on the economy and industry, key performance drivers, and interviews with management and industry experts.
- Compare financial health and projected performance with other companies in the same industry or league.
- Estimate fair value given future earnings potential relative to similar companies and general market conditions.

5.4 The Balance Sheet

The balance sheet shows the financial condition of a company **at a point in time**, typically at the financial year-end. It has three parts:

- **Assets** — the resources owned: cash, accounts receivables, inventories, motor vehicles, property, plant and equipment.
- **Liabilities** — amounts owed to banks, suppliers, vendors and other non-bank parties: accounts payables, bank loans, overdrafts.
- **Shareholders' equity** — total assets minus total liabilities; the amount financed using ordinary and preferred shares. It equals **share capital plus retained earnings minus treasury shares**.

Jurisdiction	Presentation	Equation
US	Assets on one side; liabilities and shareholders' equity summing to the same total on the other	Assets = L
British	Assets net of liabilities (net assets) on one side, equating to shareholders' equity on the other	Assets − L

Worked illustration — Singapore Airlines. SIA's financial year runs **1 April to 31 March**. The statements used are for FY2020/2021, FY2019/2020 and FY2018/2019; competitor airlines use a calendar

year. The analysis spans the COVID pandemic, which came to global prominence in **Q1 2020** and hit the airline industry severely. Government intervention differed across airlines, which complicates trend analysis and peer comparison.

SIA's statement of financial position, recast into standard US format (S$ million):

Item	2021	2020	2019
Share capital	7,180.2	1,856.1	1,856.1
Mandatory convertible bonds	3,496.1	—	—
Treasury shares	(133.2)	(156.0)	(171.5)
Other reserves	5,362.8	7,614.0	11,602.2
Equity attributable to owners	15,905.9	9,314.1	13,286.8
Non-controlling interests	372.2	418.6	396.4
Total equity	**16,278.1**	**9,732.7**	**13,683.2**
Total long-term liabilities	15,590.0	12,978.0	9,443.6
Total current liabilities	5,713.2	11,002.1	7,378.4
Total long-term assets	27,909.3	28,869.9	25,005.5
Total current assets	9,672.0	4,842.9	5,499.7
Total assets	**37,581.3**	**33,712.8**	**30,505.2**

The common size statement expresses every item as a percentage of total assets (100%), which makes balance-sheet items comparable across companies. For SIA in 2021: property, plant and equipment **62.5%** of total assets; borrowings (long-term) **28.1%**; cash and bank balances **20.7%**; total equity **43.3%**.

When examining a balance sheet, focus on:

- The **largest items** on both sides.
- The **relationship between assets and liabilities**.
- **Significant trends** in the accounts over time.

5.5 Assets

Assets convertible to cash within a year are **current assets**; the rest are **long-term or non-current assets**. The amount of current assets relative to current liabilities reflects short-term liquidity. Ratios compare current assets against current liabilities, against overall liabilities, and as a proportion of total assets.

The questions of interest are how assets are allocated for productive use to generate earnings, whether asset values will change over time, and what mix of debt and equity funds them.

5.5.1 Cash and Bank Balances

Some companies hold a large share of earnings in cash because of the liquidity, risk and operational needs of their industry. In cyclical downturns a large cash balance accumulated from past earnings lets a company keep funding the business and meet short-term obligations. Excess cash also lets it buy raw materials cheaply and invest directly in technology, property, equipment and businesses without extra debt or asset sales.

Airline	Cash / total assets (December 2020)
IAG	19.1%
Cathay Pacific	9.5%
American Airlines	0.4%
SIA (March 2021)	**20.7%**

SIA maintained a cash war chest much higher than most airlines for several years. More recently, proceeds from capital raised at the height of the pandemic in mid-2020 (rights issue and mandatory convertible bonds) were kept mostly in cash, presumably as a war chest to conserve liquidity should the downturn be protracted, or for future investment in plant and machinery when the market recovers.

Note that IAG's ”high” cash ratio results from a **reduced balance sheet** (denominator), driven largely by a decline in shareholders' equity from losses and reduced equity reserves during the pandemic.

Disadvantages of high cash. Cash not invested in the business is an **underutilised asset**, and a perpetually large balance sacrifices returns available from other financial assets. In some countries large cash balances make companies targets for acquisition. High cash and cash generation also suggest potential for high dividend payouts: management may set cash aside for larger payouts if it sees no need for additional investment. Cash-based businesses such as gaming and mature industries such as tobacco (Genting, Las Vegas Sands; Altria, BATS, Swedish Match) naturally run high cash balances and pay high dividends.

5.5.2 Inventories

For retailers, a warning signal is a **sudden build-up in inventories amid falling sales**. If products stay longer on the shelves, the company may be struggling to sell.

An inventory build-up can also be positive:

- An increase in finished goods in anticipation of greater demand next period.
- An increase in raw material inventory ahead of rising prices, or an opportunity to buy at very low prices.
- A change in the mix of products manufactured.

Investigate the underlying reason rather than concluding from the number alone. During downturns, rapid technological advance and changes in taste can make older inventory obsolete and hard to sell; excess inventory may require major price discounts and write-downs.

5.5.3 Account Receivables

Accounts receivables are amounts owed by customers for products or services delivered on credit. A **cash-based business** such as a supermarket or restaurant should not carry high receivables. Airline passenger ticket sales are largely cash-based — usually **95% or more** — so receivables are typically **less than 5%** of an airline's assets. Manufacturing companies selling on credit must watch receivables for bad debts.

Pay attention to receivables quality in downturns. Where a company depends on a few buyers, receivables may unexpectedly become bad debts if a major buyer becomes insolvent. Knowing the main buyers, the share of sales to them, and tracking their performance can give early warning. **Falling sales with a build-up in receivables during a recession** signals collection problems and likely bad debts in later quarters.

Rising receivables alongside rising sales may mean the company is offering more credit terms to distributors and retailers so its goods gain an advantage over competing products. Low cash sales suggest more credit and hire purchase sales, which generate additional interest earnings and can be lucrative (Courts in Singapore; car dealers offering credit alongside banks).

A **reduction** in receivables, for example by shortening the collection period, raises liquidity. That may indicate liquidity challenges, or simply a more efficient collection system.

5.5.4 Investments

Some companies hold sizeable investments. Investment holding companies are in the business of managing investments — Berkshire Hathaway is the prominent example. When businesses generate strong cash flows, managers may invest excess cash in financial assets rather than reinvesting entirely, or park cash in short-term securities until a better use appears. Over time the portfolio can become large; SIA's short- and long-term investments combined were no more than **S$500 million** in each of the three years.

For some companies investment income exceeds income from the core business. When investments become too large they should be outsourced to different fund managers for diversification, professional monitoring and accountability. Self-management is acceptable provided there are sufficient checks and balances.

Low-risk, yield-oriented investments give a steady income stream; riskier holdings can produce significant **mark-to-market (MTM) losses** when markets turn. Accounting best practice is to reflect investments at **market value**, but some countries allow latitude on unrealised gains and losses — for example, permitting reclassification of short-term investments into long-term investments at cost or book with no immediate MTM requirement. Representatives should be aware of this potential manipulation.

5.5.5 Plant, Property and Equipment

SIA's largest balance-sheet item is Aircraft, Parts and Engines. Together with Plant, Property and Other Equipment it accounts for **62.5%** of total assets. Other airlines may classify both under plant, property

and equipment (PPE). Large PPE balances are characteristic of **capital-intensive industries**: shipping, airlines, railways, steel manufacturing, plantations and mining.

Airline	PPE / total assets (December 2020)
IAG	57.9%
Cathay Pacific	64.5%
American Airlines	51.1%
SIA (March 2021)	**62.5%**

Four considerations when reviewing PPE:

- The **age** of plants, equipment, aircraft and vessels.
- Whether the **technology** is obsolete.
- The estimated **cost of upgrading**.
- How resources are deployed to **fund upgrades**.

SIA funds fleet and equipment renewal largely from equity capital and reserves, and renews diligently as part of its branding and value proposition. For manufacturing, the concern is that large plant commitments can become obsolete quickly given the pace of technological change. Where competing technologies exist and eventual victors emerge, obsolescence must be managed: choosing a technology that becomes obsolete too quickly can mean massive write-downs and the cost of replacement, plus staff retraining.

Some companies pass obsolescence risk to subcontractors, keeping PPE spending manageable. In **Apple's 2015/16 accounts, PPE was only 7–8% of overall assets**.

5.5.6 Property

In Singapore, Hong Kong and most of Asia, property companies — developers and owners of commercial and retail property, and REITs — are blue-chip investments. Property holdings form the bulk of their value, typically **55–70%**, functioning both as an earnings generator and a long-term store of value. Including holdings through associates and joint ventures, the proportion is even larger, **70–85%**.

Property company	Investment property, landbank and development property	Current property	Total property holdi
City Developments	37.8%	26.3%	64.1%
CapitaLand	43.2%	10.6%	53.8%
UOL	47.3%	10.2%	57.5%
GuocoLand	41.1%	31.2%	72.3%

Properties developed and completed for sale in the current period are **inventory** for a property company and sit in current assets. Undeveloped land bank, developments beyond a year, and investment property held for rental income are classified as **long-term assets**.

Property companies are significantly affected by economic slowdowns when inventory and asset values are revised down. Write-downs have occurred in major downturns, though upward revisions are more common because of re-zoning, reclassification of land use and general land price appreciation in Asia. Companies in Singapore and Malaysia that acquired industrial and agricultural property in the ordinary course of business later redeveloped it for residential use in the 1990s. **Singapore land values rose about 14 times, or about 6.8% a year, over the 40 years from 1976 to 2016.**

Because significant value is locked in physical property, lowering the productive use of capital, many property companies unlock it by injecting non-core holdings into REITs for listing, freeing capital for higher-return investments.

In downturns with high unemployment, watch the funding support for property holdings with large development commitments. Residential property under development may stay on the books much longer, incurring more interest and charges. Small developers face higher liquidity risk, and bank credit lines tighten or become costlier. With insufficient cash reserves relative to commitments, over-commitment can produce a liquidity crunch; in previous downturns several small developers were forced to liquidate choice properties at the bottom of the cycle or were acquired.

5.5.7 Intangible Assets

Intangible assets of value include **goodwill, patents, copyrights, trademarks and licences**, which have a long predetermined legal life. Many intangibles — goodwill is the notable exception — are **amortised** over time.

For companies that have made acquisitions, the common intangible is **goodwill**: the premium paid above the book value of the target. The acquirer pays a premium for brand value, proprietary technology or superior management. Goodwill represents a value premium over the fair market value of tangible assets, so large write-downs are common. Energy companies wrote down more goodwill than most industries because of the sharp oil price drop caused by COVID-19; the largest impairment in 2020 was **Baker Hughes' USD14.77 billion write-down**, associated with an oil-field services and equipment business largely acquired in its 2017 merger with GE's oil and gas division.

Many companies write down goodwill even without urgent need. A goodwill write-down is typically charged directly against **equity reserves**. Unproductive asset value leaves total assets while nothing changes operationally, so **ROA and ROE in subsequent years are higher** because the denominators are lower. Bank CEOs remunerated on ROA and ROE improvement can therefore deliver "better performance" over their tenure that reflects systematic write-downs rather than operating improvement.

5.6 Liabilities

5.6.1 Debt and Bank Borrowings

On the liabilities side, pay particular attention to bank borrowings and other debt financing such as bond issues. Four questions:

- The **size** of debt used to finance assets and relative to equity.
- The **funding mix** for the debt obligations.
- The **currency denomination** of the debt.
- The **security and terms and conditions** of the debt.

a. Size of debt and bank borrowing. The higher the debt funding the asset base, the greater the financial risk. Over-reliance on debt in a low-rate environment causes difficulty servicing it if rates rise significantly. There should be an appropriate debt-equity mix, compared against the industry norm.

Companies with larger debt tend to be those that:

- Are regularly engaged in major infrastructure or property projects (construction and engineering, property developers).
- Require large asset commitments (steel and iron foundries, power generation, ships and aircraft, mining, oil drilling).
- Are involved in leveraged mergers and acquisitions.

Perpetual refinancing is used by some, but can suddenly become difficult and costly in downturns or after project overruns. Having sufficient cash flow for immediate obligations and the ability to mobilise liquid assets to retire debt matter especially to bond investors.

b. Funding mix. Companies should maintain an appropriate mix between:

- Short-term and long-term financing.
- Funding from different banks and the capital markets.
- Funding from shareholders.

Excessive funding of long-term assets with short-term financing creates liquidity problems because short-term assets may not cover short-term obligations. In downturns, long-term borrowings can suddenly become short-term if banks deny credit extensions — documentation often contains clauses letting banks shorten the credit term. Hence the need for a mix of funding sources rather than dependence on one or two banks. For companies large enough, bond issues during low-rate cycles provide cheap financing beyond bank borrowings.

c. Currency denomination of debt. An unhedged foreign currency obligation is precarious. Companies borrowing in a foreign currency because of low rates (Japan) risk the obligation growing if that currency appreciates.

In **1997**, many Southeast Asian companies financed operations with **USD-denominated Eurobonds**, which offered lower rates than much higher local Asian rates — a carry trade of borrowing in USD and investing in local debt. When the **Rupiah and Baht fell by more than 75%** during the Asian Currency Crisis,

the USD Eurobond borrowings **quadrupled in local currency terms**, and many companies found local currency cash flows and liquid assets could not cover the USD debt. Several declared bankruptcy.

5.6.2 Accounts Payables

Accounts payables show short-term credit extended by suppliers for goods and services. The longer the payment period, the longer the effective financing received. To improve operating cash flow, a company can stretch payables while shortening the receivables collection period.

A **build-up of payables** in quarterly statements may suggest the company is trying to increase liquidity. A negative reading is that the company is struggling to pay, signalled by a build-up of payables each quarter **accompanied by falling sales**.

5.6.3 Deferred Taxes

Deferred tax expected to be paid in the current period is a **current liability**; portions payable in future years are **long-term liabilities**. When earnings rise consistently year on year and the company defers part of its tax, deferred taxes tend to rise. Constant deferral provides a form of **free capital**. If earnings fall sharply or turn negative, deferred taxes fall.

5.6.4 Pension Obligations

Less common in Singapore, pension obligations feature in countries without a national retirement fund, where a pension is set aside as a retirement benefit for long-serving employees. Some international companies, especially those with US exposure, hold pension funds for international staff. In markets such as the US, where pension obligations are monitored under the **Employee Retirement Income Security Act**, pension obligations can become a major problem if poorly managed and underfunded.

5.7 Shareholders' Equity

Shareholders' equity reflects the accumulated net worth, or accounting book value, of a company: what shareholders own after all liabilities are paid. It consists of:

- Share capital.
- Reserves.
- Provisions.
- Foreign exchange adjustments.

5.7.1 Share Capital

Share capital is the original capital provided by common and preferred shareholders. **Preferred shareholders** receive a fixed dividend and are paid before common shareholders, but do not benefit from future earnings growth. **Common shareholders'** dividends can increase with earnings over time but are paid only after preferred holders receive their fixed payments. Common shareholders are therefore **residual owners**, earning what remains after all other obligations. Share capital is the amount issued out of the total authorised share capital.

5.7.2 Treasury Stock

Treasury stock is the portion bought back and ”retired” by the company. When a company buys back stock, remaining shareholders own a larger share of earnings and dividends. Buybacks create a treasury stock account and benefit the remaining common shareholders.

5.7.3 Reserves and Hidden Reserves

Accumulated net earnings or losses over the years, after dividends, are reported in **reserves** or capital reserves. Write-downs in the value of long-term assets or investments, and provisions for losses, are sometimes charged **directly against reserves** rather than through profit and loss, because they are not seen as part of normal operating performance.

Banks make large provisions for loan losses in downturns, but the losses may not materialise. When economies recover and customers repay what was classified as bad, some banks are slow to reverse provisioning, creating

hidden reserves. Hidden reserves also arise from excessive write-downs of assets such as real estate and from provisions for extraordinary losses.

Having reserves means a company is probably worth more than its accounts state. Some companies write back provisions under profits or extraordinary profits to smooth or lift earnings in later years.

5.8 The Income Statement

The income statement, or profit and loss statement, is the main focus for most investors because it reflects sales and earnings performance, and it is accompanied by comments on the outlook. It can be recast as a **common size statement** based on sales revenue.

Worked illustration — SIA income statement, FY ended March 2021 (S$ million, with common size %):

Item	2021	2020	2019	2021 %	2020 %	2019 %
Revenue	3,815.9	15,975.9	16,323.2	100.0	100.0	100.0
Staff costs	1,160.5	2,563.6	2,816.9	30.4	16.0	17.3
Fuel costs	1,015.5	4,636.5	4,587.1	26.6	29.0	28.1
Depreciation	2,075.9	2,134.2	1,327.9	54.4	13.4	8.1
Total expenditure	6,328.4	15,916.8	15,256.1	165.8	99.6	93.5
Operating (loss)/profit	**(2,512.5)**	**59.1**	**1,067.1**	**(65.8)**	**0.4**	**6.5**
Finance charges	(267.9)	(220.9)	(116.1)	(7.0)	(1.4)	(0.7)
Impairment of aircraft	(1,734.3)	—	—	(45.4)	—	—
Impairment of goodwill	(170.4)	—	—	(4.5)	—	—
Loss before taxation	**(4,957.2)**	**(220.2)**	**868.6**	**(129.9)**	**(1.4)**	**5.3**
(Loss)/profit for the year	**(4,283.4)**	**(169.4)**	**721.6**	**(112.3)**	**(1.1)**	**4.4**
Loss per share (cents)	(115.6)	11.2	57.7			

Four inferences follow:

- Revenue fell **76%** in FY2020/2021 against FY2019/2020. Costs are more inelastic, so the $\mathbf{S}9.5 billion * *cut in operating expenses did not offset the * *S$**12 billion** revenue loss, producing an operating loss of about **S$2.5 billion**.
- Fuel, a variable cost, fell more than **75%** to roughly **S$1 billion**, but still accounted for **26%** of a reduced revenue base.
- The operating loss was $S2.5 billion but the pre-tax loss was * *S4.9 billion**, because of non-operating items: impairment of aircraft $\mathbf{S}1.7 billion * *, impairment of goodwill * *S$**170 million**, losses of associated companies $\mathbf{S}127 million * *and others * *S$**128 million**.
- The common size statement shows a **volume-driven business with small margins**: operating profit was **6.5%** of sales in FY2018/2019, a good pre-pandemic year, and net margin before tax **5.3%**. Changes in fuel prices or fare competition dramatically affect profitability.

Beyond the statements, qualitative comments at press and analyst briefings reveal management's outlook and the key variables they watch. Together with other research, these feed a forecast of future sales and earnings.

5.8.1 Sales and Revenue

Break sales revenue into two parts:

- **Unit sales growth.**
- **Pricing changes.**

Raising either increases revenue, so forecasting revenue means watching the factors that drive each.

Unit sales growth drivers	Pricing change drivers
New geographical markets	Competition (direct and indirect)
New target customer segment	Brand value development
New uses	Cost of production
New range or version of product	Buyers' financial health

Unit sales growth drivers	Pricing change drivers
Production capacity Competitors' market share Limits to competition	Government pricing restrictions

For an airline, securing new lucrative routes adds ticket sales (more units); successful service quality and branding allows premium pricing. Conversely, new players with deep financial pockets create strong competition and reduce revenue.

Segmental breakdown. Breaking revenue and profit down by line of business clarifies the contribution of each. Tracking news on the sales and performance of key businesses and new products lets the analyst anticipate the impact on overall growth. Companies also provide segmental breakdowns of assets to show resource allocation.

5.8.2 Interest and Investment Income

Cash-rich companies with strong cash flows over several years accumulate financial investments, producing large interest and dividend flows that may not be part of core activity. SIA's interest income was **S**35$million$ $anddividendincome$ **S8.4 million** in FY2020/2021.

Some companies have made speculative investments that wiped out core profits. Review the accounting notes to understand the type of investments made and the likely income volatility.

5.8.3 Costs and Expenses

Watch for changes in major cost items measured as a percentage of sales.

a. Major cost items. For manufacturers, production costs — labour, raw materials, utilities — sit in cost of goods sold, with raw materials typically significant. If raw material prices fall while finished goods prices and sales hold, margins rise. Some manufacturers separate production costs from marketing, distribution and administration, splitting costs into cost of goods sold and selling and administrative costs. Research and development is another significant cost not going directly into production. Distribution and service companies reflect cost of sales or operating cost before operating profit.

Three things to note:

- Major cost items as a percentage of sales.
- Significant rising or falling trends in cost (raw materials, fuel).
- Cost items that are **non-cash** accounting costs, which do not affect cash flow.

Major cost items differ by industry. In service industries such as banking and hospitality, staff costs dominate. In transportation, fuel, staff and depreciation dominate.

For SIA, staff costs were **30.4%** of revenue in FY2020/2021 and **16%** in FY2019/2020; fuel costs were **26.6%** and **29%** respectively.

Jet fuel prices are highly volatile. After the precipitous fall in H1 2020, the recovery was equally striking. Because higher fuel costs are hard to pass on through ticket prices, higher fuel prices cut airline profitability. Airlines hedge fuel with forwards and futures, and hedging brings its own problems: airlines that bought fuel futures for H1 2020 delivery became **over-hedged** when demand collapsed, and the unused long hedges had to be unwound at a loss when prices fell. Even without a demand collapse, airlines that hedged at forward prices become uncompetitive against unhedged airlines when fuel prices fall unexpectedly.

b. Depreciation. For capital-intensive companies depreciation can be significant. SIA's depreciation was more than **50% of sales** in FY2020/2021 under atypical conditions, and roughly **8%** in FY2018/2019. Depreciation is not a real cash outflow but an accounting method. For transport companies making large ongoing commitments to aircraft or ships, depreciation will always be large.

Depreciation can be abused. A cosmetic approach when performance is poor is to stretch depreciation over more years: a taxi fleet operator depreciating taxis over **10 years instead of 5** dramatically lowers depreciation expense and improves reported net profit, while nothing changes in cash flow.

5.9 The Cash Flow Statement

For capital-intensive industries, net earnings may not accurately show the ability to support short-term liabilities and borrowings. The cash flow statement adds back depreciation and other non-cash items, revealing the real cash earnings from operations and the level of liquidity.

It has three parts:

- Cash flow from **operating** activities.
- Cash flow from **investing** activities.
- Cash flow from **financing** activities.

Worked illustration — SIA cash flows (S$ million):

Operating activities	2021	2020	2019
Cash flow before working capital changes	149.3	3,211.3	2,791.3
Decrease in trade and other creditors	(2,621.5)	(346.2)	(28.1)
Decrease in sales in advance of carriage	(1,473.3)	(674.0)	273.3
Decrease in trade debtors	225.8	660.3	(206.1)
Net cash (used in)/provided by operating activities	**(3,292.4)**	**2,731.9**	**2,801.1**

Investing activities	2021	2020	2019
Capital expenditure	(2,695.5)	(5,103.5)	(5,562.3)
Proceeds from sale and leaseback	1,230.5	117.2	—
Net cash used in investing activities	**(1,314.3)**	**(4,964.8)**	**(5,362.4)**

Financing activities	2021	2020	2019
Proceeds from issuance of shares and mandatory convertible bonds	8,829.2	—	1,350.0
Proceeds from borrowings	4,579.0	3,985.2	2,280.4
Repayment of borrowings	(4,344.1)	(878.3)	(93.7)
Dividends paid	—	(355.5)	(449.8)
Net cash provided by financing activities	**9,737.0**	**1,935.6**	**2,931.7**
Net cash inflow/(outflow)	**5,130.3**	**(297.3)**	**370.4**
Cash and cash equivalents at end of year	7,783.0	2,685.3	2,944.0

Depreciation accounted for **63% (about S**$2 billion$)$ ** $of the negative operating cash flow in FY 2020/2021; adding it back to the$ $tax loss greatly reduced the negative operating cash flow. Under the more benign conditions of FY 2018/2019, profit before tax w$ $*S$**868 million** while operating cash flow was **S$2.8 billion** — more than three times larger.

The negative operating cash flow in FY2020/2021 was largely funded by the rights issue and mandatory convertible bonds totalling **S**$8.8 billion$ ** $. Investing cash flows show that large capital investments such as aircraft purchases were$ $*deferred** . No dividends were paid. With reduced investing and additional capital raised, SIA ended FY 2020/2021 with net ca$ $*S$**7.8 billion**.

Why cash flow matters. For most investors, and especially bondholders, the major concern is **consistently negative operating cash flow**. A poor overall net cash position can be temporarily masked by selling assets and businesses, creating positive investing cash flow. This often begins with selling peripheral and poorly performing businesses and excess assets, but a desperate company starts selling assets that would fuel future growth, or assets it can temporarily do without. Without recovery in the core business, these efforts come to nothing.

5.10 Per Share Calculations — BVPS, DPS, EPS and Price

The share count used depends on what is being measured.

Measure	Formula	Share count
Book value per share (BVPS)	Shareholders' equity / total number of shares outstanding	Shares outstanding **at t**
Earnings per share (EPS)	Net earnings / **weighted average** number of shares	Weighted average, becau
Fully diluted EPS	Net earnings / fully diluted number of shares	Assumes all derivatives a
Dividend per share (DPS)	Dividends paid / total number of shares outstanding	Shares outstanding **as a**

EPS should also be computed on a **fully diluted** basis, which is particularly relevant for EPS projections. To ensure comparability, all future EPS forecasts should be calculated on a fully diluted basis.

The share price is already per-share and is frequently compared with these computed figures. The **forward price-earnings (PE) ratio** is the price paid for future EPS and is a popular comparison method; the **price-to-book** ratio is another. Both are revisited in Chapter 6.

5.11 Ratio Analysis

Ratio analysis is an established method for interpreting financial accounts: it evaluates financial strength, the quality of earnings and performance using simple mathematical ratios.

Used in isolation, ratios are meaningless. An interest coverage ratio of two times may be low for some industries and very high for others. Findings must be compared and benchmarked against companies in the same industry. Properly applied, ratios diagnose financial health.

Question	Ratios
How profitable is the company?	Return on assets, return on equity, gross profit margin, net profit marg
How solvent is the company?	Debt-to-equity, debt-to-total assets, interest coverage, current ratio, qui
How efficiently does the company use its assets?	Total asset turnover, inventory turnover, receivables turnover

Four categories are examined: **liquidity, leverage, operating efficiency/activity, and profitability**.

5.11.1 Liquidity Ratios

Liquidity is the availability of funds at any given time and at a reasonable cost. Liquidity ratios indicate the ability to meet short-term financial obligations. Banks are governed by liquidity ratio levels specified by their local authorities. One lesson of the 2008 Global Financial Crisis is the importance of liquidity, not just for banks but for all companies. Maintaining strong liquidity can be expensive when interest rates are high.

Ratio	Formula
Current ratio	Current assets / current liabilities
Quick ratio	(Cash + marketable securities + accounts receivable) / current liabilities
Cash ratio	(Cash + near-cash) / current liabilities
Cash flow ratio	Cash flow from operating activities / current liabilities

a. Current ratio. The most common liquidity ratio, measuring the degree to which current assets cover current liabilities. Conventional wisdom suggests **2 or more**, but the cost of too much liquidity is unproductive use of assets, since current assets tend to generate low returns. Companies now manage working capital more efficiently, producing generally lower current ratios.

b. Quick ratio. Also called the **acid test ratio**. Same denominator as the current ratio, but the numerator **excludes inventories** because they are less liquid. It indicates highly liquid assets available to cover current liabilities: cash, marketable securities and accounts receivables.

Quick ratios vary by industry. A supermarket's quick ratio may be as low as **0.2** because it holds substantial inventory, but that is not a liquidity concern: its current assets are fast-moving consumer goods financed by short-term supplier credit. Quick ratios are less meaningful for non-retail and non-merchandising companies such as banks and other financial institutions, which hold no inventory.

c. Cash ratio. A more stringent test, looking at cash and cash equivalents. The numerator typically includes only cash and short-term marketable securities convertible quickly to cash; assets that cannot immediately be converted are omitted.

d. Operating cash flow ratio. Shows the cash flow from operating activities available to cover current liabilities. Some companies have a weak current ratio, with barely enough current assets to cover current liabilities, yet generate cash flow well in excess of them — typical of **cash-based service businesses** that cover short-term liabilities by rendering services for cash rather than selling inventory on credit.

Conversely, a company may have a fine current ratio but a worrying operating cash flow ratio. A company in dire straits, overcommitted to loss-making projects, may liquidate peripheral long-term assets to shore up cash for near-term obligations. If operating cash flow and income stay negative, the strategy is unsustainable and default follows later. Most defaulting Singapore corporate bond and perpetual issuers showed this pattern.

Worked illustration — SIA liquidity. Liquidity ratios rose significantly in FY2020/2021 except the operating cash flow ratio, because most funding raised through the share issue and mandatory convertible bonds in 2020 was kept as cash (numerator) while a significant portion of short-term borrowings was repaid, and sales in advance of carriage fell sharply with the downturn, reducing total current liabilities (denominator). The operating cash flow ratio turned from **0.25** in FY2019/2020 to **−0.58** in FY2020/2021 because of the trading loss.

Current ratio	2020	2019
Cathay Pacific	0.59	0.48
IAG	0.68	0.89
American Airlines	0.67	0.45
SIA (March 2021/2020)	**1.69**	**0.44**

Window dressing. Current and quick ratios at reporting dates are susceptible to manipulation. Toward year-end, companies can collect receivables earlier, reduce inventory below normal levels and delay purchases; proceeds from faster collection can pay off current liabilities.

An arithmetic trap. An equal dollar increase in both current assets and current liabilities does **not** leave the current ratio unchanged: it **lowers** the ratio if it was previously above 1, and **raises** it if it was previously below 1.

5.11.2 Leverage Ratios — Debt and Solvency Ratios

Leverage ratios show the financing mix between debt and equity, the amount of financial risk, and the capacity to meet principal and interest. They are of primary interest to **bondholders**, whose main risk is default. For other stakeholders they are more useful analysed alongside other ratios.

A principal consideration in capital structure is the extent of leverage. Higher leverage increases return on equity and reduces potential dilution of earnings, but may lower the credit rating and raise the cost of debt.

Ratio	Formula	Notes
Debt-to-equity (gearing)	Total debt / total shareholders' equity	Debt includes all short- and long-term borro
Debt-to-total assets	Total debt / total assets	Debt includes all borrowings, long- and shor
Total liabilities-to-total assets	Total liabilities / total assets	Shows how much of the asset base is funded

The greater the debt, the higher the leverage, the greater the impact on ROE, and the harder it is to generate sufficient income to repay. When reviewing debt-to-equity or debt-to-assets, **check for foreign currency-denominated debt**: if the foreign currency appreciates against the local currency, the market value of the debt becomes much larger. Companies with foreign currency bonds in countries with weak currencies are especially susceptible to ballooning debt.

Worked illustration — SIA leverage.

Ratio	2021	2020	2019
Debt/equity (gearing)	**0.70**	1.01	0.49
Debt/assets	**0.31**	0.29	0.22
Liabilities/assets	**0.57**	0.71	0.55

SIA's gearing improved in FY2020/2021 principally because shareholders' equity (the denominator) rose through the rights issue and mandatory convertible bonds. The company has always been conservative with borrowings, and its gearing is the **lowest among airlines**.

Gearing ratio	2020	2019
Cathay Pacific	1.27	1.55
IAG	11.91	2.09
United Airlines	4.49	1.26
SIA (March 2021/2020)	**0.70**	**1.01**

5.11.3 Coverage Ratios

a. Interest coverage. The times-interest-earned ratio indicates how many times fixed interest charges are covered by earnings. A higher ratio of earnings to interest indicates lower financial risk.

$$\text{Interest Coverage} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

SIA's coverage ratios were negative in FY2020/2021 because of trading losses. A higher positive ratio is good (lower financial risk); a negative ratio indicates the opposite — the loss as a multiple of financing charges. **The size of a negative ratio is also determined by the absolute amount of interest charges.** A firm with a coverage ratio of **−1** does not necessarily have lower financial risk than one at **−3**: both are incurring trading losses, but the −1 firm has **higher financing charges** (the denominator), which is not a good thing even though both made the same trading loss.

All airlines in the comparison incurred net losses in 2020 and therefore show negative coverage. Do not over-interpret negative coverage ratios when assessing financial risk — the denominator has a direct effect, so look for the reasons behind the ratio.

b. Total fixed charge coverage. Measures how well earnings cover total fixed financial charges, including fixed lease payments and preferred dividends paid out of after-tax earnings.

$$\text{Fixed Charge Coverage} = \frac{\text{Earnings Before Interest, Taxes and Lease Payments}}{\text{Interest Expense} + \text{Lease Payments} + \dfrac{\text{Preferred Dividend}}{1 - \text{Tax Rate}}}$$

5.11.4 Operating Efficiency / Activity Ratios

Efficiency ratios measure how productively management uses assets and capital to generate sales. Some measure the speed of converting inventory to sales and receivables to cash, and feed into the **cash conversion cycle** — the period between paying suppliers for inventory and collecting from customers for the inventory sold, which is the period for which short-term financing is needed.

Ratio	Formula	Notes
Total asset turnover	Sales / **average** total assets	Higher means gr
Inventory turnover	Sales / **average** inventory	Measures how m
Days to sell inventory	365 / inventory turnover, or (average inventory) / (sales / 365)	Another way to
Receivables turnover	Sales / **average** accounts receivables	How many times
Collection period	365 / receivables turnover, or (average accounts receivables) / (sales / 365)	Average days to

Total asset turnover varies substantially across industries: from about **1 for capital-intensive industries** to **over 10 for some retailing operations**. It can be affected by factors other than efficiency: a high ratio relative to the industry might imply too few assets for the potential business (foregone sales), or the use of outdated, fully depreciated assets.

Worked illustration — SIA total asset turnover.

	2021	2020	2019
Sales	3,816	15,976	16,323
Average total assets	(37,582 + 33,713)/2	(33,713 + 30,505)/2	(30,505 + 25,893)/2
Asset turnover	**0.11**	**0.50**	**0.58**

Sales/assets	2020	2019
Cathay Pacific	0.22	0.53
IAG	0.24	0.80
United Airlines	0.27	0.85
SIA (March 2021/2020)	**0.11**	**0.50**

The pandemic left many airliners idle rather than generating revenue, so turnover fell across the industry. SIA's turnover is lower than competitors', suggesting room to improve asset use — though keeping older fleets longer would raise turnover and should not be done at the expense of safety and customer experience.

Inventory turnover is more applicable to manufacturing and retail and less useful in service industries. A high ratio normally indicates efficiency, but can also indicate inadequate inventory to meet demand, leading to shortages, back orders and eventually lost sales. Too low a ratio relative to the industry may mean excess inventory and possibly obsolete stock.

Collection period should be compared with the firm's credit policy and the industry average. If the industry average is **60 days**, a collection period of **70 days** indicates slow-paying customers, tying up capital and raising bad debt risk. A figure substantially below the norm may indicate overly stringent credit terms relative to competitors, which could hurt sales.

5.11.5 Profitability Ratios

Profitability is analysed in relation to **sales** or **capital**.

Ratio	Formula	Notes
Gross profit margin	Gross profit / sales	Gross profit is profit after deducting c
Operating profit margin	EBIT / sales	Operating profit is gross profit minus
Net profit margin	Net income / sales	Net income is profit after tax but bef
Return on equity (ROE)	Net income / **average** common equity	The rate of return on shareholders' eq
Return on assets (ROA)	(Pre-tax income + interest) / **average** total assets	Measures overall efficiency in managi

EBITDA margin. Some investors add back depreciation and compute an **EBITDA profit margin**, reflecting cash earnings since depreciation and amortisation are non-cash. This can give insight into heavy manufacturing firms with large depreciation charges.

Worked illustration — SIA profitability.

	2021	2020
Operating profit margin	(4,957 + 268)/3,816 = **−122.89%**	(220 + 221)/15,976 = **0.00%**
Return on equity	−4,283 / ((16,278 + 9,733)/2) = **−32.94%**	−169 / ((9,733 + 13,683)/2) = **−1.45%**
Pre-tax ROA	(4,957 + 268) / ((37,582 + 33,713)/2) = **−13.15%**	(220 + 221) / ((33,713 + 30,505)/2) = **0**
Basic EPS	(115.6) cents	(11.2) cents
Fully diluted EPS	(115.6) cents	(11.3) cents

The ROE trap. ROE numbers must be reviewed carefully to see whether strong performance reflects better earnings or a **depletion of the equity base** in prior years.

ROE	2020	2019
Cathay Pacific	−37.5%	2.7%

ROE	2020	2019
IAG	−170.2%	25.3%
United Airlines	−80.8%	27.9%
SIA (March 2021/2020)	**−33.9%**	**−1.9%**

IAG, the merger of British Airways and Iberia, substantially reduced shareholders' equity in **2012** with losses of **EUR0.9 billion** charged to reserves; in **2013** it charged **EUR2 billion** for restatement of merger value to reserves; in **2014** it charged **EUR1.2 billion** in losses under net value in changes of hedges. United Airlines charged its losses to retained earnings in **2011 and 2012**, substantially reducing its equity base. Subsequent ROE for both looks high because equity was reduced by prior losses. Interpreting two years of ROE without reviewing the past gives an incomplete picture.

5.11.6 Quality of Financial Statements

The quality of financial statements and earnings depends on good accounting practice, conservative revenue recognition, limited reliance on external debt, understating rather than overstating asset values, and inclusion of all financial liabilities including off-balance-sheet items. Most telling is not how management reports when business is rosy, but the level of transparency and treatment of accounts and losses when conditions are poor and the industry is at its cyclical trough.

Signs of strong, high-quality financial statements:

- **Limited or lower use of debt** relative to the industry, reducing financial risk; lower debt also implies sufficient credit lines to draw on if needed.
- **Higher liquidity** than competitors and the industry.
- **Assets carried below market values.**
- A preference **not to reflect intangible assets** such as patents, goodwill, trademarks or valuable proprietary research, with costs expensed rather than deferred into amortisation schedules.
- **Funds set aside for replacement and upgrades** of plant and factories, with scheduled renewal and technological upgrade plans. Companies using obsolete or inferior technology overstate asset values, because market values are likely below original cost.
- **Conservative revenue recognition** that does not change just to reflect higher sales, conservative accounting of unrealised income by deferring recognition until realised, and adequate provisioning for losses and expenses.
- **Absence of off-balance-sheet activities.** Where such liabilities exist, account for them when estimating the true financial position.
- **Discipline in keeping to core business** and avoiding speculative financial activities.
- **Realistic recognition of bad debts** and adequate provisioning during downturns. A rising average collection period usually means bad debts may increase. A bank with an overly optimistic view of credit quality understates non-performing loans.
- **Steady, sustainable revenue and earnings growth** from existing businesses. Good-quality earnings are realistic and repeatable.

Worked illustration — SIA's response to the pandemic. The statements show decisive action to build a more resilient business:

- Raised significant equity capital (**S$8.8 billion**) through a rights issue and mandatory convertible bonds.
- Conserved cash flow by withholding dividends and deferring major capital investment.
- Used part of the capital to reduce borrowings, lowering leverage.
- Kept most of the funding in cash as a war chest against a protracted downturn.
- Ended with a highly liquid and solvent balance sheet, reduced leverage and a substantial war chest for the recovery.
- Achieved this financial resilience at the expense of short-term profitability and asset efficiency.

5.11.7 Making Comparisons and Trend Analysis

Beyond industry averages, **trend analysis** — studying past performance over time — gives valuable insight into financial health.

Business cycles affect all companies. To understand performance properly, cover **at least one full cycle** and track performance across it; otherwise the analysis may mislead, showing a company doing well when it

is simply at the crest of a cycle about to turn down. Most companies provide a **5-year or 10-year** summary of key statistics in their annual reporting, and analysts also use key operating statistics to forecast and assess performance.

5.11.8 Limitations of Ratio Analysis

- **Different accounting policies** across companies require adjustments to make figures comparable.
- **Consolidated accounts** cover groups, making it difficult to determine the performance of one company within the group.
- **Figures may not be representative** of the rest of the year — year-end inventory may be drawn down to abnormally low levels, giving a misleading impression of strength.
- **Special situations** may depress profitability, such as research and development costs for new products.
- **Inflation and general economic conditions** affect trend analysis.
- **Read the footnotes**, which contain additional information on the financial situation.
- **Comparisons require similar companies** in industry sector and size.
- **Ratios can be manipulated** — high ROE from reducing equity in prior years. Look at prior years' accounts to understand the genesis of the numbers.
- **Ratios are only meaningful over time and against peers or benchmarks.** Companies in emerging sectors (emerging technology or business models) do not lend themselves easily to ratio or financial analysis; the lack of comparable, reliable information greatly limits it.
- **Asset-light business models** with little or no hard balance sheet of operating assets, where value rests on brand perception or the founder, are especially hard to analyse by ratio.

5.12 Summary

- Company analysis divides into qualitative analysis and quantitative analysis (financial statement analysis).
- Companies establish competitive advantage by dominating and controlling key technology, research and patents; controlling or dominating key resource supply; becoming very cost efficient and the lowest-cost producer; establishing superior brand differentiation; controlling the distribution channel; focusing successfully on niche clients; developing superior management, talent and relationships; and establishing diversified core businesses with shared costs.
- Current asset values and future price movements are driven by expected future sales and earnings, not historical data.
- A company's performance is driven by qualitative factors.
- Excellent companies with sustainable profit growth potential can appear expensive in the near term but usually remain good investments for the longer term.
- The current value of a company depends on future earnings and cash flow, but historical data is still used because recent performance is the best predictor of near-term performance and provides a foundation for forecasts.
- The three major statements are the balance sheet (financial condition at a point in time), the income statement (revenue and expenses over a period) and the statement of changes in cash flow (changes in cash position over a year).
- When examining a balance sheet, focus on the largest items on both sides, the relationship between assets and liabilities, and significant trends over time.
- Some companies hold large cash balances for liquidity, risk and operational needs, but cash not reinvested represents underutilised assets and sacrifices potential returns.
- In downturns, older inventory may become obsolete and hard to sell; excess inventory may require major discounts and write-downs. A steady build-up of inventories with falling sales may indicate a problem.
- For manufacturing, large commitments to plant and factories carry the risk of rapid obsolescence.
- Some technology companies outsource manufacturing to subcontractors while retaining ownership of software, intellectual property and research.
- Falling sales with a build-up in receivables during recessions may signal collection problems and future bad debts.
- Rising receivables alongside rising sales may mean more credit terms offered to distributors and retailers, or more credit and hire purchase sales, which generate additional interest earnings.
- Best practice is to reflect both short- and long-term financial investments at market value.
- For property holdings with large development commitments, examine the funding supporting them; in downturns, properties under development may stay on the books a long time and drain liquidity.

- Significant value locked in physical property can lower the productive use of capital; injecting non-core holdings into REITs is one way to unlock it.
- Goodwill represents a value premium over the fair market value of tangible assets; write-downs hit equity directly and can artificially improve ROE in later periods.
- On the liabilities side, pay attention to the size of debt relative to assets and equity, the funding mix, the currency denomination, and the security and terms.
- Companies with larger debt tend to be those engaged in major infrastructure or property projects, those requiring large asset commitments, and those involved in leveraged mergers and acquisitions.
- Break sales growth into unit sales growth and pricing changes.
- Unit sales growth is driven by new geographical markets, new customer segments, new uses, new product ranges, production capacity, competitors' market share and limits to competition.
- Pricing changes are driven by competition, brand value development, cost of production, buyers' financial health and government pricing restrictions.
- Segmental breakdowns and model forecasts highlight the main sources of revenue and income.
- When reviewing costs, note major cost items as a percentage of sales, significant trends, and non-cash accounting costs.
- The cash flow statement shows real cash earnings from operations and the level of liquidity, in three parts: operating, financing and investing.
- Financial ratios fall into four categories: liquidity, leverage, operating efficiency and profitability.
- The quality of financial statements and earnings depends on good accounting practice, conservative revenue recognition, limited reliance on external debt, understating rather than overstating asset values, and inclusion of all financial liabilities including off-balance-sheet items.
- Financial statement analysis should always use data over several years, say five, so underlying trends can be identified.
- Ratios can be artificially manipulated; analysts should look at prior years' accounts to understand the genesis of the numbers.

Exam cues

- **Two halves of company analysis:** qualitative and quantitative. Qualitative factors **precede** the numbers.
- **Eight competitive advantages:** technology/patents, resource supply, brand, distribution, cost efficiency, niche focus, management/talent, diversified cores with shared costs.
- **Balance sheet is a snapshot; income statement and cash flow cover a period.**
- **Two balance-sheet equations:** US — assets = liabilities + equity; British — assets − liabilities = equity (net assets).
- **Common size statement** expresses every item as a percentage of total assets (balance sheet) or sales (income statement).
- **Three things to examine on a balance sheet:** largest items on both sides, the asset-liability relationship, and trends over time.
- **Cash:** high cash is a war chest in a downturn but an underutilised asset otherwise. SIA **20.7%** of total assets; American Airlines **0.4%**; IAG's high ratio reflects a shrunken denominator.
- **Receivables:** airlines are **95%+ cash-based**, so receivables are **under 5%** of assets. Falling sales + rising receivables = collection problem.
- **PPE:** capital-intensive industries run **50–65%** of assets. SIA **62.5%**; Apple only **7–8%** because it outsources manufacturing.
- **Property companies:** property is **55–70%** of assets, or **70–85%** including associates and joint ventures. Singapore land rose about **14x** (about **6.8% a year**) from 1976 to 2016.
- **Goodwill write-downs** are charged to equity reserves, lowering the denominators and **raising ROE and ROA** in later years. Baker Hughes wrote down **USD14.77 billion** in 2020.
- **1997 Asian crisis:** USD Eurobonds quadrupled in local currency terms when the Rupiah and Baht fell **more than 75%**.
- **Per-share share counts:** BVPS uses **year-end** shares; EPS uses the **weighted average**; DPS uses shares at the **ex-dividend date**; diluted EPS assumes all convertibles and derivatives are exercised.
- **Liquidity ratios:** current = CA/CL; quick = (cash + marketable securities + AR)/CL; cash = (cash + near-cash)/CL; cash flow = operating cash flow/CL.
- **Current ratio of 2** is the conventional benchmark, but lower ratios are now normal because working capital is managed better.
- **Quick ratio excludes inventory.** A supermarket at **0.2** is fine because its inventory is fast-moving and supplier-financed.

- **Window dressing:** current and quick ratios at reporting dates can be manipulated by collecting receivables early, cutting inventory and delaying purchases.
- **Ratio arithmetic trap:** an equal dollar increase in current assets and current liabilities **lowers** a ratio above 1 and **raises** one below 1.
- **Leverage ratios:** debt/equity, debt/total assets, total liabilities/total assets. Debt is carried at **face value**. Always check for **foreign currency debt**.
- **SIA gearing 0.70** in 2021 versus IAG **11.91** — the lowest in the industry, improved by raising equity, not by repaying debt.
- **Interest coverage = EBIT / interest.** A **negative** ratio's magnitude depends on the interest charge: **−1 is not better than −3** if the −1 firm has higher financing charges.
- **Fixed charge coverage** adds lease payments and preferred dividends (grossed up by dividing by 1 − tax rate) to the denominator.
- **Efficiency ratios use AVERAGE balances** in the denominator because the numerator covers a period.
- **Total asset turnover** ranges from about **1** for capital-intensive industries to **over 10** for some retailers. SIA **0.11** in 2021, **0.50** in 2020.
- **Collection period benchmark:** industry average **60 days**; **70 days** means slow payers.
- **Profitability ratios:** gross margin, operating margin (EBIT/sales), net margin, ROE (net income/average common equity), ROA ((pre-tax income + interest)/average total assets).
- **ROE trap:** high ROE can come from a **depleted equity base**, not better earnings. IAG charged EUR0.9bn (2012), EUR2bn (2013) and EUR1.2bn (2014) to reserves; United charged losses to retained earnings in 2011–2012.
- **Quality of earnings — ten signs:** low debt, high liquidity, assets below market value, few intangibles, funded replacement, conservative revenue recognition, no off-balance-sheet items, core-business discipline, realistic bad-debt provisioning, sustainable growth.
- **Cash flow has three parts:** operating, investing, financing. Depreciation is added back because it is non-cash — it was **63%** of SIA's negative operating cash flow in 2021.
- **Consistently negative operating cash flow** is the bondholder's central concern; asset sales can mask it temporarily.
- **Trend analysis must cover at least one full business cycle**, using the 5- or 10-year summary.
- **Ratio limitations:** differing accounting policies, consolidated accounts, unrepresentative year-end figures, special situations, inflation, footnotes, comparability of size and sector, manipulation, emerging sectors, asset-light models.
- **Trap:** a **higher** current ratio is not automatically better — excess current assets earn low returns.
- **Trap: ROA is normally computed before interest** to separate operating from financing decisions; using after-interest ROA across different capital structures is not comparable.
- **Trap: depreciation is not a cash outflow.** Stretching the depreciation period (5 years to 10) raises reported profit and changes nothing in cash flow.

Chapter 6 — Equity Securities

Equity is the asset class most retail investors hold, and the one where valuation judgement matters most. This chapter covers what a share is and what rights it carries, the categories of equity an adviser recommends, how indices are constructed, and the three valuation routes — net asset, discounted cash flow and relative ratios — with the arithmetic the examination tests.

Learning Objectives

The candidate should be able to:

- Describe the characteristics of equity securities, namely ordinary shares and preference shares.
- Distinguish between blue-chip, growth-oriented, income-oriented, cyclical and defensive equity investments, as well as those with high net tangible asset backing.
- Understand the differences between absolute and relative valuation approaches.
- Understand how to apply relative price valuation ratios, firm valuation ratios and yield-oriented ratios used in equity analysis — dividend yield, price-earnings, price-to-book value, price-to-sales, price-to-cash flow and EV/EBITDA — and their relevance to different types of companies.
- Describe the composition and computation of equity market indices.
- Perform calculations relating to rights and bonus issues.

- Explain the use and computation of discounted cash flow in estimating fair market value (intrinsic value).
- Apply relative valuation ratios in comparing stocks and explain how they are calculated and used.
- Understand how prospectuses for initial public offerings are read and analysed.
- Describe depositary receipts as a means of investing in foreign markets.

6.1 Introduction

Equity is possibly the most popular asset class among investors in Asia. Despite the volatility, the potential for substantial capital gains remains a major draw. This chapter covers share ownership, the characteristics of equities, strategies for investing in equities, relative pricing and valuation using cash flow models, and the composition and application of equity indices.

Shares represent ownership rights in a corporation and let investors participate in the company's performance. When companies perform well, management may pay dividends on top of price appreciation. When they perform poorly, dividends fall or disappear and the shareholder faces capital loss as the stock falls; in the worst case the price goes to zero. Many stocks worldwide, including in Singapore, have fallen to or near zero. The representative's job is to help clients select better companies and avoid lemons.

6.2 Share Ownership

6.2.1 Why Share the Ownership of a Great Business?

Why would the owners of a booming company give other investors a chance to participate? Aside from altruism, original owners may want to **share the risk of ownership**, **realise and monetise** the value of the company, and **redeploy funds** to other businesses for even greater gains. Usually, once growth has slowed from high double digits, owners of privately held companies seek to monetise part of their ownership of a still-profitable business.

For example, the owners of a 100% family-owned business could reduce their stake to **51%** by selling **49%** to common shareholders through an IPO and placement. Families may lack the right successors and prefer to professionalise management, and different family members may have different ownership agendas and motives for cashing out.

Through a well-planned IPO, ownership control can stay in the family, the business value can be partly monetised, additional capital can be raised from new shareholders, management is professionalised, and the risks and returns of the future are shared. Conversely, if the share value of a listed company falls below its perceived intrinsic value, the company is sometimes **taken private** by its major shareholders.

Owners of new growth businesses — high-tech, biotech, internet, disruptive technology — often need capital to fund growth. Founding owners may have initial support from private equity partners looking to realise the potential of the business, and raise sizeable capital through an IPO.

6.2.2 Types of Shares

Share ownership carries rights and privileges, but not all shares carry the same ones. Before listing, original owners must decide how much ownership to give up, because majority ownership usually dictates the stance on major decisions.

In Singapore there are traditionally two types of listed shares:

- **Common or ordinary shares** with voting rights.
- **Preference shares** with non-voting rights.

Typically one common share carries one vote. Since **June 2018**, SGX has allowed listing of companies with **dual class shares**: in addition to ordinary voting shares, companies can have multiple-vote shares. The intent is to encourage founders who need substantial funding to grow rapidly but want to retain majority control. Safeguards protect non-controlling shareholders: extra votes cannot be exercised on important matters where founders and minority shareholders may have conflicting interests, and to prevent founder entrenchment **each multiple-vote share can carry a maximum of 10 votes**.

In the US and Europe there are also share classes with different voting rights.

Company	Classes	Detail
Berkshire Hathaway	Class A and Class B common stock	Class A priced at **USD284,570** per share and Class B at

Berkshire Hathaway is one of the most successful equity investment holding companies in the United States. Its gain in net worth during **2016** was **USD27.5 billion**, which increased the per-share book value of both Class A and Class B stock by **10.7%**. Over the last **52 years** — since present management took over — the per-share book value has grown from **USD19 to USD172,108**, a rate of **19% compounded annually**. Warren Buffett is CEO and Chairman and owns **36.8%** of the company. | **Google (Alphabet)** | Class A, B and C | Class A: **1 vote** per share; Class B: **10 votes** per share; Class C: **no voting interest**. Class A trades at a premium to Class C; Class B is not traded but distributed internally to founders and insiders |

In Europe many companies are controlled by founding families with different share classes carrying different numbers of votes.

6.2.3 Ordinary Shareholders

In Singapore companies mainly issue ordinary or common shares for ownership interest. Holders are granted:

- **Profits, price gains and dividends.** Shareholders gain when the company makes profits, when the share price rises, and when management declares a dividend. Ordinary dividends are at management's discretion and depend on performance; common shareholders have **no right** to dividends and payouts are **not guaranteed**.
- **Voting rights.** Shareholders are legally entitled to vote on major policy decisions such as electing directors and issuing new shares. The more shares owned, the greater the voting power; minority shareholders may lack sufficient votes to pressure management.
- **Information rights.** Shareholders have a right to be informed of major events and decisions affecting the company and its value.
- **Residual claim.** In bankruptcy or liquidation, shareholders receive their share of remaining asset value only after all other creditors and claimants are paid — bondholders, suppliers, other creditors, employees and deferred taxes.
- **Limited liability.** If a corporation fails, the most shareholders can lose is their original investment. Creditors have recourse only to the corporation's assets; shareholders are not personally liable for the company's obligations.

6.2.4 Preference Shares

Technically an equity security, a preference share is a **hybrid** because it resembles both fixed income and equity. As equity it has an infinite life and pays dividends. Although most preference shares are callable by the issuer, there is typically a period of protection against calls — usually the issuer will not call within the first **5 years**. Non-callable preference shares may also be issued and carry a **lower** rate of return than callable ones, because the issuer has no option to redeem.

A preference share resembles a fixed income security with the dividend fixed and specified in advance, providing an income stream very similar to a bond. The difference is that the income continues **indefinitely** unless the shares are called or retired. Given the stipulated payments, most investors view preference shares as roughly equivalent to fixed income and as an alternative to bonds.

Dividends are specified as an annual dollar amount or as a percentage of par value. Unlike a bond, the issuer can decide **not to pay** the preference dividend if earnings are insufficient, and failure to pay does **not** constitute default.

Investors regard preference shares as **less risky than ordinary shares** because preference dividends are specified and must be paid before ordinary dividends. They are **riskier than bonds** because bondholders are paid first in liquidation. Preference shareholders have **no voting rights** except in financial distress. Companies may issue several classes of preference shares, distinguished by the method and amount of dividend payment and by associated special benefits.

Feature	What it does
Cumulative	Once common, now rare. Requires all past unpaid preference dividends to be paid before any ordinary
Callable	Most preference shares are callable at a stated redemption price, usually only after several years. Lets t
Participation	Lets the preference holder receive extra dividends when earnings are sufficient. Attractive but uncomm

Feature	What it does
Convertibility	Gives the holder the right to exchange the preference share for a specified number of ordinary shares. A

6.2.5 Rights Issue, Bonus Issue, Stock Split and Reverse Stock Split

a. Rights issue. A rights issue injects new capital by issuing new shares. Rights to subscribe are first offered to **existing shareholders**, who may buy a stated number of new shares at a specified price. This prevents dilution of their share of profit and voting interest.

Rights may be **renounceable** (tradable) or **non-renounceable** (not transferable, cannot be traded). Companies may now issue pro-rata renounceable rights shares of up to **100% of share capital**, up from the previous limit of **50%**. If an existing shareholder declines to exercise, the rights may be taken up by other investors in the market.

Worked example — Dragonfly Ltd. It is September 2022. Of an authorised capital of **500 million** shares, Dragonfly has **300 million** outstanding, trading at $2.00 **at end-September 2022. Management announces a rights issue of ** *100 million * *shares at * *$1.90** to raise capital, mainly to acquire a company developing navigation and tracking parts for commercial drones, with part of the funds for refinancing. Forecast EPS for the financial year ending December 2022 is **$0.11**.

(1) Likely price reaction. The reaction depends on the market's perception of the purpose and benefits of the capital raising and the company's prospects. If the purpose is seen as positive — the acquisition adds to future revenue, earnings and value — the price may **rise** after the announcement. If the purpose is seen as risky, or the reasons are unclear, the price may **fall below the issue price**.

(2) Theoretical ex-rights price. Before the issue, 300m shares at $2.00 *implies market capitalisation of * *600m**. After a **1-for-3** rights issue there are **400m** shares. New capital is **$190m** (100m × 1.90), *so the new theoretical market capitalisation is * *790m** ($600m + 190*m*). *The theoretical ex-rights price is * *1.975** ($790m / 400m).

(3) Weighted-average shares and revised EPS. Weighted-average shares = (300m × 9 + 400m × 3) / 12 = **325m**. Revised EPS forecast = $(0.11 \times 300m)/325m = **0.1015**$ per share.

b. Bonus issue. No additional capital is raised. It is usually a **signalling** exercise when the company has done well and expects the stock to rise on good news. A 1-for-3 bonus issue for Dragonfly gives **400m** shares, but the price is adjusted down **more sharply** because no new capital is injected. From the same $2.00 and 300m shares, the theoretical ex-bonus price is $600m/400m = **1.50**$ — sharply lower than the rights-issue price.

c. Stock split. A stock split makes shares more accessible to retail shareholders. **OCBC made a 2-for-1 split in 2005** when its price was above 13 * *; *after the split and another rights issue, its stock was * *$11.55** as of October 2017. When prices reach a level where the shareholder base is excessively skewed to institutions, a company may split to make the unit price more affordable. A split also gives majority holders more convenience in selling units. **Stock splits do not raise capital.**

d. Reverse stock split. A reverse split combines a number of shares into one or fewer. **Citibank carried out a 1-for-10 reverse split in 2011.** During the Global Financial Crisis, banks such as Citigroup that had traded above 60 * *found themselves below * *5** (Citigroup's low was **$0.97**). Certain institutional investors are governed by rules discouraging investment in "penny stocks"; the SEC defines a penny stock as one priced **below** 5 * *. *That made it difficult for institutions to own Citigroup. The* $1-for-10$ *reverse split took the shares from** **4.52** just before to 44.16** *at the close of the first trading day after. By mid-2022 Citigroup traded at * *47.10**.

6.3 Choice of Equities

Strategies depend on the return sought and the client's risk threshold. The basic strategy is to search for undervalued stocks (Section 6.5); this section covers the types of stock available.

6.3.1 Blue-Chip Companies

A simple approach for long-term investors is to buy **blue chips**: major conglomerates with great financial strength and well-established, stable, mature businesses. With a long, consistent earnings record they can generally pay steady dividends over long periods in both good and bad years.

Blue chips hold leading positions in their industries and economies, are usually pacesetters, and frequently set the standards by which others are measured. Their management has shown foresight in growing the company without jeopardising current earnings. Such companies have the advantage of size and resources for sustainable earnings. Investors seeking safety and stability, and conservative investors, usually turn to blue chips.

Blue chips had periods of high earnings growth in their early years, but growth slows with size. Many mega-growth stocks dominating today are likely to become blue chips. Because of financial strength, connections and operating capability, many Asian blue chips have diversified into new businesses or geographies to sustain growth. In smaller markets blue chips also account for a large share of market capitalisation.

Singapore blue chips include **Keppel Corporation, SembCorp Industries, City Developments, CapitaLand, DBS, OCBC, UOB and SIA**. US blue chips include **3M, Procter & Gamble, General Electric, Walmart, IBM, Hewlett Packard, Apple and Microsoft**.

Do not be complacent: the conditions fundamental to any company's success can always change, especially with technological disruption.

6.3.2 High-Growth Companies

High-growth companies usually come from industries and businesses in the new economy — at the forefront of research and development for new technologies, new business processes, renewable energy and new ways of doing business. Some of their products and services disrupt traditional ways of operating.

"Growth companies" usually means companies generating high double-digit sales and earnings growth over several years — for example **more than 20% compounded annually over several years**. There are two stages of high growth:

- High growth in sales but still reporting **negative profits**.
- High growth in **both sales and earnings**.

a. High growth, negative profits. On NASDAQ there are many concept companies with strong future sales growth potential that keep incurring higher costs and cannot break even. Many technology and biotech companies are bought for potential research breakthroughs. For many, the **burn rate** from R&D catches up before they show profit and they become bankrupt. Others persist thanks to key investors who keep funding them until profitability.

Tesla reported many years of losses but was heavily supported until its **first quarterly profit in 2017**. Twitter reported almost **12 years of losses** until Q4 2017. When it becomes clearer that high sales growth will translate into profitability, the rest of the investment community becomes interested, and such companies sometimes become acquisition targets.

b. High growth, high profits. The second group has excellent sales **and** earnings growth, part of which is reflected in the stock price. The key is identifying companies with **sustainable** high earnings growth to justify continued appreciation. These companies are usually aggressive, with strong research teams, and pay relatively **small dividends** because most earnings are retained for expansion. Key characteristics: a solidly entrenched position in an expanding market, distinctive or unique products, and high profitability, usually by pre-empting a special part of the market. Examples of phenomenal earnings growth from **2014 to 2018**: Amazon, Alibaba, Tencent, Nvidia and Facebook.

c. Characteristics. High-growth companies typically carry **high P/E multiples** because investors pay a premium for higher earnings growth relative to industry peers. Growth cannot last forever as companies get bigger; once earnings growth slows, premiums narrow and the growth stock is **re-rated** at lower ratios. These are usually **high-beta** stocks with greater market sensitivity: in a general sell-off they correct much faster. Growth-stock investors must cope with higher volatility.

6.3.3 Companies with High Dividend Yields (Income Stocks)

Companies paying higher average dividend yields appeal to retirees and others buying shares for current income. Income stocks are often sought by trust funds, pension funds, endowment funds, and charitable and educational foundations. Higher payers include mature firms generating large cash flows; being mature, they may not need cash for expansion and can afford higher dividends.

A benefit is that when markets correct, high-yield stocks usually find **price support** on the strength of their yields, provided there is confidence the dividends will be paid. For example, if the fixed deposit rate is **3%**,

stocks with dependable payouts are unlikely to fall below a dividend yield of **4–5%** before tax — the price is supported where dividend/price = 4%.

Drawbacks exist. The share may belong to a lacklustre company in a socially unpopular industry such as tobacco. With low growth potential, it pays high dividends. In some companies the high dividends are offset by a tendency to call rights issues, which can be dilutive — some REITs. And high dividends persist only if earnings can be sustained; companies in difficulty have been forced to cut dividends.

6.3.4 Cyclical Stocks

The market rises and falls with the economy, but some stocks benefit more in a boom and do worse in a downturn. These are **cyclical stocks**.

For economies driven by domestic consumption, cyclicals are among the first to benefit from increased consumer and business spending during growth and low unemployment. Automobile manufacturers (GM, Ford) are representative consumer-based cyclicals. Their earnings and prices fluctuate with the business cycle or are accentuated by it: in good conditions they enjoy strong earnings growth and price appreciation; when conditions deteriorate they are the first affected by the spending slowdown and their prices drop as earnings growth declines.

Because economies differ in composition, which stocks are cyclical differs by market. Taiwan and South Korea are driven by high-tech and electronic manufacturing, so those sectors are cyclical there. In Singapore and Hong Kong, cyclicals also come from **property and banking**, usually among the first sectors affected in a downturn.

6.3.5 Defensive Stocks

Defensive companies' earnings are more likely to withstand downturns or uncertainty. They tend to provide necessities — **pharmaceuticals, food, utilities and transportation** — and their shares are less volatile than cyclical and growth shares. **Nestlé and Coca-Cola** are typical defensive stocks. If a recession is expected, there is strong interest in "recession-proof" companies, especially if they pay a reasonable dividend yield.

6.3.6 International Equities

With internet brokers and local brokers offering execution and research on foreign stocks, international investing is now common even for retail investors. The need for diversification and exposure to growth companies forces investors to look abroad.

Route	Detail
Direct investment	Buy foreign shares on their exchanges through local brokers. Diversifies away hom
Depositary receipts (DRs)	Indirect ownership of foreign shares, letting investors buy and hold foreign securiti
Exchange-traded funds (ETFs)	Replicate a broad index such as the STI or S&P 500. Buying an ETF assumes the
Foreign shares with dual listings	Listed both in the home country and another jurisdiction. Dual listing enhances li

6.4 Equity Market Indices

The general movement of the market is measured by an equity market average or index — a group of securities supposed to reflect the entire market. The composition, the weight given to each security, and the calculation method all matter. Three weighting schemes are used:

- **Value-weighted.**
- **Price-weighted.**
- **Un-weighted or equal-weighted.**

6.4.1 Value-Weighted Series

A value-weighted series derives the initial total market value of all shares in the series (market value = shares outstanding × current price). That figure becomes the **base**, assigned an index value, normally **100**. A new market value is then computed and compared with the base to determine the percentage change, which is applied to the beginning index value.

Worked example.

Day 1	Shares	Price ($)	Market value ($)
A	1,000,000	5.00	5,000,000
B	3,000,000	3.00	9,000,000
C	5,000,000	10.00	50,000,000
		Total	**64,000,000**

Base value = index of **100**.

Day 2	Shares	Price ($)	Market value ($)
A	1,000,000	5.20	5,200,000
B	3,000,000	3.10	9,300,000
C	5,000,000	10.50	52,500,000
		Total	**67,000,000**

$$\text{New Index Value} = \frac{\text{Current Market Value}}{\text{Base Value}} \times \text{Beginning Index Value} = \frac{67{,}000{,}000}{64{,}000{,}000} \times 100 = 1.05 \times 100 = 105$$

An advantage of the value-weighted index is that it **automatically adjusts** for bonus and rights issues and other capital changes, because those actions typically do not affect market capitalisation — a fall in price is offset by a rise in shares outstanding. A constituent's importance depends on its market value, so price changes in high-capitalisation shares move the index more than those in low-capitalisation shares.

The **S&P 500** is a value-weighted index of the top 500 companies, principally by market capitalisation, accounting for around **75%** of total US stock market capitalisation. In Singapore the **STI** is also weighted by market capitalisation; it replaced the Straits Times Industrial Index, which was **price-weighted**.

6.4.2 Price-Weighted Series

A price-weighted series uses an **arithmetic average of current prices**: prices are added and divided by the number of securities. The divisor must be adjusted for capital changes (rights and bonus issues) and changes in the sample. Because the series is price-weighted, a **high-priced share carries more weight** than a low-priced one. A stock split therefore **reduces** a share's weight even though market capitalisation is unchanged.

The **Dow Jones Industrial Average (DJIA)** is a price-weighted average of **30** large, well-known industrial shares. It has been criticised for using only 30 non-randomly selected blue chips, which are not considered representative of shares listed on the NYSE.

6.4.3 Un-Weighted or Equal-Weighted Series

In an un-weighted or equal-weighted index, all shares carry **equal weight** regardless of price or market capitalisation. A $2 share is as important as a $10 share, and total market value is ignored. Changes may be based on the arithmetic or geometric average of percentage price changes.

Worked example — geometric average.

Share	Day 1 ($)	Day 2 ($)	1 + % change
A	5.00	5.20	1.04
B	3.00	3.10	1.03
C	10.00	10.50	1.05

$$\text{Geometric Average} = (1.04 \times 1.03 \times 1.05)^{1/3} = 1.040$$

Index value (Day 2) = index value (Day 1) × 1.040.

The **Value Line Composite Average** is an equal-weighted geometric average of share prices; more than **80%** of its constituents are listed on the NYSE.

6.5 Valuation Approaches

The time value material in Chapter 2 provides the foundation. There are three ways to value a listed company and its businesses:

- **Net asset valuation** — book value, liquidation value and revised net asset value.
- **Discounting of future cash flows** — the absolute valuation approach.
- **Relative valuation** — comparison against historical trends and other companies in the industry.

Net asset value approach	Absolute valuation (DCF)	Relative valuation — shareholder's perspective	Relative
Book value	Constant growth DDM	Price-to-earnings	Enterpri
Revised net asset value (RNAV)	Multi-stage DDM	Price-earnings growth	Enterpri
Liquidation value (salvage value)	Discounted free cash flow	Price-to-book value	Enterpri
		Price-to-RNAV	
		Dividend yield	

6.6 Net Asset Valuation

The net asset valuation approach is a basic balance-sheet approach: net assets are valued at book value, and revised to market value or liquidation value depending on circumstances. It is particularly relevant for companies whose value is predominantly determined by tangible assets on the balance sheet — **loans of banks, premiums of insurance companies, real estate holdings of property companies**.

6.6.1 Book Value

The book value of equity reflects total assets as stated on the balance sheet, net of all liabilities. It is one of the key numbers investors and acquirers examine. Book value divided by shares outstanding gives **book value per share**.

When conditions are severe — a market crash — or when a company is making losses with a challenging outlook, book value is one way to assess the value of the equity. Book value also represents the accumulation of retained earnings over the years. In bad times, if shareholders' equity has not been wiped out by losses, book value often represents the **floor value**.

The problem with book value in a crisis is assessing the deterioration in asset values, which can be **understated**. When that happens, book value must be revised downwards.

6.6.2 Liquidation Value (Salvage Value)

Liquidation value assumes the company will be discontinued and viable operations sold to other companies. Assets and businesses are priced for **ease of sale** so money can be recovered quickly to pay creditors. The price obtained may be much lower than market or book value, so the residual liquidation value can be much lower than equity book value. Listed companies have encountered severe business failures and unexpected losses leading to eventual liquidation.

6.6.3 Revised Net Asset Value

When the value of assets or liabilities has changed significantly before the end of the reporting period, book value may be revised to reflect actual value, from which **revised net asset value per share (RNAVPS)** is calculated in the same way as BVPS.

The real estate industry is the standard example. For property companies with substantial land bank and property holdings, professional real estate consultants provide a valuation to mark property asset values to market. But real estate prices can appreciate or fall rapidly within a financial year, so before year-end analysts may have to revise net asset value to reflect major appreciation or correction.

When revising asset values to market, **also check whether debt and other liabilities need revising**. The market value of USD-denominated debt may have appreciated in the interim; if so, the market value of liabilities should be revised too.

6.7 The Discounted Cash Flow (DCF) Approach

For the CM-EIP examination, representatives are expected to know how to **explain** the process of valuing stocks and the variables used, but will **not** be asked to compute and derive fair value.

The investment community mostly uses the DCF method to derive fair value, so this module focuses on DCF. The dividend discount model (DDM), used only occasionally for companies with high dividend payouts (**DPS/EPS above 80%**), is set out in Section 6.13.

6.7.1 Introduction to the DCF Approach

The key principle is the **time value of money**: money received in the future is worth less than money received today. Under DCF, a company's current fair value is estimated by **discounting** future free cash flows to present values.

Several DCF methods exist depending on assumptions, and their results can vary significantly. This module focuses on the **Free Cash Flow to Firm (FCFF)** approach, using the **Weighted Average Cost of Capital (WACC)** as the discount rate.

- **Free cash flow to firm (FCFF)** — a commonly used measure of projected cash flows. Derive the after-tax cash flow from operations, deduct cash set aside for capital expenditure and to sustain business activities, then add back depreciation and other non-cash expenses to reflect free cash flow available to the company.
- **Terminal value** — project FCFF over a reasonable period, for example **10 years**. Beyond that it makes more sense to assume a terminal value at the end of year 10 encompassing all future cash flows thereafter, indefinitely if the company is a going concern. The terminal value must also be discounted to present value.
- **Discount rate** — depends on how the company is financed, by equity, debt or both. It is therefore the **WACC** of the cost of equity and cost of debt used to generate future cash flows.
- **Enterprise value** — discounting future FCFF and terminal value at the WACC gives the current value of the company, the enterprise value: the sum of all discounted future FCFF and the discounted terminal value.

If the market value is **below** intrinsic value by a significant percentage (undervalued), buy. If **above** (overvalued), sell.

6.7.2 Calculating the Weighted Average Cost of Capital

FCFF is discounted at the WACC because the firm's value and future cash flows are generated with funds from both debtholders and shareholders. The returns these investors demand are the **cost of capital** to the company — the cost of raising funds through debt (bonds) and/or equity (share issues). WACC is a weighted average of the cost of equity and the cost of debt.

a. Cost of equity. The Capital Asset Pricing Model (CAPM) is applied:

$$R = R_f + \beta(R_m - R_f)$$

where

$$R$$

is the cost of equity,

$$R_f$$

the risk-free rate,

$$R_m$$

the market return,

$$R_m - R_f$$

the **equity market risk premium**, and

$$\beta$$

the sensitivity of the stock price to the market.

The typical steps in computing firm and equity values:

- Calculate the WACC.
- Estimate FCFF over a more predictable period, typically **10 years**.
- Estimate the long-run terminal value at the end of the estimation period (the price at year 10).
- Sum the present values of the FCFFs and the terminal value to obtain **enterprise value**.
- Subtract the market value of net debt and preferred shares from enterprise value to get the estimated **market value of equity** (fair value of equity), the estimated market value of the common shares.
- Divide by shares outstanding to get the **intrinsic share value** (fair value per share).

Worked illustration — Procter & Gamble.

Input	Value and source
Risk-free rate R_f	**2.33%** — the US 10-year Treasury rate at the time. For Singapore stocks, the 10-year SGS
Required market return R_m	The average price return of the S&P 500 over the last 20 years, including the GFC and dot
Beta β	**0.68** (providers quote 0.65–0.68 over five years; the higher figure is used, assuming more ma
Cost of equity	$R = 2.33 + 0.68(9.28 - 2.33) = 7.056\%$

b. Cost of debt. The effective interest rate the company pays on debt. From the balance sheet, P&G's effective interest cost (interest expense / borrowings) was **1.4%** in 2016 and about **1.1%** for the first nine months of 2017. Another way is to identify the interest cost of each debt item from the notes. The conservative approach uses the higher **1.4%**, since rates were expected to rise with Fed hikes.

c. Weighting. Inputs:

Input	Value
Market value of debt	**USD30.414 billion** — the book value of debt as at 8 July 2017. Where there are no significant
Market value of equity	**USD224.17 billion** — always use market value: **\$87.65 × 2.5576 billion shares**
Effective tax rate	**23%** (22.6% for the last nine months, 25% in 2016)
Total capital	Debt + equity = 30.414 + 224.17 = **254.584**

$$\text{WACC} = (1-0.23)(0.014)\left(\frac{30.414}{254.584}\right)+(0.07)\left(\frac{224.17}{254.584}\right) = (0.0108\times 0.1195)+(0.07\times 0.8805) = 0.0629 = 6.29\%$$

That **6.29%** is the discount rate for all future cash flows. With the low-rate environment, P&G had locked in long-term debt priced very cheaply relative to its cost of equity.

6.7.3 Estimating Free Cash Flows to the Firm

Most analysts build projections of the **pro-forma income statement**, balance sheet and cash flow statement, for about **10 years**. In primary research the analyst interviews management about prospects and expenditure needs, and earnings estimates come from projections of sales, other income and cost items.

Analysts also rely on historical ratios and relationships, reviewing past major expenditure patterns and asking management about likely major capital outlays. Alternatively, FCFF is approximately the **sum of cash flow from operating activities and cash flow from investing activities**, since the underlying numbers are often the same.

6.7.4 Calculating the Enterprise Value

Discount the estimated FCFF and terminal value over the chosen period:

$$\text{PV of Free Cash Flows} = \frac{E(FCFF_1)}{(1+\text{WACC})^1} + \frac{E(FCFF_2)}{(1+\text{WACC})^2} + \frac{E(FCFF_3)}{(1+\text{WACC})^3} + \cdots + \frac{E(FCFF_{10}) + TV}{(1+\text{WACC})^{10}}$$

The terminal value uses the estimated FCFF in year 11 divided by (WACC −

$$g_c$$

), where

$$g_c$$

is the **terminal growth rate** at which FCFF are expected to grow indefinitely:

$$\text{PV of Terminal Value} = \frac{\frac{E(FCFF_{11})}{\text{WACC} - g_c}}{(1 + \text{WACC})^{10}}$$

Worked illustration — P&G. Cash flows from operating and investing activities in 2016 are the base. P&G's cash flows grew at an average of **1.5%** over 2006–2016 and about **3%** over the last five years. The analysis applies **2.5%** growth for 2017–2021, slowing to **2%** for 2022–2026, settling at a terminal growth rate of **1.5%** — consistent with the past pattern. WACC is **6.29%**. The resulting firm value is **USD260,391 million**.

6.7.5 Calculating the Fair Value of the Stock

Enterprise value is the sum of the estimated market value of common shareholders, preferred shareholders and debt. To derive the fair value of the common stock, subtract the estimated market value of preferred shares and debt from enterprise value, leaving the estimated market value of common shareholders (fair value of common equity). Divide by shares outstanding for the **fair value per share** or **intrinsic share value**.

Worked illustration — P&G.

Item	USD million
Enterprise value	260,391
Less market value of debt	(30,514)
Less value of preference shares	(1,038)
Estimated market value of share capital	**228,839**
Intrinsic share value (228,839 / 2,557.6m shares)	**USD89.47**

Against P&G's price of **USD87.65** on 7 July, the estimate implies about **2% upside** — fairly valued. The intrinsic value can serve as the target price accompanying the qualitative research report.

6.7.6 Criticism of the DCF Model

DCF involves many assumptions and estimates. Different research teams arrive at different target prices. Determining fair or intrinsic value by DCF is **not an exact science**; it provides an estimate based on assumptions. Cynical observers suggest the estimate is open to "fudging" until the assumptions produce a price range aligned with the qualitative story. The model depends heavily on the **terminal value**, which can vary significantly with different assumptions.

6.7.7 Actual Market Practice for Cash Flow Models

The industry discounts free cash flows to estimate intrinsic value or a target price, typically discounting 10 years of cash flows and the terminal value at year 10. DCF is used **in conjunction with relative valuation** to reach a more accurate view of whether a stock is over- or undervalued.

The multi-stage DDM is **seldom applied day to day**. On occasion it is used for companies with extremely high dividend payouts, such as tobacco stocks like British American Tobacco.

6.8 Relative Valuation Ratios

DCF is an **absolute** valuation approach: intrinsic value comes from projected cash flows. A **relative** approach derives value by comparison with companies in the same industry.

Relative valuation can be made from either:

- A **shareholder's perspective** — for example the price-earnings ratio.
- A **firm-wide perspective** — for example EV/EBITDA, which takes both debtholders and shareholders into account.

In practice multiple ratios are used together for a more holistic assessment.

6.8.1 Price-Earnings (P/E) Ratio

The P/E ratio indicates how much investors are willing to pay for a dollar of earnings: current price per share divided by EPS.

$$\text{P/E Ratio} = \frac{\text{Market Price Per Share}}{\text{Earnings Per Share (EPS)}}$$

Form	Definition
Trailing (current) P/E	Current market price divided by the most recent **four quarters** of EPS. This is what news
Leading (prospective) P/E	Current market price divided by the **consensus forecast** earnings for the next year, as pro

Questions to ask when applying P/E:

- How much are you paying for the future earnings generated by the company?
- How does this compare with the company's historical P/E in past years?
- How does this compare with the P/E of other similar companies?
- Are there qualitative reasons for this company to remain at a premium or discount?

Worked example — semiconductor comparison. INTC closed at **USD34.68**. FY2016 EPS was **USD2.72**, with consensus forecasts of **USD2.86** for FY2017 and **USD2.97** for FY2018.

Measure	Calculation	Result
Trailing P/E for 2016	34.68 / 2.72	**12.8x**
Prospective P/E for 2017	34.68 / 2.86	**12.1x**
Prospective P/E for 2018	34.68 / 2.97	**11.7x**

An investor pays just **USD12.1** for a dollar of FY2017 earnings from INTC, against **USD16.7** for TSMC and **USD22.1** for UMC. INTC's lower leading P/E across all three projected years implies paying less for future earnings. But it is **not** as simple as buying the lowest P/E and selling the highest.

Reading the historical range. The historical P/E chart is constructed from 12 months of prices, with the last month being three months before the financial year end — for example, October 2012 to September 2013 prices divided by actual 2013 EPS. All three companies have December year ends, so if expected earnings are close to actual, the chart reflects what the market decided to trade at given its expectations.

- **INTC:** the prospective FY2018 P/E of **11.7x** is at the **lower end** of its normal range of **10.3x to 16.0x**. INTC traded in tandem with TSMC historically but has traded at a larger discount to TSMC since 2016 — check whether that discount is justified. If INTC should trade at **14x**, that implies **18–20% upside**, a possible **BUY** signal.
- **TSMC:** to sustain its premium over INTC it must maintain stronger EPS growth, and consensus forecasts show it does. That justifies the premium; if TSMC does not deliver, its price falls relative to INTC. Its prospective 2017 P/E of **16.7x** is beyond its normal range, and FY2018's **14.9x** is at the high end. The justification is that TSMC was vying with Samsung to be the main supplier for Apple's A11 chips for the new iPhone, so FY2018 earnings are partly tied to that phone's success. Combining P/E analysis with the qualitative picture, the premium appears justified — but on the prospective FY2018 P/E alone, TSMC looks fairly valued.
- **UMC:** prospective FY2017 and FY2018 P/Es of **22.1x** and **18.7x** look expensive against a historical range of **11–15x**. Its EPS growth is higher than TSMC's and INTC's, but from a much lower absolute EPS base, and there is no major qualitative information suggesting more sustainable growth beyond the US economic recovery. Without justification, the price may weaken.

b. Factors affecting P/E ratios. Investment is not simply buying low P/E and selling high P/E; examine the qualitative rationale. Companies remain at a perpetual premium or discount so long as they have stronger or weaker earnings potential than peers. Beyond projected earnings:

- **Low transparency and predictability of earnings.** When risks are high and predictability low, even high earnings growth justifies a **lower** multiple.
- **Impending events** that may affect earnings, such as renewal of major contracts or pending legal decisions. Analysts must make major assumptions and the P/E may be lower for the uncertainty.
- **Uneven growth trajectory.** Companies with potential exponential long-term growth but very low short-term earnings may show very high prospective P/Es in the first few years. The **PEG ratio** helps assess this.
- **Market inefficiency.** In less liquid markets where the stock is controlled by major shareholders, the market may not price efficiently, distorting the P/E through illiquidity.

c. Growth versus mature companies. Growth stocks typically command **higher** P/E ratios than the market average because investors pay more for higher expected earnings growth.

Advantages and disadvantages of the P/E ratio.

Advantages	Disadvantages
Widely used and easily understood	Earnings can be negative, making P/E meaningless
Easily used for comparison across companies and sectors	Accounting differences affect earnings and P/E
	Earnings can be volatile or lumpy, making P/E hard to interpr
	Earnings forecasts can be quite inaccurate
	P/E does not distinguish between valuation and expected grow

Shiller's inflation-adjusted P/E ratio. Robert Shiller used inflation-adjusted P/E ratios to track periods when the S&P 500 was overvalued. According to Shiller, the S&P 500's P/E ratio passes the **30x** mark during periods of overvaluation — what former US Federal Reserve Chairman Alan Greenspan called **irrational exuberance**. Although the ratio has flirted with that level in recent years, there were only two earlier occasions when the inflation-adjusted S&P 500 P/E ratio was above **30x**: the Great Depression of **1929** and the **2000** dot-com bubble.

P/E ratios and interest rates. The P/E ratio is related to the required rate of return, which is in turn related to interest rates. As interest rates increase, required rates of return on all securities generally increase. When interest rates rise, the fall in price valuations is greater than the fall in earnings, so P/E ratios fall. Conversely, when interest rates fall, price valuations are more significantly affected than earnings, so P/E ratios rise.

6.8.2 Price-Earnings Growth (PEG) Ratio

A criticism of P/E is that it ignores expected future growth: a low P/E may mean undervaluation **or** simply poorer expected growth, and a high P/E may mean overvaluation **or** stronger expected growth. Strictly, P/E ratios should only be compared between companies with similar expected growth. The **PEG ratio** addresses this: the P/E ratio divided by the expected annual earnings growth in per cent, measured by EPS.

$$PEG = \frac{P/E}{\%\text{ Growth in EPS}}$$

where the growth is typically the prospective EPS growth compounded over **3 years**; when growth potential lies further out, a **5-year** projected growth rate may be used.

By controlling for expected earnings growth, PEG indicates whether the current price is under- or overstated relative to another company with different growth prospects. PEG is most widely used for **high-growth** companies and industries.

A **low PEG indicates an undervalued share**. As a rule of thumb a PEG **below 1** is attractive but typically very hard to find: such stocks may be relatively new, not actively covered, and carry higher risk because the earnings growth may not materialise. They are typically in high-tech, biotech or disruptive technologies in high demand. Once noticed, more investors become interested and the price rises, pushing PEG above 1.

Worked illustration. Among global high-growth tech companies, **Alibaba** had the highest prospective P/Es — **44.5** in 2017 and **33.8** in 2018 — yet its PEG was among the **lowest at 1.6**, reflecting its high **28%** projected annual growth over the next few years.

A caution on projections: when markets are bullish and the economy strong, the analyst community tends to **overestimate** the sustainability of high earnings growth.

6.8.3 Price-to-Book Value (P/BV) Ratio

$$\text{P/BV Ratio} = \frac{\text{Market Price Per Share}}{\text{Book Value Per Share}}$$

P/BV is best suited to companies whose value is tied directly to asset holdings. Book value matters more for **investment companies, banks, insurance companies and property companies**, because many of their assets are reflected in the books and can be liquidated.

a. Financial institutions. Because P/BV uses book rather than market value, it can mislead when market values differ substantially from book. If a bank's loans become non-performing, their market value may be near zero while book value is unchanged. Since the Global Financial Crisis several banks have seen book value eroded by proprietary trading losses, excessive risk-taking on distressed debt and credit derivative losses. With adequate provisions and proper recapitalisation, book value can be adjusted accurately; with inadequate provisions, additional write-downs follow over time.

The lack of transparency on loan book and investment position quality is a problem when applying P/BV to banks. The ratio has also become less dependable because banks continue to encounter credit losses after the crisis — most recently provisioning for non-performing loans to oil-related companies and commodities trading losses.

b. Property companies. P/BV is actively applied to property companies because the bulk of asset values is well reflected on the balance sheet. Property companies and REITs with sizeable commercial, office and residential holdings may enjoy significant book value appreciation when the property market rises, and suffer substantial falls in a downturn. Accountants rely on independent professional real estate consultants to revalue property assets annually.

During major market and economic downturns, stock prices can fall much more sharply than book value. Companies with diffused ownership can then become good takeover targets: with the price below book value, control of the assets can theoretically be acquired cheaply, and investors with deep pockets can acquire distressed property companies with hard assets. After the **1997 Asian Currency Crisis**, some small listed property developers caught in a liquidity crunch were acquired by larger developers at prices below book value.

c. Hidden reserves in book value. Market values of land bank and property can differ significantly from book value — when land is rezoned for a different use (agricultural to commercial or residential), or when land becomes more accessible through better road and rail links. Such land may remain at book value while worth much more at market value; where accounting limits allow this, a **hidden reserve** is created. Hidden reserves also arise when provisions made in prior years are no longer needed but are not written back.

In such cases book value is **understated**, so the true P/BV is **lower** than calculated, making undervaluation more likely.

Advantages and disadvantages of P/BV.

Advantages
Book value is cumulative and usually positive, unlike EPS
Book value is usually more stable than EPS, which can be volatile
Well suited to companies whose value is closely tied to tangible asset holdings — banks, insurers, real estate companies

6.8.4 Price-to-Sales (P/Sales) and Enterprise Value-to-Sales (EV/Sales) Ratios

a. Price-to-sales. A feasible comparison for companies generating very low, zero or negative earnings. It indicates the number of times the market is willing to pay for a firm's sales revenue per share.

$$\text{P/Sales Ratio} = \frac{\text{Current Market Price Per Share}}{\text{Sales Per Share}}$$

Sales per share is annual sales divided by ordinary shares outstanding. P/Sales is useful where sales are strong but earnings are lacking or negative — some high-tech and biotech companies generate sales but remain unprofitable. Proponents argue that consistent sales indicate a strong company and that sales are a fundamental requirement for earnings.

Use it with caution. It does not present a complete picture because it ignores expenses and liabilities, and a **lower** P/Sales is not necessarily positive — it may simply indicate the company is unprofitable. Conceptually the ratio is flawed: the numerator (price) relates to shareholders only, while sales are generated from funds supplied by both shareholders and debtholders, so numerator and denominator are inconsistent.

b. Enterprise value-to-sales. Some investors consider EV/Sales more appropriate because both numerator (EV) and denominator (sales) are consistent — both debt and equity financing generate sales. Enterprise value became widely adopted in the **1980s** and is a total firm value concept.

c. P/Sales versus EV/Sales. P/Sales is easier and more convenient, and intuitively asks "how much am I paying for the projected or current sales?" But it ignores risks from debt financing, especially where large debt generates sales, so a highly indebted company can appear inexpensive on a low P/Sales. Because EV encompasses both debt and equity, EV/Sales accounts for the value of debt and is more appropriate for cross-company comparison.

Advantages and disadvantages of EV/Sales.

Advantages	Disadvanta
Sales figures are generally less subject to manipulation than earnings	High sales r
Sales are positive, unlike EPS which can be negative — useful for start-ups and companies with no earnings	Sales reven
Sales are more stable than EPS, which can be volatile	Expenses a
Applicable to almost all companies	
EV/Sales directly reflects the impact of higher debt on relative valuation, which P/Sales does not	
Conceptually correct, since numerator and denominator both relate to the total enterprise	

6.8.5 Enterprise Value-to-Cash Flow (EV/CF) Ratio

EV/CF is used because of concerns about earnings manipulation: cash flows are much less prone to manipulation. **Operating cash flow** (or free cash flow) is usually used — the total cash generated through core operations, found in the statement of cash flows under "cash generated from operations".

Worked example — Fulton Corporation. Market price $18 **per share, **1 billion **shares outstanding at 31 December 20
million of preference shares and 400 *million ** of borrowings. No associates, minority interest or unfunded pension obligation
***2,210 million** and cash and cash equivalents at year end are **$8,000 million**.

$$\text{EV/CF} = \frac{\text{Market Cap} + \text{Debt} + \text{Preferred Shares} - \text{Cash}}{\text{Cash Flow from Operations}} = \frac{(18 \times 1{,}000) + 400 + 10 - 8{,}000}{2{,}210} = \frac{10{,}410}{2{,}210} = 4.71$$

A **lower** EV/CF is desirable when comparing companies, because it implies more cash flow generated from the capital supplied.

6.8.5.1 Advantages and Disadvantages of Enterprise Value-to-Cash Flow from Operations

Advantages	Disadvantages
Cash flows are less likely to be manipulated than earnings	Cash flows can be from oper
Cash flows are independent of accounting methods	
EV/CF considers the impact of debt, since debt funding also generates operating cash flow	

6.8.6 Enterprise Value-to-EBITDA Ratio

EV/EBITDA is also called the **enterprise multiple** or EBITDA multiple and is actively used in relative valuation. It is **more popular** than EV/Sales or EV/operating cash flow.

a. EBITDA versus net earnings and operating cash flow. Four reasons EBITDA is preferred:

- EBITDA is closer to **cash-based earnings**: unlike net earnings it does not subtract non-cash depreciation and amortisation. Adding back D&A to EBIT gives the bulk of cash-related earnings.
- EBITDA is a **firm value concept**: it adds back interest cost because it considers total earnings available to creditors and shareholders. Net earnings are a shareholder concept because interest has been subtracted.
- EBITDA is preferred over operating cash flow because the latter swings widely with **working capital** changes, which may not reflect business performance since they are influenced by inventory, payables and receivables decisions.
- It is **easier to forecast** EBITDA than operating cash flow.

b. Why enterprise value is preferred over price. EV considers the impact of debt and other capital on total firm value, while price relates only to equity. Hence EV/EBITDA is preferred over P/E. EV/EBITDA became actively used in the **1980s** because P/E did not directly reflect the value of debt. Unlike P/E, EV/EBITDA **cannot be manipulated by changes in capital structure**, allowing fair comparison of companies with different capital structures.

$$\text{EV/EBITDA} = \frac{\text{Market Cap} + \text{Debt} + \text{Preferred} + \text{Minorities} + \text{Pension} - \text{Associates} - \text{Cash \& Near Cash}}{\text{Earnings Before Interest, Taxes, Depreciation and Amortisation}}$$

c. Application. Typically the current EV is used in the numerator and projected annual EBITDA for one to three years ahead in the denominator, asking: "what total capital, net of cash, is deployed to generate the cash earnings for a given year?" All else equal, a **lower** EV/EBITDA means less capital is deployed to generate cash earnings. Some companies will have perpetual EV/EBITDA discounts or premiums for the reasons in 6.8.1(b). EV/EBITDA appears in Bloomberg and most brokerage equity research.

Advantages and disadvantages of EV/EBITDA.

Advantages
The numerator (EV) captures the impact of debt used to generate earnings, enabling comparison across companies with d
The denominator (EBITDA) reflects cash-based earnings unaffected by working capital swings

6.8.7 Absolute versus Relative Valuation

	Advantages
Absolute (DCF) method	Conceptually relates intrinsic value to the earnings and cash flows due to the firm over tim
Relative valuation	Allows easy comparison across companies; intuitive and easy to implement

6.8.8 Dividend Yield

Dividend yield matters especially to income-oriented investors. Companies with stable, non-cyclical businesses paying a constant stream of healthy dividends have **downside support** for their share prices. Dividend yield is the income component of a share's return, stated as a percentage of the current price.

$$\text{Dividend Yield} = \frac{\text{Dividend Per Share}}{\text{Market Price Per Share}} \times 100$$

Worked example. A share priced at $2 * *with an annual gross dividend of * *\mathbf{0.10}$ per share has a dividend yield of **5%** (0.10 / 2.00 × 100).

The **dividend payout ratio** is dividends to earnings, indicating the percentage of earnings paid out as cash dividends. A company earning $0.40 * *per share and paying * *\mathbf{0.10}$ has a payout ratio of **25%**. The **retention ratio** is 1 minus the payout ratio, indicating the percentage of current earnings retained for reinvestment.

Worked example — expected dividend yield. KW Corp, an engineering, property and manufacturing company, was expected by ABC & Partners to pay a **1.5 cent** dividend in 2017, rising to **3.0 cents** in 2018. A 3-cent dividend on a share price of **31 cents** implies a high dividend yield of **9.7%**. The analyst expects **yield compression**: as the price rises, the yield falls. With a dividend yield expectation of **7.5%** on a 3-cent dividend, the implied share price is 3 cents / 7.5% = **40 cents**.

REITs can be valued using a dividend discount model because they are required to pay out **90% of distributable income** and their earnings are relatively predictable.

6.8.9 Cash Per Share and Valuations

Related to dividend is **cash per share**. When a company holds a significant proportion of cash, there is potential for large dividend payments, and such companies are potential acquisition targets.

Consider two companies with the same P/E of **10x**:

	Company A (cash rich)	Company B (cash poor)
Price	2.50	3.00
EPS	0.25	0.30
Net cash per share	**0.50**	Net debt

Company A holds excess idle cash of **$0.50** per share even after paying off its debt; Company B is in net debt. Company A is **undervalued**: if acquired at 2.50 and the idle cash of $0.50 per share is netted out, the buyer is paying a P/E of only **2.00 / 0.25 = 8x**, lower than Company B's 10x.

6.8.10 Strategic Value and Break-up Value of a Listed Company

In corporate acquisitions, consider the **strategic** and **break-up** value. An acquirer hopes the acquisition produces more earnings and higher combined value than the offer price.

Strategic value. When perceived strategic value exceeds the offer price (which is usually at or above market price), both the acquirer's and target's share prices may rise. If perceived strategic value is **less** than the offer price, the acquirer's stock is likely to **fall**.

Break-up value. Some acquisitions expect the sum of the parts sold to exceed the target's value as a going concern — the company is undervalued because of how it is managed. In the **1970s** in the US there were many hostile takeovers carried out expecting to sell parts of the businesses and assets at a higher total value than the takeover market value (share price × shares outstanding).

6.8.11 Earnings Yield

The earnings-to-price (E/P) ratio measures earnings against share price; expressed as a percentage it is the **earnings yield**. It is the **inverse of the P/E ratio**.

Earnings yields are usually compared against **bond yields** to assess whether share prices are over- or undervalued. The difference between the two is the **yield gap**. If the S&P 500 earnings yield is still **1.5–2.0%** above US Treasury bond yields, there is still room for the S&P to rise. As at October 2017 the US Treasury bond yield was **2.45%** against an S&P 500 earnings yield of about **4%**.

After the Global Financial Crisis of 2007/08 there was a sharp drop in earnings yields, but the extremely low rates through most of 2009–2016 supported earnings recovery, and earnings yields rebounded to as high as **6.0–7.0%** immediately after the crisis.

6.9 Initial Public Offerings

In Singapore, listing on the **SGX Mainboard** requires fulfilling listing requirements set by the exchange. Listing on **Catalist**, SGX's secondary board, is supervised by a **sponsor** (the equivalent of an Issue Manager for a Mainboard listing) and **no quantitative requirements** are needed: sponsors decide on the suitability of the candidate. Sponsors are qualified professional firms with experience in corporate finance and mergers and acquisitions.

In any IPO the issuer prepares to obtain SGX approval to list on the Mainboard or the secondary board. The issuer engages an investment bank or an SGX member company as **lead manager** or sponsor, and **underwriters** such as brokerages and other financial institutions are involved. The lead manager earns **management fees** and the underwriters earn **underwriting fees**. Lawyers draft the prospectus and related documents, and public accountants prepare the audited financial reports.

Once SGX approves the draft prospectus, it can be lodged on the MAS **Offers and Prospectuses Electronic Repository and Access (OPERA)** website. The issuer distributes the prospectus to institutional and retail investors; with pre-placement activity, institutions may view the offering before retail investors. IPOs are usually distributed in two tranches:

- A **private placement tranche**, typically placed with institutional and accredited investors, usually preferred customers of brokerages and banks.
- A **public retail tranche** of retail investors.

Usually the private tranche is a small group taking a major portion of the offering, while the retail tranche is smaller and much more widely distributed.

The public offering begins from the date the prospectus is registered with MAS and typically opens for at least **2 market days**, excluding the date of commencement. Applications are allocated or balloted if oversubscribed, within a few days. Trading begins the next business day after allocation, on a "ready" basis — bought or sold for settlement **2 days** after the transaction date (**T+2**).

Risks to watch when subscribing:

- **Vendor share sales** — the sale of the issuer's own holdings rather than new shares. The risk is that the issuer is offloading shares because it knows the outlook is poor. Watch for a **lock-up option**, a moratorium on the sale of the issuer's shares for a stipulated period, which is the issuer's expression of confidence.
- **Deferred shares or share options** awarded before the IPO may **dilute** new investors' earnings, because EPS is depressed once they are exercised.
- The **over-allotment option** (the **Greenshoe option**) gives the issuer the right to increase the offering through an over-allotment of shares. Underwriters can use it to stabilise the price if the shares perform poorly on initial trading by buying back the additional shares. Not all IPOs carry it.
- The issuer's right to **claw back** — reallocation of shares between institutional and retail investors. If institutional take-up is poorer than expected, the claw-back may be invoked and retail investors get a higher allocation of a share institutions probably avoided for good reason.

Beyond these, investors must do their own due diligence by studying the prospectus, available on the OPERA Public Portal at www.mas.gov.sg. When evaluating the prospectus, note:

- **Underwriters** — they generally purchase any remaining unsubscribed shares.
- **Business model** — examine the model, strategy and industry competition, usually in the Management's Discussion, Business Overview and Competition sections.
- **Use of proceeds** — if proceeds repay a large amount of debt, little may be left for expansion; if they buy out existing shareholders, it may be a bail-out at new investors' expense.
- **Management** — examine the experience and background of key appointment holders such as the CEO and CFO to validate their track record.
- **Financial analysis** — analyse the balance sheet, income statement and cash flow statement using the ratios covered earlier. Pay special attention to overly rapid expansion of sales, inventory growing faster than sales, an unusual fall in free cash flows, or aggressive acquisitions.
- **Timing** — note the **lock-up period**, the moratorium during which pre-IPO investors and promoters may not sell.
- **Conflicts of interest** between existing owners and management.
- **Secondary offerings** — after the IPO, shares can be issued through secondary offerings, typically as a financing exercise. This may be a **rights issue**, renounceable or non-renounceable, offering existing shareholders additional shares in proportion to their holdings. To avoid dilution they must take up the rights, which means putting up more money. In a rights issue the company lodges an **Offer Information Statement (OIS)** rather than a prospectus with MAS.
- A secondary offering can also be a **compliance placement** made to meet regulatory requirements on share distribution.

6.10 Valuing Companies in Initial Public Offerings

Both DCF and relative valuation are used to identify a target price or price range for IPOs. DCF follows Section 6.7: discount the free cash flow to derive firm value, subtract debt to obtain equity value, and divide by total shares outstanding for the target IPO price.

Worked example — Q-Star Pharmaceuticals. A sovereign wealth fund owns Q-Star, a pharmaceutical and biotech company preparing over **2–3 years** for a major listing on NASDAQ or the NYSE with a secondary listing on an Asian exchange. The owners want an estimate of current value.

Input	Value
Sales	USD7.65 billion
EBITDA	USD2.20 billion
Assets	USD18.30 billion
Shares post-issue	750 million
Debt	USD11.0 billion

The relative valuation uses EV/Sales, EV/EBITDA and EV/Assets of the top five pharmaceutical companies. To be conservative, the **lowest** multiple is selected in each category, and a further **20% discount** is applied because Q-Star lacks the patent pipeline and brand image of the top pharmaceutical companies.

Selected ratio	Lowest (times)	After 20% discount	Q-Star base (USD bn)	Firm value (USD bn)	Market cap (USD
EV/Sales	4.42	3.53	Sales 7.65	27.00	16.00
EV/EBITDA	12.33	9.86	EBITDA 2.20	21.69	10.69
EV/Assets	1.40	1.12	Assets 18.30	20.50	9.50

Multiplying Q-Star's sales, EBITDA and assets by the discounted multiples gives possible enterprise values. Netting off the **USD11 billion** debt leaves market capitalisation; dividing by **750 million** post-IPO shares gives a target range of **USD12.67 to USD21.33** per share.

Some prefer to apply the discount to the target price range rather than to the multiples, which produces a **higher** target range. A more common approach is to compare P/E and P/BV of similar companies and use a selected average or discounted multiple. The eventual price is a function of market conditions and the owners' expectations.

6.11 Depositary Receipts

6.11.1 Depositary Receipts (DRs)

DRs are equity securities that trade in an investor's home market and represent an underlying security in the foreign issuer's home market. They overcome obstacles to issuing and investing overseas — inefficient cross-border settlement, imperfect information flow and unfamiliar market practices.

For issuers, DRs offer:

- Broader and more diversified investor exposure, which may improve liquidity.
- Enhanced visibility for the company's products or services in an overseas market.
- A flexible mechanism for raising capital.

A variant is the **non-voting depositary receipt (NVDR)**, which facilitates foreign investment in an issuer's home market. The main difference from direct investment is that NVDRs carry **no voting rights**, so foreign investors cannot participate in corporate decisions. That feature enables foreign investment in firms that would otherwise be prohibited by regulatory restrictions on foreign ownership.

6.11.2 American Depositary Receipts (ADRs)

ADRs are the most common type of DR and trade on the NYSE, NASDAQ or the US OTC market. An ADR is a non-US company's equity security that trades in the US as though it were a US equity security.

Level	Where it trades	Capital raising
Level 1	The traditional US OTC market; not listed on any US securities exchange	No
Level 2	Listed on a US securities exchange such as the NYSE or NASDAQ	No
Level 3	A listed ADR programme used to offer **new shares** to US investors	**Yes**

Sponsored Level 2 and 3 ADRs are issued by a depositary bank on behalf of, and under a deposit agreement with, the non-US company. Level 2 ADRs do not involve capital raising; Level 3 ADRs allow new capital to be raised. **Unsponsored** ADRs are issued by depositary banks without a formal agreement with the non-US company.

Global Depositary Receipts (GDRs) are generally issued offshore from the US and are commonly listed on the **London** and **Luxembourg** stock exchanges.

6.11.3 Risks of Investing in DRs

- **Currency risk** — from the currency the DR is denominated in, the currency the share is denominated in, and the investor's home currency.
- **Price risk** — fluctuations in the underlying share price. ADRs often trade at a **premium** to their underlying shares because of regulatory restrictions in the issuer's home market on foreign holdings.
- **Counterparty risk** — financial or operational failure of the depositary bank or its custodian in processing DR obligations.
- **Single-listed ADRs** — surrendering ADRs whose underlying shares are not listed in their home markets leaves the investor holding common shares that are not traded and have no transactional liquidity. Converting ADRs into unlisted shares is not advisable.
- **Fungibility** — not all DRs have **two-way fungibility**, meaning DRs can be converted into underlying shares and underlying shares into DRs. All DRs have underlying shares backing them; the DRs are issued and traded outside the issuer's country, while the underlying shares are lodged in custody with a custodian in the issuer's country.

6.12 Summary

- Valuation can use relative valuation and discounted cash flow methods to produce an estimated value or range of values.
- The two most important characteristics of an ordinary share are its **residual claim** and **limited liability**. Residual claim means ordinary shareholders are paid last in liquidation; limited liability means losses are limited to the original investment.
- Legally a preference share is an equity security, but it is often viewed as a **hybrid** because it has characteristics of both fixed income and equity. Different classes are distinguished by the method and amount of dividend payment and by associated special benefits.
- Preference shares rank above ordinary shares for dividends and in liquidation, but below bonds. Preference dividends can be omitted without default, and preference holders have no vote except in distress.
- Rights issues raise new capital from existing shareholders and prevent dilution; bonus issues raise no capital and are a signalling device; stock splits make shares more accessible and raise no capital; reverse splits consolidate shares, often to escape penny-stock restrictions.
- Equity indices are value-weighted, price-weighted or equal-weighted. Value-weighted indices adjust automatically for capital changes; price-weighted indices give high-priced shares more weight; equal-weighted indices ignore price and size.
- Three valuation approaches exist: net asset value (book, liquidation, revised net asset value), absolute (DCF), and relative (ratios).
- Net asset valuation suits companies whose value is dominated by tangible assets — banks, insurers, property companies.
- DCF discounts projected free cash flow to the firm at the WACC, adds the discounted terminal value, subtracts debt and preferred shares, and divides by shares outstanding.
- DCF is sensitive to the terminal value and to the WACC and growth assumptions, so it is used alongside relative valuation.
- Relative valuation ratios are either shareholder-perspective (P/E, PEG, P/BV, P/RNAV, dividend yield) or firm-perspective (EV/Sales, EV/EBITDA, EV/CF).
- P/E must be read against history and peers, and against the qualitative reasons for a premium or discount.
- PEG controls for expected growth and is most useful for high-growth companies; a PEG below 1 is attractive but rare.
- P/BV suits banks, insurers and property companies, but is distorted by intangibles, inadequate provisions and hidden reserves.
- P/Sales is useful where earnings are negative but ignores expenses and debt; EV/Sales is conceptually consistent because both numerator and denominator relate to the whole enterprise.
- EV/CF uses cash flow because it is harder to manipulate than earnings.
- EV/EBITDA is preferred over P/E because it is unaffected by capital structure and closer to cash earnings.
- Dividend yield is the income component of return; the payout ratio is dividends to earnings and the retention ratio is 1 minus the payout ratio.

- Cash per share matters because excess cash supports dividends and makes a company an acquisition target.
- Strategic value and break-up value matter in acquisitions.
- The earnings yield is the inverse of P/E and is compared with bond yields; the difference is the yield gap.
- IPOs on the SGX Mainboard must meet listing requirements; Catalist listings are sponsor-supervised with no quantitative requirements.
- IPO risks include vendor share sales, deferred shares and options, the over-allotment option and claw-back provisions.
- Depositary receipts give access to foreign markets; ADRs have three levels, and only Level 3 raises new capital.
- DR risks include currency, price, counterparty, single-listing illiquidity and limited fungibility.

6.13 Additional Reference Materials

The dividend discount model is used only occasionally, for companies with high dividend payouts (**DPS/EPS above 80%**). It is set out here in full because the examination tests the formulas and the sensitivity of the result.

6.13.1 Dividend Discount Model

The intrinsic value of an asset can be estimated by the dividend discount approach, which determines the present value of a security. Under the DDM, the intrinsic value of a common share is the **present value of all future dividends**:

$$P_0 = \frac{D_1}{(1+R)} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \cdots + \frac{D_\infty}{(1+R)^\infty}$$

where

$$P_0$$

is the intrinsic or theoretical value of the share today,

$$D_t$$

is the dividend expected at the end of period

$$t$$

, and

$$R$$

is the required rate of return (discount rate) on the share.

The difficulty with implementing this as a valuation model is that investors must forecast and discount an **infinite** stream of dividends — both insurmountable tasks. The model can be modified to account for a subsequent sale of the shares:

$$P_0 = \frac{D_1}{(1+R)} + \frac{D_2}{(1+R)^2} + \cdots + \frac{D_n}{(1+R)^n} + \frac{P_n}{(1+R)^n}$$

where

$$n$$

is the holding period and

$$P_n$$

the expected selling price at the end of period

$$n$$

.

Worked example. A 3-year holding period, dividends of 1.00**, **1.05** and 1.10**$at the end of each year, an expected selling$ *10** at the end of year 3, and a **12%** required return:

$$P_0 = \frac{1.00}{1.12} + \frac{1.05}{(1.12)^2} + \frac{1.10}{(1.12)^3} + \frac{10.00}{(1.12)^3} = 0.8929 + 0.8371 + 0.7830 + 7.1178 = \$9.63$$

Compare the estimated value with the price to decide whether to buy.

6.13.2 Constant Growth Dividend Discount Model

When dividends grow at a constant rate

$$g$$

, the model generalises to:

$$P_0 = \frac{D_0(1+g)}{(1+R)} + \frac{D_0(1+g)^2}{(1+R)^2} + \frac{D_0(1+g)^3}{(1+R)^3} + \cdots + \frac{D_0(1+g)^\infty}{(1+R)^\infty}$$

where

$$D_0$$

is the current dividend and

$$g$$

the constant growth rate. This simplifies to:

$$P_0 = \frac{D_1}{R-g}$$

where

$$D_1$$

is the dividend expected at the end of period 1. The model suits the valuation of **very large, broadly diversified companies** and is often called the **Gordon model**, after Myron J. Gordon.

Three assumptions:

- Dividends grow at a **constant rate**.
- The constant growth continues **indefinitely**.
- The required rate of return

  $$R$$

  is **greater than**

  $$g$$

  . If not, the model does not apply.

Rearranging gives the required rate of return:

$$R = \frac{D_1}{P_0} + g$$

Worked example. Alpine Limited pays dividends per share of **\$0.20**, expected to grow at **5%** a year for the foreseeable future; investors require **12%** for investments of similar risk.

$$P_0 = \frac{0.20 \times 1.05}{0.12 - 0.05} = \frac{0.21}{0.07} = \$3.00$$

The current dividend must be compounded for one period because the formula uses the dividend expected one period from now.

Sensitivity. A small change in either estimate has a large effect:

Change	Calculation	Value
Base case	0.21 / (0.12 − 0.05)	**\$3.00**
R rises from 0.12 to 0.13	0.21 / (0.13 − 0.05)	**\$2.63**

Change	Calculation	Value
g rises from 0.05 to 0.06	0.212 / (0.12 − 0.06)	**$3.53**

A **1%** change in either

$$g$$

or

$$R$$

produces a large difference in estimated value. The crucial relationship is the **spread** between

$$R$$

and

$$g$$

: anything that narrows the spread raises the estimated value, and anything that widens it lowers the value.

6.13.3 Estimating Expected Growth Rate of Dividends

A starting point is the company's **sustainable growth rate** — the maximum growth achievable while maintaining a constant debt-equity ratio and without external equity financing:

$$g = \text{Retention Rate} \times \text{Return on Equity} = RR \times ROE$$

A firm can raise sustainable growth by increasing its retention rate (reducing the payout ratio) and reinvesting at its historical ROE, or by maintaining the retention rate while raising ROE.

Worked example. A firm retains **60%** of earnings with an ROE of **15%**:

$$g = 0.6 \times 0.15 = 0.09 = 9\%$$

.

Applying the model to P&G. For 2016, P&G's EPS was **3.80** and DPS **2.66**.

Required return. The average price return of the S&P 500 over the last 20 years, including the GFC and dot-com bubble, was **+7.36%**; the annual dividend yield was about **1.92%**, giving a total return of about **9.28%**. So

$$R_m = 9.28\%$$

.

Cost of equity by CAPM. With

$$R_f = 2.33\%$$

and

$$\beta = 0.68$$

:

$$R_{P\&G} = 2.33 + 0.68(9.28 - 2.33) = 7.056\%$$

Growth rate. Average ROE for 2016 was net income / average shareholders' equity = 10,508 / 60,516.5 = **17.36%**. The payout ratio was DPS/EPS = 2.66/3.80 = **0.70**. So:

$$g = 0.1736 \times (1 - 0.70) = 0.0521 = 5.21\%$$

As a consistency check, annual DPS growth over the last five years was

$$(2.66/1.97)^{1/5} - 1 = 0.0619$$

.

Applying the constant growth model.

$$P_0 = \frac{2.66 \times 1.0521}{0.07056 - 0.0521} = \frac{2.7986}{0.0185} = \$151.28$$

That is almost **twice** P&G's price of **$87.65** at the close of 7 July 2017. The wide difference highlights the model's problem: the estimated value swings widely with small changes in

$$R$$

and

$$g$$

. When

$$R$$

equals

$$g$$

the value goes to **infinity**, and the model is invalidated when the required return is less than the growth rate.

6.13.4 Multi-Stage Dividend Discount Growth Model

Multi-stage growth describes a situation where expected dividend growth must be described using **two or more growth rates**. A two-stage model assumes near-term growth at a given rate for some period, typically **2 to 10 years**, followed by a different sustainable long-term rate.

a. Two-stage DDM. Many companies grow rapidly for several years and then slow to a stable rate; others pay no dividends for several years during early growth. The constant growth model is inappropriate for these firms because growth is not constant throughout.

The two-stage model forecasts a growth rate for the near term, when dividends are more predictable, and assumes a long-term rate close to long-run GDP growth, possibly **1–2 percentage points** above it:

$$P_0 = \sum_{t=1}^{n} \frac{D_0(1+g_1)^t}{(1+R)^t} + \frac{\dfrac{D_n(1+g_c)}{R-g_c}}{(1+R)^n}$$

where

$$g_1$$

is the growth rate when dividends are more predictable,

$$g_c$$

the constant growth rate when they are less predictable,

$$n$$

the number of periods of supernormal or subnormal growth, and

$$D_n$$

the dividend at the end of the abnormal growth period.

The first term discounts the dividend stream for the first

$$n$$

periods at

$$R$$

. The second term is the constant-growth version, dividing the dividend expected at the end of period

$$n+1$$

by

$$R - g_c$$

; because the result is a present value at the end of period

$$n$$

, it must be discounted back to time zero.

Worked example — P&G. Assume confidence in projecting dividends for the first **five years**, with long-run constant growth thereafter.

$$D_0 = \$2.66$$

,

$$R = 7\%$$

,

$$g_1 = 5\%$$

,

$$g_c = 3.5\%$$

(above the US long-run GDP growth of 2%).

Projected dividends:

Year	Calculation	Dividend
1	2.66×1.05	\$2.7930
2	2.66×1.05^2	\$2.9327
3	2.66×1.05^3	\$3.0793
4	2.66×1.05	\$3.2333
5	2.66×1.05	\$3.3950

The expected price at the end of year 5, when long-run growth of 3.5% begins:

$$D_6 = 3.3950 \times 1.035 = 3.5138 \qquad P_5 = \frac{3.5138}{0.07 - 0.035} = 100.3943$$

Summing the present values:

$$P_0 = \frac{2.7930}{1.07} + \frac{2.9327}{1.07^2} + \frac{3.0793}{1.07^3} + \frac{3.2333}{1.07^4} + \frac{3.3950}{1.07^5} + \frac{100.3943}{1.07^5}$$

$$= 2.6103 + 2.5615 + 2.5136 + 2.4667 + 2.4206 + 71.5797 = \$84.15$$

This comes close to P&G's current price, so on this model P&G is slightly **overvalued**. The model still suffers the same sensitivity: using **3%** instead of 3.5% for long-run growth reduces the valuation by about **10%** to **\$74.90**. Because about **85%** of the value (71.5797 / 84.15) comes from the terminal value, anything affecting the terminal value affects the model significantly.

b. Three-stage DDM. The model can be extended by assuming a different growth rate for years 6 to 10. This is useful for gradually tapering dividend growth to the eventual terminal rate: Stage 1 (years 1–5) supernormal growth, Stage 2 (years 6–10) a gradual slowdown, Stage 3 (year 11 onwards) a long-run sustainable level.

c. Practical application. The DDM is best applied when dividend payouts are **much higher** and more of the estimate can rest on the more predictable early years. If the early years account for a larger proportion of estimated fair value, the model depends less on the terminal constant-growth assumption. Accuracy and effectiveness are better for companies in the **mature phase** with very high payouts — **70–120% of EPS**, where DPS can exceed EPS if the company pays out of retained earnings and has sufficient cash. The DDM therefore tends to be more useful for gaming companies, tobacco companies and other mature companies with huge payouts.

d. Intrinsic value or estimated fair value. Compare the estimated fair value with the prevailing price. If intrinsic value **exceeds** price, the share is undervalued and should be purchased. If price **exceeds** intrinsic value, the share is overvalued and should be avoided or sold.

6.13.5 Zero-Growth Model

The owner of a preference share receives a fixed dividend for an infinite period. Because the dividend is a **perpetuity**, the share value is the stated annual dividend divided by the required rate of return:

$$P_0 = \frac{\text{Dividend}}{\text{Discount Rate}}$$

Worked example. A preference share with a $100 * *parvaluepays * *\mathbf{7}$ a year. At a required return of **8%**, the intrinsic value is $7/0.08 = **87.50**$.

Exam cues

- **Two defining features of an ordinary share: residual claim** and **limited liability**.
- **Claim order in liquidation:** creditors (including bondholders) → preference shareholders → ordinary shareholders.
- **Preference shares are a hybrid:** fixed dividend, infinite life, no vote except in distress, dividends omissible without default, ranked above ordinary but below bonds.
- **Preference share features:** cumulative (rare now), callable (usual, with about **5 years** of call protection, priced at a premium to par, higher yield), participation (uncommon), convertibility (uncommon, lowers the yield because the issuer has sold an option).
- **Rights issue raises capital; bonus issue does not.** Rights: 300m shares at $2.00 plus 100m at $1.90 \rightarrow 400mshares, theoreticalex-rights**1.975**$. Bonus 1-for-3: 400m shares, theoretical ex-bonus **$1.50**.
- **Weighted-average shares for EPS:** (300m × 9 + 400m × 3)/12 = **325m**; revised EPS = $(0.11 \times 300m)/325m = **0.1015**$.
- **Renounceable rights are tradable; non-renounceable are not.** The pro-rata renounceable limit rose from **50% to 100%** of share capital.
- **Stock split raises no capital.** OCBC 2-for-1 in **2005** at above $13. * *Reversesplit : * * Citigroup1 - for - 10in * *2011 * *, from * *4.52$ to $44.16 * *$; $SECpennystockthresholdis * *5**$.
- **Dual class shares on SGX since June 2018**, with a maximum of **10 votes** per multiple-vote share.
- **Berkshire Hathaway:** Class A **USD284,570**, Class B **USD189.39**; 1 Class A converts into **1,500** Class B; Class B has **1/200th** the votes. **Alphabet:** Class A 1 vote, Class B 10 votes, Class C no votes.
- **Equity categories:** blue chip (safety, steady dividends), high growth (above **20%** compounded, high P/E, high beta, small dividends), income (high yield, price support at **4–5%** when deposits pay **3%**), cyclical (autos; property and banks in Singapore and Hong Kong), defensive (pharmaceuticals, food, utilities, transport — Nestlé, Coca-Cola).
- **Index weighting:** value-weighted (S&P 500, STI), price-weighted (DJIA, 30 shares), equal-weighted (Value Line, geometric average, over **80%** NYSE-listed).
- **Value-weighted worked example:** base 64,000,000 = 100; new 67,000,000 → **105**.
- **Equal-weighted worked example:** $(1.04 \times 1.03 \times 1.05)\hat{}(1/3) = \mathbf{1.040}$.
- **Value-weighted indices adjust automatically for bonus and rights issues**; price-weighted indices need a divisor adjustment, and a split reduces a share's weight.
- **Three valuation approaches:** net asset value, absolute (DCF), relative.
- **DCF steps:** WACC → project FCFF for **10 years** → terminal value at year 10 → discount → enterprise value → subtract debt and preferred → divide by shares.
- **CAPM:**
$$R = R_f + \beta(R_m - R_f)$$
. P&G: $2.33 + 0.68(9.28 - 2.33) = \mathbf{7.056\%}$.
- **WACC worked example:** $(1 - 0.23)(0.014)(30.414/254.584) + (0.07)(224.17/254.584) = \mathbf{6.29\%}$.
- **Terminal value:**
$$E(FCFF_{11})/(\text{WACC} - g_c)$$
, then discounted 10 years. The discount rate must **exceed** the growth rate.
- **P&G DCF result:** enterprise value **USD260,391m**, less debt 30,514 and preferred 1,038 → equity **228,839m**, divided by 2,557.6m shares = **USD89.47** against a price of **USD87.65** (about **2%** upside).
- **P/E worked example:** INTC at 34.68 with EPS 2.72 → trailing **12.8x**; 2017 forecast 2.86 → **12.1x**; 2018 forecast 2.97 → **11.7x**. INTC's normal range is **10.3x–16.0x**; a move to 14x implies **18–20%** upside.

- **PEG:** P/E divided by expected EPS growth. Alibaba's P/E of **44.5** with **28%** growth gives a PEG of **1.6** — among the lowest. A PEG **below 1** is attractive but rare.
- **P/BV suits banks, insurers and property companies.** Hidden reserves make book value understated, so the true P/BV is **lower** than calculated.
- **P/Sales is conceptually flawed** because price relates to shareholders while sales are generated by debt and equity; **EV/Sales is consistent**.
- **EV/CF worked example:** $(18 \times 1{,}000 + 400 + 10 - 8{,}000) / 2{,}210 = \mathbf{4.71}$. A **lower** ratio is better.
- **EV/EBITDA is preferred over P/E** because it is unaffected by capital structure and closer to cash earnings; it became widely used in the **1980s**.
- **Dividend yield:** \$0.10 on a \$2.00 price = **5%**. Payout ratio $0.10/0.40 = \mathbf{25\%}$; retention ratio **75%**.
- **Yield compression:** a 3-cent dividend at a 7.5% required yield implies a price of **40 cents**.
- **REITs must distribute 90% of distributable income**, which is why a DDM applies.
- **Cash per share:** Company A at 2.50 with 0.50 net cash and EPS 0.25 is effectively **8x** after netting cash, versus Company B at **10x**.
- **Earnings yield is the inverse of P/E.** The **yield gap** is the earnings yield minus the bond yield; **1.5–2.0%** above Treasuries leaves room to rise. October 2017: Treasury **2.45%**, S&P 500 earnings yield about **4%**.
- **SGX Mainboard requires quantitative listing requirements; Catalist is sponsor-supervised with none.**
- **IPO tranches:** private placement (institutions and accredited investors, usually the larger portion) and public retail (smaller, widely distributed). Offer opens at least **2 market days**; trading starts the next business day on a **T+2** basis.
- **IPO risks:** vendor share sales, deferred shares and options, the **Greenshoe** over-allotment option, and **claw-back**.
- **Rights issues after listing lodge an Offer Information Statement (OIS)**, not a prospectus.
- **ADR levels:** Level 1 OTC (no listing, no capital raising), Level 2 exchange-listed (no capital raising), Level 3 exchange-listed **with** capital raising. **GDRs** are issued offshore from the US, commonly London and Luxembourg.
- **DR risks:** currency, price (ADRs often trade at a **premium**), counterparty, single-listing illiquidity, and limited **two-way fungibility**.
- **Trap:** a **bonus issue** and a **stock split** both raise no capital, but a bonus issue is a capitalisation of reserves while a split simply divides existing shares — the price adjustment differs.
- **Trap:** a **low P/E** is not automatically cheap and a **high P/E** is not automatically expensive; the difference may be expected growth, transparency, impending events or illiquidity.
- **Trap: P/BV** uses book value, not market value — for a bank with non-performing loans the market value of assets may be far below book.
- **Trap:** the **terminal value** usually dominates a DCF; in the P&G two-stage example about **85%** of value came from the terminal value.

Chapter 7 — Deposits and Fixed-Income Securities

This is the largest chapter in the syllabus and the most calculation-heavy after Chapter 2. It runs from cash and deposit insurance through money market instruments, bond types and features, credit ratings and spreads, the full set of fixed-income risks, bond pricing, duration and convexity, yield measures, the term structure, and convertible bond analysis.

Learning Objectives

The candidate should be able to:

- Discuss the different ways we use cash.
- Describe various basic money market instruments.
- Describe the characteristics of a fixed-income instrument such as a bond.
- Define and explain the terminology used in fixed-income securities.
- Explain the credit risk in holding bonds.
- Describe the issuers and their motivation in issuing bonds.
- Describe the primary and secondary bond markets.
- Explain how prices are quoted in bond trading.
- Discuss various special features in bond issuance and explain their significance to investors.
- Describe high-yield bonds and their implied bond rating.

- Discuss asset-backed fixed-income instruments.
- Describe features of the internal and external bond markets, and the Eurobond market.
- Explain the differences among the various grades of debt rating by rating agencies.
- State the various risks a fixed-income investor is exposed to.
- Calculate the value of a bond.
- Calculate the bond equivalent yield for bonds paying interest more often than once a year.
- Explain the relationship between coupon rate, market yield, required yield and bond price.
- Explain why bond prices converge to par on maturity.
- Explain duration and convexity and their implications for a bond's price risk.
- Understand price value of a basis point and how it monitors price risk in a bond portfolio.
- Calculate yield measures such as current yield and yield to maturity.
- Explain reinvestment risk and how it is affected by coupon rate and maturity.
- Discuss the term structure of interest rates in relation to the expectations and liquidity preference theories.
- Analyse convertible bonds, including conversion parity price and conversion premium.

7.1 Introduction

This chapter covers the basic interest-bearing asset classes:

- **Cash.**
- **Money market instruments.**
- **Fixed income instruments.**

Cash is interesting because physical cash as a medium of exchange is gradually being replaced by digital forms of storage and disbursement. It is often classified as a non-volatile, safe asset class — and for that reason pays the **lowest return**. Keeping excessive cash from **2008 to 2017** was non-optimal because of opportunity cost and loss of purchasing power.

Beyond cash, money market instruments provide better returns, followed by fixed income instruments of various credit risks and tenures. Most money market instruments are short-term, usually **3–6 months**. For a wider range of tenures, risks, returns and payoff profiles, investors use fixed income instruments — from sovereign notes and bonds to corporate bonds, and from plain vanilla bonds to bonds with embedded options and structured elements.

7.2 Cash Deposits and Cash-Equivalents

All investments begin with, and can be liquidated to, cash. During downturns and crises, cash deposit balances rise as other financial assets are sold down. Cash may be in local or foreign currency, and deposits generally provide interest through savings or term deposits, making them the most basic form of fixed income investment.

Because cash provides the lowest return, institutional investors and fund managers generally minimise cash balances, holding only what is needed for redemptions. But if a major downturn is anticipated they may **increase** cash holdings, giving them funds to take advantage of opportunities. Corporate entities must also manage excess cash and liquidity: excess cash needs to be invested to optimise returns, and for very large conglomerates even the short-term cash float must generate a reasonable return before remittance. For some banks, cash management services for corporates have become a lucrative activity.

For individuals, income and monies received fall into four categories:

- Money for **consumption**.
- **Reinvestment** into the business (for entrepreneurs).
- **Reserves** set aside for family needs, life's risks and unfortunate eventualities.
- **Excess cash**.

Reserves and excess cash can be invested in personal assets such as property and in financial assets, or held as cash and cash equivalents for liquidity.

7.2.1 Currency Deposits

a. Local currency deposits. The most widely known cash-equivalent investments are savings and time deposits at banks and finance companies. **Time deposits** are held for a specified period — 3, 6, 12 months or longer, up to 36 months. The longer the period, the higher the interest, but a portion of interest may

be forfeited as a penalty for premature withdrawal. Although low risk, the disadvantage is **low yield**: from **July 2009 to July 2017**, money in savings or fixed deposits accumulated returns of at most **5–6%**, while the **S&P 500 appreciated more than 133%** over the same period, and average Asian housing prices rose significantly.

Time deposits therefore do not provide a good inflation hedge and can be "costly" by forgoing other investments.

b. Foreign currency deposits. Foreign currency deposits with Singapore banks can offer potentially higher interest but carry exchange rate risk. In Singapore, foreign currency deposit accounts are **not covered** by the Deposit Insurance Scheme.

Local currency deposits reduce direct exposure to currency fluctuations, but an investor may be better off in foreign currencies when the local currency continually depreciates. If the domestic economy has a poor outlook with weak GDP growth and worsening political stability, local currency weakness can be expected. To mitigate sustained depreciation, investors can hold a portion of deposits in a stable, relatively stronger currency. For growing companies and wealthy individuals, holding part of assets in foreign currency has become integral to diversification and risk management.

c. Deposit insurance in Singapore. The DI Scheme was introduced on **1 April 2006** and reviewed and enhanced by MAS and the Singapore Deposit Insurance Corporation (SDIC) in **May 2011**. All full banks and finance companies in Singapore are members, except those exempted by MAS. The scheme protects small depositors if a member fails.

Coverage limit	Effective
S$50,000	Original
S$75,000	**1 April 2019**
S$100,000	**1 April 2024**

Coverage extends to individuals and other non-bank depositors — sole proprietorships, partnerships, companies, and other unincorporated entities such as charities, associations and societies. It applies **only to Singapore dollar-denominated deposits** with a DI member, including:

- Savings account deposits.
- Fixed deposit account deposits.
- Current account deposits.
- Monies under the CPF Investment Scheme (CPFIS).
- Monies under the CPF Retirement Sum Scheme (CPFRS).
- Monies under the CPF Supplementary Retirement Scheme.
- Murabaha, as prescribed by MAS.

Not insured by SDIC:

- Foreign currency deposits.
- Structured deposits.
- Investment products such as unit trusts, shares and other securities.

Monies under CPFIS and CPFRS are aggregated and **separately** insured up to **S$100,000**.

A payout may be made on MAS's decision where a court order has been made to wind up a DI member, or where MAS determines a member is insolvent, unable or likely to become unable to meet its obligations, or about to suspend payments.

Worked example 1 — individual accounts, one bank. An individual places **S$650,000** across accounts, all with a single bank.

Account	Balance (S$)	Insured	Not insured
Individual current account	10,000		
Individual savings account	10,000		
Individual fixed deposit account	280,000		
Subtotal, S$ deposits	**300,000**	**100,000**	**200,000**
CPF deposits (CPFIS)	100,000	**100,000**	—

Account	Balance (S$)	Insured	Not insured
Individual USD currency investment	250,000	—	**250,000**
Total	**650,000**	**200,000**	**450,000**

For accounts with a single bank, the scheme covers only up to $\mathbf{S}100,000 **of the aggregate balance across the different account *S**100,000**. So although the individual has S$650,000 of deposits, only **S$200,000** is insured.

For deposits above S$100,000, the scheme allows some benefit from **joint accounts** and **multiple accounts**. Partners in a joint account are separately insured for their share.

Worked example 2 — different banks and joint accounts. A man and his wife hold accounts with three banks. At Bank A he has **USD250,000** in deposits, $\mathbf{S}30,000 **in an individual current account, **S**40,000** in an individual savings account, and $\mathbf{S}80,000 **in a joint account with his wife. At Bank B he has **S**100,000** in a joint fixed deposit with his wife and $\mathbf{S}50,000 **in a CPF IS deposit. At Bank C he has another **S**100,000** in an individual account. All three banks are SDIC members.

The insured amount is computed **per bank** for S$ deposits, with joint account balances split between the partners, and CPF monies insured separately.

7.3 Fixed-Income Securities

Fixed income securities are debt instruments providing periodic payments at an interest rate known in advance, with eventual return of principal at maturity.

Money market instruments (1 year)	Capital market instruments (> 1 year)
Treasury bills	Government bonds
Commercial paper	Agency debts
Negotiable certificates of deposit	Municipal and provincial government debts
Banker's acceptances	Corporate bonds
Repurchase agreements	Convertible bonds
Trade bills	Other securities
	Asset-backed securities — auto loans, student loans, credit-card debts
	Collateralised debt obligations (CDOs)

One major classification is by **credit risk**, splitting securities into **investment-grade** and **high-yield** bonds. Investors in short-term money market instruments look for low risk, so the majority of those instruments are investment grade.

7.4 Money Market Instruments

Money market instruments are debt instruments issued by governments, financial institutions and corporations with maturities of **less than one year**. Because of the short maturity they carry relatively low risk, but the **minimum denomination is usually large — S$250,000 and above**. Examples include Treasury bills, negotiable certificates of deposit or banker's acceptances, commercial paper, repurchase agreements and bills of exchange.

For institutional portfolios, a portion of funds may be placed in money market instruments for **liquidity** — the ability to convert an investment into cash quickly with little or no loss in value. As near-cash instruments, their primary function is to provide a pool of reserves for emergencies or to accumulate funds for specific purposes.

Investment in money market instruments generally **increases** when investors are uncertain about other asset classes, or when stocks and bonds are expected to perform poorly. Holdings fall when a bull market is expected.

Professional fund managers avoid allocating too much beyond liquidity requirements because of the low yield. Although money market instruments carry almost no risk of principal loss, they are subject to **reinvestment risk** — the risk that proceeds cannot be reinvested at equivalent rates, possibly because rates have fallen. Short-term instruments carry **more** reinvestment risk because proceeds must be reinvested each time the instrument matures.

7.4.1 Treasury Bills

Short-term government securities with maturities of **one year or less**, issued to the investing public to fund government expenditure. Buying a T-bill is equivalent to lending to a sovereign government.

T-bills issued by a creditworthy government are the safest investments and generally considered **risk-free**. "Risk-free" here means no risk of defaulting on interest or principal when due.

T-bills are sold on a **discount basis**: investors pay less than face value and receive face value at maturity. The difference is the interest earned and is used to calculate the yield. The T-bill yield is usually the benchmark **risk-free rate**. T-bills can be readily sold and converted to cash before maturity at modest cost.

7.4.2 Banker's Acceptances and Certificates of Deposit

Both are money market instruments issued by financial institutions.

A **banker's acceptance** facilitates international commercial trade: it represents a claim on the issuing bank for a specific amount on a specific date. It is a **negotiable** security issued on a **discounted** basis.

A **certificate of deposit (CD)** is issued by a bank indicating that a specific sum has been deposited. It carries a maturity date and a specific interest rate, generally on a **compounding** basis, with interest payable at maturity.

7.4.3 Commercial Papers

Short-dated **unsecured** promissory notes issued by a corporation. Because they are unsecured, only corporations with **very strong credit ratings** can issue them. Like T-bills and banker's acceptances, commercial paper is issued at a **discount** to face value.

7.4.4 Repurchase Agreements

Secondary trading in money market instruments can take place through a **repurchase agreement (repo)**: the sale of a money market instrument with a commitment by the seller to buy it back at a specified price on a future date. Effectively it is a **collateralised short-term loan**, with the money market instrument as collateral. The yield is the difference between the sale and purchase prices.

7.4.5 Bills of Exchange

Widely used to finance international trade, important because of the time lag in trade transactions and the distance between buyers and sellers. A bill may be drawn payable **on sight or demand**, or payable after a period — commonly **91 days** — in which case it is a **term bill**. Bills of exchange are **negotiable** and generally transferable by endorsement and delivery.

7.5 Bonds

A **coupon bond** is a debt security on which the issuer is obligated to make regular interest payments (coupons) on specified dates and repay principal at maturity. The coupon rate is usually fixed for the life of the bond, giving rise to the term "fixed income".

The debt securities market divides into three maturity segments by original maturity:

- **Money market instruments** — short-term, maturing within one year.
- **Medium-term notes** — more than one year and up to **10 years**.
- **Long-term bonds** — longer than **10 years**.

7.5.1 Benefits and Drawbacks of Investing in Bonds

Benefits. Bonds suit investors interested in:

- **Principal preservation.**
- **A steady stream of periodic income.**

Fixed income securities have long been regarded as good vehicles for current income: under normal circumstances the investor receives cash flows with reasonable certainty. For capital gains, bond values can appreciate from improvements in the issuer's credit quality or when market yields fall. Active bond traders look for undervalued bonds likely to enjoy credit rating improvements and upgrades.

Drawbacks. A major disadvantage of fixed coupons and principal repayment is that these securities **do not adjust for inflation**. Inflation erodes the purchasing power of the cash flows and can lead to swings in interest rates, producing price fluctuations and substantial capital losses if the securities are not held to maturity.

A second disadvantage is the **lack of activity and liquidity** in smaller secondary markets — those other than the major bond markets of the US, Europe and Japan. Investors must be confident of holding relatively illiquid bonds until maturity.

7.5.2 Terminology

Par value (face value) or principal is the redemption value — the amount repaid at maturity. For most bonds the par value is **$1,000**. Par value is **not** the same as market value, which rises or falls with factors affecting required return, such as interest rate movements.

The **coupon** is the periodic interest payment over the life of the bond. The **coupon rate** multiplied by the principal gives the dollar coupon payment. Coupons are typically paid **annually or semi-annually**.

In Singapore, Government bonds are **scripless**: ownership of Singapore Government Securities is reflected as electronic book entries. Otherwise bonds may be **bearer** or **registered**, but since most markets have moved to electronic trading and settlement, few bearer bonds are issued today, given the high security risk.

Term-to-maturity is the number of years remaining to maturity. Two types exist:

- **Term bonds** — a single maturity date; most corporate bonds.
- **Serial bonds** — a series of maturity dates; essentially bundles of bonds issued at the same time with different maturities.

The longer the maturity, the more volatile the bond price. Conservative investors choose shorter maturities to reduce interest rate and default risk. Price volatility is not a concern if the bond is held to maturity, but it is if the investor is forced to sell early.

7.5.3 Issuers

Bond issues are classified as:

- **Government.**
- **Government-related bodies.**
- **Corporations.**

Governments issue bonds to pay for projects such as infrastructure and essential services like healthcare and education. In some countries, such as the US, government agencies are active issuers. In Singapore, **government-linked corporations** tap the bond market rather than sourcing directly from the government.

Corporations raise funds through bonds as well as equity and bank loans. The corporate sector provides the most diverse issues in type and quality.

7.6 Price Quotations

For pricing, fixed income securities are classified as **discount** or **coupon** securities. Securities with maturities of **1 year or less** — T-bills, commercial paper — are discount securities; those with maturities **over one year** are coupon securities.

7.6.1 Discount Basis

T-bills are quoted on a **bank discount basis**:

$$Y_d = \frac{D}{F} \times \frac{360}{t}$$

where

$$Y_d$$

is the annualised yield on a bank discount basis,

$$D$$

is the dollar discount (face value minus price),

$$F$$

is the face value, and

$$t$$

is the number of days remaining to maturity.

Worked example. A T-bill with **90 days** to maturity, face value $100,000 * *, sellingfor * *$**98,000**:

$$D = 100{,}000 - 98{,}000 = 2{,}000 \qquad Y_d = \frac{2{,}000}{100{,}000} \times \frac{360}{90} = 8.0\%$$

Although widely used, the bank discount yield is **not a meaningful measure of return** from holding a T-bill, because it is calculated on **face value** rather than the actual amount invested.

7.6.2 Price Basis

Prices of fixed income securities with maturities over one year are quoted as a **percentage of par value**. With a par value of $1,000, aquoteof * *98.50 * *represents * *985**, and **95¾** represents **$957.50**.

US Treasury bonds and notes are quoted in **thirty-seconds (1/32)** of a percentage point, while corporate bonds are quoted in **eighths (1/8)**. A Treasury quote of **90-24** means **90 24/32**.

All fixed income securities trade on an **accrued interest basis**: the buyer pays the price plus interest earned since the last coupon date. For an **8%** bond paying semi-annual interest on 1 June and 1 December, sold on **1 September**, the seller is entitled to accrued interest of **$20 * *(80 \times 3/12)$ for the three months held.

7.7 Bond Markets

The bond market divides into the **internal bond market** and the **external bond market** (popularly the Eurobond market). The internal market further divides into the **domestic** and **foreign** bond markets.

In the Singapore context, the domestic bond market comprises **Singapore Government Securities (SGS)** and corporate bonds issued by Singapore companies. The foreign bond market refers to foreign issuers who have issued bonds in the Singapore market.

Global bond portfolios invest **40% or more** of assets in foreign bonds. Some follow a conservative approach favouring high-quality developed-market bonds; others own lower-quality bonds from developed or emerging markets. Some invest exclusively outside the US; others invest in both.

7.7.1 The Singapore Bond Market

The market comprises Government securities (T-bills and bonds) plus corporate bonds issued by quasi-government statutory organisations, local companies and foreign entities.

a. Singapore Government Securities. SGS were initially issued to meet banks' need for a risk-free asset in their liquid asset portfolios. The Singapore Government does not need to finance expenditure through bonds, because it operates a **balanced budget policy** and often enjoys surpluses. SGS issuance therefore serves primarily to:

- Build a **liquid SGS market** to establish a **benchmark yield curve** for pricing Singapore dollar private debt securities.
- Encourage an active **secondary market**, for both cash transactions and derivatives, for risk management.
- Attract domestic and international issuers and investors to the Singapore bond market.

SGS auctions of T-bills and bonds are conducted regularly according to an annual calendar. MAS introduced **"mini-auctions"** of SGS bonds in **2015** as a regular feature to address unexpected instances of strong demand.

b. Singapore corporate bond market. Despite challenges in the offshore energy market, which triggered some defaults, and the global slowdown, overall corporate bond issuance rose in **2017**. Corporate debt issued totalled **SGD259 billion**, while total debt outstanding rose to **SGD386 billion**. Several new issues came from **green bonds**, which fund projects delivering environmental benefits.

7.7.2 Eurobond Market

The Eurobond market is an international bond market. A **Eurobond** is a bond that is:

- Underwritten by an **international syndicate**.
- Offered at issuance **simultaneously to investors in a number of countries**.
- Issued **outside the jurisdiction of any single country**.

Eurobonds derive their name from originally being sold in Europe, typically London. The market is divided by the currency of denomination: USD-denominated Eurobonds are **Eurodollar bonds**; yen-denominated are **Euroyen bonds**.

7.7.3 Primary and Secondary Bond Markets

a. Primary market. Bonds are issued to the public in the primary market. SGS are sold at **auctions through Primary Dealers**. A company raising funds appoints an investment bank to help.

b. Secondary market. Once issued, bonds trade in the secondary market — either the exchange where the bond is listed (for retail bond issues) or the **OTC market**. Beyond being a place to buy and sell, the secondary market performs the important function of **benchmarking market values and yields** of traded bonds. That information is needed to value bond portfolios and to establish benchmark yield curves for bonds of different credit ratings. Of particular interest is the **spread over the yield on Government issues of similar maturity** — the difference between the corporate bond yield and the Government bond yield.

7.8 Types of Bonds

7.8.1 Floating Rate Bonds

Instead of a coupon fixed for the bond's life, the coupon on a floating rate security is **periodically reset** against a predetermined benchmark such as **SIBOR**. For example, the coupon may reset every six months to a spread of **100 basis points** over the six-month Treasury bill rate. Floating rate securities appeal to some institutional investors because the income stream more closely matches the floating nature of some of their liabilities.

7.8.2 Zero-Coupon Bonds

Zero-coupon bonds make **no periodic coupon payments**. They are sold at prices far below par — at **deep discounts** — and the investor receives par at maturity. The difference between maturity value and purchase price generates the effective interest rate of return.

The most attractive feature is that there is **no reinvestment risk** within the bond's life, because there is no coupon to reinvest, so the rate of return is locked in when held to maturity.

Disadvantages:

- **Call features** potentially limit capital appreciation when rates decline, because issuers are likely to redeem.
- **High price volatility** — a rise in rates hits a zero-coupon bond much harder than a coupon bond. Interim volatility is not a concern if held to maturity.
- **Liquidity may be limited.**

7.8.3 Convertible Bonds

Bonds convertible, at the holder's option, into ordinary shares of the issuing company within a specified period. Yields are generally **lower** than comparable straight bonds because of the conversion privilege, which the holder effectively pays for through the lower yield. There is potential capital appreciation from a rise in the underlying share price, a fall in interest rates, or an improvement in the issuer's credit rating.

Convertibles benefit the issuer because they allow **refinancing at a lower rate** and **defer potential dilution**. At issue, the conversion rate is set so the effective price paid per share on conversion is **above** the current share price, making conversion unattractive. If the share price appreciates, conversion may become worthwhile, and the amount converted determines the extent of dilution.

7.8.4 Covered Bonds

A covered bond is a **highly rated** paper backed by **bank assets**, typically costing less to issue than unsecured bonds. Covered bonds have a long history in Europe dating to **1769**; though many banks have come and gone, these bonds have **never experienced a default**. Partly for that appeal, and in response to the Global Financial Crisis of 2007–2008, covered bonds are now issued in many countries. MAS released guidelines in **December 2013** allowing local banks to sell covered bonds. A bank in acquisition mode might consider a covered bond offering as an additional, more attractive way to raise funds.

7.8.5 High-Yield Bonds

Bonds rated **below BBB** are **high-yield** or **junk** bonds. They trade primarily on their **creditworthiness** rather than the level of interest rates. Their high yields reflect factors such as the small size of the company or a lack of credit history.

Complex junk structures exist, particularly for **leveraged buyout (LBO)** financing and recapitalisations. To reduce heavy interest payments, firms issue bonds with **deferred coupon structures** that let the issuer avoid cash interest for **three to seven years**.

7.8.6 Deferred-Coupon Bonds

Four structures:

Structure	Detail
Deferred-interest bonds	The most common deferred-coupon structure. Sell at a **deep discount** and pay no i
Step-up bonds	Pay coupon interest, but the rate is **low initially** and then steps up to a higher rate
Payment-in-kind (PIK) bonds	Give the issuer the option to pay cash at a coupon date or give the holder a **similar**
Perpetual bonds	**No maturity.** Pay a higher coupon to compensate for the indefinite holding period

7.9 Bond Features

Unlike a plain vanilla or straight bond — a basic fixed income security with no special features — some issues carry features that **augment risk**.

7.9.1 Call Provisions

A **call provision** gives the **issuer** the right to retire the debt, fully or partially, before maturity. A non-callable bond gives no such right.

The call is attractive to issuers because it gives flexibility to control financing costs if rates fall, and can help circumvent restrictions in the indenture.

It is unattractive to investors, who risk losing a high-coupon bond when rates decline. To protect the investor there may be a **call protection period** after issue. The prospect of a call also **limits upside price potential** in a falling-rate environment. Because of these disadvantages, callable bonds carry **higher yields** than equivalent non-callable bonds.

The **call price** is normally **above par**. The difference between call price and par is the **call premium**, which tends to be **higher early in the bond's life** and declines gradually toward par as maturity approaches.

7.9.2 Sinking-Fund Provision

Corporate indentures may require the issuer to retire a certain amount of debt each year. The issuer may buy the bonds in the open market if the price is below par, or pay the trustee, who calls bonds for redemption by **lottery**. The purpose is to **reduce credit risk**.

Some sinking-fund provisions embed options letting the issuer accelerate repayment. The **doubling option** grants the right to retire **twice** the amount the sinking fund requires, supposedly to reduce credit risk.

7.9.3 Put Provisions

Puttable bonds give the **bondholder** the right to sell the bonds back to the issuer **at par** on designated dates. A put provision **limits downside risk**: when rates rise and the price falls, the investor can put the

bond back at par. As with a convertible, the investor has effectively bought a put option from the issuer and pays for it with a **lower yield**.

7.9.4 Security for Bonds

The issuer's promises and the investors' rights are specified in a **bond trust deed**. A **trustee** is legally appointed to look after bondholders' interests and ensure all provisions are met, including timely payment of interest and principal.

Failure to pay principal or interest is **legal default**, and court proceedings can enforce the contract. Bondholders, as creditors, have a **priority claim** over ordinary and preferred shareholders on both income and assets.

Bond type	Claim
Secured (senior) bonds	Backed by a legal claim on specified property in the event of default — collateral, as fo
Debenture	Any debenture stock, bond, note or other debt security within the meaning of the Secu
Subordinated (junior) bonds	Claim on income and assets **subordinated** to secured debt and after main creditors
Income bonds	Junior debt: the issuer can **omit or delay** interest if earnings are too low

When an asset is pledged, the **title** is held by the lender until the loan is paid; the borrower still owns and can use the asset, but the documents of ownership are held by the lender.

7.9.5 Guarantees and Credit Insurance

Loans and bonds issued by subsidiaries may carry **guarantees from the parent**, adding to credit standing. Where that is lacking, the company can improve credit quality by securing **credit insurance** for a fee.

7.10 Asset and Mortgage-Backed Securities

Asset and mortgage-backed securities are created for sale to investors by lenders pooling and securitising illiquid asset cash flows. The primary difference between **asset-backed securities (ABS)** and **mortgage-backed securities (MBS)** is the underlying asset: MBS are created from pooling **mortgages**, while ABS evolved from MBS and are created from pooling **non-mortgage** assets.

ABS represent an interest in pooled assets including:

- Auto loans and leases.
- Consumer loans.
- Credit card receivables.
- Royalty payments.

The sub-prime crisis. Before the 2007 crisis, the **Federal Home Loan Mortgage Corporation (Freddie Mac)** and the **Federal National Mortgage Association (Fannie Mae)** issued large amounts of MBS. Numerous mortgages were promoted and approved despite poor credit quality — **sub-prime** loans — because loan officers were rewarded on commission. This lending occurred amid a property bubble building from **2001 to 2007**. Freddie Mac and Fannie Mae purchased and invested in huge numbers of mortgages and MBS with their funding advantage, having **lower capital requirements** than other regulated institutions.

When the bubble burst, mortgage defaults began and MBS fell in value. Several institutions that purchased these loans suffered massive losses and became insolvent. Their default and demise triggered further losses and defaults in other investments, putting banks globally at risk. Defaulting MBS were at the root of the sub-prime crisis in 2007 and the global financial crisis that followed.

Singapore. The asset securitisation market was active from its commencement in **2006** but became very quiet after the 2007 crisis. Examples include commercial MBS structured and sold by **DBS Bank**, dealing with pre-sales of properties under construction, where progress payments were securitised to provide construction finance. Another deal, structured and lead-managed by **HSBC for Courts Asia**, used two distinct **special purpose vehicles (SPVs)** — one in Malaysia and one in Singapore — for a multi-currency, multi-jurisdiction transaction, with Courts' loan receivables as the SPV assets.

7.10.1 Collateralised Debt Obligations (CDOs)

A CDO is structured by an investment manager or bank that pools cash flow-generating assets (bonds or loans), which are repackaged and sold to investors. Investors purchase the CDOs, typically issued by an **SPV** and secured and backed by the acquired assets.

CDO type	Backed by
Collateralised bond obligation (CBO)	A pool of **bonds**
Collateralised loan obligation (CLO)	A pool of **loans**

CDOs are effectively a type of asset-backed security. A fund manager looks after a portfolio of assets, and the proceeds from issuing the CDOs finance the acquisition. A CDO is therefore a vehicle for an investment manager to gather assets to generate management fees, or to arbitrage — earning the yield between the return on the assets and the rate paid to CDO investors.

During the sub-prime crisis, **40–50%** of US CDOs were made up of MBS. Those MBS were mixed with other investment grade securities and, with insurance purchased on them, re-sold as CDOs with **higher credit ratings**.

Some banks selling CDOs included **credit default swaps (CDS)** to protect buyers against default. An insurance company found the CDS premiums attractive and began selling CDS to different parties **even for the same MBS** — guaranteeing payment to several parties if the same MBS defaulted. As long as the CDOs and MBS did not default, the insurer earned premiums from multiple investors on the same asset. On default, it was obligated to pay all the CDS buyers, and when defaults occurred in 2007/8 it had to pay **many times over**.

7.11 Credit Risk

Credit risk or **default risk** is the risk that the issuer cannot make timely principal and interest payments as agreed. Most corporate bonds pay a higher yield than comparable US Treasury securities, which are regarded as free of credit risk.

7.11.1 Credit Risk Rating

Credit risk is rated primarily by **S&P Global, Fitch Ratings and Moody's Investors Service**. Ratings range from **AAA** (highest) to **D** (in default).

Moody's	S&P	Fitch	Credit worthiness
Aaa	AAA	AAA	**Extremely strong** capacity to meet financial
Aa1 / Aa2 / Aa3	AA+ / AA / AA−	AA+ / AA / AA−	**Very strong** capacity; differs from the highest
A1 / A2 / A3	A+ / A / A−	A+ / A / A−	**Strong** capacity but somewhat more susceptib
Baa1 / Baa2 / Baa3	BBB+ / BBB / BBB−	BBB+ / BBB / BBB−	**Adequate** capacity; adverse conditions more l
Ba1 / Ba2 / Ba3	BB+ / BB / BB−	BB+ / BB / BB−	**Less vulnerable** in the near term than other
B1 / B2 / B3	B+ / B / B−	B+ / B / B−	**More vulnerable** than BB, but currently has
Caa	CCC	CCC	**Currently vulnerable**, dependent on favoura
Ca	CC	CC	**Currently highly vulnerable**
—	C	C	**Currently highly vulnerable** to non-payme
C	D	D	Has **failed to pay** one or more financial obliga
—	SD	RD	**Selective default** on a specific issue or class,
NR	NR	NR	No rating requested, or insufficient information
e, p	pr	Expected	**Preliminary** ratings pending final documenta
WR	—	—	Rating **withdrawn** — maturity, calls, puts, co
Unsolicited	Unsolicited	Unsolicited	Initiated by the agency, not requested by the is

Short-term issuer credit ratings (S&P).

Category	Definition
A-1	**Strong** capacity; highest category. A **plus sign (+)** indicates extremely strong capacity
A-2	**Satisfactory** capacity, but somewhat more susceptible to adverse changes than the highest category
A-3	**Adequate** capacity; adverse conditions more likely to weaken it
B	**Vulnerable** with significant speculative characteristics; currently has capacity but faces major ongoing uncert
C	**Currently vulnerable** to non-payment resulting in SD or D; dependent on favourable conditions
R	Under **regulatory supervision** owing to financial condition; regulators may favour one class of obligations o
SD & D	Failed to pay one or more financial obligations when due. An obligor is in default unless S&P believes paymer

7.11.2 Credit Spreads

A **credit spread** is the rate difference, usually in basis points, between two bonds of the **same maturity** but **different credit quality**. It directly measures the cost of risk — the extra yield investors demand to take on a given default risk. **As the issuer's default risk increases, its bond price falls and its spread widens.**

Worked example. A 10-year Treasury at **2%** and a 10-year corporate bond at **4%** means the corporate offers a **200 basis point** spread over the Treasury. A **AAA** investment grade industrial bond traded at roughly a **60 basis point** spread over the 10-year USD Treasury Note.

Comparisons can be made between any bond type and a benchmark yield, with the spread monitored over time. A US Financials bond rated **AA** offered a **higher** yield than a US Industrial bond rated **A** — because since the Global Financial Crisis and the European Sovereign Debt Crisis, financial institutions' credit ratings remain weak. **Credit spreads widen against the benchmark during crises and narrow when conditions and risk appetite improve.**

7.11.3 Non-Rated/Unrated Bonds

Some Asian companies choose not to be rated, believing the main investors know their business and history — an Indonesian corporate seeking funding mainly from local institutions familiar with the company, for example. An unrated company is not necessarily of poor credit or afraid of being rated; it is simply **not rated**.

Regulators are generally concerned about unrated debt because of lower transparency and higher investor risk. Local and regional investors are attracted by the **higher yields**, which reflect the higher risk. Some rely on the company's brand or its association with the government; others depend on the credit analysis of the institutions underwriting and marketing the bonds. Those institutions must carry out **due diligence** and ensure no conflicts of interest.

7.11.4 Mitigating Poor Credit Quality of the Bonds

Poor credit quality raises the cost of financing. Companies can improve it by:

- Creating a dedicated **sinking fund** setting aside income to sustain interest payments and eventually repay the debt.
- Offering specified assets, such as property, as **collateral or security**.
- Offering **parental guarantees**, backing the issue with the parent's or major equity holder's credit standing.
- Working with a financial institution to **structure and repackage** lower-quality bonds with higher-quality bonds, pooling them as the underlying for a higher-rated security (a CDO).
- Getting a financial institution to offer **credit insurance or credit default swaps**, guaranteeing to cover losses if the issuer defaults.

7.11.5 Credit Default Swaps

A CDS is the most commonly used form of credit derivative. In simple terms it is a **contract of insurance** guaranteeing a buyer of debt protection against default. CDS let investors exchange the risk of default for a premium.

- The **buyer** pays a premium to be insured against default.
- The **seller** receives regular payments and must make a **lump-sum payment** (the principal) if default occurs.

CDS were conceived to let financial institutions transfer credit exposure and free up regulatory capital. Besides protecting against default of high-risk bonds, sovereign and corporate debt, investors use CDS to hedge the credit risk of MBS and CDOs. In the bond context, CDS mitigate risk by transferring it from one party to another **without transferring the underlying bond**.

The derivatives market was largely **unregulated and lacked transparency until 2009**, after the Global Financial Crisis. Issuers of CDS saw the market's lucrativeness given the absence of regulation ensuring the seller had the money to pay if the bond defaulted, and there was no clarity on which sellers ultimately held the exposure. Many institutions, including insurers, sold CDS excessively for premiums while heavily **undercapitalised and unhedged**. CDS gave buyers a **false sense of security**, so they bought riskier debt believing they were protected. The system worked while there were no defaults and no expectation that all debts would come due at once.

When US mortgagees began defaulting, MBS could not make payments and defaulted. Institutions that purchased the securities came under distress and some went bankrupt — **Lehman Brothers**. Insurers that sold multiple CDS on the same MBS could not provide the protection they sold — **the bailout of AIG**. The CDS market fell apart overnight.

Today CDS are the engine of the credit derivatives market, with enhanced regulatory structure, letting investors hedge or gain credit exposure and arbitrage.

7.11.6 Red Flags for Credit Analysis and Distressed Debts

Financial statements are often released too late and only confirm perceptions. Fluctuations in the value of credit assets tend to precede the release of financials, leaving investors reactive. To manage a debt portfolio proactively, focus on variables that affect current and future revenues and costs **before** they appear in the accounting numbers.

Category	Red flags
Projected revenue falls	Sudden, significant and sustained falls in sales or prices from oversupply or stru
Projected cost rises or stickiness	Key cost items rise and operating costs spike; revenues fall from industry chang
Regulatory restrictions	New regulation restricts old forms of revenue creation; new regulation drasticall
Projected operating cash flow falls	If current and projected operating cash flow stays negative, the only ways to se
Departure of key officers	Regular changes in financial controllers, accountants, CEO, COO, CFO, CIO a

Window-dressing of financials is another red flag, but it typically does not allow timely mitigation. A "wait-and-see" approach until the company has difficulty meeting debt and interest obligations is too risky. The primary objective should be to **minimise risk** while still achieving the desired return.

7.11.7 Investing in Distressed Debts

A company facing distress may not default. With credit deterioration the debt may be priced below par, but that does not mean the company will fail as a going concern.

At that point, re-examine projected operating cash flow and asset disposals to assess whether the company can meet immediate obligations and whether assets are available to fulfil maturing debt. If it can, with sufficient buffer, the debt may be a worthwhile investment because it is priced at a discount, giving a higher yield.

Some hedge funds, private equity firms and higher-risk-threshold investors see distressed debt as an opportunity. There is potential for recovery and substantial gains, especially where the challenges relate to a **short-term credit squeeze or cyclical downturn** with recovery when key variables normalise. A distressed bond priced at 0.30–**0.40 on the dollar** may recover a significant portion of value.

If, however, the likelihood of revenue recovery is low — structural industry change — and peripheral asset values are unavailable or insufficient to support the debt, and there is insufficient cash flow to tide over, investors should salvage what they can. Several major institutions sustained massive losses from distressed debt during the Global Financial Crisis in 2008.

7.12 Other Sources of Risk

7.12.1 Interest Rate Risk

Bond prices are most vulnerable to interest rate changes. There is an **inverse relationship**: as rates rise, bond prices fall, and vice versa. This is also called **delta risk**. For an investor holding to maturity, intermediate volatility is not a concern; for one who plans to sell before maturity, a rate rise means capital loss.

7.12.2 Reinvestment Risk

Interest income is usually reinvested, but in a changing rate environment the reinvestment rate is uncertain. There is a risk that the rate at which interim cash flows can be reinvested will **decline**, lowering the eventual return. Reinvestment risk is **greater for higher coupon bonds and longer holding periods**.

Interest rate risk and reinvestment risk make **opposite** predictions. A rate rise pushes the bond price down (the price must fall to give the higher yield, since the coupon is fixed), while the higher rate at which coupons can be reinvested pushes the eventual return up. These are **not fully offsetting**: generally the reinvestment effect only slightly mitigates the interest rate effect.

7.12.3 Timing or Call Risk

Callable bonds carry the risk of redemption before maturity. Three disadvantages for the investor:

- Issuers are likely to redeem when rates decline, exposing the investor to **reinvestment risk** at lower yields.
- **Capital appreciation is limited**, because the price will not rise much above the call price.
- The promised **cash flow pattern is not known with certainty**.

7.12.4 Inflation or Purchasing Power Risk

Because promised cash flows do not change, rising general prices reduce their real value. Inflation risk is the uncertainty about how much lower the real return will be than the nominal return. It concerns **unanticipated** inflation: if inflation is anticipated and factored into the price, investors are adequately compensated.

7.12.5 Marketability or Liquidity Risk

The ease with which a security can be sold at or near fair value in the secondary market. An indication is the **size of the bid-offer spread**: the greater the spread, the greater the marketability risk. For an investor holding to maturity, marketability risk matters less.

7.12.6 Exchange Rate or Currency Risk

The SGD cash flows of a foreign currency bond depend on prevailing exchange rates when payments are received. If the foreign currency **depreciates** against the SGD, fewer SGD are received — currency risk. If it **appreciates**, the investor enjoys currency gains.

7.12.7 Volatility Risk

The price of a bond with an embedded option depends on the level of interest rates and other factors influencing the option's value, one of which is the **expected volatility of interest rates — vega risk**. The value of an option rises when expected rate volatility increases. The risk that a change in volatility adversely affects the price is volatility risk.

7.12.8 Political Risk

A government may declare withholding taxes on a bond, declare a tax-exempt bond taxable, or impose capital controls on remittance of funds. These actions can adversely affect value. The possibility of political or legal actions adversely affecting a security's value is **political or legal risk**.

7.12.9 Event Risk

Occasionally the issuer's ability to pay interest and principal is seriously impaired by a natural catastrophe, regulatory change, takeover or corporate restructuring. These are **event risks** and result in a rating downgrade.

7.12.10 Sector Risk

Sector risk applies only to a particular industry or sector: the stocks of many companies in one sector fall at the same time because of an event affecting the whole industry. Investors can reduce sector risk by **diversifying**.

7.13 Bond Valuation

The value of a bond equals the **present value of its expected cash flows** — the periodic interest payments plus the principal repayment at maturity — discounted at the required rate of return.

For annual coupons:

$$P = \frac{C_1}{1+k} + \frac{C_2}{(1+k)^2} + \frac{C_3}{(1+k)^3} + \cdots + \frac{C_n}{(1+k)^n} + \frac{M}{(1+k)^n} = \sum_{t=1}^{n} \frac{C_t}{(1+k)^t} + \frac{M}{(1+k)^n}$$

For semi-annual coupons:

$$P = \sum_{t=1}^{2n} \frac{C_t}{(1+k/2)^t} + \frac{M}{(1+k/2)^{2n}}$$

where

$$P$$

is the present value today,

$$C_t$$

the coupon payments for period

$$t$$

,

$$M$$

the maturity value,

$$n$$

the number of years to maturity, and

$$k$$

the required market yield — the yield currently available for a similar security of this maturity.

The discount rate

$$k$$

the investor wants is the **required rate of return**. The **required yield** is the yield offered by comparable bonds of the same maturity and credit quality.

Worked example. A **10-year** bond with par **\$1,000** and coupon rate **8%**, with a required market yield of **6%**.

a. Annual coupons.

$$\text{Price} = \frac{80}{1.06} + \frac{80}{1.06^2} + \frac{80}{1.06^3} + \cdots + \frac{80}{1.06^{10}} + \frac{1{,}000}{1.06^{10}} = \$1{,}147.20$$

Calculator inputs: **N = 10**, **PMT = 80**, **FV = 1,000**, **I/Y = 6**; compute PV.

b. Semi-annual coupons.

$$\text{Price} = \frac{40}{1.03} + \frac{40}{1.03^2} + \cdots + \frac{40}{1.03^{20}} + \frac{1{,}000}{1.03^{20}} = \$1{,}148.77$$

The number of periods is **doubled** and the discount rate **halved**.

7.13.1 Valuation of a Zero-Coupon Bond

A zero-coupon bond has only the maturity value as a cash flow, so its price is the present value of the maturity amount:

$$P = \frac{M}{(1 + k/2)^{2n}}$$

Worked example. A zero-coupon bond maturing in **15 years** with par **$1,000**, at a required return of **9%**:

$$P = \frac{1{,}000}{(1.045)^{30}} = \$267.00$$

Even without coupons, the semi-annual rate of **4.5%** and the half-year period are used, to stay consistent with semi-annual coupon bonds. Calculator inputs: **FV = 1,000**, **N = 30**, **I/Y = 4.5**; compute PV.

7.14 Relationship between Bond Price and Market Yield

There is an **inverse relationship**: bond prices move opposite to market yields. The price is the present value of the cash flows, so as yields rise the present value falls with the higher discount rate. Given expectations of a series of yield increases through monetary policy such as Fed hikes, existing bond prices and portfolios fall in value as yields rise. The reverse also holds.

Required yield	Bond value (% of par)
10%	**112.46%**
12%	**100.00%**
14%	**89.41%**

This is for a **10-year, 12% semi-annual coupon** bond. As the required yield rises, the bond value declines.

7.15 Relationship between Coupon Rate, Market Yield and Price

When a bond is issued, the coupon rate is set at approximately the prevailing market yield. When the market yield **equals** the coupon rate, the price is **par** (100%). If yields rise above the coupon rate, the investor must be compensated through a **lower price** — the bond sells at a **discount**. If yields fall below the coupon rate, the price rises — the bond sells at a **premium**.

- Coupon rate **equals** market yield → bond trades at **par**.
- Coupon rate **lower** than market yield → bond trades at a **discount**.
- Coupon rate **higher** than market yield → bond trades at a **premium**.

7.16 Relationship between Market Yield and Other Bond Features

7.16.1 Yield and Maturity

The longer the maturity, the greater the impact of a rate change on price, because longer-term cash flows are more sensitive. When yields rise, longer maturities suffer greater price declines; when yields fall, longer maturities rise more.

An investor whose horizon matches the bond's maturity, and who is unconcerned about marking price fluctuations to market, would be **indifferent** to interest rate risk, and would also have the opportunity for capital gains should yields fall. For bond fund managers and traders, however, portfolios reflect gains or losses each year with yield changes, so active management entails adjusting maturity and duration.

7.16.2 Yield and Coupons

Current yield is the rate of return given the current price; the **coupon rate** is the rate based on par value, usually the price at first issue.

A bond with **smaller coupons** has a larger percentage of its value attributable to the long-term principal repayment. Since longer-term cash flows are more sensitive to rate changes, **bonds with smaller coupons are more severely affected by a rise in yield** than bonds with larger coupons. For two bonds of equal maturity, the one with the **larger coupon** is less affected by rising yields and priced slightly higher.

7.16.3 Yield and Callable Features

A call feature gives the issuer the right to redeem when cheaper refinancing is available. When rates fall, the call benefits the issuer, which can refinance at the lower rate without paying market price to call. For the buyer it is a disadvantage, because the buyer is exposed to **reinvestment risk** if the bond is called when rates fall.

7.17 Duration and Convexity

7.17.1 Duration

Duration and convexity are central to bond portfolio risk management. Duration is affected by **coupon rate, yield and remaining time to maturity**. It is a **linear** measure of the approximate price sensitivity of a bond to **small** interest rate changes — not more than **1% or 100 basis points**.

All else equal, duration is **higher** for bonds with **lower coupons, lower yields and longer maturities**.

Modified duration calculates price changes from small yield changes. A portfolio's overall duration is calculated to monitor profit and loss sensitivity to small yield changes. **Portfolio duration is the weighted-average duration** of all bonds in the portfolio.

7.17.2 Convexity

Duration is an imperfect measure because it approximates price sensitivity linearly, whereas the price-yield relationship is **convex**.

- With **positive convexity**, duration increases as yields fall. For large rate changes, price **appreciation exceeds** price depreciation. This favours the investor, so a higher-convexity bond has a **higher price** than a lower-convexity bond.
- With **negative convexity**, price appreciation is **less** than price depreciation for large rate changes.

Convexity is directly related to the coupon rate and remaining term: **the lower the coupon, the higher the convexity**. **Zero-coupon bonds have the highest convexity.**

7.17.3 Duration and Convexity: The Price-Yield Relationship

Duration and convexity are the two metrics used to assess how a bond's price responds to rate changes, and are the most prevalent measures for quantifying market risk across rate environments.

Duration matters when capitalising on rate sensitivity. A higher duration implies greater price movement up or down. Anticipating **falling** rates, a portfolio's duration should be **increased** to maximise sensitivity. Anticipating **rising** rates, duration should be **decreased** to minimise sensitivity.

Convexity is the **second derivative** of the price function, measuring the rate of change in duration — the curvature of the price-yield relationship. **Regardless of whether rates are rising or falling, investors should seek greater convexity.**

Interest rate	More duration	More convexity
Rising	Price falls **more**	Price falls **slower**
Falling	Price rises **more**	Price rises **faster**

A more convex bond outperforms a less convex bond **both** when rates rise and when they fall.

7.17.4 Present Value of a Basis Point (PVBP or PV01)

PVBP, or PV01, is a risk management tool: the **change in present value of the entire bond portfolio for a one basis point movement in interest rates**. Typically a limit is set for the total PVBP effect, given parallel rate movements of **100, 200 and 300 basis points**.

7.18 Time Path of a Bond

Holding all else constant, a bond's price must **converge to par** as maturity approaches, because it will be redeemed at par. Over time the price of a **discount** bond **increases**, and the price of a **premium** bond **declines**. A bond selling at par stays at par as it approaches maturity.

7.19 Yield Measures

The most common measures quoted by bond dealers are **current yield** and **yield-to-maturity**.

7.19.1 Current Yield

The ratio of the interest payment to the current price:

$$\text{Current Yield} = \frac{\text{Annual Interest Payment}}{\text{Bond Price}}$$

Worked example. A **10-year, 6%** coupon bond with par **$1,000** priced at **115%** (that is, $1,150):

$$\text{Current Yield} = \frac{60}{1{,}150} = 5.22\%$$

Current yield considers only the annual interest payment and **does not account for capital gain or loss**.

7.19.2 Yield-to-Maturity (YTM)

YTM is the rate of return earned if the bond is purchased at a given price and held to maturity. It is the **discount rate that makes the present value of the bond's cash flows equal to its current market value**.

Worked example. A price of $1,148.80 * *for a * *10 - year * *bond with par * *1,000** and coupon **8%**, paid semi-annually:

$$1{,}148.80 = \frac{40}{1+k/2} + \frac{40}{(1+k/2)^2} + \cdots + \frac{40}{(1+k/2)^{20}} + \frac{1{,}000}{(1+k/2)^{20}}$$

Solved by trial and error, or with a calculator: **N = 20**, **PV = −1,148.80**, **PMT = 40**, **FV = 1,000**; compute I/Y = **3.00%** semi-annually.

Market convention annualises by **doubling** the semi-annual yield, giving **6%** — the **bond-equivalent yield**.

The proper way to annualise is the effective annual yield:

$$\text{Effective annual yield} = (1 + \text{semi-annual rate})^2 - 1 = (1.03)^2 - 1 = 6.09\%$$

Approximation formula. Relates net annual effective cash flow to the average amount invested, assuming the price gradually converges to par:

$$\text{Approximate YTM} = \frac{\text{Annual Interest Payment} \pm \text{Amortised Gain or Loss}}{(\text{Current Market Price} + \text{Par Value})/2}$$

For the bond above, the amortised loss per year is $(1{,}148.80 - 1{,}000)/10 =$ **$14.88**:

$$\text{Approximate YTM} = \frac{80 - 14.88}{(1{,}148.80 + 1{,}000)/2} = \frac{65.12}{1{,}074.40} = 6.06\%$$

Zero-coupon YTM. Substituting zero coupons into the present value model and solving for

$$k$$

:

$$\frac{k}{2} = \left[\frac{\text{Maturity Value}}{\text{Price}}\right]^{1/2n} - 1$$

Worked example. A **15-year** zero-coupon bond with maturity value $1,000 * *selling for * *$**300**:

$$\frac{k}{2} = \left[\frac{1{,}000}{300}\right]^{1/30} - 1 = 0.0409 = 4.1\%$$

The number of periods is **30 semi-annual periods**, double the years. The **bond-equivalent yield is 8.2%**.

YTM versus current yield. YTM is more meaningful because it considers both coupon income and any capital gain or loss realised on holding to maturity, and it accounts for the timing of cash flows.

The investor realises the stated YTM only if:

- Coupon payments can be **consistently reinvested at the YTM**.
- The bond is **held to maturity**.

The risk that future reinvestment rates differ from the YTM at purchase is **reinvestment risk**. If the bond is not held to maturity, it may be sold at a price different from par and at a different time, producing a return different from the YTM. The risk of selling at a loss because yields have risen is **interest rate risk** or **price risk**.

7.19.3 Reinvestment Risk

Maturity and coupon rate affect the degree of reinvestment risk.

- For a given coupon rate, **the longer the maturity, the greater the reinvestment risk**, because more of the total dollar return depends on compounding to realise the YTM.
- For a given maturity, **the higher the coupon rate, the greater the reinvestment risk**, because there is more to reinvest.

A **zero-coupon bond has no reinvestment risk** if held to maturity, because there is nothing to reinvest.

7.19.4 Taxable Equivalent Yield (TEY)

Some bonds are tax-exempt, so their stated rate is lower than comparable taxable bonds. To compare, compute the TEY:

$$\text{Taxable Equivalent Yield} = \frac{\text{Tax-Exempt Yield}}{1 - \text{Marginal Tax Rate}}$$

Worked example. An investor in the **28%** marginal bracket holds a 5-year tax-exempt bond with a YTM of **7%**:

$$\text{TEY} = \frac{0.07}{1 - 0.28} = 9.7\%$$

7.20 Term Structure of Interest Rates

The relationship between yield-to-maturity and term to maturity is the **maturity structure** or **term structure** of interest rates — the **yield curve**.

7.20.1 Types of Yield Curves

Shape	Description
Rising (most common)	Yields on short-term issues are low and rise consistently with maturity
Declining	Short-term yields are **higher** than long-term yields
Humped	Intermediate-term yields are higher than short-term, and long-term rates decline below short-te
Flat	Short-term and long-term yields are the **same**

7.20.2 Theories of Interest Rates

Three theories explain the different yield curve shapes.

a. Expectations theory. The shape is determined by investors' **interest rate expectations**. When the market expects rates to rise, lenders prefer to lend short, forcing short-term rates down, while borrowers prefer to borrow long, forcing long-term rates up — creating an **upward-sloping** curve. If the market expects rates to fall, lenders prefer the long end (forcing long rates down) and borrowers the short end (forcing short rates up), producing a **downward-sloping** curve. When the market is ambivalent, the curve tends to be **flatter**.

Worked example. A 1-year T-bill yields **6%**, and after one year a new 1-year T-bill is expected to yield **8%**. An investor with a 2-year horizon has two options:

- Invest in 1-year T-bills on a rolling basis, giving an average yield of **7%**:

$$[(1.06)(1.08)]^{1/2} - 1 = 0.07$$

.

- Invest in a 2-year security, requiring a yield of **7%** — the average of the 6% realised in year one and the 8% expected in year two.

b. Liquidity preference theory. Argues the yield curve should **always be upward sloping**, and any other shape is a temporary aberration. Because long-maturity securities are more price-volatile, investors require a **higher return** to hold them. Short-term securities are more easily converted to cash without large price movements, so investors pay a higher price and receive a lower yield. From the lender's perspective, higher yields are required to induce long-term lending because liquidity is sacrificed for longer. This again justifies an upward slope.

c. Market segmentation theory. Presumes investors have their own **maturity preferences dictated by the nature of their liabilities**. Life insurance companies with long-term liabilities prefer long-term bonds; banks prefer short-term liquid securities to match deposits. The yield for each maturity is determined **solely by supply and demand in that segment**. Long-term borrowers' and lenders' activities set long-term rates; short-term trades set short-term rates.

7.21 Analysis of Convertible Bonds

A **convertible bond** gives the holder the right to convert into a predetermined number of ordinary shares of the **issuer** within a specified period. **Exchangeable bonds** grant the right to exchange for ordinary shares of a firm **other than the issuer**. A major reason for adding the convertible feature is to increase the issue's attractiveness, ensuring a market for it.

Convertibles are issued as part of fundraising. The feature is a **sweetener**, letting the company issue at **lower coupon rates** than comparable straight bonds, lowering financing costs.

Worked example — Genesis convertible bond.

Feature	Value
Maturity	10 years
Coupon rate	4%, paid semi-annually
Conversion ratio	**40** (1 bond converts to 40 shares)
Par value	$1,000
Market price of the convertible	**$1,100**
Market price of the common share	**$24**
Annual dividend	**$0.35** per share

7.21.1 Minimum Value of a Convertible Bond

A convertible is a "double" security: a **straight bond** plus a **call option** on the shares. Two values must be considered.

a. Investment value. The bond value, also called the **straight value**, is the value without the conversion feature, calculated from the cash flows as if there were no option. It requires the market yield on a comparable straight bond. With a comparable yield of **8%**:

$$\text{Bond Value} = \sum_{t=1}^{20} \frac{C_t}{(1+k/2)^t} + \frac{\text{Par Value}}{(1+k/2)^{20}} = \$728.19$$

b. Conversion value. The total market value of the shares the bond can be exchanged for:

$$\text{Conversion value} = \text{Conversion Ratio} \times \text{Market Share Price} = 40 \times 24 = \$960$$

The minimum value of a convertible bond is the greater of its conversion value and its investment value. With a conversion value of $960 * *andaninvestmentvalueof * *\mathbf{728}$, the minimum value is **$960**.

If the bond sold below $960 — say at $728 — arbitrageurs would buy it, convert, and sell the shares at $24, making * *232^{**}$ per bond. If the investment value were higher than the conversion value, the bond would trade at its investment value; if it traded below, the yield would exceed comparable straight bonds, and investors would bid the price up until yields equalised.

7.21.2 Conversion Ratio and Conversion Price

$$\text{Conversion Ratio} = \frac{\text{Par Value of CB}}{\text{Conversion Price}}$$

For Genesis the ratio is **40**, meaning a $1,000 par convertible exchanges for 40 ordinary shares.

$$\text{Conversion Price} = \frac{\text{Par Value of CB}}{\text{Conversion Ratio}} = \frac{1{,}000}{40} = \$25$$

7.21.3 Conversion Parity Price

What the investor effectively pays per ordinary share if the convertible is purchased and converted:

$$\text{Conversion Parity Price} = \frac{\text{Market Price of CB}}{\text{Conversion Ratio}} = \frac{1{,}100}{40} = \$27.50$$

The conversion parity price can be considered a **share price breakeven point**. Any further increase in the share price beyond this point produces the same percentage increase in the convertible's value.

7.21.4 Conversion Premium

An investor buying the convertible instead of the share pays a premium of **3.50 * *(27.50 − $24), or **14.6%**, over the current share price. The conversion premium is the difference between the convertible's market price and its conversion value, expressed in absolute or percentage terms.

$$\text{Conversion premium (\$)} = \text{Market Value of CB} - \text{Conversion Value} = 1{,}100 - 960 = \$140$$

$$\text{Conversion Premium (\%)} = \frac{\text{Market Value of CB} - \text{Conversion Value}}{\text{Conversion Value}} = \frac{140}{960} = 14.6\%$$

Normally a convertible also trades at a premium to its **straight value**, because it embeds an option equivalent to a call on the share at a strike of **$25** (one $1,000 bond converts to 40 shares). The price differential between the convertible and its straight value is effectively the value of the **embedded call option**. The investor pays the higher price for the potential capital gain if the share appreciates, while downside risk is limited to the straight value.

7.21.5 Breakeven Time

Convertibles are often issued with a **current yield higher than the dividend yield** on the shares. Investors compare attractiveness by how quickly the conversion premium is recovered from the income advantage of holding the convertible.

$$\text{Breakeven Time} = \frac{\text{Conversion Premium}}{\text{Bond Income} - \text{Dividend Income}}$$

For Genesis, the conversion premium is **$140**.

Item	Calculation	Amount
Bond income	0.04 × $1,000	**$40**
Dividend income	40 × $0.35	**$14**
Differential	40 − 14	**$26**
Breakeven time	140 / 26	**5.38 years**

Breakeven time ignores the time value of money, but is a useful indicator of relative attractiveness.

7.21.6 Downside Risk

The investment value is usually the **floor price** of a convertible. Downside risk is measured by comparing the market price with the investment value:

$$\text{Premium Over Straight Value (\%)} = \frac{\text{Market Price of CB}}{\text{Investment Value}} - 1$$

For Genesis: $1,100 / $728 − 1 = **51%**.

Despite its use in practice, this measure is **flawed** because the straight value changes as the discount rate changes.

7.21.7 Investing in Convertible Bonds

Convertibles offer a unique combination of an income stream from a fixed income security and participation in the capital appreciation of the underlying share — **upside potential and downside protection**. The yield is usually **higher than the dividend yield** of the underlying share but **lower than a comparable plain vanilla bond**.

When the share price rises above the conversion strike, the convertible's price is mainly influenced by the share. If the share price in the example rose to $50 * *, the bond would have a minimum value of * *$**2,000**, and conversion would be clearly profitable. When trading at a reasonable conversion premium, the convertible's value is influenced by interest rates, the share price and the share's price volatility. When trading near investment value, the embedded option is almost worthless because the share price is substantially below the strike, and the convertible trades like a straight bond.

7.22 Summary

- For individuals, income falls into four categories: consumption, business reinvestment, reserves for family needs and life's risks, and excess cash.
- All full banks and finance companies in Singapore are DI Scheme members except those exempted by MAS. Effective **April 2024** the coverage limit is **S$100,000**, covering individuals, charities, sole proprietorships, partnerships, companies and other unincorporated entities.
- Money market instruments are debt instruments issued by governments, financial institutions and corporations with maturities under one year: T-bills, negotiable CDs or banker's acceptances, commercial paper, repos and bills of exchange.
- A fixed income security provides periodic payments at a rate known in advance, with eventual return of principal at maturity.
- A bond is a fixed income security on which the issuer makes regular interest payments on specified dates and repays principal at maturity.

- The bond market divides into the internal and external bond markets.
- In Singapore, the domestic bond market comprises SGS and corporate bonds issued by Singapore companies; the foreign bond market refers to foreign issuers issuing in Singapore.
- The Eurobond market is divided by the currency of denomination; USD-denominated issues are Eurodollar bonds.
- The secondary market benchmarks market values and yields of traded bonds.
- Special bond types include callable, puttable, convertible, sinking-fund, zero-coupon, floating rate, deferred-coupon and perpetual bonds.
- Convertibles benefit issuers by allowing financing at a lower rate and deferring potential dilution.
- Bonds rated below **BBB** are high-yield or junk bonds.
- ABS may be secured against auto loans and leases, consumer loans, commercial assets and credit card receivables.
- A CDO is structured by an investment manager or bank pooling cash flow-generating assets, repackaged and sold to investors.
- Credit risk is rated primarily by S&P Global, Fitch and Moody's, from **AAA** to **D**.
- Deferred-interest, step-up and payment-in-kind bonds have deferred-coupon structures.
- The credit spread measures the cost of risk; as default risk increases, bond price falls and spread widens.
- A CDS guarantees against bond default and transfers credit exposure. The buyer pays a premium and receives a lump sum on default; the seller receives premiums and pays the insured amount on default.
- Beyond financial ratios, rely on swings in projected revenues, costs and cash flows, regulatory policy and changes in key management.
- Bonds are most vulnerable to credit and interest rate risk. Other risks include reinvestment, call, inflation, liquidity, currency, volatility, political, event and sector risk.
- The value of a bond is the present value of its future cash flows: coupons plus principal for a coupon bond, principal only for a zero-coupon bond.
- There is an inverse relationship between bond prices and required market yield.

Exam cues

- **Cash pays the lowest return.** From **July 2009 to July 2017**, deposits returned at most **5–6%** while the **S&P 500 rose more than 133%**.
- **Deposit insurance:** introduced **1 April 2006**, enhanced **May 2011**; limit **S**$50,000 \rightarrow S$**75,000 (1 April 2019)** $\rightarrow$ **S**$100,000(1April2024) **.Covers**S$ **deposits only** — not foreign currency deposits, structured deposits, unit trusts, shares or other securities. CPFIS and CPFRS monies are aggregated and **separately** insured up to S$100,000.
- **DI worked example:** S$650,000across accounts at one bank \rightarrow only * *S$200,000** insured (S$100,000 for S$ deposits plus S$100,000 for CPF).
- **Money market instruments:** maturity **under 1 year**, minimum denomination **S$250,000 and above**, low risk but **high reinvestment risk** because of the short maturity.
- **Discount instruments:** T-bills, commercial paper, banker's acceptances. **Interest-bearing:** certificates of deposit.
- **Commercial paper is unsecured**, so only very strong credits can issue it.
- **Repo = collateralised short-term loan.** The yield is the difference between sale and purchase prices.
- **Bill of exchange:** payable on sight or demand, or after a period — commonly **91 days** (a term bill). Negotiable by endorsement and delivery.
- **Maturity segments:** money market **1 year**; medium-term notes **> 1 to 10 years**; long-term bonds **> 10 years**.
- **Par value is usually $1,000** and is not market value. **Term bonds** have one maturity date; **serial bonds** have several.
- **Bank discount yield:**

$$Y_d = (D/F)(360/t)$$

. Worked: 90 days, face 100,000, price 98,000 $\rightarrow$ **8.0%**. It is **not a true return** because it uses face value, not the amount invested.
- **Price quotes:** percentage of par. **98.50 = $985**; **95¾ = $957.50**. Treasuries in **1/32** (90-24 = 90 24/32); corporates in **1/8**.
- **Accrued interest:** an 8% semi-annual bond sold 3 months after the last coupon owes **20 * *(80 $\times$ 3/12).
- **SGS exist to build a benchmark yield curve**, not to fund the government, which runs a balanced budget. **Mini-auctions** began in **2015**. Corporate debt issued **SGD259 billion**, outstanding

SGD386 billion in 2017.

- **Eurobond:** underwritten by an international syndicate, offered simultaneously in several countries, issued outside any single jurisdiction.
- **Zero-coupon bonds have no reinvestment risk** but the **highest price volatility** and **highest convexity**.
- **High-yield = below BBB.** Deferred-coupon structures: deferred-interest (**3–7 years**), step-up, PIK (**5–10 years** choice), perpetual (no maturity, redeemable typically **3–5 years** after issue, ranks **before equities**).
- **Call provision:** issuer's right; call price above par; call premium **higher early**, declining to par. Callable bonds carry **higher yields**.
- **Put provision:** holder's right to sell back **at par**; lowers the yield because the investor has bought a put.
- **Sinking fund** reduces credit risk; the **doubling option** lets the issuer retire twice the required amount.
- **Claim order:** secured (senior) → subordinated (junior) → income bonds; bondholders rank above preference and ordinary shareholders.
- **ABS vs MBS:** MBS pool **mortgages**; ABS pool **non-mortgage** assets. **CBO** = bonds; **CLO** = loans. **40–50%** of US CDOs were MBS before the crisis.
- **Rating scale:** AAA/Aaa down to D. **Investment grade is BBB−/Baa3 and above**; below that is high-yield. **SD** = selective default; **R** = under regulatory supervision; **WR** = withdrawn.
- **S&P grace period rule:** any stated grace period longer than **five business days** is treated as five business days.
- **Credit spread:** same maturity, different credit quality. 10-year Treasury at **2%** vs corporate at **4%** = **200bp**. Spreads **widen in crises** and **narrow** when risk appetite improves.
- **CDS:** buyer pays premium, receives lump sum on default; seller receives premiums, pays on default. **Lehman** failed as a CDS buyer; **AIG** was bailed out as a CDS seller.
- **Distressed debt:** a bond at 0.30–**0.40 on the dollar** may recover substantially if the problem is a short-term squeeze rather than structural decline.
- **Ten fixed-income risks:** interest rate (delta), reinvestment, timing/call, inflation, marketability/liquidity, currency, volatility (vega), political/legal, event, sector.
- **Interest rate risk and reinvestment risk move in opposite directions** and are **not fully offsetting** — reinvestment only slightly mitigates rate risk.
- **Bond pricing:**
$$P = \sum C_t/(1+k)^t + M/(1+k)^n$$
. Worked: 10-year, 8% coupon, 6% yield → $1,147.20 * *annual*, **$**\mathbf{1{,}148.77}$ semi-annual.
- **Zero-coupon pricing:**
$$P = M/(1+k/2)^{2n}$$
. Worked: 15 years, 9%, par 1,000 → **$267.00**.
- **Price-yield table:** 10-year 12% semi-annual bond at 10% yield = **112.46%**; at 12% = **100.00%**; at 14% = **89.41%**.
- **Coupon vs yield vs price:** coupon = yield → **par**; coupon < yield → **discount**; coupon > yield → **premium**.
- **Duration:** higher for **lower coupons, lower yields, longer maturities**. Linear approximation valid only for changes up to **100 basis points**. Portfolio duration is the **weighted average**.
- **Convexity:** lower coupon → higher convexity; **zero-coupon has the highest**. Positive convexity means appreciation exceeds depreciation. **Always prefer more convexity**, whether rates rise or fall.
- **Duration vs convexity table:** rising rates — more duration falls more, more convexity falls slower; falling rates — more duration rises more, more convexity rises faster.
- **PVBP/PV01:** change in portfolio present value for a **one basis point** move; limits set at **100, 200 and 300bp** parallel shifts.
- **Time path:** discount bonds **rise** to par; premium bonds **fall** to par; par bonds stay at par.
- **Current yield:** annual interest / price. Worked: 6% coupon at 115% → **5.22%**. It ignores capital gain or loss.
- **YTM:** the discount rate equating present value to price. Worked: price 1,148.80, 8% semi-annual, 10 years → semi-annual **3.00%**, bond-equivalent **6%**, effective annual **6.09%**.
- **Approximate YTM:** (annual interest ± amortised gain/loss) / ((price + par)/2). Worked: (80 − 14.88)/1,074.40 = **6.06%**.
- **Zero-coupon YTM:** (1,000/300)^(1/30) − 1 = **4.1%** semi-annual, bond-equivalent **8.2%**.
- **YTM is realised only if** coupons are reinvested at the YTM **and** the bond is held to maturity.
- **Reinvestment risk rises with maturity and with coupon.** A zero-coupon bond has **none**.

- **TEY:** tax-exempt yield / (1 − marginal rate). Worked: 7% at 28% → **9.7%**.
- **Yield curve shapes:** rising (most common), declining, humped, flat.
- **Three theories:** expectations (shape reflects rate expectations), liquidity preference (always upward sloping, premium for long maturities), market segmentation (each maturity set by its own supply and demand).
- **Expectations worked example:** 6% now and 8% expected next year → 2-year yield **7%**.
- **Convertible worked example (Genesis):** conversion value $960 * *; investmentvalue * *$**728**; minimum value $960 * *; conversionprice * *$**25**; conversion parity price $27.50 * *; conversionpremium * *$**140 or 14.6%**; breakeven time **5.38 years**; premium over straight value **51%**.
- **Trap:** the **bank discount yield** overstates the true return because it divides the discount by **face value** rather than the price paid.
- **Trap: current yield** ignores capital gain or loss, so it understates the return on a discount bond and overstates it on a premium bond.
- **Trap:** a **negative** convexity bond underperforms in both directions; a **positive** convexity bond outperforms in both.
- **Trap:** the **premium over straight value** is a flawed downside measure because the straight value itself moves with the discount rate.

Chapter 8 — Portfolio Management

Portfolio management is where the earlier chapters converge: risk and return measurement, correlation, the pricing models that link risk to expected return, the process that turns a client's objectives into an asset mix, and the measures that judge whether the manager added value. This chapter covers all four parts.

Learning Objectives

The candidate should be able to:

Efficient Market Hypothesis (EMH)

- State the different levels of market efficiency under the EMH in the context of the current market environment.
- Explain how active and passive portfolio management can be applied by investors.

Portfolio Theory

- Appreciate how investment portfolios are constructed and managed.
- Explain the investment management principles behind diversification and portfolio risk.
- Distinguish between systematic and non-systematic risks, particularly in the hedging of each.
- Compute the correlation coefficient and use it to determine the extent of diversification of risks in a portfolio.
- Calculate portfolio return and risk.
- Show how portfolio risks can be diversified internationally.
- Understand and explain the Capital Asset Pricing Model (CAPM) and the risk-free rate.
- Explain the Capital Market Line for the portfolio's risk-return trade-off.
- Explain the Security Market Line for the portfolio's risk-return trade-off.
- Use the Security Market Line to determine overvalued and undervalued securities.
- Explain the Arbitrage Pricing Theory and how it compares with the CAPM.

Portfolio Management Process

- Describe and explain the portfolio management process.
- Explain how investment policies are established.
- List and explain the investment objectives in the investment policy statement.
- Describe the contents of a typical investment policy statement.
- Discuss and explain the investment constraints in portfolio management.
- Explain asset allocation and the difference between strategic and tactical asset allocation.

Performance Measurement

- Compute and measure investment returns across different cash flow assumptions and movements in the portfolio.
- Use the dollar-weighted rate of return to measure performance.

- Use the time-weighted rate of return to measure performance.
- Compare the advantages and disadvantages of dollar-weighted versus time-weighted rates of return.
- Describe the Modified Dietz method, assuming a constant rate of return over the measurement period.
- Appreciate the Global Investment Performance Standards.
- Explain risk-adjusted measures of performance such as the Sharpe ratio, Treynor ratio and Jensen's alpha.
- Describe performance attribution as a means to measure a portfolio against an established benchmark.

8.1 Introduction

The chapter has four parts.

- The **Efficient Market Hypothesis** — its variants and its relevance to portfolio management, plus portfolio management strategies in light of it.
- **Harry Markowitz's portfolio theory** — the foundation of portfolio management: diversification and correlation, systematic and non-systematic risk, and **William F. Sharpe's CAPM**, which describes the risk-return relationship by evaluating investments in a portfolio context.
- The **portfolio management process** — constructing a portfolio: the determinants of investment policy (objectives and constraints), and strategic and tactical asset allocation.
- **Performance** — how returns are measured, risk-adjusted measures (Sharpe, Treynor, Jensen's alpha), and performance attribution.

8.2 The Efficient Market Hypothesis

The EMH states that an efficient market is one where **security prices always fully reflect available information**. Market participants act collectively, with prices adjusting quickly to all known information about the firm. The direct implication is that stocks always trade at fair value and it is **impossible to outperform the overall market consistently on a risk-adjusted basis** through expert stock selection or market timing, because all available information is already in the price.

After more than half a century of research the validity of the EMH remains **inconclusive**. Possibly this is because there are **varying degrees** of market efficiency, allowing a minority with an advantage to exploit pockets of informational inefficiency for abnormal gains over certain periods.

There are three forms: weak, semi-strong and strong.

8.2.1 Weak Form of the EMH

Under the weak form, investors **cannot consistently make abnormal profits using historical price and volume information**. All past price information is fully reflected in the current price. Future prices cannot be predicted by analysing past prices and therefore follow a **random walk** without serial dependency. This implies investors cannot consistently achieve excess returns using **technical analysis**.

Some participants do not behave as if markets are weak-form efficient. Technical analysis remains used to detect unusual volume increases and price trends signalling buying or selling interest, and to trade with momentum and direction. Extensive **high-frequency trading (HFT)** seeking to exploit price trends and momentum is also inconsistent with weak-form efficiency. Institutions operating **dark pools** obtain confidential client trading information that could provide an unfair advantage — as in the **SEC enforcement action against two major banks** that supplied dark pool trading information to external high-frequency traders.

Despite this behaviour, the empirical evidence indicates technical analysis **generally does not yield abnormal returns**, supporting the weak form.

8.2.2 Semi-Strong Form of the EMH

Under the semi-strong form, **all publicly available information** is fully reflected in the current price, and prices react rapidly to new public information such as earnings announcements and management changes. This implies **neither fundamental nor technical analysis** allows investors to consistently beat the market.

As with weak-form efficiency, many participants do not behave as if markets are semi-strong efficient. There are many anecdotes of value-oriented investors with consistently successful track records, though it has been argued this can occur by **pure chance** given the sheer number of investors, and that observed excess returns may simply be compensation for an **unidentified risk**.

8.2.3 Strong Form of the EMH

Under the strong form, **all information, public or private**, is fully reflected in the current price. Investors cannot make consistent abnormal profits on any type of information, including insider or private information — even insider trading would give no advantage.

The evidence suggests real-world markets are **not** strong-form efficient. The history of insider trading in the US and numerous high-profile cases continue to suggest inside information does provide an advantage. Hence the need for regulatory oversight by the SEC and equivalent bodies worldwide. In less developed markets with fewer enforcement and monitoring capabilities, the advantage conferred by inside information could be even higher.

8.2.4 Interpreting and Applying the EMH

Supporters of market efficiency highlight that the **average fund manager does not outperform the market**. Despite claimed expertise, the majority underperform their benchmarks **even before fees**. Even for managers who outperform, the outperformance **does not generally persist**.

Markets are therefore probably quite efficient, making it difficult to beat the benchmark consistently. But there may be **periods of inefficiency** in which irrational behaviour and uneven distribution of price-sensitive information let more astute professionals outperform.

For those who believe in efficient markets, a **passive** strategy makes sense: find an efficient low-cost fund that replicates a benchmark index. For those who believe markets can be inefficient, they can:

- Try to outperform by trading actively.
- Adopt a longer-term approach, selecting and holding investments with strong fundamentals.
- Identify and choose an active portfolio manager.

8.3 Passive versus Active Management Strategies

Institutional and corporate investors choose between passive and active management. For individuals, the choice depends on additional factors beyond belief in market efficiency:

- Funds available for investing (size of investment).
- Desire for personal involvement in trading or investment.
- Amount of time available to manage investments.
- Investment expertise.
- Market structure.

8.3.1 Passive Strategy

The EMH implies it is not possible to outperform through active share selection and analysis. For investors who believe in efficient markets, frequent trading generates large brokerage commissions without enhancing expected performance. **Passive management**, such as a **buy-and-hold** strategy, is therefore recommended. Passive investors believe such a strategy will, over the long term, produce results just as good as active management because of market efficiency.

Alternatively they can invest in an **equity index fund** or portfolio. Index funds track a selected market index, and the manager is judged on **how well they track the target index**. Expenses — research costs, management fees, brokerage commissions — are kept to a minimum.

Investors who:

- Can only invest small amounts,
- Lack the time or expertise to monitor and manage investments, and
- Do not wish to pay fees,

should probably adopt a passive approach and rely on index funds for diversification.

8.3.2 Active Strategy

The existence of numerous asset management firms suggests most investors do not subscribe to the EMH and favour an active approach. Valuation models and analysis of the economy, industries and individual companies are all elements of active management, whose objective is to **outperform the market on a risk-adjusted basis**.

Active investors believe they have advantages over other participants — superior analytical skills, or access to information not available to others. Three active strategies:

a. Security selection. Value-oriented specialists believe they can outperform by selecting securities they believe are **undervalued**, typically using fundamental analysis and valuation models such as the dividend discount or P/E ratio model. They tend to look for **low P/E shares** or shares trading below intrinsic value.

b. Sector rotation. Shifting sector weights to take advantage of sectors expected to outperform while avoiding or underweighting those expected to underperform. Under the Industry Classification Benchmarks the market is classified into **10 industries**, further subdivided into super sectors and sectors; a **sector is the smallest subset of an industry**. It is common to categorise ordinary shares into broad sectors — interest-sensitive, cyclical and defensive — each expected to perform differently through the business cycle:

- **Interest-sensitive** shares are expected to **underperform during periods of high interest rates**.
- **Cyclical** shares do well in an upturn and badly in a downturn.
- **Defensive** shares are not hurt as badly during a downturn.

Effective sector rotation depends heavily on **accurate assessment of current economic conditions**.

c. Market timing. When equities are expected to do well, timers shift holdings from cash equivalents to equities; when equities are expected to do poorly, they liquidate equity holdings. Market timing strategies therefore involve **active asset allocation**.

8.4 Portfolio Theory

8.4.1 Introduction to Portfolio Theory

Portfolio theory discusses how investors create **optimal portfolios** that maximise expected return while minimising risk. A simple optimal portfolio can be built from two asset classes — S&P 500 equities and US Treasury bonds.

Over the last 90 years, portfolio returns for different equity-bond weights were plotted against portfolio standard deviation. Although stock-bond correlation over the last decade was mostly negative (averaging about $= -0.20$), since **August 2017** correlation has turned positive again, so a positive correlation of $= 0.1$ is applied in the illustration.

The **efficient frontier** represents all combinations of the two asset classes generating the **highest return at a given level of risk**. Every combination on the frontier is an optimal portfolio, but for clients with different risk profiles: portfolios with higher equity weights suit higher risk appetites, with the highest-risk portfolio being **100% equity**; portfolios with higher bond weights suit lower risk appetites.

Investors should **not** pick portfolios that lie **below** the efficient frontier — for example a portfolio at **5.2% return and 8.1% volatility**, or one at **4.7% return and 8.5% volatility** — because at that level of risk better choices offer higher returns.

Most banks present potential optimal allocations diagrammatically on the efficient frontier, with lower risk and return portfolios at one end and higher at the other. The frontier is a **curve, not a straight line**, because the two assets' returns are not perfectly correlated.

If adding alternative investments — hedge funds, derivatives, private equity, commodity funds, real estate funds — can either **add return per unit of risk** or **lower risk per unit of return**, they are beneficial and move the efficient frontier toward the **upper left** of the graph.

But simply adding new assets does not necessarily increase return per unit of risk: some assets increase portfolio risk without adequately improving returns. Review the impact on expected volatility and returns before accepting any proposed addition.

8.4.2 Portfolio Diversification

Diversification is the financial equivalent of "don't put all your eggs in one basket". Studies show that increasing the number of securities from different industries reduces portfolio risk. Further diversification comes from different asset classes and selective international investment.

Diversification across asset classes is recommended because it produces a better risk-adjusted return, but it should not be done in an ad hoc or blanket fashion. Buying assets with a **lower return per unit of risk** and **high correlation** with the existing portfolio is unlikely to improve diversification. Identifying assets

with the potential to generate higher return per unit of risk requires a well-thought-out process of reviewing markets, sectors and companies to overweight or underweight, either as long-term allocation or tactically.

Worked illustration — diversification within an asset class. A portfolio holding only one security (Apple) faces three broad levels of risk:

- **General economic environment** — business cycles, interest rates, currency movements.
- **Industry-specific risk** — the IT/consumer electronics industry: changes in consumer preferences, technological obsolescence.
- **Firm-specific risk** — new product launches, research and marketing success, Apple's management.

Adding a second stock (Monsanto, a producer of genetically modified grains) reduces portfolio risk **to the extent that firm-specific influences differ**. It does not matter whether the factors affecting Monsanto are diametrically opposed to those affecting Apple; all that matters is that correlation is **less than perfectly positive**. As long as correlation is not 1, adding Monsanto has the potential to diversify risk.

As more securities are added, the effect of firm-specific factors on volatility is further limited, so portfolio volatility continues to decline but **at a slower rate**. Studies show that around **half of portfolio standard deviation is eliminated by the time there are 10 securities**. The benefits then diminish, with marginal risk reduction small.

Theoretically, a portfolio containing **all** shares in a market (a stock market index) has completely diversified away all non-systematic risk, leaving only **systematic risk** — risks common to and affecting all securities. Because all shares are affected by the business cycle to varying degrees, business cycle risk cannot be avoided however diversified the portfolio.

The risk remaining after extensive diversification is **market risk** or **systematic risk**, attributed to market-wide sources. The risk eliminated by diversification is **firm-specific risk** or **non-systematic risk**.

8.4.3 Systematic and Non-systematic Risks

$$\text{Total risk} = \text{Systematic risk} + \text{Non-systematic risk}$$

Systematic risk (undiversifiable risk) results from factors affecting all firms, such as interest rate risk and inflation risk. It is caused by changes in the economic, political and sociological environment that affect the prices of all marketable securities, and **cannot be eliminated by diversification**.

Non-systematic risk (firm-specific risk) can be mitigated through diversification. It is the portion of total risk peculiar to a firm or industry, arising from factors affecting a company's earnings potential — company strategies, management errors, new product developments and innovations. Because it is caused by factors affecting specific firms or industries, it can be minimised by investing in a broad number of different firms or industries. It can theoretically be **reduced to zero** with a well-diversified portfolio of uncorrelated shares.

Four further points:

- **Systematic risk cannot be reduced through diversification**, but depending on exposure and asset class, investors can partly or almost fully **offset** it through **hedging** with options and futures.
- **Non-systematic risk can be reduced by diversification.** The larger the number of assets, the lower the overall asset-specific risk.
- **Systematic risk is reflected in beta ( )** — the sensitivity of the portfolio to market movement.
- **Realised total risk** is captured by the realised standard deviation of the portfolio, which includes systematic risk, non-systematic risk, and an **error term** capturing the deviation between predicted and actual standard deviation.

8.4.4 Correlation Coefficient

The correlation coefficient (, "rho") measures the extent to which the returns of two securities move together. **Correlation denotes association, not causation.**

Value	Meaning
+1.0	**Perfect positive correlation** — returns have an exactly direct linear relation; identical return patterns
0	Returns have **no linear relation**
−1.0	**Perfect negative correlation** — exactly inverse linear relation; when one return is high the other is negative

For efficient diversification, include securities whose returns are **not highly correlated**. If returns are highly correlated, risk cannot be reduced significantly.

- Combining securities with **= 1** provides **no reduction** in portfolio risk; the portfolio risk is simply the weighted average of the individual risks.
- Combining two securities with **= 0** brings **some reduction**; adding more securities with lower correlation to the portfolio achieves significant risk reduction.
- Combining two securities with **= −1** can **eliminate portfolio risk altogether** — the principle behind hedging strategies.

As long as correlation is less than perfectly positive, risk can be reduced. Typically returns have some correlation but are not perfectly positive, so portfolio risk can almost always be reduced but usually **not completely eliminated**.

8.4.5 Correlation is Not Stable

Assumptions about correlation must be adjusted from time to time because correlations between asset groups **change over time**. Correlations between S&P 500 returns and US 10-year Treasury yields fluctuated between negative and positive territory from **1997 to 2016**.

8.4.6 Correlation Rises During Crisis

One concern about diversification is the potential for asset correlations to **move toward 1 during major financial crises**. Though rare, extremely high correlation occurred in the **2008 Global Financial Crisis**.

Systematic risk was higher because financial institutions and the entire banking sector were threatened. As credit dried up and redemptions became rampant, fund managers had to sell assets to meet redemptions. Even companies less related to the crisis had to sell strong-performing assets to cover margin positions on directly affected assets. The **contagion effect** rippled through global markets because counterparty risk was at an all-time high as institutions had to be bailed out. Almost all assets corrected downwards, and correlations gravitated toward 1. The silver lining was that correlation remained highly positive during the **2009 rebound**.

There was nowhere for funds to hide in 2008 except **cash and government bonds**. The reverse happened in 2009, with recovery across most asset classes except listed real estate, which took longer.

8.4.7 Calculating Portfolio Return and Risk

The expected return of a portfolio is a **weighted average** of the expected returns of its individual securities:

$$R_p = \sum_{i=1}^{n} w_i r_i$$

where

$$R_p$$

is the expected portfolio return,

$$w_i$$

the portfolio weight for security

$$i$$

,

$$r_i$$

the expected return on security

$$i$$

,

$$n$$

the number of securities, and

$$\sum w_i = 1.0$$

.

Portfolio risk is NOT a weighted average of the individual securities' risk. For two assets:

$$\sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2}$$

where

$$w_1$$

and

$$w_2$$

are the portfolio weights,

$$\sigma_1$$

and

$$\sigma_2$$

the standard deviations, and

$$\rho_{1,2}$$

the correlation coefficient.

Worked example.

Security	Expected return	Standard deviation	Weight
A	18%	20%	40%
B	9%	10%	60%

Correlation coefficient between A and B is **0.8**.

$$\text{Expected return} = (0.4)(0.18) + (0.6)(0.09) = 0.072 + 0.054 = 12.6\%$$

$$\sigma_p = \sqrt{0.4^2 \times 0.2^2 + 0.6^2 \times 0.1^2 + 2(0.4)(0.6)(0.8)(0.2)(0.1)} = \sqrt{0.0064 + 0.0036 + 0.00768} = \sqrt{0.01768} = 0.1330 = 13.30\%$$

The effect of correlation:

Correlation	Portfolio risk
$= 1$	**14%**
$= 0$	**10%**
$= -1$	**2%**

Portfolio risk falls steadily from **14% to 2%** as correlation declines from +1.0 to −1.0.

8.4.8 International Diversification

Portfolio risk can also be reduced through international diversification, because share price movements in different countries are fairly uncorrelated.

Inherent in international diversification is **foreign exchange risk**. One way to remove it is to **hedge using forward exchange contracts**.

International diversification does **not always** lead to higher risk-adjusted returns. It makes sense to add securities from new markets only if the additions have the potential to raise the portfolio's risk-adjusted return; adding the wrong securities and markets can **lower** it.

Top-down fundamental research on the economy, industry and company is important, as are reviews of political and currency risk. **Back-testing simulations** can show how securities or markets performed in different periods, including under stress.

8.4.9 The Capital Asset Pricing Model

Derived by **Sharpe, Lintner and Mossin** in the mid-1960s, CAPM describes the equilibrium relation between **systematic risk** and **expected return**.

Assumptions:

- Investors evaluate portfolios on the basis of risk and return.
- Investors are rational and maximise return for a given level of risk, and minimise risk for a given level of return.
- Individual assets are **infinitely divisible** — fractional shares can be bought or sold.
- Investors can **borrow or lend at the risk-free rate**.
- There are **no taxes or transaction costs**.
- There is **no inflation**, or any change in interest rates, or inflation is fully anticipated.
- All investors have the same **one-period time horizon**.
- Information is **freely and instantly available** to all investors.
- Investors have **homogeneous expectations** of risk, return and covariances.
- Capital markets are in **equilibrium**.

Because all investors have the same expectations, they arrive at the same optimal portfolio of risky assets — the **market portfolio (M)**. Being completely diversified, it comprises only systematic risk, which cannot be eliminated because it results from macroeconomic factors affecting all securities. All assets are included in portfolio M **in proportion to their market value**.

In theory the market portfolio should include all risky assets worldwide, financial and real. In practice the portfolio of all ordinary shares on the S&P 500 is used to represent the universe, with the **S&P 500 Composite Index** as the proxy.

CAPM introduces the **risk-free asset**, which has no risk and provides the risk-free rate (

$$R_f$$

). Its return variance is **zero** and its covariance with any risky asset is **zero**.

Two important relations follow:

- The **capital market line**, specifying the equilibrium relation between expected return and **total risk** for efficiently diversified portfolios.
- The **security market line**, specifying the equilibrium relation between expected return and **systematic risk**. It applies to individual securities as well as portfolios.

a. Uses of the CAPM. CAPM evaluates investments against the market benchmark risk-return relation to see whether an investment provides a fair return given its systematic risk. Despite its shortcomings it is still used to offer insight into the market risk-return relation and to determine the cost of capital for securities.

b. The Capital Market Line. The CML shows the risk-return trade-off for **efficient portfolios**. It is the efficient frontier formed from combinations of the market portfolio with the risk-free asset. In equilibrium all investors should hold a portfolio somewhere on the CML.

$$E(R_p) = R_f + \frac{E(R_m) - R_f}{\sigma_m}\sigma_p$$

where

$$E(R_p)$$

is the expected return on any efficient portfolio P on the CML,

$$R_f$$

the risk-free rate,

$$E(R_m)$$

the expected return on the market portfolio,

$$\sigma_m$$

the standard deviation of market returns, and

$$\sigma_p$$

the standard deviation of portfolio P's returns.

$$E(R_m) - R_f$$

is the **market risk premium**.

The **slope** of the CML,

$$[E(R_m) - R_f]/\sigma_m$$

, is the **market price of risk** for efficient portfolios: the additional return the market requires for a **1% increase** in the portfolio's standard deviation. The expected return for any portfolio on the CML equals the risk-free rate plus the product of the market price of risk and the quantity of portfolio risk.

c. The Security Market Line. While the CML applies only to efficient portfolios, the SML depicts the risk-return trade-off for **all assets**, including individual securities and inefficient portfolios.

The relative systematic risk of a security or portfolio is the volatility of its return relative to the market portfolio's volatility, multiplied by the correlation coefficient. This is **beta ( )**. The market portfolio's beta is **1**.

- $>$ **1** — proportionately higher systematic risk than the market, requiring a higher expected return.
- $<$ **1** — lower systematic risk than the market, with a lower expected return.

A security with beta **1.5** has, on average, **1.5 times** as much systematic risk as the market. A risk-free asset has a beta of **zero**, but an asset with beta zero need not be risk-free — it can have unsystematic risk; either way both should yield the risk-free rate.

$$E(R_i) = R_f + \beta_i(R_m - R_f)$$

where

$$E(R_i)$$

is the expected return on security or portfolio

$$i$$

,

$$R_f$$

the risk-free rate,

$$R_m$$

the expected market return,

$$\beta_i$$

the systematic risk relative to the market, and

$$R_m - R_f$$

the market risk premium.

Securities with betas greater than 1 should have larger risk premiums than the market and therefore require higher rates of return, because beta measures undiversifiable risk and greater systematic risk should be accompanied by greater expected return. Securities with betas below the market's are less risky and should have lower required returns.

Worked example. Growth Portfolio A has a beta of **1.20**, the risk-free rate is **2.5%** and the expected market return is **14%**:

$$E(r) = 2.5 + 1.2(14 - 2.5) = 16.3\%$$

d. Generating positive alpha. Alpha is the actual outperformance of the portfolio above what is predicted. If the actual return exceeds the predicted CAPM return, the manager is generating positive alpha — the ability to outperform the predicted CAPM return (or whatever model return is used as benchmark).

For Growth Portfolio A, an actual return of **17%** implies alpha of **17 − 16.3 = 0.7%**. At **15%**, alpha is **−1.3%**.

Investors should buy portfolios generating positive alpha and sell or short those generating negative alpha. Demand for positive alpha raises its price; as the price rises, expected return falls, reducing and ultimately eliminating the alpha that created the excess demand. Conversely, a drop in demand for a negative-alpha fund lowers its price and raises its expected return, pushing alpha back to zero. Such buying and selling pressure should leave most securities with **zero alpha most of the time**, and positive or negative alphas are unlikely to persist for many securities. **Unless analysis shows otherwise, alpha should be assumed to be zero.**

e. Overvalued and undervalued securities. The SML identifies over- and undervaluation.

- Securities **above** the SML provide **above-average expected returns** for their betas and are **underpriced**.
- Securities **below** the SML offer **lower returns** than predicted for their risk and are **overpriced**.

In a perfectly efficient market all securities should lie **on** the SML. If they do not, the market mechanism forces them back: a security above the SML attracts demand, pushing its price up and reducing its expected return; a security below attracts selling, pushing its price down and raising its expected return. Buying and selling continues until equilibrium, when all securities lie on the SML.

f. Expensive or cheap markets. The SML also reflects the value of markets at a point in time.

- A **steep** SML indicates a large incremental return for a small increase in systematic risk. The market may be **excessively risk-averse** and therefore **cheap**.
- A **flat** SML indicates little incremental return for additional systematic risk. The market appears to be **ignoring risk** and may be **overvalued**.

8.4.10 Arbitrage Pricing Theory

A main criticism of CAPM is the **inability to observe or properly measure the return on the market portfolio**, which consists of all risky assets. Moreover, CAPM's assumptions are largely unrealistic. CAPM is essentially a **single-factor model** where price fluctuations are explained only by the market factor.

In response, **Stephen Ross** developed the **Arbitrage Pricing Theory (APT)** in the early 1970s — a **multifactor equilibrium pricing model**.

APT shares these assumptions with CAPM:

- Investors have **homogeneous beliefs**.
- Investors are **risk-averse utility maximisers**.
- Markets are **perfect**, and frictions such as transaction costs are irrelevant.
- Returns are generated by a **factor model**.

But APT does **not** assume:

- A single-period investment horizon.
- Absence of taxes.
- Borrowing and lending at the risk-free rate.
- Investors selecting portfolios on the basis of expected return and variance.

A factor model holds that underlying **risk factors** affect realised and expected returns. These are not firm-specific factors but broad economic forces with three characteristics:

- The risk factor must have a **pervasive influence** on share returns. Firm-specific events are not APT risk factors.
- The risk factor must **influence expected return**.
- At the beginning of each period the risk factor must be **unpredictable by the market as a whole**. The **rate of inflation is not an APT risk factor** because it is partially predictable; **unexpected inflation** — the difference between actual and expected inflation — **is**.

Economic factors might include real economic growth, unexpected inflation, changes in interest rates and political uncertainties.

$$R_i = E(R_i) + b_{i1}f_1 + b_{i2}f_2 + \cdots + b_{in}f_n + e_i$$

where

$$R_i$$

is the actual return on security

$$i$$

in a period,

$$E(R_i)$$

the expected return,

$$f$$

the deviation of a systematic factor from its expected value,

$$b_i$$

the sensitivity of security

$$i$$

's returns to a factor, and

$$e_i$$

a random error term unique to security

$$i$$

.

The actual return equals the expected return if the factors are at their expected levels and the error term is zero.

The problem with APT is that the factors are **not well specified**, at least for the future. Most empirical studies suggest **three to five factors** influence security returns. **Chen, Roll and Ross (1986)** identified four systematic factors:

- Unanticipated changes in **inflation**.
- Unanticipated changes in **industrial production**.
- Unanticipated changes in the **default-risk premium**.
- Unanticipated changes in the **term structure of interest rates**.

Investors can better understand returns by identifying the risk factors affecting asset returns. A manager can structure a portfolio to increase exposure to one or more factors, or hedge against them.

8.5 Portfolio Management Services

Portfolio management services vary by client type.

Dedicated and personalised services are available to institutional funds, sovereign wealth funds, corporate funds and funds of accredited individual investors. With dedicated services, funds (for example **USD50m AUM**) may be separately managed from other funds, and the manager provides regular monthly updates and one-to-one performance reviews.

Retail investors, who generally have smaller amounts, can use unit trust funds traded on global exchanges and private funds sold by banks. **Berkshire Hathaway** is a notable listed investment company solely involved in managing investments. Some funds are country-specific, others offer regional or global diversification; there are also sector-specific and thematic funds. By pooling investor monies, retail investors benefit from diversification through economies of scale and shared resources.

8.5.1 Investment Approach and Style

Portfolio management services are not suitable for all investors. Some do not trust anyone else to manage their funds and prefer a hands-on approach. Others do not wish to incur management fees, which average **1.0%–1.5%**. Some prefer a hybrid approach: managers run the core conservatively while the investor directly manages a portion.

Regardless of preference, both individual and institutional investors benefit from a **disciplined** portfolio management approach.

Clients who self-manage	Clients with both self-managed and professionally managed	Clients
Likes trading ideas and strategies	Manages some investments directly and trades occasionally	No tim

Clients who self-manage	Clients with both self-managed and professionally managed	Clients
Wants good research	Has a proportion of funds managed externally	Wants
Believes in their own ability	May use external managers as well as trade directly	Wants
Wants a cost-effective, efficient execution platform	Long-term oriented but will take short-term opportunities	Willing
Has time to manage own portfolio		
Can trade listed diversified and thematic funds		

a. Advisory portfolio management. For investors who wish to remain actively engaged in decisions. Banks provide such services only for **affluent (accredited) clients** willing to place significant amounts — for example **USD10 million** — to justify the added expertise and resources. Less affluent clients rely on their relationship manager or internet portal for research, advice and asset allocation recommendations. At the lowest end, retail banking clients may receive no support services when purchasing funds.

b. Discretionary portfolio management. For most investors buying portfolio investment mandates, the general practice is to give the fund manager **discretion**. The manager decides asset allocation and securities selection **without obtaining permission for each transaction**, guided by the client's investment profile and policy statement.

The portfolio management process has three layers:

- **Style of management** — quantitative vs qualitative; fundamental vs technical; use of derivatives and hedging, currency overlay, alternative investments; use of computer models and trading algorithms; in-house vs external research.
- **Management process** — active management vs rigid model limits; asset allocation discipline; the amount of cash held on average and in crisis; policy on redemptions; rebalancing frequency.
- **Audit and governance** — resource capabilities; performance review and reward; performance attribution; fees for performance.

These sit between **economic and market conditions** (what the markets will allow) and **client objectives and specifications** (what the client wants). The process runs: research outlook → expected return and risk targets for the model portfolio → portfolio construction (strategic asset allocation, securities selection, implementation and execution, currency overlay or hedging) → create the **investment policy statement** → portfolio monitoring and rebalancing (tactical asset allocation, profit taking and risk management) → performance measurement and review (against client expectations, market benchmark, peer managers, and performance attribution) → performance reporting and review with the client.

8.5.2 Portfolio Management Process for Retail Investors

Portfolio management evaluates investment risk and return to construct an optimal portfolio, taking account of the investor's objectives, risk profile, and any unique constraints and circumstances.

Understanding personal goals helps the representative determine objectives, identify a suitable fund or service, and recommend strategies. Conservative strategies may suit money set aside for retirement or children's education. Individual clients generally have **shorter time horizons** than institutional clients, but an investor setting aside dedicated funds through a trust for grandchildren may have a long enough horizon for the higher volatility of a growth-oriented fund.

The process for individual investors is **dynamic, continuous and systematic**:

- **Setting investment objectives** — analyse objectives, requirements, preferences and constraints to define an acceptable level of risk and a reasonable expected return for that risk.
- **Developing and implementing strategies** — set guidelines for asset allocation and define the circumstances in which the allocation may vary.
- **Monitoring market conditions** — track market conditions and changing relative values, and periodically re-evaluate the investor's needs, circumstances and objectives.
- **Reviewing and adjusting the portfolio** — make adjustments to reflect changes in the market or the investor's requirements.
- **Measuring portfolio performance** — measure performance and evaluate it relative to a benchmark.

Objectives, constraints and strategy should always be stated in a **written document**. A well-drafted investment policy provides the discipline for efficient management, minimising **whipsaw reactions** to temporary price swings or protracted volatility.

8.5.3 Determinants of the Investment Policy Statement (IPS)

The IPS, or **investment mandate**, **belongs to the client, not the portfolio manager**. The manager must therefore spend sufficient time with the client to define investment goals and acceptable risk levels, and to note constraints and preferences.

The IPS typically covers:

- **Client description and circumstances.**
- **Investment objectives** — return requirement and risk tolerance.
- **Investment constraints** — liquidity, time horizon, tax, regulatory and legal, and unique needs or preferences.
- **Asset allocation** — strategic asset allocation ranges, and a rebalancing policy. Investors often specify a **policy portfolio** (conservative, balanced or aggressive) as part of their objectives.

8.5.4 Investment Portfolio Objectives — Risk and Return Requirements

In defining objectives, emphasis should be placed on determining the client's **sensitivity to risk**. Return requirement and risk tolerance are usually established through an interview and discussion session, and a **risk profiling questionnaire**.

Understanding the client's profile goes beyond regulatory compliance: it embodies taking a genuine interest in the client's needs and financial well-being.

The life cycle approach is a useful framework for analysing an individual's risk appetite.

Stage	Circumstances
1st — early career	Priorities are saving for liquidity, buying a home, obtaining life insurance, then investments. A lar
2nd — mid-career	Can establish a serious investment programme; liabilities substantially lower. The horizon remains
3rd — late career	Near retirement, with greater emphasis on income needs and capital preservation. Assets significa

Each individual's risk tolerance is unique. The risk profile is also affected by net worth, health, family situation, future financial commitments, familiarity with investments, job stability and emotional temperament.

8.5.5 Investment Constraints

a. Liquidity requirements. These vary considerably and must be clearly communicated. If funds are for the next mortgage payment or the coming semester's tuition, **immediate liquidity is essential** and financial assets are preferred. Such needs should be met with high-quality, short-term debt instruments that can be easily sold — **Treasury bills and commercial paper**. If assets held have poor marketability, such as real estate or shares in a closely held business, liquidity needs must be monitored carefully.

In practice few investment companies hold a liquidity reserve, because most financial assets are highly liquid: shares and bonds can generally be sold within minutes at a price close to market value. Shortfalls are normally met by selling financial assets.

b. Time horizon. The investment period, which affects the nature of investments selected and the strategies employed. Investors with a **long** horizon can absorb and smooth out the ups and downs of risky assets such as ordinary share portfolios, and can build riskier portfolios. Investors with a **short** horizon tend to have less volatile portfolios, typically more bonds than shares, with shorter duration bonds and lower-volatility shares.

c. Tax considerations. The manager must consider:

- The client's **income tax bracket**.
- Different tax treatments of different assets (tax-exempt or not).
- Different components of total return (dividend or interest income versus capital gains).

Investors in **high tax brackets** should consider tax-exempt fixed income securities if the **taxable equivalent return** exceeds the return on comparable taxable bonds. For such investors, equities with a large **capital gain** component may be more attractive than high dividend income-yielding shares.

d. Regulatory and legal constraints. Institutional investors such as insurance companies are regulated with regard to their investments, and are generally more regulated than private individuals. An individual

using personal savings has greater freedom, but if using **CPF** money the investment is governed by the **CPF Investment Scheme**, which specifies the types and quantum of permissible investments.

e. Clients' unique needs or preferences. It is the **fiduciary duty** of fund managers to ensure client instructions are adhered to. Some clients specify a limit on equity investments or foreign currency exposure. These preferences and instructions must be clearly spelt out in the investment policy **in writing** to avoid misunderstandings.

8.5.6 Asset Allocation

The purpose of asset allocation is to **maximise portfolio return at a level of risk consistent with the investor's objective**. Whether asset allocation matters more than securities selection has been vigorously debated. A study by **Brinson, Hood and Beebower** showed that **91.5% or more** of a portfolio's return comes from asset allocation, leaving **less than 10%** attributable to security selection. Asset allocation also has a much greater impact on reducing total risk exposure than picking an investment in any single asset category.

At the asset allocation stage, economists and market strategists provide input on macroeconomic expectations and market trends, because these affect the returns of the various asset classes.

a. Strategic asset allocation (SAA). How a portfolio's funds are divided among asset classes, given the client's risk profile and risk-return requirements, and the manager's long-term forecasts of expected returns, variances and covariances. SAA is the **"policy" or "long-run" asset mix**, normally expressed as percentages of total portfolio value.

Asset class	Allocation range
Equities	45–55%
Bonds	35–45%
Cash equivalents	5–15%
Total	**100%**

Such an approach is a **constant-mix strategy**. Maximum and minimum limits can be set for each asset class to accommodate different risk profiles and market conditions. Permissible asset classes must be clearly defined by the nature of the assets (cash, bonds, shares) and by sector, industry and geographical region or country, with possible sub-categories.

The optimal asset mix depends on the investor's return objectives and risk tolerance: **the lower the tolerance for risk, the more conservative the asset mix**. A conservative allocation relies heavily on bonds and short-term securities for stable returns, emphasising current income and capital preservation. An aggressive portfolio seeking capital gains has the highest exposure to ordinary shares and less to bonds and short-term securities, increasing expected return and risk.

SAA is not expected to change often. But for an individual, the long-run attitude to risk can change — through family circumstances, health, business circumstances, or a negative impact on wealth that makes the client much more risk averse. The SAA should then be adjusted. **Annual reviews** should check whether the client's view of risk-taking has changed.

b. Tactical asset allocation (TAA). For a given SAA mix, the manager is typically allowed to reduce or increase the allocation to an asset class to exploit market opportunities, **as long as the allocation stays within a certain range**. TAA is action taken to **overweight or underweight** asset classes over the **short run** in response to changing economic and market conditions, to optimise returns given perceived risks. It also seeks to exploit relative inefficiencies in security prices across asset classes.

Worked example. If the outlook for equities is favourable:

Asset class	SAA range	Original TAA	New TAA
Equities	45–55%	45%	**55%**
Bonds	35–45%	50%	**40%**
Cash equivalents	5–15%	5%	5%
Total		**100%**	**100%**

TAA models fall into three groups: **valuation approaches**, **cyclical considerations**, and a combination.

- The **valuation approach** identifies the relative value of an asset class: expensive classes are sold and cheap ones bought. The most common is the **risk premium or spread approach**: if the expected return on equities relative to bonds is **above average**, allocate more to equities; if **below average**, allocate less. This is the **equity-bond yield gap**.
- The **cyclical consideration approach** holds that cycles in share and bond prices are closely related through macroeconomic factors. Equity market movements typically **precede** changes in economic activity, so managers try to identify components of economic activity that are good predictors of market movements.

8.6 Performance Measurement

Two questions must be answered:

- How well or badly is the manager performing relative to a benchmark?
- What factors contributed to the success or failure of the portfolio?

8.6.1 Measuring Portfolio Returns

A proper measure of total return captures both the **income** and the **capital gain or loss** component. Performance is typically measured over a regular period, monthly or quarterly. Return measurement is simple with no cash flow movements, and more complicated when the client deposits or withdraws.

a. No deposits or withdrawals.

$$R_p = \frac{V_E - V_B}{V_B}$$

where

$$V_E$$

is the end value and

$$V_B$$

the beginning value.

Worked example. A portfolio worth $30 million **at the start of the quarter and **$35 million** at the end:

$$R_p = \frac{35 - 30}{30} = 16.7\%$$

b. Deposits or withdrawals at the end of the period. The deposit must be **deducted from the ending value** first.

$$R_p = \frac{(V_E + \text{Withdrawals} - \text{Deposits}) - V_B}{V_B}$$

Worked example. A portfolio worth $50 million **at the start; the client deposits **$5 million** just before the end of the quarter; the end value is **$53 million**:

$$R_p = \frac{(53 - 5) - 50}{50} = -4.0\%$$

c. Deposits or withdrawals at the beginning of the period. Adjust the **beginning value**: increase it by a deposit, decrease it by a withdrawal.

$$R_p = \frac{V_E - (V_B + \text{Deposits} - \text{Withdrawals})}{V_B + \text{Deposits} - \text{Withdrawals}}$$

Worked example. If the $5 million deposit was received at the **beginning** of the quarter:

$$R_p = \frac{53 - (50 + 5)}{50 + 5} = -3.6\%$$

d. Dollar-weighted rate of return (DWR). When cash flows occur **between** the beginning and end of the period, the DWR can be used. It is equivalent to the **internal rate of return**: it equates the present value of all cash flows, including the ending market value, with the beginning market value.

Worked example. If the $5 million deposit was made in the **middle** of the quarter, solve for

$$r$$

:

$$50 = \frac{-5}{(1+r)} + \frac{53}{(1+r)^2}$$

$$r = -1.92\%$$

(semi-quarterly return)

The quarterly return is

$$(1 - 0.0192)^2 - 1 = -3.8\%$$

. In the first half the loss was 1.92%, leaving 0.9808 of the original value; in the second half there is another 1.92% loss on that 0.9808, so

$$(0.9808)^2$$

of the original remains.

e. Time-weighted rate of return (TWR). An alternative when cash flows occur within the period. Calculating TWR requires the portfolio value **before each cash flow**.

Worked example.

Item	Value
Portfolio value at beginning of quarter	$50 million
Portfolio value just before deposit	$49 million
Deposit in middle of quarter	$5 million
Portfolio value immediately after deposit	$54 million
Portfolio value at end of quarter	$53 million

First half:

$$(49 - 50)/50 = -2.0\%$$

. Second half:

$$(53 - 54)/54 = -1.85\%$$

.

Quarterly return by linking:

$$[(1 - 0.02)(1 - 0.0185)] - 1 = -3.81\%$$

.

With frequent cash flows, many calculations may be needed to derive the TWR.

f. Annualising returns. From quarterly returns:

$$\text{Annual Return} = [(1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)] - 1$$

Over several years, the annualised geometric mean return — the **compound annual growth rate (CAGR)** — is used.

Worked example. A portfolio grew from **$100 to $250 in 5 years**:

$$250 = 100(1+r)^5 \implies (1+r)^5 = 2.5 \implies r = 20.1\%$$

g. Dollar-weighted versus time-weighted return. The two methods can produce different results, sometimes substantially. The **DWR is inappropriate for evaluating a portfolio manager**, because it is strongly influenced by the **size and timing of cash flows**, over which the manager has no control. The **TWR is more appropriate**, because the objective is to measure the manager's performance independent of the client's actions. Fund managers are evaluated using the TWR because they have little or no control over inflows and outflows.

h. Modified Dietz method. A true TWR requires valuing the portfolio just before each cash flow. In practice cash flows often occur intra-month when valuations are unavailable. The Dietz method overcomes this by assuming a **constant rate of return** during the period. The original Dietz method assumes all cash flows occur at the **midpoint**; the **modified Dietz** method weights each cash flow by the amount of time it is held.

$$R_p = \frac{V_E - V_B - CF}{V_B + WCF}$$

where

$$CF$$

is the net cash flow over the period and

$$WCF$$

the weighted cash flow:

$$WCF = \sum_i CF_i \frac{n - t_i}{n}$$

where

$$CF_i$$

is the cash flow on day

$$t_i$$

,

$$t_i$$

the number of days from the beginning of the period until

$$CF_i$$

, and

$$n$$

the number of days in the period. The numerator assumes the cash flow occurs at the **end of the day**; if assumed at the beginning, use

$$(n - t_i + 1)$$

.

Worked example. A portfolio worth $10 million$ **at the start of a month. Cash inflows of** $**100,000$ **on day 12** and $200,000$ *on day* 18 **. *End-of-month value* $**11 million**.

$$WCF = 100{,}000 \times \frac{30-12}{30} + 200{,}000 \times \frac{30-18}{30} = 60{,}000 + 80{,}000 = 140{,}000$$

$$R_p = \frac{11{,}000{,}000 - 10{,}000{,}000 - 300{,}000}{10{,}000{,}000 + 140{,}000} = 6.90\%$$

Advantage: no portfolio valuation is needed on the date of each cash flow. **Disadvantage:** it is a **less accurate** estimate of the true TWR, and can be affected by cash flows, especially when they are large relative to the portfolio or occur during periods of high volatility.

i. Global Investment Performance Standards (GIPS). Before **February 2005**, the Association of Investment Management and Research (AIMR) issued minimum standards for presenting portfolio performance. On **4 February 2005** the CFA Institute Board of Governors approved the revised GIPS, creating a **single global standard** for performance reporting and raising minimum standards worldwide. GIPS replaced country-specific standards and are widely accepted internationally.

Fundamental principles:

- **Total return**, accounting for realised and unrealised capital gains and income, must be used.
- **TWR must be used.**
- Portfolios must be valued (marked to market) **at least monthly**, and returns **geometrically linked** — CAGR for annualised returns.
- Performance must be calculated **after deducting actual trading expenses** (brokerage, commissions).
- Dividends, interest income and other time-based income must be **accrued**.
- **Annual returns for all years must be shown.** If the firm is more than **10 years old**, a **10-year** record must be presented; otherwise, the record since inception.
- **Disclose** whether results are gross or net of investment management fees.
- The use of **leverage and derivatives** must be expressly disclosed.

8.6.2 Risk-Adjusted Measures of Performance

Evaluating a manager on returns alone is inadequate without considering the risk taken. Three generally recognised methods of measuring risk-adjusted return — the **composite measures** of portfolio performance:

a. Sharpe ratio. Developed by William Sharpe, it relates the portfolio's **excess return** (return above the risk-free rate, also called the risk premium) to the portfolio's **total risk**, measured by standard deviation.

$$\text{Sharpe ratio} = \frac{R_p - R_f}{\sigma_p}$$

where

$$R_p$$

is the average portfolio return,

$$R_f$$

the risk-free rate, and

$$\sigma_p$$

the standard deviation of portfolio returns.

The question it answers: "How much return above the risk-free rate am I getting for the portfolio volatility I am bearing?" It is the excess return per unit of **total risk**. **The higher the Sharpe ratio, the better the performance.**

Worked example. Risk-free rate **6%**.

Portfolio	Annual return	Standard deviation	Sharpe
A	0.10	0.15	$(0.10 - 0.06)/0.15 = \mathbf{0.27}$
B	0.16	0.20	$(0.16 - 0.06)/0.20 = \mathbf{0.50}$
Market	0.13	0.18	$(0.13 - 0.06)/0.18 = \mathbf{0.39}$

Portfolio A has the lowest excess return per unit of total risk and underperforms both B and the market. Portfolio B performs better than A and the market.

b. Treynor ratio. Relates the same average excess return to **systematic risk**, measured by the portfolio's beta.

$$\text{Treynor ratio} = \frac{R_p - R_f}{\beta_p}$$

This is the excess return per unit of **systematic risk**. The Treynor measure implies systematic risk is the proper measure for evaluating performance. The question it answers: "How much return above the risk-free rate am I getting for the market sensitivity the portfolio exposes me to, as captured by ?"

Worked example. Risk-free rate **6%**.

Portfolio	Annual return	Beta	Treynor
A	0.14	0.9	$(0.14 - 0.06)/0.9 = \mathbf{8.9\%}$
B	0.18	1.1	$(0.18 - 0.06)/1.1 = \mathbf{10.9\%}$
Market	0.20	1.0	$(0.20 - 0.06)/1.0 = \mathbf{14\%}$

c. Sharpe versus Treynor. The main difference is the **definition of risk**: Sharpe uses **total risk** (standard deviation), Treynor uses **systematic risk** (beta).

If portfolios are **perfectly diversified** — correlation between portfolio and market returns is **1.0** — the two measures give **identical rankings**, because total risk is only systematic risk once all unsystematic risk is diversified away. As portfolios become **less diversified**, the possibility of differences in rankings increases.

The Sharpe measure therefore accounts for the extent of diversification during the measurement period. If a portfolio is **inadequately diversified**, its ranking using the **Treynor** ratio is likely to be **higher** than with the Sharpe measure, because Treynor ignores non-systematic risk, which is present in an inadequately diversified portfolio.

d. Jensen's alpha. Based on CAPM, measuring risk-adjusted return as alpha. The Jensen measure uses CAPM to determine the expected return on a portfolio given the market's performance, the risk-free rate and the portfolio's beta:

$$R_e = R_f + \beta_p(R_m - R_f)$$

The portfolio's alpha is the difference between the actual return and the CAPM-expected return:

$$\alpha_p = R_{\text{actual}} - R_{\text{predicted}} = R_{\text{actual}} - [R_f + \beta_p(R_m - R_f)]$$

A **positive alpha** reflects superior performance; a **negative alpha** inferior performance.

e. Information (appraisal) ratio. Looks at the amount of excess return relative to the excess risk. When comparing two portfolios with positive alpha, it compares alpha per unit of excess risk. A higher information ratio suggests the manager generates more outperformance per unit of excess risk taken.

$$\text{Information ratio} = \frac{\alpha_p}{\sigma_p}$$

The higher the information ratio, the better.

f. Sortino ratio. The Sharpe ratio applies to portfolios whose returns are expected to be **normally distributed**, and includes both positive and negative deviation in the standard deviation. But when managers protect downside risk with derivatives, short securities and the market, and enjoy the effects of leverage on positive volatility, they can create **positively skewed** portfolios. It would be unfair to penalise a manager for high positive deviation and a positively skewed distribution. The Sortino ratio therefore focuses on **downside risk**, with the denominator being an average of **downside volatility** for the period. This is particularly useful for comparing **hedge funds**.

$$\text{Sortino ratio} = \frac{R_p - R_f}{\sigma_{\text{negative}}}$$

The Sortino ratio calculates the return above the risk-free rate per unit of **negative volatility**.

8.6.3 Performance Attribution

Performance attribution determines **why** a manager did better or worse than a properly constructed benchmark. The purpose is to decompose total performance into specific components associated with specific decisions.

Attribution often begins from the broadest asset allocation choices and progressively focuses on finer details. The difference between the managed and benchmark portfolio performance can be expressed as the sum of contributions from decisions made at each level of portfolio construction. One common system decomposes relative performance into three components:

- **Asset allocation** choices across equity, fixed income and money markets.
- **Industry (sector) choice** within each market.
- **Security choice** within each sector.

Part of the analysis is identifying a performance benchmark — the **bogey** — designed to measure returns if a **passive** strategy were adopted. Weights assigned to the bogey depend on the investor's risk tolerance. Any difference between the portfolio's results and the passive benchmark must be due to **asset allocation bets** or **security selection bets**.

a. Characteristics of good benchmarks. Useful benchmarks should be:

- **Unambiguous** — the asset class, securities and composition clearly stated.
- **Investable** — practically and realistically investable as a passively managed portfolio.
- **Measurable** — returns obtainable from the market and performance measurable.
- **Consistent** with the investment manager's philosophy.
- **Bought in** by the manager as an acceptable benchmark.

Worked example — bogey portfolio.

Managed portfolio	Weight	Actual return	Weighted return
Equities	0.70	7.50%	5.25%
Bonds	0.10	2.00%	0.20%
Cash	0.20	0.50%	0.10%
			5.55%

Bogey portfolio	Benchmark weight	Benchmark return	Weighted return
Equities	0.60	6.00%	3.60%
Bonds	0.30	1.50%	0.45%
Cash	0.10	0.50%	0.05%
			4.10%

Excess return = 5.55% − 4.10% = 1.45%.

b. Asset allocation decision. Compare the weights of the managed and bogey portfolios, multiplied by the **benchmark** return:

Asset class	Actual weight	Benchmark weight	Excess weight	Benchmark return	Contribution
Equities	0.70	0.60	0.10	6.00%	**0.60%**
Bonds	0.10	0.30	−0.20	1.50%	**−0.30%**
Cash	0.20	0.10	0.10	0.50%	**0.05%**
					0.35%

Asset allocation contributed **0.35%** of the 1.45% excess return. The major factor was the higher equity weighting, which provided a 6% return.

c. Sector and security selection. The remaining **1.10%** is attributed to selection, using the **actual** weights and the **excess** returns:

Asset class	Actual return	Benchmark return	Excess return	Actual weight	Contribution
Equities	7.50%	6.00%	1.50%	0.70	**1.05%**
Bonds	2.00%	1.50%	0.50%	0.10	**0.05%**
					1.10%

Summary:

Component	Contribution
Asset allocation	**0.35%**
Sector and security selection — equities	1.05%
Sector and security selection — bonds	0.05%
Selection subtotal	**1.10%**
Excess return of portfolio	**1.45%**

8.7 Portfolio Management and Fund Sales

Depending on whether the representative works in a fund management team, a bank, an insurer or another segment, fund or portfolio sales and after-care may be integral to fund management or a separate function. Operationally there will be differences, but there must be proper communication between portfolio managers and the originators of products and services.

The sales force must ensure the right message is conveyed and that there is **no misrepresentation** about:

- What the portfolio is targeting to do — what it can and cannot do.
- The underlying assets and instruments used.
- The risks the client would be allowing the fund manager to assume.
- The level of service and after-care rendered by portfolio managers and service teams.
- Reliance on third-party vendors who receive payments, such as **retrocessions**, for promoting the fund.

Often institutions have strong expertise within the portfolio management team but not within the sales force. **Mis-selling** can result from poor communication and a lack of enforcement of required practices when marketing portfolio management services.

8.8 Summary

Efficient Market Hypothesis

- **Weak form** — all past price information is factored into the current price; past price and volume data have no relation to future direction; technical analysis cannot consistently produce excess returns.
- **Semi-strong form** — all available market and non-market public information is factored in; neither fundamental nor technical analysis reliably produces excess returns.
- **Strong form** — all market, non-market and inside information is factored in and no one has monopolistic access; no one can earn excess returns.

Passive versus active

- EMH subscribers favour **passive** management such as buy-and-hold, believing it produces results as good as active management over the long term.
- Non-subscribers favour **active** management seeking to outperform on a risk-adjusted basis, believing they have superior analytical skills or better information access. Examples: security selection, sector rotation, market timing.

Portfolio theory

- Portfolio risk is reduced as the number of securities increases.
- Two sources of risk: **systematic**, attributed to macroeconomic factors and not eliminable by diversification; **non-systematic** or firm-specific, controllable and minimisable through diversification.
- Correlation between assets changes over time, and during major crises correlations between financial asset classes **rise**.
- For successful diversification, seek securities that are **uncorrelated**; otherwise risk cannot be significantly reduced. Portfolio risk is lowest when returns are **negatively correlated**.

- Portfolio expected return is a weighted average of constituent returns, but portfolio risk is **not** a weighted average of constituent risks.
- The **efficient frontier** represents optimal portfolios; adding less-correlated assets can shift it toward the upper left.
- **CAPM** relates expected return to systematic risk through beta. The **CML** applies to efficient portfolios and uses total risk; the **SML** applies to all assets and uses beta.
- **APT** is a multifactor model that does not require a single-period horizon, absence of taxes, risk-free borrowing and lending, or mean-variance portfolio selection.
- The **IPS** belongs to the client and records objectives, constraints and asset allocation.
- **SAA** is the long-run policy mix; **TAA** is short-run overweighting and underweighting within SAA ranges.
- **DWR** is the IRR and is influenced by cash flow size and timing; **TWR** is preferred for evaluating managers.
- **Modified Dietz** approximates TWR without needing valuations at each cash flow.
- **GIPS** requires total return, TWR, at least monthly valuation, geometric linking, and disclosure of fees, leverage and derivatives.
- **Sharpe** uses total risk; **Treynor** uses systematic risk; **Jensen's alpha** is the excess over the CAPM-predicted return.
- **Performance attribution** decomposes excess return into asset allocation and selection components against a bogey.

Exam cues

- **Three EMH forms:** weak (past prices), semi-strong (all public information), strong (all information including private). Evidence supports weak and semi-strong; strong form is **not** supported, which is why insider trading is regulated.
- **Weak form implies technical analysis fails; semi-strong implies fundamental analysis also fails; strong implies even insider information fails.**
- **The average fund manager underperforms the benchmark even before fees**, and outperformance does not persist.
- **Passive** suits EMH believers: buy-and-hold or index funds, judged on **tracking** the index, with minimal expenses.
- **Active** strategies: **security selection** (undervalued, low P/E), **sector rotation** (10 industries; interest-sensitive, cyclical, defensive), **market timing** (active asset allocation).
- **Efficient frontier:** highest return for a given risk. Portfolios **below** the frontier are inefficient. Adding less-correlated assets shifts the frontier **up and to the left**.
- **Diversification:** about **half** of portfolio standard deviation is eliminated with **10 securities**; benefits then diminish. A market index eliminates all non-systematic risk.
- **Total risk = systematic + non-systematic.** Systematic cannot be diversified but can be **hedged** with options and futures. Non-systematic can be reduced to **zero** by diversification.
- **Correlation:** $= +1$ gives **no** risk reduction (portfolio risk is the weighted average); $= 0$ gives **some**; $= -1$ can **eliminate** risk. Correlation is **not stable** and **rises toward 1 in a crisis**.
- **Portfolio return** is a weighted average; **portfolio risk is not**. Two-asset formula:

$$\sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2}$$

  .
- **Portfolio worked example:** A (18%, 20%, 40%) and B (9%, 10%, 60%), $= 0.8 \rightarrow$ return **12.6%**, risk **13.30%**. At $= 1$ risk is **14%**; at $= 0$ it is **10%**; at $= -1$ it is **2%**.
- **International diversification** reduces risk because country returns are fairly uncorrelated, but introduces **FX risk**, which can be hedged with forwards. It does **not always** raise risk-adjusted return.
- **CAPM assumptions (ten):** risk-return evaluation, rationality, infinite divisibility, risk-free borrowing and lending, no taxes or transaction costs, no inflation or fully anticipated, single-period horizon, free and instant information, homogeneous expectations, equilibrium.
- **Market portfolio M** contains all assets in proportion to market value and has only systematic risk. In practice the **S&P 500** is the proxy.
- **CML:**

$$E(R_p) = R_f + [(E(R_m) - R_f)/\sigma_m]\sigma_p$$

  . Applies to **efficient portfolios** only; uses **total risk**. Slope is the **market price of risk**.

- **SML:**
$$E(R_i) = R_f + \beta_i(R_m - R_f)$$
. Applies to **all assets**; uses **beta**. Market beta = **1**.
- **CAPM worked example:** R_f = 1.20, R_m = 2.5%, = 14% → **16.3%**. Actual 17% → alpha **+0.7%**; actual 15% → alpha **−1.3%**.
- **Above the SML = under-priced (undervalued); below the SML = overpriced (overvalued).** In equilibrium all securities lie on the SML.
- **Steep SML = cheap market** (large return for small risk increase); **flat SML = overvalued market** (risk being ignored).
- **Alpha should be assumed zero** unless analysis shows otherwise, because arbitrage eliminates it.
- **APT:** multifactor, by **Stephen Ross**, early 1970s. Does **not** assume a single-period horizon, no taxes, risk-free borrowing and lending, or mean-variance selection. Factors must be **pervasive**, **influence expected return**, and be **unpredictable**. **Unexpected** inflation is a factor; the inflation **rate** is not.
- **Chen, Roll and Ross (1986)** four factors: unanticipated inflation, industrial production, default-risk premium, term structure.
- **APT's weakness:** the factors are **not well specified**; empirical work suggests **three to five** factors.
- **IPS belongs to the client**, not the manager. It records objectives (return requirement, risk tolerance) and constraints (liquidity, time horizon, tax, regulatory and legal, unique needs).
- **Life cycle:** early career — high risk growth; mid-career — still growth but reduce risk; late career — lower risk, income and capital preservation.
- **Liquidity needs** are met with **T-bills and commercial paper**. **CPF** money is governed by the CPF Investment Scheme.
- **Asset allocation explains 91.5% or more** of portfolio return (Brinson, Hood and Beebower); security selection less than **10%**.
- **SAA** is the long-run policy mix, a **constant-mix** strategy, e.g. equities 45–55%, bonds 35–45%, cash 5–15%. **TAA** overweights and underweights within those ranges.
- **TAA approaches:** valuation (risk premium or **equity-bond yield gap**) and cyclical (equity markets lead economic activity).
- **Return with no cash flow:**
$$(V_E - V_B)/V_B$$
. Worked: 30 → 35 = **16.7%**.
- **Deposit at end of period:** deduct from the ending value. Worked: 50 → 53 with a 5 deposit = **−4.0%**.
- **Deposit at start of period:** adjust the beginning value. Worked: **−3.6%**.
- **DWR = IRR.** Worked: mid-quarter deposit → semi-quarterly **−1.92%**, quarterly **−3.8%**.
- **TWR** links sub-period returns. Worked: −2.0% and −1.85% → **−3.81%**.
- **CAGR:** 100 → 250 in 5 years = **20.1%**.
- **DWR is inappropriate for judging managers** because it is driven by client cash flows; **TWR is preferred**.
- **Modified Dietz:**
$$R_p = (V_E - V_B - CF)/(V_B + WCF)$$
, with
$$WCF = \sum CF_i(n - t_i)/n$$
. Worked: WCF **140,000**, return **6.90%**.
- **GIPS:** approved **4 February 2005**; requires **total return**, **TWR**, **monthly** valuation, geometric linking, after-trading-expense returns, accrued income, **10-year** record for firms over 10 years old, and disclosure of fees, leverage and derivatives.
- **Sharpe:**
$$(R_p - R_f)/\sigma_p$$
. Worked: A **0.27**, B **0.50**, market **0.39**.
- **Treynor:**
$$(R_p - R_f)/\beta_p$$
. Worked: A **8.9%**, B **10.9%**, market **14%**.

- **Sharpe vs Treynor:** identical rankings only when correlation with the market is **1.0**. An **inadequately diversified** portfolio ranks **higher on Treynor** than on Sharpe.
- **Jensen's alpha:**

$$R_{\text{actual}} - [R_f + \beta_p(R_m - R_f)]$$

.
- **Information ratio:**

$$\alpha_p/\sigma_p$$

. **Sortino:**

$$(R_p - R_f)/\sigma_{\text{negative}}$$

— useful for hedge funds and positively skewed returns.
- **Attribution worked example:** managed **5.55%** vs bogey **4.10%** → excess **1.45%**; asset allocation **0.35%**, selection **1.10%** (equities 1.05%, bonds 0.05%).
- **Good benchmarks are unambiguous, investable, measurable, consistent with the manager's philosophy, and accepted by the manager.**
- **Trap:** portfolio **risk** is not the weighted average of constituent risks — only portfolio **return** is.
- **Trap:** the **CML** uses total risk and applies only to efficient portfolios; the **SML** uses beta and applies to all assets. Do not mix them.
- **Trap:** a **beta of zero** does not make an asset risk-free — it can still carry unsystematic risk.
- **Trap: DWR** and **TWR** answer different questions. Use DWR for the client's actual experience, TWR for the manager's skill.

Chapter 9 — Exchange Traded Funds (ETFs), Unit Trusts, Real Estate Investment Trusts (REITs) and Insurance

This chapter covers the pooled vehicles an adviser recommends: ETFs, unit trusts, REITs, business trusts, stapled securities, UCITS, the alternative investments (hedge funds, private equity, commodity and real estate funds), and insurance and annuities. The examination tests the mechanics, the fee structures and the suitability judgements.

Learning Objectives

The candidate should be able to:

- Describe exchange-traded funds (ETFs).
- Understand the role of ETFs in investments.
- Delineate the advantages and risks of investing in ETFs.
- Define a unit trust and describe its features.
- Distinguish between unit trusts and investment trusts.
- Define the advantages of investing in unit trusts.
- Describe the major types of unit trusts.
- Discuss the evaluation of unit trusts.
- Evaluate the charges and fees in unit trusts.
- Explain how the bid and offer prices of unit trusts are determined.
- Explain what a Real Estate Investment Trust (REIT) is.
- Compare REITs and business trusts and explain their differences.
- Explain what stapled securities are.
- Understand and describe alternative investment instruments: hedge funds, insurance and annuities.
- Describe the CPF Investment Scheme.

9.1 Introduction

In Singapore, most pooled investments are **Collective Investment Schemes (CIS)** or **unit trusts**, governed by the **Code on Collective Investment Schemes (Code on CIS)** issued by MAS.

Three main structures currently allow a wide group of investors to participate in asset managers' capabilities:

- **Unit trusts.**
- **Investment holding or dealing companies.**
- **Limited partnerships.**

These structures are sometimes less than ideal for investment funds. A company can only pay dividends out of **profits**, and there are restrictions on return of capital. The **Variable Capital Companies Act** was therefore passed in **October 2018** to govern a new structure, the **Variable Capital Company (VCC)** — essentially a company with two key features:

- A VCC can **freely issue and redeem shares and pay dividends using its capital**.
- A VCC can adopt an **umbrella structure with multiple sub-funds**, increasing operating efficiency.

This encourages funds with similar structures in other jurisdictions to **re-domicile** in Singapore and can help Singapore enlarge its share of the global asset management market.

The chapter proceeds from conventional CIS — ETFs and unit trusts, which invest primarily in listed equity and fixed income securities — to alternative CIS focused on niche asset classes and strategies (REITs, business trusts, hedge funds, private equity), and finally to insurance products, which pool not only investment funds but **mortality and longevity risks** as well.

9.2 Exchange-Traded Funds

ETFs are funds designed to **track the performance of specific indices, or fixed baskets of stocks or debt securities**. They differ from most open-end and closed-end funds because they involve **no active management** of securities. Underlying assets may be local or international, and beyond stocks an ETF may be based on fixed income, commodities, money market instruments or currency. Through ETFs, investors gain access to a wide range of markets and asset classes.

Over the last two decades ETFs have proliferated on major bourses. Singapore has numerous SGX-listed ETFs tracking major indices and selected asset classes, mostly long ETFs. Popular examples include the **S&P 500 SPDRs** ("Spiders") tracking the S&P 500 Index, and the **DJIA DIAMONDS** tracking the Dow Jones Industrial Average.

Worked illustration — SPDR Straits Times Index ETF. Singapore's first locally created ETF, designed to track the STI. Originally listed as "Street TRACKS STI", it has traded on SGX-ST since **April 2002**. Its objective is to replicate the STI's performance as closely as possible, before expenses.

Feature	Detail
Trading name	STI ETF 100
Pricing	About **1/100th** of the STI level. If the STI is at 1,700, each unit is priced at **$17.00**
Minimum bid size	$0.01
Board lot size	**100 units**
Dividend yield	Expected to approximate the STI yield; dividends paid **twice a year**
Annual costs	**0.3%** (management fee, trustee fee and other fund expenses)

The SPDR STI ETF is like a mutual fund tracking the STI, investing in component stocks in proportion to their index weights. ETF units can be bought and sold **during trading hours** like any listed security — unlike mutual funds, which can only be bought or sold at the **end of the day** with the fund manager.

ETFs are **passive** instruments seeking to replicate the price movement of a benchmark index. If replication is good, the ETF's price movement follows the benchmark.

NAV versus market price. The ETF's **Net Asset Value (NAV)** reflects the value of the units owned and is determined at the **daily close of trading** in all markets where the ETF is listed, using a process explained in the Trust Deed or Prospectus. NAV per unit represents **fair value**. But the ETF's **market price can deviate from NAV per unit**, because price is also determined by investor sentiment affecting supply and demand.

ETFs are designed to **minimise commissions and expense ratios** compared with an average mutual fund. Annual fees are lower because ETFs are passively managed. The main costs are the **total expense ratio** and **brokerage commissions** on buying or selling. Instead of buying on the exchange, investors can buy from **primary or participating dealers**, who purchase ETF shares from the sponsor and offer them to the market.

Creation and redemption. Unlike a typical unit trust that creates units for investors when they purchase, units of the STI ETF can only be created for, or redeemed by, **participating dealers** in sizes of **100,000 units**. To create an ETF share, the participating dealer deposits a specified portfolio of stocks closely

approximating the index's underlying securities with the trustee. To redeem, ETF shares are delivered to the trustee in exchange for the underlying basket.

Arbitrage. ETFs are quoted at bid and ask prices that track but do not necessarily reflect NAV. If the listed price deviates sufficiently from NAV, **arbitrage** reduces the differential. When the traded price is **above** NAV, an arbitrageur sells ETF shares and buys the underlying portfolio. When the traded price is **below** NAV, the arbitrageur sells the underlying portfolio and buys ETF shares.

9.2.1 Types of ETF Structures

ETFs are either **physically** or **synthetically** structured to replicate the basket of stocks or bonds making up the index. Physically replicated ETFs hold the actual securities in index proportions. Synthetically replicated ETFs hold statistically correlated securities that together mimic the index's price behaviour, and may use swaps or derivatives.

a. Cash or direct replication (physical ETFs). More common. Two methods:

- **Full replication** — the ETF purchases **all** listed stocks in the exact index proportions.
- **Representative sampling** — the ETF invests in a **subset** of the index comprising the more dominant stocks. Used when index constituents are not liquid.

b. Synthetic replication (synthetic ETFs). Uses swaps and derivatives to mimic the index. **Advantage:** reduces the ETF's **tracking error**. **Disadvantage: counterparty risk**, because the derivative counterparty might default.

An example is an ETF using a **total return swap** — a contractual arrangement to swap interest or return obligations. Swap-based ETFs usually **do not pay dividends**; instead the counterparty delivers periodic payments of income and capital gains return on the reference basket, and the ETF pays a fee to the counterparty contractually obligated to deliver the index returns. There is therefore a risk to investors if the counterparty cannot fulfil its payments.

9.2.2 Investment Strategies with ETFs

- **Long-term buy-and-hold.**
- **Short-term tactical trading**, because of price transparency and liquidity.
- **Core-satellite allocations** — a core mimicking a broad index, with satellites of higher-risk securities offering potentially higher return.
- **Hedging** against other risks in the portfolio.
- **Cash equitisation** — converting cash to equities through ETFs to minimise the **cash drag** on a portfolio.

9.2.3 Securities Borrowing and Lending for ETFs

Securities lending is where ETFs pay agents to lend out shares in their portfolios to other investors, earning interest. For example, an ETF can transfer stocks to another investor over the lending period (say **30 days**) in exchange for collateral such as cash or other securities equivalent to, for example, **105%** of the stocks' value. If the ETF needs to sell the lent stocks, it does a reverse transaction with the borrower.

Borrowers are usually investors who have **sold the stock short** in anticipation of a fall and need to borrow for delivery, intending to buy back lower and return the stock.

If the borrower cannot fulfil its obligations, the ETF faces a loss equal to the value of the lent stocks. **Securities borrowing and lending therefore creates counterparty risk for the ETF.**

9.2.4 Advantages of Investing in ETFs

- **Diversification** — buying ETF shares gives exposure to a diversified portfolio.
- **Small capital outlay** — diversification with a small amount.
- **Low costs** — annual management fees normally **less than 1%**, against **1–2%** for actively managed unit trusts, with **no initial charges or redemption fees**.
- **Liquidity** — ETF shares trade at any time during exchange hours at prevailing prices; unit trusts transact only at the **daily closing NAV**.
- **Transparency** — investors always know the basket of securities in an ETF, whereas unit trusts usually disclose holdings only **quarterly, semi-annually or annually**.

9.2.5 Risks of Investing in Standard ETFs

Physical and synthetic ETFs differ in their risks:

Risk	Detail
Tracking error	**Greater in physical ETFs** than synthetic, principally because synthetic ETFs achieve mor
Counterparty risk	Synthetic ETFs carry counterparty risk from swap and derivative arrangements. Physical ET
Market and sector risk	When underlying securities fall, the ETF's value suffers. Sector funds may be hurt by the spe
Foreign exchange risk	ETFs with constituents traded in foreign currencies are subject to FX risk affecting overall re
Taxation	Non-US investors holding US-established ETFs may be subject to **US withholding taxes** o

9.2.6 Smart Beta ETFs

A **smart beta ETF** seeks better risk-return performance than an index of selected stocks. Selection and weighting are based on criteria relating to better-performing stocks — improvements in cash flow, dividends, revenue/earnings ratios.

In contrast to a standard ETF following market-capitalisation weighting, a smart beta ETF **selects and weights securities on positive performance criteria**. To some extent smart beta simply entails good fundamental stock selection, weighting top performers more heavily using quantitative criteria. Other smart beta ETFs exploit **market anomalies** or overlay technical measures such as **momentum and volatility**.

9.2.7 Volatility (VIX) Driven ETFs

A volatility index (VIX) or VIX futures ETF is meant for **very short-term trading**, not investing.

In a bullish market volatility generally falls. Traders may try to buy long volatility ETFs on days when volatility spikes, but this is difficult to predict and time consistently. Holding a long volatility position waiting for an upward spike can be **deadly** as volatility trends downward: a **1–2% fall each day** soon produces massive losses over time. High volatility generally does not last, and the beneficiaries tend to be those who **shorted volatility** — invested in the inverse of the VIX.

Code	Popular volatility ETF or ETN	Total assets ($m)	YTD return	Average volume	1 July 202
VXX	iPath S&P 500 VIX Short-Term Futures ETN	1,009.7	**−56.85%**	17,964,400	$29.15
SVXY	ProShares Short VIX Short-Term Futures	585.3	**+35.26%**	2,794,200	$55.88
VXZ	iPath S&P 500 VIX Mid-Term Futures ETN	36.3	−17.29%	12,000	$25.36
VIXM	ProShares VIX Mid-Term Futures ETF	92.5	−17.11%	45,900	$30.39
XVZ	iPath S&P 500 Dynamic VIX ETN	12.9	+1.78%	4,800	$37.90

Under bear market conditions volatility is more significant and frequent, benefiting long volatility positions. But conditions change dramatically and rapidly, so the trader must be **fleet-footed and disciplined** to realise target gains and limit losses.

9.2.8 High Risk Leveraged ETFs and Long-Short ETF Pairs

A **leveraged ETF** uses financing and derivatives such as options and futures to **drastically amplify** potential returns from tracking an index. Such ETFs are usually leveraged **2x or 3x** the index value, and some track the **inverse** movement. Leveraged ETFs are often issued in **pairs** — long and short a particular index.

Large speculators with enough financial muscle sometimes try to drive the index one way or the other by moving key index securities, to make **2x–3x** leveraged profits on their ETF position. The practice is not exactly illegal unless there is proof of collusion to manipulate the index. Such manipulation tends to be greater for **smaller stock indices** such as the NASDAQ Biotech Index or the NYSE Arca Gold Miners Index. **In general, leveraged ETFs are not suitable for retail investors.**

Leveraged ETFs amplify the **daily** percentage price change. A **2–4% daily change** is normal, and a **±20–25%** change is possible in a single session, so there is potential for massive gains and losses.

Singapore has several inverse ETFs, such as the **Deutsche Bank x-trackers S&P 500 Inverse Daily UCITS ETF**, which traders use at specific points in the US market cycle. Its value was **USD15.99 in**

November 2017, less than a quarter of its **USD79.72** at first trading. Betting against a bullish market is difficult and traders need to be nimble.

Long position	YTD % 1 Jul 21	Short position	YTD % 1 Jul 21	Daily performance of
LABU (3x)	−21.89	LABD (−3x)	−28.53	S&P Biotechnology Select Industry Index
NUGT (2x)	−16.43	DUST (−2x)	−3.62	NYSE Arca Gold Miners Index
YINN (3x)	−9.22	YANG (−3x)	−16.52	FTSE China 50 Index
TQQQ (3x)	+34.83	SQQQ (−3x)	−39.79	NASDAQ 100 Index

Inverse or short ETFs can be issued independently but are often introduced as part of a pair to long positions, letting traders profit or hedge when an index falls. They can offset loss-making long positions during a trading day or over a short period.

Because of sharp movements, issuers of leveraged ETFs often carry out **reverse splits** to reset prices at higher levels, so regular price resetting is common.

Some ETFs are **not meant for the average retail investor** or even institutional portfolio investment. They are intended for **speculative traders** willing to take substantial risk and volatility, with the time and emotional demeanour to manage the positions.

9.3 Unit Trusts

A **unit trust** is a CIS that pools funds from different investors to be invested and managed by one or more fund managers under the care of an **independent trustee**. In Singapore the Code on CIS sets out best practice on management, operation and marketing that trustees and managers are expected to observe.

Jurisdiction	Term
United States	**Mutual fund** — may not be issued in the form of a trust
United Kingdom	**Unit trust**
Germany	**Investment fund**
Switzerland and Luxembourg	**Sicaf** (Société d'investissement à capital fixe) for a closed-end investment company; **S**

The naming convention matters because the bulk of offshore investment funds are domiciled in **Luxembourg**.

In Singapore the unit trust is the most common form of CIS, pooling small investors' resources to invest in securities under professional management. The cash and investments are registered in the name of, and held by, the **trustee**, whose job is to safeguard investors and ensure managers operate within the **trust deed**. That document, an agreement between trustee and managers, sets out the aims and objectives, the charges for management and the basic rules of operation. For some offshore funds, redemption and purchase are handled directly by the asset manager's servicing team.

9.3.1 Fund Domicile

The **domicile** is where the fund is based and set up. A fund can be registered in multiple jurisdictions but can only have **one place of domicile**.

There is heavy competition between Singapore and other jurisdictions to be the fund domicile. Singapore benefits from being a fund management hub, with a business-friendly and well-respected regulatory framework, strong communication and transport infrastructure, and existing tax treaties. By launching new structures such as the **VCC**, Singapore hopes to encourage more alternative investment funds to be domiciled and managed there.

9.3.2 Local and Offshore Unit Trusts

Singapore allows marketing of offshore funds regardless of whether they are managed by Singapore managers, so Singapore-domiciled funds compete for investor attention with offshore funds actively marketed by foreign banks.

Under Singapore law there are three ways to offer participation in offshore funds:

- A **public offering on the local exchange** — for example foreign-domiciled ETFs with Singapore agents.
- A **private placement**.
- Invoking an exception under **Division 5A of Part IV of the Companies Act (Cap 50)**, which allows offshore funds to be marketed to **Institutional Investors, Expert Investors and Accredited Investors** in Singapore.

The presence of both offshore and locally domiciled funds benefits Singapore as a global wealth and asset management hub.

9.3.3 Open-End and Closed-End Unit Trusts

Feature	Open-end	Closed-end
Market	Over-the-counter	Exchange traded
Counterparty	Issuer	Listed exchange or issuer
Exit	Sale or redemption through trustee	Sale in the market
Issued shares/units	**Continuous issuance**	**Fixed number of units**
Price determined by	**NAV** (fund performance net of management fees)	**Market demand and supply** as well as

In Singapore most unit trusts are **open-end**: not listed on the exchange, offered to retail investors. The open-end feature describes the availability of **fresh units** the manager can sell to retail investors. **Closed-end** funds do not issue fresh units — the number is fixed — and are usually sold to accredited investors via private placements.

A unit trust divides a fund into equal portions called **units**, each representing the same proportion of the value of the trust's assets. If a trust's investments are worth $10 million * *and there are * *5 million * *units issued, each unit is worth * *2$. Most unit trusts' investments are valued **daily**, and each day the previous day's valuation prices are published in newspapers and serve as the opening dealing prices. The value of a unit trust is its **NAV** — the value of assets minus liabilities.

9.3.4 Role of Trustee

The managers **neither hold the investments nor issue or redeem units**. That is done by the **trustee**, who holds the trust's cash, shares and other investment assets. In an open-end trust the trustee creates new units on purchases and redeems units on liquidation, so the capitalisation is "open" with units outstanding changing **daily**.

A **prospectus** is typically issued with details of the investment objectives, focus and approach, performance and fee charges. Unit trusts are more suitable as **medium to long-term** investments. Foreign funds not incorporated in Singapore may be offered to the public if they are **recognised funds** for retail distribution.

9.3.5 Advantages of Investing in Unit Trusts

- **Diversification and divisibility** — pooling lets holders own fractional shares of many securities, shielding them from large losses.
- **Small capital outlay** — an initial investment of **$1,000 to $5,000** is affordable, giving access to a portfolio otherwise out of reach.
- **Professional management** — full-time analysts and portfolio managers who respond quickly to market changes.
- **Flexibility** — holders can **switch** between a "family of funds" within the same family, responding to changes in plans and markets. If the plan changes from low to moderate risk, the holder could switch from an income fund to an equity fund.
- **Liquidity** — holders can sell to the trustee, who must buy back at NAV less relevant charges. Selling shares in 30 companies individually would be more time-consuming.
- **Lower transaction costs** — trading large blocks achieves savings on brokerage and commissions.
- **Security** — assets are always legally held on investors' behalf by an **independent trustee**, not the fund manager.

9.3.6 Disadvantages of Investing in Unit Trusts

- **Sales charges** — relatively high charges on buying in, and high transaction costs with frequent buying and selling.
- **Management fees** — annual fees paid to the manager **regardless of performance**. Compare fees across funds, because high fees affect returns.

9.3.7 Major Types of Unit Trust Funds

Unit trusts are classified by the assets they invest in.

Traditional funds — core asset classes	Diversified asset allocation strategies	Alternative investments
Money market	By risk profile (yield, income, balanced, growth equity)	Hedge funds
Equity	By geography (global, country)	Commodity funds
Fixed income	By industry or sector	Real estate funds and F
	By theme	Private equity
		Quantitative and algori

Fund characteristics to distinguish: OTC vs exchange-traded (open vs closed-end); passive vs actively managed; with or without derivatives; leveraged; inversed; load vs no-load; fund of funds (with or without retrocession).

Examples in Singapore:

- **Money market funds** hold short-term (typically under one year) fixed income instruments — bank certificates of deposit, commercial paper, Treasury bills. Because of the large denominations in direct dealing, these funds give access to competitive money market rates. Interest is earned and credited **daily**. Investors normally pay **no sales or redemption charge** but do pay a management fee.
- **Fixed income funds** invest in fixed income securities, sometimes restricted to particular types: corporate bond funds, US government bond funds, convertible bond funds, high-yield bond funds.
- **Equity funds** invest in equity securities; some specialise in particular sectors or regions.
- **Balanced funds** hold a mix of equities and bonds.
- **Asset allocation funds** shift the mix dynamically.
- **Index funds** replicate a benchmark.
- **Capital guaranteed funds** invest most capital in low-risk assets to preserve capital at maturity, with the balance in riskier assets such as options linked to equities, funds or indices, or direct equities, for upside. Maturity is determined by the expiration date of the guaranteed portion, and the guarantee usually applies **only if held to maturity**.
- **Capital preservation funds** differ from capital-guaranteed funds: they aim to provide some level of capital preservation, the language is **less onerous** for the issuer, and **no guarantor is required**.
- **Hedge funds** employ derivatives, short selling and arbitrage, focusing on **absolute returns** rather than relative performance against an index. In a **market-neutral** fund each long position is offset by an equal-value short, reducing correlation with the market. The name is a **misnomer** because hedge funds are not necessarily hedged — a manager may take directional bets and be net long or short. Managers often invest their own money and are paid a **performance fee**.
- **Quant funds** use price and market modelling based on complex mathematical algorithms. The best-known example is **Long-Term Capital Management**, whose board included **Myron S. Scholes and Robert C. Merton**, who shared the 1997 Nobel Memorial Prize in Economic Sciences for a "new method to determine the value of derivatives".
- **High frequency trading (HFT) funds** rely on algorithms and high-speed data transmission. Not all HFT funds are hedge funds, but many hedge funds use quantitative models and algorithms, some reporting significant profits from HFT.
- **A family of funds or umbrella fund** is a set of funds with different objectives offered under a single unit trust. Umbrella funds generally permit switching within the family at **little to no cost**, letting the investor change strategy inexpensively.

Investors in Singapore can use **CPF** funds to invest in products under the **CPF Investment Scheme (CPFIS)**. To protect CPF members, managers of CPF monies must report vital information **quarterly** to the CPF Board, and a **funds-tracking company** is appointed to publish information on performance and risks of CPFIS-included funds.

9.3.8 Fund Documentation — Prospectus, Product Highlights Sheets, Monthly Fund Fact Sheets, Annual Reports

With access to asset managers' websites, information on a fund is easily obtained from its **prospectus** and **product highlights sheet**. For new launches these provide details about the asset managers, the investment approach, the types of investments selected, the risk exposures, and the different types of fees and how they are calculated.

Monthly fund fact sheets and **annual reports** provide regular updates on performance and summary information. Summary information on the monthly fact sheet includes:

- Fund objective.
- Key information on fees.
- Performance data and ratios.
- Benchmark comparison.
- Top holdings.
- Country allocation.
- Risk statistics.
- Sector allocation.

9.3.9 Fund Objectives

The statement of fund objectives indicates the **target outcomes** and should also convey the **investment approach** taken to achieve them — for example a diversified portfolio across asset classes, a diversified portfolio of global equities for long-term capital growth, diversification within a single sector, or investment in a non-diversified portfolio.

9.3.10 Fees and Expenses

a. Load versus no-load funds. A **no-load fund** imposes neither an initial sales charge nor redemption charges: units are priced at NAV and the bid and offer prices are the same. A **load fund** with a **front-end load** is sold at NAV plus a sales commission. Most load funds are sold by brokers, financial planners and insurance company employees; the load often serves as commission for these distributors and is usually **1.5% to 5% of NAV**.

When holders sell, they usually receive NAV multiplied by units sold. Some funds impose a **realisation charge or redemption fee** — a **back-end load** — which may be waived if units have been held beyond a stipulated period (for example **60 days**). Its purpose is to **discourage selling right after buying**, avoiding the transaction costs of frequent securities sales to meet redemptions.

b. Management fee. All unit trusts charge annual management fees, typically **0.5% to 2.0% of NAV**, accrued **daily** and debited directly from the trust's asset value.

c. Various operational costs. Under an umbrella structure a **switching fee of about 1%** may apply when switching between sub-funds, so frequent switching is expensive. Other charges include annual **trustee, custodian and audit fees**, accrued daily, monthly or as incurred and deducted from assets — usually **0.1% to 0.15% of NAV**.

There are also set-up costs including legal and marketing, charged once or amortised over years. Beyond the front-end load, investment companies may charge an annual **distribution fee** paying for advertising, promotion and selling, charged directly against NAV. Some costs appear ”waived” but end up as expenses charged on the portfolio.

d. Performance fee. Some funds pay an additional incentive charge for substantially exceeding benchmark targets. Measures ensure the manager does not unfairly benefit from a fund recovering after a sharp fall — a **high watermark** is set so the manager must generate a certain level of profit before the bonus can be taken from NAV.

The **total expense ratio** measures total expenses (including management fees) against NAV. It is often reviewed when comparing funds to assess a manager's **operational efficiency**.

9.3.11 Performances and Benchmarks

Past performance must be considered when selecting a unit trust. While ”past performance is no guarantee of future performance”, it remains a useful guide. A unit trust is usually managed **relative to a benchmark**,

so the investor can compare the fund's return against it: if the fund's return is higher than the benchmark, the manager has done well.

Performance is typically charted against the benchmark **since inception**, with a table showing performance over the last **3 or 5 years**. Check whether returns are shown **net of management fees** — most fact sheets indicate net returns.

Where a fund has **dual currency tranches**, returns may also be reflected in the alternate currency: a USD return may differ substantially when reflected in SGD or EUR because of exchange rate movements.

For traditional funds with a **relative return** target, the benchmark is usually an index; a few use a weighted combination of indices from two or more asset classes. For funds with **absolute return** targets, an interbank rate such as **LIBOR + 1%** might be the benchmark to beat.

9.3.12 Net Asset Value

For funds sold OTC, the fund's value is the **NAV**:

$$\text{NAV} = \frac{\text{Fund securities} + \text{Other fund assets} - \text{Liabilities (including management fees)} - \text{Debt used for leveraging}}{\text{Total fund units outstanding}}$$

For open-end funds the outstanding units change with issuances and redemptions. NAV rises and falls with the value of investments and should be reflected **net of management fees**.

9.3.13 Bid-Offer Price

There is a difference between NAV and the price at which an investor can buy and sell. The principle is that a holder buying new units should pay a sum sufficient to purchase a proportionate amount of the underlying securities, **including the expenses necessary for that purchase**. Existing holders should not unfairly bear costs from a transaction they did not create.

To achieve equity between incoming and outgoing holders, valuations show two prices.

Offer price — the price at which an investor can **buy** units, representing the cost of creating new units equal in value to existing units based on market prices at the latest valuation:

- Determine NAV per unit by assessing the market value of the trust's securities.
- **Add** expenses related to purchasing securities, such as stockbrokers' commission and clearing fees.
- **Add** cash held by the trust and accrued income.
- **Add** the manager's initial charge.

Bid price — the price at which an investor can **sell** units, equivalent to NAV per unit if all securities were sold and proceeds distributed in cash:

- Determine NAV per unit by assessing the market value of the trust's securities.
- **Deduct** expenses related to the sale of securities.
- **Add** cash held by the trust and accrued income.

9.3.14 Top Holdings and Concentration Risk

Many funds do not allow a single security to exceed **5%** of total assets, but some managers allow exposure to rise to **8–9%** in an attempt to outperform. Top-holding choices significantly affect returns, concentration risk and volatility. Under- or over-weighting securities that dominate benchmark indices — Tencent, Alibaba, Apple in technology — can materially affect performance.

9.3.15 Allocation by Asset Class, Market and Sectors

The fund fact sheet shows how the fund is invested by asset class (for multi-asset funds), by market and by sector. Reviewing allocation shows how much exposure is favoured in one country or sector over others, which some investors prefer.

9.3.16 Risk-Adjusted Return and Other Risk Ratios

Beyond benchmarked returns, review risk-adjusted returns and risk using:

- **Sharpe ratio.**
- **Treynor ratio.**
- **Beta.**
- **Jensen's alpha.**
- **Information ratio.**
- **Sortino ratio** — for alternative investment funds with asymmetrical return distributions.

Generally, investors should pick funds generating **higher returns above the risk-free rate per unit of risk taken**.

9.3.17 Evaluating Suitability of Unit Trusts and Foreign Funds Sold in Singapore

All the client-understanding aspects from Chapter 8 apply. Building a client profile and using the fund fact sheet lets representatives select funds meeting the client's needs.

Client profiling consideration	Fund suitability
Risk tolerance Return expectations Specific objectives	The selected fund should match the client's risk appetite and return expectation, alo
Investment time horizon Reference functional currency Key preferences and constraints Investment style and specific needs	Note that when the fund is only a small part of the overall portfolio, it may have a c

a. Investment objectives and time horizon. Knowing the investor's objectives and horizon is essential. Generally unit trusts suit investors with **medium to long-term** horizons, the exception being certain **leveraged ETFs and highly volatile commodity ETFs**, which are meant more for traders.

The risk continuum runs from **money market funds** (least risky) to **specialised funds** (riskiest).

To meet best practice, some fund houses provide statements highlighting the type of client the product suits.

Product suitability statements — who is the product suitable for?

Fund A: Conservative Fund. Only suitable for investors who:

- Seek long-term capital growth.
- Seek regular income distribution in respect of the Distribution Class.
- Are looking for exposure to companies listed on SGX-ST and are comfortable with the volatility of a **single-country equity fund**.

Fund B: Aggressive Fund. Only suitable for investors who:

- Seek capital appreciation with **no need for income**.
- Seek access to dynamic emerging market countries.
- Are comfortable that the portfolio may generate potentially higher returns but comes with a **higher risk profile** than a diversified Global Equity Fund.

b. Diversification. A common motivation is a diversified portfolio through the fund's securities. Depending on objectives, some funds are narrowly focused: a unit trust investing in China-related large-cap companies has **less diversification** than one investing in global large-caps.

c. Expenses. The **expense ratio** indicates the cost and efficiency of the fund.

d. Fund size and capacity. A small fund can be hurt because fixed expenses have a material impact on performance. A **SGD10,000** legal expense billed to an **SGD8 million** fund is **0.125%** of the fund; billed to an **SGD1 million** fund it is **1%**.

Conversely, very large unit trusts are difficult to manage. A fund with **SGD10 billion** is huge relative to the market; the manager cannot be as nimble and may struggle to add value. When a unit trust reaches its **optimal capacity**, it typically **stops accepting fresh subscriptions**.

9.3.18 Retrocessions

Retrocessions are fees received by banks and other financial institutions for distributing third-party products to clients. When fund recommendations are made to new clients there can be a **conflict of interest**, compromising the independence of the adviser, and clients are often not fully aware of the practice. Retrocession fees paid by fund houses to banks are a **cost offset against fund performance**, negatively affecting NAV for existing clients.

While the practice has been **arrested by the courts in Switzerland**, it is still widely practised in Asia by some institutions.

Similar concerns apply to **funds of funds** selecting external funds: if the fund of fund house receives retrocessions, the objectivity of the selection process is potentially diminished. Investors also worry about the **dual layer of fees** — at fund level and again at fund-of-fund level. Retrocessions may result in significant overall effective costs, though some investors are less concerned if the overall return compensates.

9.4 Real Estate Investment Trusts

9.4.1 Overview of REITs

A **REIT** is a CIS pooling individual investors' funds to invest in real estate assets. In Singapore, REITs are governed by **Appendix 6 — Property Funds (the Property Funds Appendix)** in the Code on CIS. A REIT is a unit trust predominantly invested in real estate and real estate-related assets: the Code requires a minimum of **at least 75% of the fund** to be invested in **income-producing real estate**. A simple way to understand a REIT is as a **passive investment vehicle** engaged in real estate investment for rent collection.

In many respects a REIT is like any other unit trust, but three differences matter:

- A REIT requires a **wider range of specialists** to manage it.
- A REIT typically has a **Sponsor**, usually with a stake, that sources the properties injected into the initial portfolio and may continue to provide a **pipeline** of assets.
- **Valuations** of underlying assets are less current: unlike equities and bonds where market values are readily available, the Code on CIS requires REIT properties to be valued only **once a year**.

Unit trusts that are not REITs are typically **not listed**, whereas REITs are listed on the **SGX Mainboard**. Unlisted property trusts, while constituted the same way as listed REITs, are not publicly traded.

REITs can be **sector-specific** (retail malls) or hold investments across property sectors — healthcare, hospitality, industrial, offices, retail. REITs are **long-term investments** subject to property cycle fluctuations.

9.4.2 Advantages of REITs

Compared with direct investment in real estate for rental income:

- Buying into a REIT is much more **affordable** and achieves **greater diversification** than purchasing a single property.
- A REIT has significantly greater **liquidity**, trading easily and quickly on the exchange without substantial transaction costs.
- REITs distributing at least **90% of taxable income** each year enjoy **favourable tax treatment**, and many do; there is **no corresponding tax advantage** for direct investment.
- REITs come with **professional property management** without separately contracting for it.

9.4.3 Risks of Investing in REITs

Risks vary with the REIT's characteristics, including the geographical location of properties. Investors should consider not only the expected distribution yield but also the **quality and lease tenor** of underlying properties, **tenant quality and concentration**, and the **weighted-average lease expiry**.

Risk	Detail
Market risk	REITs are listed and traded, so price depends on supply and demand. Prices reflect the outlook fo
Income risk	REITs typically distribute at least **90% of income** each year, but may be unable to distribute if
Concentration risk	REITs afford some diversification but may be vulnerable if a substantial portion of income comes
Liquidity risk	Although REIT units trade on the exchange, the **underlying properties** may not be easily liqui

Risk	Detail
Leverage risk	In Singapore REITs are subject to a **single-tier leverage limit of 45%**. Servicing acquisition d
Refinancing risk	Debt financing exposes the REIT to interest rate increases on refinancing. Because REITs distribu

9.5 Business Trusts

9.5.1 Overview of Business Trusts

Business trusts (BTs) are governed by the **Business Trusts Act 2004**, passed in September 2004 to regulate this form of business structure. Like unit trusts, a BT is established under a **trust deed**. A BT is a **hybrid** with characteristics of both companies and trusts, but unlike a company it is **not a separate legal entity**. BTs have become a popular form of SGX listing, with participation from port infrastructure and pay television.

In Singapore, BTs are a **single responsible entity** organisation: the **trustee-manager** is vested with the roles of both trustee and manager of the trust assets. The trustee-manager has **legal ownership** of the BT's assets and is responsible for looking after unit holders' interests and managing the business. Investors subscribe for units and are paid distributions from the BT's income.

9.5.2 Advantages of Business Trusts

- **Limited liability** of unit holders.
- **Distributions can be paid out of operating cash flow** instead of accounting profits.
- **Investment restrictions applicable to CIS do not apply** to BTs.

The main attraction is effectiveness in **unlocking value from fixed assets that generate cash flow**. Infrastructure providers and asset-backed businesses find the structure attractive: such businesses usually have **depreciating assets that reduce accounting profits despite strong positive cash flows**, so paying distributions from operating cash flow is valuable.

Like REITs, BTs pool investors' money and give fractional ownership of underlying business assets, facilitating diversification. But unlike REITs, **BTs can actively engage in any business area**, whereas REITs are required by law to invest in real estate for rental income.

9.5.3 Risks in Business Trusts

BTs are subject to the same risks as unit trusts and REITs — market, liquidity, income, concentration and refinancing risk. On refinancing, note that while there is **no cap on the debt limit**, debt can still become a problem if the BT uses that freedom for **indiscriminate borrowing**, producing an unsustainable debt structure.

Given the wide range of businesses allowed, investors should analyse the **contractual details of the BT's assets** to determine the consistency and stability of income sources. For example, existing port infrastructure providers on traditional shipping routes are potentially at risk as global warming opens new, more efficient and economical routes, creating demand for new port locations.

9.5.4 Comparison between Listed REITs and Business Trusts

	Listed REIT
Entity	Constituted by a trust deed
Ownership	Unit holders do not have individual legal ownership of assets but have **bene**
Independence of trustee and manager	Trustee and manager are **separate and independent**
Trust objective	**Passive** investments
Distribution	**Mandatory distribution of a minimum of 90% of annual income** to
Gearing	**45% limit**
Restrictions on investments	Minimum **90%** to be generated by **rental income**
Regulations	Securities and Futures Act, Code on CIS
Taxation	**Tax exempt** for individual unit holders

9.6 Stapled Securities

A **stapled security** is created when an issuer lists **two or more securities traded as a single instrument** on the exchange. The stapled security shares the same investor. For example, several Singapore REITs in the hospitality sector are stapled to a business trust for listing.

Stapled securities are popular with issuers because they combine a **passive income security** with an **active income security**, enhancing the issue's attractiveness. From the investor's perspective they provide some **diversification of business risks**.

In a stapled security the issuer is bound by the regulations specific to **each** security being stapled. If a REIT and a business trust are stapled, the issuer must pay the minimum **90% of annual income** to REIT unit holders to qualify for favourable tax treatment, but is **free from paying dividends** on the business trust. Such stapled securities **cannot be marketed as REITs**.

9.7 Undertakings for Collective Investment in Transferable Securities

UCITS is a **European Union directive**, applicable in the UK since **2002**. The original objectives were to remove barriers to cross-border distribution and marketing of funds by allowing funds to invest in a wider range of financial instruments. UCITS provides the **"passport"** for funds to be sold internationally.

UCITS funds are attractive to retail investors globally because of **common standards of disclosure** and **consistent investment guidelines**. Under **UCITS III**, funds can invest in a wide range of assets beyond equities and bonds — for example using derivatives and index tracking products to limit risk or increase returns.

Since **2006**, MAS has allowed UCITS funds to be distributed and sold in Singapore if they meet local regulatory conditions. The number registered for sale in Singapore has increased significantly over the last 5–10 years, reflecting the trend in Asia.

9.8 Alternative Investments

Beyond REITs and business trusts, alternative investments include:

- **Hedge funds.**
- **Private equity funds.**
- **Commodity funds.**
- **Real estate funds.**

The main benefit is the ability to **further diversify** using asset classes **less correlated** with traditional equities and fixed income, potentially increasing risk-adjusted returns.

Unfortunately, during major financial crises such as 2008, alternative investments do **not necessarily provide much of a buffer**, because they are also affected by a widespread massive sell-off. Under normal conditions, however, they have been found to be less correlated with equities and bonds.

9.8.1 Overview of Hedge Funds

Unlike traditional funds, hedge funds can take **short positions** using derivatives. Depending on the mandate, they adopt either:

- A **single strategy** approach, following one approach.
- A **multi-strategy** approach, emphasising one approach over others given the opportunities in a particular period.

9.8.2 Hedge Funds versus Conventional Funds

Like conventional funds, hedge funds can take long positions to exploit positive market movements and are exposed to downturns. Unlike conventional funds, they can use **short selling or derivatives** to reduce or hedge their portfolios' **beta sensitivity** to the market.

Hedge funds often have **absolute return targets** and can rely on **leverage** to significantly augment returns and produce alpha. But some strategies require a **longer time to unwind**, and hedge funds can be **less transparent** about pricing and positions. Because unwinding takes time, the manager only knows the actual price when trades execute — hence the lack of pricing transparency.

Hedge fund managers often receive **incentive fees** on top of higher management fees.

Hedge funds	Conventional funds
Capture **alpha** (freedom to hedge away or reduce beta)	Capture **alpha + beta** (limited ability to reduce b
Performance depends mostly on the manager's skills and strategies	Performance depends mostly on market factors and
Absolute return targets	**Index-based benchmark** targets
Can be **leveraged significantly** to augment the impact of strategy	**Limited** ability to use debt and leverage
Require **longer to unwind** and are **less transparent** on pricing	More transparent on investments and pricing
Management fees **and performance incentives**	Management fees

9.8.3 Risks of Investing in Hedge Funds

- **Risky investment strategies** — speculative strategies combined with leverage can produce outsized losses. Leveraged positions are sometimes hedged with derivatives, bringing **basis risk** (the risk remaining from an imperfect hedge because a complete offsetting position may not exist) and **counterparty risk** (the other party failing to meet its contractual obligation).
- **Illiquidity** — hedge funds can invest in highly illiquid assets and may not unwind at fair prices. An unanticipated downturn can produce **margin calls** on leveraged positions and unusually large redemptions, forcing liquidation at unfavourable prices. Many hedge funds impose **lock-up periods** during which investors cannot redeem, and some have **gate provisions** letting the manager limit withdrawals on a given date to a percentage of net assets.
- **Poorer investor protection** — hedge funds are more lightly regulated and less transparent than conventional funds.

9.8.4 Hedge Funds — Relative Value Arbitrage Strategies

a. Long-short equity. A relative value approach: the manager goes long on a stock believed undervalued and short on a stock believed overvalued within the same sector. For example, going long TSMC and short Intel on the view that TSMC's premium over Intel will widen.

For such strategies it is useful to set a target for closing the positions. Positions can be rolled over, but there must be justification for further widening. If there is no additional news that would cause the premium to widen further, the spread position should be closed.

If the manager has strong conviction, they can take a **net long** position by investing more in the long leg than is short-sold — but then the strategy is **no longer market-neutral** and has partial exposure to a sector decline.

b. Convertible arbitrage. Also a relative value approach, but the analysis is between the **convertible securities and the common stock of the same company**. The strategy purchases convertibles while shorting the underlying stock. The rationale is that convertibles are usually **slower to reprice to fair value** than the underlying stock, creating arbitrage opportunities. The strategy is not riskless: the stock price can rise while the convertible falls.

Other, less widely used relative value strategies include **fixed income arbitrage** (exploiting yield differences between bond types), **credit long-short** and **volatility arbitrage**.

9.8.5 Hedge Funds — Event-Driven Strategies

Event-driven strategies emerge when specific events present themselves, and hedge funds opportunistically exploit them. Events include **financial distress** (bankruptcies, liquidation), **acquisitions**, **capital restructuring** and **consolidation**.

a. Distressed securities. Some hedge funds — sometimes called **vulture funds** — look selectively for opportunities in bonds priced at distressed levels. Depending on the severity of the company's crisis, distressed bonds can be priced at **10% to 99% of par**. Hedge funds buy distressed debt and borrowings they believe undervalued.

When there was a global downturn in oil and gas, the bonds of many oil-related companies, including several in Singapore, were priced at distressed levels. Some investors and banks were willing to sell at a discount, and these were bought by investors and hedge funds willing to bear the risk for potential gains.

Hedge funds typically have three options:

- **Resell** the debt at a much higher price when the credit situation improves.
- **Exchange** debt for equity and sell the equity for a profit, or combine the stake with a private equity firm and other investors to make a takeover bid.
- **Allow time** for the company to recover and pay off the full principal at maturity, yielding substantial gains above the discounted price.

Investing in distressed debt is very risky and should be managed by experts. A well-known **Swiss bank** was forced to sell its billion-dollar portfolio to a distressed-debt specialist at a huge loss in **2016** when it decided to exit the high-risk business.

b. Corporate events arbitrage. Event-driven funds also look for mergers, acquisitions, takeovers and liquidations. Depending on valuation and the deal process, they may:

- Buy the **target** and sell the **acquirer**, when the acquirer is overvalued.
- Buy **both** acquirer and target, when the acquisition is believed to create greater value for the merged entity through synergies.
- Buy shares of the **target** in a hostile takeover with competing offers.

Not every corporate finance deal offers mispricing opportunities, so such funds may face **dry periods**.

9.8.6 Hedge Funds — Global Macro Strategies

Global macro strategies take **directional views** on financial products and derivatives in anticipation of market reactions to significant economic or political changes in one or more countries. The manager decides to go long or short based on how macroeconomic trends will affect interest rates, currencies, sovereign credit, commodities, equity indices, equities and bonds.

Approaches range from very **qualitative fundamental** to **quantitative computer-based models**, or a blend. Because of the size of their positions and risks, they need to invest in fairly **liquid** assets to unwind or hedge.

Examples. Shorting the pound sterling and British stocks on the London Exchange during the **Brexit Referendum** was an effective global macro strategy. Conversely, a strategy that shorted US equities expecting a decline after the Republican victory in the US elections would have had to reverse quickly to avoid losses when US equities rallied shortly after. Global macro funds must be **nimble** to unwind positions and cut losses.

9.8.7 Hedge Funds — Directional Computer-Based Quantitative Trading Strategies

Directional quantitative strategies combine **technical analysis, market data and funds flow** to make decisions, mainly following technical price trends and momentum. As long as the trend is intact, the fund maintains its position until a reversal is observed.

Momentum trading. A fund following momentum continues to buy options and futures on indices in line with upward trend momentum — the philosophy that "the trend is your friend". Once the model signals a slowdown or reversal, cumulative gains are realised at once and the model awaits confirmation of the next trend. If indicators signal further correction, shorting of futures and options begins until the next reversal point, at which shorts are closed with purchases.

Such strategies benefit from **clear and persistent trends over weeks or months**. With disruptive reversals or a high frequency of reversals over very short periods, transaction costs are likely too high to make profits.

9.8.8 Funds of Hedge Funds — Multi-Strategies

Funds of hedge funds adopt different strategies, allocating more to some over others in a given period, because in any period some strategies outperform others due to market conditions.

The fund of hedge funds approach attempts to provide the expertise and due diligence to ensure:

- **Adequate diversification** between hedge funds, reducing concentration risk.
- That monies can be **actively reallocated** to strategies that do well under certain market conditions.

The service comes with **additional management cost**, on top of the performance and incentive fees paid to hedge fund managers. With diversification, returns typically fall to the **average** hedge fund return, which may not fit some investors' expectations.

9.8.9 Overview of Private Equity Funds

Private equity funds are **long-term investments sold OTC**, mainly to accredited individuals, corporate clients, institutional clients and sovereign wealth funds to complement their overall portfolio. They invest in businesses at different stages of their growth cycles.

Private equity provides businesses with capital to grow to a particular size before being **listed through an IPO** or **sold privately** to major corporations. It therefore gives founders a viable **exit option** and investors the opportunity to realise gains.

9.8.10 Stages of Private Equity Investment

a. Venture capital. Venture capitalists co-invest with an entrepreneur to provide **seed and/or start-up capital** in the early stages. Many newly founded tech and biotech companies are at the conceptual or research stage where products are not yet developed, and may go through several years of R&D and losses before potential revenues and profits. In return for the higher risk, venture capitalists expect a **higher-than-average return**.

b. Growth capital. Funds for companies looking to **scale up** and take advantage of high growth or expansion. Growth capital tends to focus on **later-stage** investments — companies already established but still requiring funds or co-sharing of expansion risk when scaling product range, production volume, distribution and market reach.

c. Buyout funds. Invest in **mature companies with stable cash flows**, established products and developed personnel. The buyout may occur **pre-IPO** for owners wanting to exit before the IPO process, or as part of the IPO process with an agreement to transfer shares between major shareholders **1–2 years after** the IPO to facilitate smooth running.

d. Special situations.

- **Going private** — when major shareholders believe the market undervalues the company. Buying the company and taking it private lets them benefit from a larger share of profits. Private equity vehicles can act as intermediary, or major shareholders can raise private capital among friendly parties to make an offer to remaining shareholders. When the time is ripe the business can be re-listed or consolidated with other businesses for listing.
- **Distressed businesses** — private equity firms work with private parties to take over companies facing financial distress. In downturns, good businesses may succumb to financing and liquidity issues and be forced to sell; private equity can work with creditor banks to take them over.

9.8.11 The Private Equity J-Curve

Private equity investments tend to have **negative initial returns and cash flows** for a relatively long period, so cash flows follow a **J-curve**.

The first **5–6 years** are the **investment period**. If an investor commits **$10 million**, it is not all invested at the outset: the manager makes several **cash calls** and draws down the capital as good opportunities arise. If many good opportunities arise early, the drawdown is faster. The **harvest period** typically begins from the **6th year**, when the fund starts making payouts.

The incentive is that the expected **internal rate of return** is typically in the double digits (**10–15%**). The downside is that funds are **locked up for several years**, and early redemption or withdrawal before the 6th year typically results in **losses**. To address suitability, it must be highlighted that private equity is **long-term** in nature: the investor must be willing to sacrifice liquidity to secure the targeted returns, the first few years involve only drawdowns, returns usually accrue only from the 6th year, and early redemptions result in losses.

9.8.12 Commodity Funds and Real Estate Funds

Commodity and real estate funds are largely similar to commodity ETFs and REITs. The main difference is that commodity and real estate funds are sold **OTC**, while commodity ETFs and REITs are **exchange-traded**. Some OTC funds distinguish themselves by being **more transparent** about their investment approach and types of investments.

a. Commodity funds. Offer indirect exposure to a portfolio of commodity-related stocks, bonds, futures and funds. They can focus on pure sectors (metals, agriculture, energy) or be a fund of funds diversified

across commodity types, possibly overweighting one sector. The more diversified the fund, the less volatile it should be, though exposure to the futures market makes some volatility unavoidable.

- **Agriculture** funds suffer the same problem as their underlying commodities: successful production often outstrips demand, causing long downward cycles of overhanging supply.
- **Energy, base metal and ferrous metal** sectors are more **cyclical**: they do well when global economic activity picks up, driving underlying demand, and poorly when activity slows.
- **Precious metals** funds typically do well in times of uncertainty and volatility in traditional asset classes, and can be heavily influenced by major currency movements and interest rates.

Agriculture funds	Energy funds	Metals funds
Grains	Crude oil	Precious metals
Soft commodities	Natural gas	Base metals
Livestock	Refined products	Ferrous metals
Dairy	Petrochemicals	
Biofuel	Electricity	

b. Real estate funds. Not too different from REITs. OTC real estate funds can cover agricultural, industrial, residential, commercial, medical and even undeveloped land banks. Although focused on generating stable, high rental income, OTC real estate funds sold through private banks can be **less liquid**.

As with REITs, the main concern is the **mismatch between unit liquidity and the illiquidity of underlying assets**. In a property bubble, asset prices can fall very sharply once the bubble bursts, leading to redemptions. When redemptions mount, funds without deep pockets that struggle to unwind property holdings may be caught in a **liquidity crunch** and be unable to meet redemptions. Selling underlying property at market lows forces managers to realise losses, so in such situations real estate funds typically seek to **delay redemptions**.

9.9 Insurance

Some events are unpredictable as to when they occur but have severe financial repercussions when they do. Risk management processes must mitigate the effects of such unforeseen events.

Some insurance products also serve as investment instruments, and representatives should understand them to advise clients well. The two main types of life insurance policies that can also serve as savings or investment instruments are **whole life** and **endowment** insurance.

9.9.1 Whole Life Insurance Policy

Premiums for a whole life policy provide a combination of **life cover and investment**, and therefore have **cash or surrender values**. The **sum assured** is payable on the death of the life insured, whenever that occurs. Policy owners pay fixed monthly or annual premiums and beneficiaries receive fixed death benefits. Meanwhile, reserves and assets backing future liabilities earn interest and grow, building **cash values** that can be withdrawn through **surrender** (cancellation) of the policy, subject to its terms.

Alternatively, if owners do not want to surrender, they have access to accumulated policy savings: they can normally obtain **policy loans** from the insurer up to a certain percentage of the policy's cash value, the limit being insurer- and policy-specific.

9.9.2 Endowment Insurance Policy

Premiums also provide a combination of life cover and investment, but endowment policies usually **mature after a fixed period** — for example **10, 15 or 20 years** — and the sum assured is payable on the **maturity date or on death, whichever is earlier**. As with whole life, the sum assured is fixed.

Endowment policies carry **higher premiums** than conventional whole life and term insurance, but may be useful for special lump-sum needs such as college expenses or buying a retirement home. The **guaranteed cash values received may be less than the sum of premiums paid**, because part of the premium pays for insurance protection while the rest is invested and subject to investment risk.

9.9.3 Using Insurance Policies for Investments

Both whole life and endowment policies are effectively investments payable at a future date, either on death or earlier. Policies structured to provide a fixed death benefit only are **non-profit (non-participating)** policies, in contrast to **with-profits** policies.

The return of insurance policies can be linked to the insurer's investment performance either through a **with-profits** policy, where benefits are **indirectly** affected, or through an **investment-linked** policy, which has a **direct** link.

a. With-profits policies. Each year the insurer values the assets and liabilities of its life fund. This normally reveals a **surplus**, part of which can be allocated to the with-profits policy owner as an addition to the sum assured. This addition — a **bonus** or profit — is usually **reversionary**, meaning it is payable only at the same time as the sum assured.

Under a with-profits policy the link between benefits and the insurer's investment and operating performance (mortality experience, operating expenses) is **not direct** and depends on the annual valuation of fund assets and liabilities, where many other factors are considered.

Allocation of the surplus as a reversionary bonus ultimately depends on the **board of directors**, based in part on the appointed actuary's recommendations. Bonuses are therefore only **loosely linked** to investment and operating performance. Allowances must be made for guarantees on the basic sum assured, so the bonus system does not directly reflect the value of the underlying assets. Bonuses are also declared **yearly**, so they cannot match daily fluctuations in asset values.

b. Investment-linked life insurance policies. Offer policies whose values are **directly linked** to the investment performance of the underlying instruments, usually by formally linking policy values to **units in a fund** run by the insurer or by external fund managers. Such funds usually consist of a mixture of equity, fixed income, managed and other funds. The investment element varies with the underlying assets and **fluctuates daily** with their performance.

9.10 Annuities

An **annuity** is a series of payments guaranteed for a number of years or over the lifetime of the party receiving the payments (the **annuitant**).

9.10.1 Types of Annuities

- **Immediate annuity** — payments begin **as soon as it is purchased**. Always purchased with a **lump sum (single premium)**.
- **Deferred annuity** — payments begin at some **future date**, specified in the contract or at the annuitant's option. The amount received depends on contributions, the interest earned, the annuitant's **gender** and the annuitant's **age** when payments begin. May be purchased with a single premium or periodic premiums.

Examples of **non-insurance annuities** include monthly payments on home mortgages and fixed-rate coupon payments on a bond.

9.10.2 Annuity Calculations

Present value of an annuity:

$$PV_{\text{annuity}} = C \times \left[\frac{1-(1+i)^{-n}}{i}\right]$$

where

$$C$$

is the cash flow per period,

$$i$$

the interest rate and

$$n$$

the number of payments.

Worked example. \$1,000 paid every year for **5 years**, at an annual discount rate of **3%**:

$$PV = 1{,}000 \times \left[\frac{1-(1.03)^{-5}}{0.03}\right] = \$4{,}579.71$$

Calculator: **PMT = −1,000**, **N = 5**, **I/Y = 3**; compute PV.

Future value of an annuity:

$$FV_{\text{annuity}} = C \times \left[\frac{(1+i)^n - 1}{i}\right]$$

Worked example. \$1,000 paid every year for **8 years**, at **5%**:

$$FV = 1{,}000 \times \left[\frac{(1.05)^8 - 1}{0.05}\right] = \$9{,}549.11$$

Calculator: **PMT = −1,000**, **N = 8**, **I/Y = 5**; compute FV.

9.10.3 Uses of Annuities

a. Insurance against outliving one's income. In its pure form a **life annuity** is a contract whereby, for a cash consideration, the insurer agrees to pay the annuitant a stipulated sum periodically **throughout life**. The consideration is fully earned by the insurer immediately on the annuitant's death. The purpose is to protect against **loss of income from excessive longevity** — exactly the opposite of life insurance, which protects against loss of income from **premature death**.

b. Investment vehicle or savings instrument. Most annuities are savings instruments designed to **accumulate funds and then systematically liquidate** them, usually during retirement. The annuitant can pay a lump sum or regular premiums; at the annuity starting date they receive a guaranteed income for life, the amount depending on the cash value and the payment option selected.

Annuities are attractive for someone who has not yet accumulated an estate but wants financial independence in old age. **Professional entertainers and athletes**, who enjoy very large income for a limited period, find annuities particularly attractive as a savings medium, purchasable through flexible periodic premiums or a single premium when large sums arrive.

c. Hedge against adverse financial developments. Wealthy individuals who have accumulated an estate purchase annuities as a hedge against adverse financial developments. Large estates can be wrecked through unwise investments, and individuals once wealthy can depend on annuity payments purchased when they were more affluent.

d. Comparing life insurance and annuities. Life insurance guards against living **too short** a life; life annuities guard against living **longer than expected**, supporting post-retirement living expenses for as long as the annuitant lives. To provide sufficient retirement income, life annuity products are often structured as **savings and investment products before payments begin**. The savings and asset-building phase is crucial to retirement planning and an integral part of many life annuity products, where regular premiums are paid into the retirement savings fund for investment and growth.

9.11 Risk Management for Retirement Funds

For investment of **CPF** funds ultimately meant for retirement, the CPF Board has provided guidance aligned with risk management best practice:

- Investors should be made aware that funds with **higher equity proportions** are generally riskier and more volatile.
- Investment providers should highlight **focused or concentration risk** — geographical, sector or specific investment-related.
- For a fund to be included under **CPFIS** and eligible to take CPF monies, it must meet the **CPFIS funds admission criteria**, including being in the **top quartile of the global peer group** and benchmarking **total expense ratios to the median of CPFIS funds** in the respective risk class.

Diversification and dollar-cost averaging are recommended to spread risk and potentially improve overall returns.

9.11.1 Dollar Cost Averaging Versus Market Timing

a. Market timing. The technique of shifting investments in and out of risky asset classes to:

- Capture the upside for risky asset classes during bull markets.
- Exit before downturns, avoiding the downside during bear markets.

Market timing is an **active** investment approach.

b. Dollar cost averaging. A practice for investors who do not want to undertake market-timing risk: invest **steady, equal amounts over a period of time**. Because equal dollar amounts are invested, **fewer shares are accumulated when the price is high and more when the price is low**. Executing it accurately will likely leave the investor with **odd lots**.

Worked example. An investor with $60,000 * *isconsideringasecuritycurrentlypricedat * *\mathbf{5}$.

- Invest all **$60,000** at the start: **60,000 / 5.00 = 12,000 units**.
- Buy **$5,000** worth evenly over the next 12 months.

Month	Price	Quantity	Cost
1	5.00	1,000	5,000.00
2	4.80	1,042	5,000.00
3	4.60	1,087	5,000.00
4	4.60	1,087	5,000.00
5	4.70	1,064	5,000.00
6	4.50	1,111	5,000.00
7	4.50	1,111	5,000.00
8	4.50	1,111	5,000.00
9	4.30	1,163	5,000.00
10	4.30	1,163	5,000.00
11	5.00	1,000	5,000.00
12	4.80	1,042	5,000.00
Total	**4.622**	**12,980**	**60,000.00**

Investing all at the start gives an average price of **$60,000 / 12,000 =** $5 * *.Dollarcostaveraginggives * *$**60,000 / 12,980 =** $4.622 * *.Thisisalsoalwaysmathematicallylowerthanthesimpleaverageofthetwelvemonthlyprices(* *4$ but that does not mean much: what matters is whether the average cost is below the **final** price at the end of the 12 months, which depends on the price trend.

c. Missing the best days. Advocates of passive investing cite empirical evidence that most investors fail to use market timing to make a positive impact sustainably. Not only are the odds against them, the "punishments" are devastating if the best trading days are missed.

Consider the best **25 days** of the S&P from **1970 to 2016** (47 years). An investor who started at 18 and retired 47 years later in 2016, but **missed the best 25 days**, would see the return fall from **1,910% to just 371%** — from **6.7% to 3.0% annually**. That is even lower than the return on a **1-month US T-bill** rolled over regularly, which would generate an average yield of **4.9%** over the period. Passive buy-and-hold would have done much better.

Two further considerations:

- The best trading days usually occur **immediately after some of the worst** trading days, so the chance of missing the best days is high because many investors sell in panic.
- Best trading days tend to occur in **close proximity** to each other, further increasing the risk of missing most of them.

It is almost impossible to predict the best and worst trading days, because daily movements are affected by many highly unpredictable events; they are usually identified only in hindsight. Rather than trying to identify good and bad days, a better strategy may be **dollar cost averaging**.

d. Missing the best and worst days. Market timers may miss some of the best days, but they may also miss some of the worst. Removing the **worst 25 days** of S&P 500 performance over the same period, the return rebounds to **2,750% (7.6% annualised)** — better than the **1,910% (6.7% annualised)** earned by the buy-and-hold investor.

These results do **not** necessarily show that market timing is superior to passive management. Because market timers are not always invested, their non-participation means they are likely to miss some of the best **and** some of the worst days. Direct evidence that market timers can beat a passive strategy is needed.

Unfortunately, most working adults lack the time, expertise or emotional temperament to be disciplined active managers of their own portfolios, and **70–80% of active fund managers** have been found to underperform the market each year. Dollar cost averaging as a passive form of investing for retirement may therefore be a good approach.

9.12 Summary

- In Singapore, pooled investments are CIS or unit trusts, governed by the Code on CIS. The VCC structure, introduced by the Variable Capital Companies Act in **October 2018**, allows free issuance and redemption of shares, dividends paid from capital, and an umbrella structure with sub-funds.
- ETFs track indices or fixed baskets and involve no active management. They trade during exchange hours, unlike unit trusts which transact at the daily closing NAV.
- Physical ETFs use full replication or representative sampling; synthetic ETFs use swaps and derivatives, reducing tracking error but adding counterparty risk.
- ETF advantages: diversification, small capital outlay, low costs, liquidity, transparency. Risks: tracking error, counterparty, market and sector, FX, and taxation.
- Smart beta ETFs select and weight securities on performance criteria rather than market capitalisation.
- VIX ETFs are for very short-term trading; leveraged and inverse ETFs are generally unsuitable for retail investors.
- A unit trust pools funds under an independent trustee. Open-end trusts issue units continuously at NAV; closed-end trusts have a fixed number of units priced by market demand and supply.
- Unit trust advantages: diversification, small outlay, professional management, flexibility, liquidity, lower transaction costs, security. Disadvantages: sales charges and management fees.
- Fees include front-end load (**1.5–5%**), back-end load, management fee (**0.5–2.0%** of NAV), operational costs (**0.1–0.15%** of NAV), switching fee (about **1%**) and performance fee subject to a high watermark.
- The offer price adds purchase expenses, cash and accrued income and the manager's initial charge; the bid price deducts sale expenses and adds cash and accrued income.
- Retrocessions create conflicts of interest and reduce NAV.
- A REIT must invest at least **75%** in income-producing real estate, distributes at least **90%** of taxable income, and is subject to a **45%** leverage limit. Properties are valued only **once a year**.
- Business trusts are governed by the Business Trusts Act 2004, are not separate legal entities, use a single trustee-manager, may distribute from operating cash flow, have **no gearing limit** and **no investment restrictions**.
- Stapled securities combine two or more securities traded as one instrument, and cannot be marketed as REITs.
- UCITS is an EU directive providing a passport for cross-border fund sales; MAS has allowed UCITS distribution in Singapore since **2006**.
- Alternative investments diversify because they are less correlated with equities and bonds, but do not necessarily buffer during crises.
- Hedge fund strategies include relative value (long-short equity, convertible arbitrage), event-driven (distressed securities, corporate events), global macro, and directional quantitative trading.
- Private equity stages: venture capital, growth capital, buyout, and special situations (going private, distressed). Cash flows follow a **J-curve**, with the investment period of **5–6 years** and harvest from year 6.
- Whole life insurance provides life cover plus cash value; endowment insurance matures after a fixed period. With-profits policies link benefits indirectly; investment-linked policies link directly.
- Immediate annuities begin payments at purchase; deferred annuities begin later. Annuities protect against longevity risk, the opposite of life insurance.
- CPFIS admission requires top-quartile performance and total expense ratios benchmarked to the CPFIS median.
- Dollar cost averaging invests equal amounts over time, accumulating more units when prices are low.

9.13 Additional Reference Materials

9.13.1 Key Features of Variable Capital Companies

The **Variable Capital Company (VCC)** is a legal entity structure for investment funds introduced to offer asset managers **greater flexibility and lower costs**, encouraging fund managers to co-locate fund domiciliation with their substantive fund management activities in Singapore and deepen the asset servicing ecosystem.

Key features:

- A VCC can be used for both **traditional and alternative funds**, and for both **open- and closed-end** funds.
- A VCC can house a **standalone fund or a collection of sub-funds** under an umbrella structure, with **segregation of assets and liabilities** of the sub-fund. An umbrella structure lets asset managers achieve cost efficiencies by consolidating administrative functions at the umbrella level.
- Shares of the VCC can be **redeemed without shareholders' approval**, and **dividends are payable from capital**, offering greater flexibility than existing investment companies. Shares are redeemed and issued at **NAV**.
- A VCC must be managed by a fund manager that is **registered, licensed or exempted by MAS**.
- For taxation, a VCC is a **company and a single tax entity**.
- A VCC **does not need to publicly disclose its shareholder register**, offering investor privacy.

9.13.2 CPF Investment Risk Classification System

The CPF Board has classified unit trusts included under the **CPF Investment Scheme (CPFIS)** under different risk categories. The **Risk Classification System** splits the investment risk of a unit trust into **two axes**: **equity risk** and **focus risk**.

i. Equity risk relates to the proportion of "riskier" investments in the unit trust. The greater the proportion invested in stocks, the higher the equity risk; the greater the proportion in bonds and cash, the lower the equity risk. Four categories:

Category	Fund type
Higher risk	Equity funds
Medium to high risk	Balanced funds
Low to medium risk	Bond funds
Lower risk	Cash funds

ii. Focus risk reflects how focused the investments are in one particular geographical region, country, industry or individual company. Two categories:

- **Broadly diversified** — investments spread across relatively more regions, countries, industries and companies.
- **Narrowly focused** — investments concentrated in particular regions, countries, industries or companies.

In general, a unit trust has a **higher** level of focus risk when it is relatively **less diversified**. A unit trust investing in shares of companies worldwide has **lower** focus risk than one investing only in Japanese corporations; the Japan-only trust has higher focus risk because it invests in one foreign country only.

Within the focus risk category, CPFIS-included unit trusts are further divided into sub-groups:

- **Regional** — regional exposure, for example Europe, Asia or Emerging Markets.
- **Sector** — specific industry exposure, for example healthcare and biotechnology.
- **Country** — country-specific, for example Japan or China.

These sub-groups make it easier for CPF members to compare unit trusts that are roughly similar in geographic focus.

The system works alongside the CPFIS admission criteria: a fund must be in the **top quartile of its global peer group** and have its **total expense ratio benchmarked to the median of CPFIS funds** in the respective risk class before it can accept CPF monies.

Exam cues

- **VCC Act passed October 2018.** Two features: free issue and redemption of shares with dividends from capital, and an umbrella structure with sub-funds.
- **SPDR STI ETF:** priced at about **1/100th** of the STI level; board lot **100 units**; dividends **twice a year**; annual costs **0.3%**; creation and redemption only through participating dealers in **100,000-unit** blocks.
- **ETF arbitrage:** traded price **above** NAV → sell ETF, buy the basket; **below** NAV → sell the basket, buy the ETF.
- **Physical vs synthetic:** physical uses full replication or representative sampling; synthetic uses swaps and derivatives. **Synthetic has lower tracking error but higher counterparty risk.**
- **ETF advantages:** diversification, small outlay, low cost (**under 1%** vs **1–2%** for active unit trusts), liquidity, transparency.
- **ETF risks:** tracking error (greater in **physical**), counterparty, market and sector, FX, and **US withholding tax** for non-US investors in US ETFs.
- **Securities lending** creates counterparty risk; collateral is typically about **105%** of the lent stock's value.
- **Smart beta** weights on performance criteria rather than market capitalisation.
- **VIX ETFs are for trading, not investing.** VXX returned **−56.85%** while SVXY (short VIX) returned **+35.26%** in the same period.
- **Leveraged ETFs** are **2x or 3x**, with normal daily moves of **2–4%** and possible **±20–25%** in a session. **Not suitable for retail investors.** Issuers use **reverse splits** to reset prices.
- **Unit trust terminology:** US = mutual fund; UK = unit trust; Germany = investment fund; Luxembourg = **Sicaf** (closed-end) and **Sicav** (open-end). Most offshore funds are domiciled in **Luxembourg**.
- **Open-end vs closed-end:** open-end issues units continuously at **NAV**; closed-end has a **fixed** number of units priced by **market demand and supply**.
- **Trustee holds the assets and issues or redeems units**; the manager does neither.
- **Unit trust advantages:** diversification and divisibility, small outlay (1,000–**5,000**), professional management, flexibility (switching within a family of funds), liquidity, lower transaction costs, security.
- **Unit trust disadvantages:** sales charges and management fees paid regardless of performance.
- **Front-end load 1.5–5% of NAV; management fee 0.5–2.0% of NAV; operational costs 0.1–0.15% of NAV; switching fee about 1%; back-end load often waived after 60 days.**
- **High watermark** prevents a performance fee being paid twice on the same recovered value.
- **Offer price** = NAV + purchase expenses + cash and accrued income + manager's initial charge. **Bid price** = NAV − sale expenses + cash and accrued income.
- **Concentration:** many funds cap a single security at **5%**; some allow **8–9%**.
- **Fund size:** a **SGD10,000** expense is **0.125%** of an **SGD8m** fund but **1%** of an **SGD1m** fund. Very large funds lose nimbleness and stop accepting subscriptions at **optimal capacity**.
- **Retrocessions** create conflicts of interest, reduce NAV, and are **banned by the courts in Switzerland** but still practised in Asia.
- **REIT:** at least **75%** in income-producing real estate; distributes at least **90%** of taxable income; **45%** leverage limit; properties valued **once a year**; listed on **SGX Mainboard**.
- **REIT advantages over direct property:** affordability, diversification, liquidity, **90% distribution tax advantage**, professional management.
- **REIT risks:** market, income, concentration, liquidity, leverage, refinancing.
- **Business trust:** Business Trusts Act **2004**; **not a separate legal entity**; **single trustee-manager**; distributions from **operating cash flow**; **no gearing limit**; **no investment restrictions**; **single-tier taxation**.
- **REIT vs BT:** REIT has separate trustee and manager, passive investments, mandatory **90%** distribution, **45%** gearing cap, **90%** rental income requirement, tax-exempt for individuals. BT has a single trustee-manager, active business, no minimum distribution, no gearing cap, no investment restriction, single-tier tax.
- **Stapled securities** combine a passive and an active income security; the issuer must still meet the REIT's **90%** distribution on the REIT component; **cannot be marketed as a REIT**.
- **UCITS:** EU directive, in the UK since **2002**; **UCITS III** allows derivatives and index tracking; MAS permitted distribution in Singapore since **2006**.
- **Hedge funds vs conventional:** alpha vs alpha+beta, absolute vs index targets, leverage permitted vs limited, less transparent and slower to unwind, performance fees.
- **Hedge fund risks:** risky strategies (basis and counterparty risk), illiquidity (lock-ups and gates), poorer investor protection.

- **Hedge fund strategies:** long-short equity, convertible arbitrage, fixed income arbitrage, credit long-short, volatility arbitrage; event-driven (distressed, corporate events); global macro; directional quantitative (momentum).
- **Distressed debt** can be priced at **10–99% of par**. A Swiss bank sold a billion-dollar portfolio at a huge loss in **2016**.
- **Private equity J-curve:** investment period **5–6 years**, harvest from **year 6**, expected IRR **10–15%**, early redemption typically results in losses.
- **Private equity stages:** venture capital, growth capital, buyout, special situations (going private, distressed).
- **Commodity funds:** agriculture suffers overhanging supply; energy and metals are cyclical; precious metals do well in uncertainty.
- **Whole life** pays on death whenever it occurs and builds cash value; **endowment** matures after a fixed period (**10, 15 or 20 years**) and pays on maturity or death, whichever is earlier. Endowment premiums are **higher**, and guaranteed cash values may be **less than premiums paid**.
- **With-profits** links benefits **indirectly** through annual valuation and a **reversionary bonus** decided by the board; **investment-linked** links **directly** to unit values and fluctuates daily.
- **Immediate annuity** begins payments at purchase with a **single premium**; **deferred annuity** begins later and depends on contributions, interest, **gender** and **age**.
- **Annuity formulas:**

$$PV = C[1 - (1 + i)^{-n}]/i$$

;

$$FV = C[(1 + i)^{n} - 1]/i$$

. Worked: $1,000 for 5 years at 34,579.71**; $1,000 for 8 years at 59,549.11**.
- **Annuities protect against longevity risk; life insurance protects against premature death** — exact opposites.
- **CPFIS admission:** top quartile of the global peer group, total expense ratio benchmarked to the CPFIS median in the risk class.
- **Dollar cost averaging worked example:** $60,000 over 12 months at prices averaging $4.633 gives an average cost of * *4.622** and **12,980 units**, against **12,000 units** at **$5** if invested at the start.
- **Trap:** a **REIT** must distribute **90%** of income, which is why it cannot easily build cash reserves — the source of its **refinancing risk**.
- **Trap:** a **business trust** has **no gearing limit**, but that is not a licence for indiscriminate borrowing.
- **Trap: capital guaranteed** funds guarantee only if held to **maturity**; **capital preservation** funds have no guarantor and use less onerous language.
- **Trap:** a **hedge fund** is not necessarily hedged — it may be net long or short.
- **Trap: retrocessions** are paid by the fund house to the distributor and reduce the fund's NAV, so they harm existing investors as well as creating a conflict.

Chapter 10 — Warrants

Warrants are the smallest chapter in the syllabus but one of the most calculation-heavy per page. This chapter covers what a warrant is, how company warrants differ from structured warrants, and the four valuation measures the examination tests: intrinsic value, conversion price, warrant premium and gearing ratio.

Learning Objectives

The candidate should be able to:

- Describe the features of stock warrants.
- Explain the valuation of warrants using intrinsic value, conversion price and warrant premium.
- Describe the gearing ratio in the context of warrant pricing.
- Understand the determinants of the minimum and maximum values of warrants.
- Illustrate how the warrant premium is influenced by the underlying stock price, the life of the warrant, exercise price, price volatility of the underlying stock, dividend on the underlying stock, leverage, interest rates and size of the warrant issue.

10.1 Definition and Terminology

10.1.1 Definition of Warrants

A **warrant** is a derivative instrument giving the holder (buyer) the right, but **not the obligation**, to buy or sell a specified number of securities — usually the issuer's common share — at a **fixed price** (the exercise or strike price) within a **specified time period**.

Warrants issued by companies trade on the **Singapore Exchange Securities Trading Ltd (SGX-ST)** platform and are known as **company warrants**. They may be considered **long-dated equity call options**: while call options expire within a few months, warrants typically have maturities of **a few years**.

Warrants are generally issued by corporations as a **sweetener**, attached to an offering of bonds or rights issues. This lets the issuer obtain financing at a **lower interest rate**, reducing its financing cost. Warrants can be **detached** from the host security and traded separately. Because they are issued as an incentive, there is **no price paid per se** for the warrant by the buyer of the host security.

When warrants are exercised, **new shares are issued**, increasing shares outstanding and **diluting** the issuer's earnings among shareholders. This differs from equity call options, which cause **no dilution** because no new shares are issued on exercise.

10.1.2 Company Warrants versus Structured Warrants

Two types of warrant are available on SGX.

Company warrant. Issued by companies to their shareholders. Maturity is usually measured in **years** from the issue date. If exercised, the company **issues new shares**, so the number of shares in the market increases.

Structured warrant. Issued by a **bank, financial institution or other investment entity** — usually a third-party issuer, not the underlying company. The issuer is effectively selling the holder an option to buy or sell the underlying company's shares at a predetermined price at a later date. The **underlying company does not endorse or guarantee** such warrants.

The terminology for structured warrants is more complex. The example **CapitalaMBeCW181009 (COWW)**, issued by Macquarie Bank, decomposes as:

Term or code	Attribute	Detail
Capitala	Underlying asset (security)	CapitaLand Limited shares
MB	Issuer	Macquarie Bank Limited
e	Exercise style	**European** style warrant
CW	Type of warrant	**Call Warrant** (**PW** represents a Put Warrant)
181009	Expiry date (YYMMDD)	Expires **9 October 2018**
COWW	Warrant code or symbol	A unique 4-digit identification code ending with the letter **"W"**, assigned by

Company-issued warrants	Structured warrants
Issued by the company on its **own shares**	Issued by a **third party** such as a financial institution
Less liquidity	**More** liquidity
One direction only — call warrant	**Both call and put** warrants can be issued
Usually **one** exercise price	Can have warrants on the **same underlying** with **different exercise price**
Maturity typically measured in **years**	Shorter-term maturity, usually **6–12 months**
New shares issued upon exercise	**No new shares** issued upon exercise

10.1.3 Basic Features of Warrants

- **Underlying security** — the security that can be purchased on conversion of the warrant.
- **Exercise price** — the price at which the holder can buy the underlying security. Typically, when warrants are issued, the exercise price is set **above** the prevailing share price.
- **Conversion ratio** — the number of warrants required for conversion to **one** share.
- **Expiration date** — the last date a warrant can be exercised.

When warrants are **not** converted into the underlying shares, holders **do not receive dividends**.

10.2 Warrant Valuation

The worked example throughout is **Duty Free International Limited (DF Int'l)**. One DF Int'l warrant converts into **one** DF Int'l share by paying an exercise price of $0.430 * *. On the date the warrants were listed (* * 18 May 2017 * *), the stock traded at * * **0.38**. By **17 January 2018**, the share price had fallen to $0.26 * * and the warrants traded at * * **0.030**.

10.2.1 Intrinsic Value

When the share price is **greater** than the exercise price, the warrant is **in-the-money** and intrinsic value is the difference between the current share price and the exercise price:

$$\text{Intrinsic Value} = \frac{SP - EP}{n}$$

where

$$SP$$

is the share price,

$$EP$$

the exercise price, and

$$n$$

the conversion ratio.

When the share price is **below** the exercise price, the warrant is **out-of-the-money** and intrinsic value is **zero**.

Since issuance, DF Int'l's share price has remained below the **$0.430** exercise price, so intrinsic value is **zero** because the difference is negative.

Worked example. If the share price rises to **$0.48**, the warrant is in-the-money and intrinsic value is:

$$\frac{0.480 - 0.430}{1} = \$0.05$$

10.2.2 Conversion Price

The **total price** the investor must pay to convert warrants into one underlying share:

$$\text{Conversion price} = n \times WP + EP$$

where

$$WP$$

is the warrant price. With a warrant price of **$0.03**:

$$\text{Conversion price} = 0.030 + 0.430 = \$0.46$$

10.2.3 Warrant Price and Warrant Premium

The **warrant premium** is the additional amount spent by purchasing warrants and paying the exercise price to convert to one share, as opposed to buying the share outright in the market. The term "premium" here does **not** carry the same meaning as in conventional options, where the premium is the option price.

$$\text{Warrant premium (\$)} = n \times WP + EP - SP$$

In percentage terms:

$$\text{Warrant Premium (\%)} = \frac{n \times WP + EP - SP}{SP} \times 100$$

Worked example — two scenarios.

	Scenario 1 (17 January 2018)	Scenario 2 (today)
Underlying stock price	\$0.26	\$0.480
Intrinsic value	**Zero** per warrant	$(0.480 - 0.430)/1 =$ **\$0.05** per warrant
Warrant price	\$0.03 per warrant	**\$0.08** per warrant (assumed price rise)
Warrant premium (\$)	$0.030 + 0.430 - 0.26 =$ **\$0.20**	$0.080 + 0.430 - 0.480 =$ **\$0.030**
Warrant premium (%)	$0.20 / 0.26 \times 100 =$ **76.92%**	$0.030 / 0.480 \times 100 =$ **6.25%**

Leverage. Investors are interested in warrants mainly because of the **leverage effect**: the potential for greater percentage gains and losses than the underlying share for a given change in the underlying's price. Comparing the two scenarios, the DF Int'l share price rose **85%** from \$0.26 to \$0.480, but the warrant price rose **167%** from \$0.03 to \$0.08.

10.2.4 Gearing Ratio

The **gearing ratio** compares the price of warrants needed to convert to one share relative to the share price at a point in time. Gearing gives investors the ability to participate in the underlying share's performance at a **lower price**.

$$\text{Gearing ratio} = \frac{SP}{n \times WP} = \frac{0.265}{0.030} = 8.8x \text{ (approximately)}$$

10.2.5 Minimum Value and Maximum Value

Warrants should trade within well-specified price bounds in an **arbitrage-free** market.

For a warrant with a conversion ratio of **1**:

- The **maximum value** is the **underlying share price**. Most warrant prices do not trade at their maximum, and prices fall as warrants approach expiration.
- The **minimum value** is the **intrinsic value**:

$$\text{Minimum value} = SP - EP \text{ if } SP > EP; \qquad = 0 \text{ if } SP \leq EP$$

In the DF Int'l example, with a share price of $0.26 * *and exercise price of * *\mathbf{0.43}$, the share price is below the exercise price and the warrant is **out-of-the-money**. If the share price rises to **\$0.48**, the warrant is **in-the-money**. If the share price **equals** the exercise price, the warrant trades **at-the-money**.

Reading the price bounds. When the share price is below the exercise price, the minimum value is **zero**, represented by the horizontal line. As the share price rises above the exercise price, the minimum price becomes positive and rises.

As the underlying share price increases beyond a point, the **premium of the warrant price over intrinsic value decreases**, because the leverage potential declines — a given share price increase becomes an increasingly smaller percentage of the share price.

The line representing the **maximum price** is at a **45-degree angle** to the x-axis, meaning the maximum price of a warrant equals its underlying share price.

10.3 Factors Affecting the Price of a Company Warrant

Company warrants are typically attached to bonds or rights issues as a sweetener but can be detached and traded independently. Once detached, the price is directly influenced by eight factors.

a. Underlying share price. Warrants are affected by the underlying share price. Unlike structured warrants, which may be call or put, **company warrants are all call warrants**, so they are worth **more** if the underlying share price rises.

b. Life of the warrant. Other things equal, the **longer** before expiry, the **greater** the warrant price. An out-of-the-money warrant may become attractive later as the share price appreciates. As the life shortens, the warrant **loses its speculative value**. On the day of expiration, warrants are worth their **intrinsic value**, after which they are **worthless**.

c. Exercise price. The **lower** the exercise price, the **greater** the warrant price. The ability to convert by paying a lower exercise price increases intrinsic value and hence the warrant's price.

d. Price volatility of the underlying share. The **more volatile** the underlying share, the **higher** the warrant price, because warrants benefit from price upside but limit price downside. Volatility can change over time.

e. Dividend on the underlying share. Because warrant holders are **not entitled to dividends**, the declaration of dividends **reduces** the call warrant's price slightly, since call warrants "lose out".

f. Leverage. Warrants with **greater leverage** opportunities command **higher** prices.

g. Interest rates. The **higher** the interest rate, the **higher** the call warrant price, because of interest savings from the lower capital needed to buy warrants compared with an equivalent long stock position. For completeness, the higher the interest rate, the **lower** the **put** warrant price, because more interest must be paid to buy warrants whereas an equivalent short stock position gains more interest on the short-sale proceeds.

h. Size of warrant issue. The more warrants outstanding, the larger the potential **earnings dilution** of existing shares. If the warrants were deep in-the-money, conversion would increase shares outstanding and reduce the value of each share. There is therefore a **negative** relation between the size of the warrant issue and the warrant price.

Factor	Effect on warrant price
Remaining life of warrant ↑	↑
Exercise price ↑	↓
Underlying share price ↑	↑
Volatility ↑	↑
Dividend expectations ↑	↓
Leverage opportunities ↑	↑
Interest rates ↑	↑
Issue size of warrant ↑	↓

10.4 Summary

- Company warrants are **call** warrants issued only by the company. Both call and put structured warrants are issued by a third party, usually a bank. Company warrants are settled by **delivery of shares**; structured warrants are **cash-settled**, with no delivery of shares.
- A company warrant gives the investor an option to buy a stated number of common shares at a specified price within a specified period. Unlike common shares, warrants **expire**, and warrant holders are **not entitled to dividends**.
- Beyond the fundamental value of the underlying share, the warrant investor must consider **intrinsic value, warrant premium, gearing ratio and expiration date**.
- The **maximum value** of a warrant is the underlying share price. The **minimum value** is the intrinsic value — zero or the difference between the share price and exercise price, whichever is larger.
- The price investors will pay depends on the underlying share price, the life of the warrant, the exercise price, the price volatility of the underlying share, dividend expectations, leverage opportunity, the interest rate level, and the size of the warrant issue.

Exam cues

- **A warrant is a right, not an obligation**, to buy or sell at a fixed price within a specified period. Company warrants are **long-dated equity call options** — maturities of **years**, not months.
- **Warrants are issued as a sweetener** on bonds or rights issues, letting the issuer finance at a **lower interest rate**. There is **no price paid per se** by the buyer of the host security.
- **Exercise dilutes earnings** because **new shares are issued** — unlike equity call options, which cause no dilution.
- **Company vs structured warrants:** company = own shares, less liquid, **call only**, one exercise price, **years**, **new shares issued**. Structured = third party, more liquid, **call and put**, multiple exercise prices on the same underlying, **6–12 months**, **no new shares**.

- **Structured warrant code decomposition:** CapitalaMBeCW181009 = underlying **CapitaLand**, issuer **Macquarie Bank**, **e** = European style, **CW** = call warrant (**PW** = put), **181009** = expiry **9 October 2018**, **COWW** = SGX code ending in **W**.
- **Four basic features:** underlying security, exercise price (set **above** the prevailing share price at issue), conversion ratio (warrants per **one** share), expiration date.
- **Warrant holders receive no dividends.**
- **Intrinsic value**
$$= (SP - EP)/n$$
when in-the-money; **zero** when out-of-the-money. DF Int'l at $0.48 with EP $0.43 and $n = 1 \rightarrow **0.05**$.
- **Conversion price**
$$= n \times WP + EP$$
. DF Int'l: 0.030 + 0.430 = **$0.46**.
- **Warrant premium ($)**
$$= n \times WP + EP - SP$$
; **premium (%)** divides by
$$SP$$
. Scenario 1: 0.20/76.92**0.030 / 6.25%**.
- **Gearing ratio**
$$= SP/(n \times WP)$$
. DF Int'l: 0.265/0.030 = **8.8x**.
- **Leverage worked example:** share price rose **85%** ($0.26 → 0.480) $while the warrant rose **1670.03$ → $0.08).
- **Maximum value = underlying share price** (the 45-degree line). **Minimum value = intrinsic value** (zero or SP − EP).
- **Premium over intrinsic value falls** as the share price rises well above the exercise price, because leverage potential declines.
- **Eight price factors:** underlying share price ↑ → price ↑; life ↑ → ↑; exercise price ↑ → ↓; volatility ↑ → ↑; dividends ↑ → ↓; leverage ↑ → ↑; interest rates ↑ → ↑; issue size ↑ → ↓.
- **Interest rates raise call warrant prices** (interest saved on the smaller capital outlay) but **lower put warrant prices**.
- **Issue size has a negative relation** to warrant price because of potential dilution.
- **On expiry day** a warrant is worth its intrinsic value; afterwards it is **worthless**.
- **Trap:** the "premium" in **warrant premium** is not the option price — it is the extra cost of the warrant route versus buying the share outright.
- **Trap:** company warrants are **call only**. Only structured warrants can be puts.
- **Trap:** a warrant's **maximum** value is the share price, not the intrinsic value — the intrinsic value is the **minimum**.

Chapter 11 — Technical Analysis and Quantitative Analysis

Technical analysis is the alternative to fundamental analysis: it studies past price and volume to predict future direction, and it is not concerned with fair value. This chapter covers its philosophy and assumptions, the challenges to it, the indicators, the chart patterns, Elliott Wave Theory, and the role of quantitative analysis.

Learning Objectives

The candidate should be able to:

- Explain the philosophy and assumptions in the application of technical analysis to investment decisions.
- Discuss the challenges to technical analysis.
- Describe the various technical indicators.
- Distinguish between support and resistance levels.
- Describe share price and volume analysis techniques.
- Explain chart patterns such as head and shoulders, double tops and bottoms, consolidation rectangles, triangles, flags and pennants, wedges and gaps in identifying price trends.
- Explain Elliott Wave Theory as a technical analysis approach.
- Describe the essentials of quantitative analysis.

11.1 Overview

11.1.1 What is Technical Analysis?

Besides fundamental analysis, another approach to security evaluation is **technical analysis**. Though controversial, many traders employ technical trading rules or indicators to make investment decisions.

11.1.2 Basic Philosophy and Assumptions

Technical analysis is the methodology of **forecasting fluctuations in securities prices through an examination of historical market data**. It can be applied to individual securities, the entire market, and any asset class — foreign exchange, gold or bonds.

Technical analysis uses **charts, technical trading rules and indicators**. It is sometimes called **market or internal analysis** because it studies demand and supply conditions based on market data such as price changes and trading volume. It differs from fundamental analysis, which focuses on factors affecting a company's business profitability — economic, industry and company-specific factors.

Technical analysis holds that prices are determined by **demand and supply**, which are affected by both **fundamental** factors (macroeconomic forces) and **non-fundamental** factors (market sentiment and speculative forces). Technical analysts study past price and volume behaviour to identify shifts in supply and demand so as to predict future price behaviour. **They are not concerned about a security's fair value.**

The five underlying assumptions:

- A security's **market value is determined only by its supply and demand**.
- Supply and demand are affected by both **rational and irrational** factors — not only economic variables but also hopes, fears, opinions, moods and guesses of potential buyers and sellers. The market constantly assesses all these factors.
- Prices of individual securities and the overall market **exhibit trends**, which tend to persist over an appreciable length of time.
- Trends develop in reaction to **changes in the supply-demand balance**, and such changes can be detected by studying market actions.
- **Chart patterns tend to repeat themselves.**

Where analysts agree and disagree. Generally, technical and non-technical analysts accept the first two assumptions: at any point in time price is determined by supply and demand, and supply and demand is governed by many variables. The key difference concerns **irrational factors** — technical analysts expect them to **persist**, while fundamental analysts expect only a **temporary** effect, with rational beliefs prevailing in the long run.

Opinions differ on the third assumption, about the **speed of adjustment**. Technical analysts believe new information is priced by the market **over a period of time** rather than at one point, because there are different sources of information and some investors have earlier access. Prices therefore adjust gradually to the gradual flow of information, and because the adjustment is gradual, prices exhibit **trends that persist over long periods**.

The fourth assumption implies that as prices move from one equilibrium to a new one, the change can be detected sooner or later in charts. Technical analysts hope to benefit by being among the **first to detect a shift in trend**.

11.1.3 Challenges to Technical Analysis

The **Efficient Market Hypothesis** presents the greatest challenge. Studies testing weak-form market efficiency generally support the notion that prices adjust **rapidly** to new information, leaving little opportunity for technical analysts to exploit market data.

According to the **random walk theory**, share prices follow a random walk, meaning price movements are **independent over time**. Whether past price data can predict future prices is therefore questionable.

Another criticism: many technical analysts are familiar with chart patterns and act on them **at the same time**. If a popular rule is followed by many participants, some act first, causing prices to change quicker than expected, so only those who act earliest benefit. The popularity and resulting competition **eventually negate the value** of the rule. That said, chart reading is a **very subjective** exercise, and several interpretations of the same pattern are not unusual, so there is always an element of doubt and disagreement.

11.2 Price-based Technical Indicators

Price-based technical indicators attempt to mine information from current and historical market prices to "beat" the market. They typically **measure market momentum and the relative state of the market** — whether it is **overbought or oversold**.

11.2.1 Moving Average

A moving average is the average of a security's closing price over a given period. Typically each day is given the **same weight**: in a 100-day moving average, each day has a weight of **1/100**. A more sophisticated moving average gives **higher weights to more recent prices**. The moving average is **more stable than the daily price** because it includes past prices, making it more useful for detecting trends. If the current market price starts to **diverge directionally** from the moving average, this may indicate a **trend reversal**.

11.2.2 Advance-Decline Line (Breadth Index)

Daily index movements can be misleading because an index is based on selected shares that may not represent the overall market. The **advance-decline (A/D) line**, an index covering **all securities traded**, overcomes this by capturing the **breadth** of the market.

The A/D line measures, on a **cumulative daily basis**, the net difference between the number of shares with price advances and those with price declines in a period (usually a day or a week). The net advance or decline is the number of advances minus the number of declines, and the line plots a **running total** over time.

The A/D line is compared with an index such as the **Straits Times Index (STI)** and normally rises and falls in line with it. If both are rising (declining), the market is **technically strong (weak)**.

- In a **rising** market, any **divergence** between the index and the A/D line is of concern, because it means most shares are not participating — indicating a **market peak**.
- If the A/D line is **rising** while the index is **declining**, it signals a **reversal of the falling trend**.

11.2.3 Moving Average Convergence Divergence (MACD)

MACD is a **trend-following momentum indicator** showing the relationship between two moving averages of a security's price. It reveals changes in the **strength, direction, momentum and duration** of a trend.

The MACD indicator is a collection of **three time series** calculated from historical price data, most often the closing price:

- The **MACD series** is the difference between a **"fast"** (short-period) **exponential moving average (EMA)** and a **"slow"** (longer-period) EMA of the price series.
- The **signal or average series** is an EMA of the MACD series itself.

An exponentially weighted moving average reacts **more significantly to recent price changes** than a simple moving average (SMA), which applies equal weight to all observations.

The indicator therefore depends on **three time parameters** — the time constants of the three EMAs. The notation **MACD (a,b,c)** denotes the indicator where the MACD series is the difference of EMAs with characteristic times

$$a$$

and

$$b$$

, and the average series is an EMA of the MACD series with characteristic time

$$c$$

. Parameters are usually measured in **days**, and the most common values are **12, 26 and 9** — MACD (12,26,9).

A fast EMA responds more quickly than a slow EMA to recent price changes. By comparing EMAs of different periods, the MACD series indicates changes in trend:

- When the MACD **crosses and falls below** the signal line, it is a **bearish** signal — possibly time to sell.

- When the MACD **crosses and rises above** the signal line, it is a **bullish** signal, suggesting upward momentum.

11.2.5 Bollinger Bands

Bollinger Bands are a type of **price envelope** developed by **John Bollinger**. Price envelopes define upper and lower price range levels. Bollinger Bands are plotted at a **standard deviation level above and below a simple moving average**. Because the distance is based on standard deviation, the bands **adjust to volatility swings** in the underlying price.

Calculation:

- Compute the **simple moving average**, typically a **20-day SMA**. The first data point averages the first 20 closing prices; the next drops the earliest price, adds day 21 and averages, keeping the data set at 20.
- Obtain the **standard deviation** of the price over that data set.
- Multiply the standard deviation by **two** (a commonly used multiplier) and both **add and subtract** that amount from each point along the SMA, producing the upper and lower bands.

The purpose is to provide a **relative definition of high and low prices**. By definition prices are high at the upper band and low at the lower band. Many traders believe the closer prices move to the **upper** band, the more **overbought** the market; the closer to the **lower** band, the more **oversold**. An overbought or oversold market may be primed for a **trend reversal or price correction**.

Individual indicators are typically **not used in isolation** but in conjunction with others. Moving averages and the relative strength index are also common.

11.3 Trends, Support and Resistance Levels

Technical analysts rely on **trend lines** and **trend channels** to identify trends.

- An **uptrend** is a rising trend line connecting a series of **ascending lows**.
- A **downtrend** is a falling trend line connecting a series of **descending highs**.

An uptrend is in place when **current highs are above previous highs and current lows are also above previous lows**. When current highs are no longer above previous highs and, more importantly, current lows fall below previous lows, a **trend reversal** may be occurring.

A downtrend occurs when current lows are lower than previous lows and current peaks are lower than previous peaks. A potential change may occur when this pattern is no longer intact.

The trading range between the top and bottom trend lines is the **trend channel**, which may be rising, flat or declining. If share prices stay within a rising channel, the technician would **hold the share and ride the trend**.

Support and resistance.

- A **support level** is a price range where a **substantial increase in demand** is expected. At this level prices are relatively attractive to buyers who missed the opportunity earlier.
- As prices increase, a **resistance level** develops where buyers refuse to pay higher prices, while sellers may be attracted by higher prices to take profits. This supply is the market **"overhang"**.

A movement through and beyond either level is a **breakout**, signalling strong confirmation and continuation of the primary trend. A breakout **above resistance** is **bullish**; a breakout **below support** is **bearish**.

Technical analysts are also alert to **reversals**. A penetration below a support line is a **sell signal**, indicating a reversal of the uptrend. In the standard illustration, the move to point **C** creates a new low past the previous low at **A**, and the move to **D** is lower than the previous high at **B** — confirming the trend has reversed downwards.

11.4 Significance of Volume

Volume — the number of securities transacted — shows the **intensity of excess demand or supply** at a point in time and reflects changes in investor attitudes.

- If prices advance on **low** volume, the enthusiasm implied is **not as strong** as when a rise is accompanied by very high volume.

- A price increase with **heavy** volume relative to normal is an indication of **bullish** activity.
- A price decline with **heavy** volume is **very bearish**, reflecting a strong and widespread desire to sell.
- In a sustained bull market, if prices **correct downward on declining volume**, this may indicate a **lull** followed by continuation of the rally.
- A rally that develops on **contracting** volume could signal a possible **trend reversal**.

11.5 Share Price and Volume Techniques

Three techniques: the Dow Theory, the Moving Average Line, and the Relative Strength Index.

11.5.1 The Dow Theory

Named after **Charles Dow**, the oldest and most popular method for identifying **long-term trends** in the equity market. It is based on the movements of the **Dow Jones Industrial Average (DJIA)** and the **Dow Jones Transportation Average (DJTA)**.

Three movements:

Movement	Duration	Character
Primary	Several months to several years	The market trend, either **bullish** or **bearish**. Historically, **bear**
Secondary	Several weeks to months	Shorter than the primary and **opposite in direction**. Usually r
Daily fluctuations	Day to day	Considered **random**, with no forecasting value for long-term inve

Dow Theory forecasts are based on the **primary and secondary** movements. Its purpose is to determine **where the market is and where it is going, but not the magnitude** of the movements.

In practice: if the swings of the market index are **successively higher** and the successive lows are also higher, the trend is **up** and investors are in a bull market. If successive highs and successive lows are **lower**, the direction is **down** and investors are in a bear market.

Confirmation. A new primary trend is confirmed only when **both the DJIA and DJTA give the same signals**. When the two averages **diverge**, the Dow Theory assumes the **prior trend is maintained**.

Five weaknesses:

- It relies too heavily on the **DJIA**, which contains a limited selection of shares — old, large "blue chip" companies, with aggressive high-tech companies **under-represented**.
- It is **slow** in identifying a major trend and may miss an important price move or be **"whiplashed"** — caught by a short-term reversal even though the long-term trend is intact but not yet visible.
- It uses the **Transportation Average**, comprising mainly railway shares that no longer have a major impact on the economy.
- It is designed to identify the **major** trend and is therefore **not useful for trading the short trend**.
- It **does not help identify which shares to buy or sell**.

11.5.2 Moving Average Line

Moving averages smooth out erratic price changes so the underlying trend can be clearly observed, and are used to **confirm the direction** of price trends. Popular averages are the **200-day** and **50-day**; the 200-day identifies **long-term** trends while a **20- to 30-day** average identifies **short- and intermediate-term** trends.

Trend changes are identified not by a change in the direction of the moving average, but by a crossover between the moving average and the index itself.

- If the overall trend has been negative, the moving-average line lies **above** current prices. If the price **reverses and breaks upwards through the moving-average line from below**, accompanied by **heavy volume**, most technical analysts consider this a very positive change and expect a **reversal of the downtrend**.
- In an uptrend the moving-average line is rising but **below** current prices. If prices **break downwards through the line from above** with heavy volume, this is a **bearish** pattern signalling a reversal of the long-run uptrend.

Because the moving average is an average of past prices, if prices keep rising the line also rises but at a **slower rate** than current prices. If current prices (which lead the averages) fall, the line follows the downtrend. The price line therefore naturally cuts the moving average when a trend reversal occurs.

The signal is **stronger** if:

- The price cuts **both** short- and long-term moving averages, and
- The **short-term average cuts the long-term average in the same direction**.

11.5.3 Relative Strength Index (RSI)

The RSI is a **momentum indicator** measuring the degree of buying or selling interest in a security. Developed by **J. Welles Wilder**, it measures the **speed of change** of price movements.

$$RSI = 100 - \left[\frac{100}{1 + \dfrac{\text{Average of upward price change}}{\text{Average of downward price change}}} \right]$$

The index ranges from **zero to 100**. An RSI **above 70** is considered **overbought**; below **30** signals an **oversold** condition.

A better way to use the RSI is to track the **past peaks and troughs** of the underlying and examine the typical RSI levels for that market and security, to calibrate the overbought and oversold levels.

The RSI is useful in situations of **surprises and shocks**:

- **Emotional selling** under unwarranted conditions produces oversold conditions. For the contrarian, a low RSI of **20–30 or lower** could signal a **buying opportunity**.
- In situations of **overly optimistic demand**, a strong RSI of **70–80 or higher** could signal a **profit-taking opportunity**.

The RSI is **less useful when trends are steady and consistent**. Where constant demand for a growth stock keeps its RSI persistently high, the level should **not** be read as a sell signal; instead look for an opportunity to accumulate when the RSI is not as high.

In practice, periods of overselling and overbuying can last **one to two weeks**, during which prices can continue to rise or fall. Technical analysts therefore usually **wait for a few days** of overbuying before recommending profit-taking, and of overselling before recommending accumulation.

11.6 Charts and Chart Patterns

Charts are the technical analyst's main tool. Types include:

- **Line charts.**
- **Bar charts.**
- **Candlestick charts.**
- **Point and figure charts.**

Most show the **high, low, opening and closing** prices for each trading day.

There are two types of price pattern:

- **Reversal patterns** indicate that an important reversal in trend is occurring.
- **Continuation patterns** suggest the market is only taking a pause, possibly to correct a short-term overbought or oversold condition, after which the existing trend is expected to resume.

11.6.1 Head and Shoulders

The head-and-shoulders pattern is a **reversal** pattern and probably one of the more reliable indicators of major reversals. The **upright** formation occurs at **market peaks** and consists of a left shoulder, a head and a right shoulder.

Component	Description
Left shoulder	First strong price rally, normally accompanied by **heavy volume**, followed by a decline on **considera**

Component	Description
Head	Prices advance again on **high volume** to a level **higher than the top of the left shoulder**, then fa
Right shoulder	A third rally on **less volume** than during either the left shoulder or the head. Prices **fail to reach th**

A breakout is confirmed if the price **closes below the neckline by 3% of the share price**. Once penetrated, the potential magnitude of decline is measured by the **distance between the head and the neckline, projected downward from the neckline**.

Formation time varies: with rapid price reactions it can take **a few weeks**; otherwise **a few months or years**. Formation tends to be quicker for **currencies** and slower for **commodities**.

Inverse head and shoulders indicates a **market bottom**.

Component	Description
Left shoulder	Begins with a significant **selloff on high volume**, followed by a brief rally, sometimes supported by s
Head	A second wave of selling as participants in the initial rally give up their positions. Heavy volume drive
Right shoulder	Profit-taking by shorter-term traders and those who did not sell earlier. This sell-off is **lighter and sh**

When demand picks up and price breaks past the neckline, **volume typically rises**. Once fully formed, the pattern implies the prior downtrend is over and an **uptrend reversal** is likely.

11.6.2 Double and Triple Tops and Bottoms

A **double top** is a **reversal** pattern indicating a **market top**. It consists of two tops separated by a valley. The main feature is that the **second top forms with distinctly lower volume** than the first. A double top is **confirmed when the neckline is broken on the way down**.

There can be more tops rising to a resistance level, resulting in a selloff. Once resistance is reached, the **volume of each subsequent rise falls** and the number of buyers pushing prices toward the peak dwindles, forcing a correction as buying interest is overwhelmed by selling.

A **double bottom** occurs in a **bear market** and signals a potential reversal. The **first bottom** is normally accompanied by **heavy volume**, while the **second bottom is usually dull** with little volume. **Volume increases significantly on the breakout.** Some patterns extend to **triple** tops or bottoms. As with the double top, a double bottom is **confirmed if the neckline is broken on the way up**.

11.6.3 Consolidation Rectangles

A consolidation rectangle forms when prices move **sideways within a narrow range** for an extended period, usually as a **continuation** pattern. A breakout is technically significant: **upwards** indicates a bullish trend, **downwards** signals further weakness.

During formation it is **difficult to predict** which way price will break, so it is assumed the **prevailing trend continues until proven otherwise**.

11.6.4 Triangles

Triangles form with **four points of contact** — two support and two resistance points. Three types:

Type	Boundaries	Signal
Symmetrical	Upward-sloping support line and downward-sloping resistance line	A breakout may occur on **eit**
Ascending	Upward-sloping support line and an **almost horizontal** resistance line	**Bullish.** Buyers are more ag
Descending	Almost horizontal support line and a **downward-sloping** resistance line	**Bearish.** Sellers are more ag

11.6.5 Flags and Pennants

A **flag** looks like a flag on the chart — a **parallelogram** of price fluctuations. It usually occurs after a **sharp, almost vertical** rise or fall, followed by **declining volume**. As the flag completes, prices **break out in the same direction** they were moving before.

- In a **rising** market the flag usually forms with a **slight downtrend**. Volume is **very heavy just before** formation, then **dries up** as the pattern develops, only to **explode** as price works out of the formation. Flags can take as few as **5 days** to as many as **3–5 weeks**. They represent a period of **controlled profit taking** in a rising market.
- In a **falling** market the flag usually forms with a **slight upward bias**, also on declining volume. Rising prices with declining volume is essentially **bearish**. When price breaks out downwards, the almost vertical slide resumes. Volume tends to pick up but **does not have to be explosive**, unlike the breakout from a flag in an upward move.

A **pennant** forms after a sharp rise or fall with lower volume during formation. The difference from a flag is that a pennant is formed by a set of **converging**, instead of parallel, lines.

11.6.6 Wedges

A **wedge** is formed by a series of **converging peaks and troughs**, with both converging lines moving in the **same direction**.

- A **rising wedge** has both boundary lines slanting **upwards**. It usually develops on a **falling** price trend. After completion, prices usually **fall very sharply**, especially if volume is high. Rising wedges are characteristic of **bear market rallies**.
- A **falling wedge** is bounded by converging lines slanting **downwards**. It represents a **temporary interruption of a rising trend**. When prices move out of a falling wedge they are likely to **drift sideways** before beginning to rise.

11.6.7 Gaps

A **gap** is a price range where **no transactions occur**. Gaps occur when the lowest (highest) traded price for a period is higher (lower) than the highest (lowest) traded price in the previous period. On a chart it appears as an **empty vertical space** between adjacent trading periods. **Daily gaps are more common** than weekly or monthly gaps.

A gap is **closed or covered** when price retraces the entire range of the gap. A gap may be covered in a few days, weeks or months, **if at all**. Although the general principle is that gaps will eventually be covered, this is **not always** the case, so trading strategies should not be implemented solely on the assumption that a gap will be filled in the immediate future.

Gap type	Where it appears	Chara
Breakaway	When price breaks out of a price pattern	Emph
Runaway (continuation)	During rapid, straight-line rises or falls when trends accelerate and emotions run high	Occu
Exhaustion	The **last** in a series of runaway gaps	Repre

Generally: **breakaway gaps develop at the beginning** of a move, **runaway gaps in the middle**, and **exhaustion gaps at the end**.

11.7 Elliot Wave Theory

In his study of equity prices in **1938**, **R.N. Elliott** proposed the **Elliott Wave Theory**: prices move in **cycles of waves and sub-sets of waves** and exhibit the **Golden Ratio, 1.618**. Each cycle comprises a **5-wave advance and a 3-wave retreat**.

In Elliott Wave charts, markets move in 5-wave and 3-wave patterns, and the turning points creating these patterns are often at **Fibonacci support and resistance levels** (61.8, 161.8, and so on).

The Fibonacci series is **0, 1, 1, 2, 3, 5, 8, 13, 21, ...**, where each number is the sum of the two preceding numbers. As the series progresses, the ratio of consecutive numbers (for example 21 ÷ 13) approaches the **Golden Ratio of 1.618**.

11.8 Quantitative Analysis

Quantitative analysis involves the **abstraction of market reality** — the economy, financial markets and their environment — into relations incorporated into a **financial model**. The objectives are to identify

price anomalies and arbitrage possibilities and, lately, to increase the **speed of access to market information** that gives an advantage.

Its significance is evidenced by the rapid growth in the use and development of quantitative techniques. Quantitative analysis uses **statistics and calculus** for pricing, algorithmic trading, asset allocation, risk management, trading strategy modelling, portfolio optimisation, and identifying trading and investment opportunities.

Quantitative analysts are typically called **"quants"** or **"rocket scientists"**. Within an organisation they may work in the **front office** developing trading strategies with complex mathematical tools, or in **risk management** measuring the risks bundled into a transaction or product.

Quant funds extensively use quantitative analysis in fund management decisions. This can be seen as an outcome of markets becoming more sophisticated in understanding and measuring risk. While less sophisticated traders look only at price changes, quants **decompose** those changes so specific risk profiles can be created to benefit from them: instead of looking only at the **first order** of price change (delta), quants look at **second and third order derivatives**.

There is controversy about the usefulness and fairness of such models. **Long Term Capital Management** was a hedge fund relying extensively on quantitative analysis; despite the elegance and apparent mathematical robustness of its strategies, it incurred **huge losses**.

High frequency trading, which relies on physical transmission speeds and complex price discovery algorithms, is under **intense scrutiny from regulators**.

Given rapid advances in computing and increasing global competition, quantitative analysis will continue to develop as a tool necessary to enhance industry standards in fund and risk management.

11.9 Summary

- Technical analysis, also called market or internal analysis, studies past prices and volume to predict future price direction for individual securities or the aggregate market.
- It assumes market prices discount everything and prices move in trends, and that shifts in trends can be detected by studying charts.
- The greatest challenge is the **Efficient Market Hypothesis**, which argues prices adjust rapidly to new information, leaving little opportunity. Popular technical rules are followed by many participants, and the resulting competition eventually negates their effectiveness.
- Indicators include the **A/D line** (breadth index), which measures the aggregate movement of all securities traded, and **moving averages**, which smooth erratic price changes so the trend is clearer.
- **MACD** is a trend-following momentum indicator showing the relationship between two moving averages. **Bollinger Bands** are envelopes plotted at a standard deviation level above and below a simple moving average; closer to the upper band is more overbought, closer to the lower band more oversold.
- Under the **Dow Theory** there are three movements: primary, secondary and daily fluctuations. Its objective is to identify the **primary trend**.
- The **moving average line** smooths erratic movements, and **crossovers** between the moving average and the index identify trend reversals.
- The **RSI** identifies overbought or oversold securities.
- Chart types include line, bar, candlestick, and point and figure.
- There are two types of chart pattern: **reversal** (head and shoulders, double tops and bottoms) and **continuation** (consolidation rectangles, triangles, flags, pennants, wedges).

Exam cues

- **Technical analysis is not concerned with fair value.** It studies supply and demand through price and volume.
- **Five assumptions:** price determined only by supply and demand; supply and demand affected by rational **and irrational** factors; prices exhibit **persistent trends**; trends develop from changes in the supply-demand balance and can be detected; **chart patterns repeat**.
- **The key disagreement** with fundamental analysts is the **persistence of irrational factors** and the **speed of adjustment** — technical analysts believe information is priced **gradually**, producing trends.
- **Greatest challenge: the EMH.** Weak-form efficiency and the **random walk theory** both undermine technical analysis. Popular rules are self-defeating because many act at once.

- **Moving average:** equal weights by default; more stable than daily price; **divergence** from price may signal a reversal.
- **A/D line (breadth index):** cumulative advances minus declines across **all** securities. Index rising with A/D falling = **market peak** warning; A/D rising with index falling = **reversal of the downtrend**.
- **MACD:** three EMAs; **MACD (12,26,9)**; MACD crossing **below** the signal line is **bearish**, **above** is **bullish**.
- **Bollinger Bands: 20-day SMA ± 2 standard deviations**. Near the upper band = **overbought**; near the lower band = **oversold**.
- **Uptrend:** current highs above previous highs **and** current lows above previous lows. **Downtrend:** the reverse.
- **Support** = substantial increase in demand expected. **Resistance** = buyers refuse higher prices while sellers take profits — the **overhang**. Breakout **above resistance is bullish**; **below support is bearish**.
- **Volume:** price rise on heavy volume = **bullish**; price decline on heavy volume = **very bearish**; a correction on declining volume in a bull market may be a **lull**; a rally on contracting volume may signal **reversal**.
- **Dow Theory:** three movements — **primary** (months to years), **secondary** (weeks to months, **opposite** direction, retraces **one-third to two-thirds**), **daily** (random). **Bear markets last a shorter period than bull markets.** Confirmation requires **both DJIA and DJTA** to agree; divergence means the prior trend is maintained.
- **Dow Theory weaknesses:** over-reliance on the **DJIA** (blue chips, high-tech under-represented), **slow** and prone to **whiplash**, the **Transportation Average** is railway-dominated, useless for **short-term** trading, and it does not say **which shares** to trade.
- **Moving average crossovers:** price breaking **up** through the line from below on heavy volume = **bullish reversal**; price breaking **down** through the line from above on heavy volume = **bearish reversal**. Stronger if the price cuts **both** short- and long-term averages and the short-term average crosses the long-term in the same direction.
- **RSI:** ranges **0**–**100**; **above 70 overbought**, **below 30 oversold**. Contrarians buy at **20**–**30** and take profit at **70**–**80**. Less useful in steady trends.
- **Head and shoulders: reversal** at a **peak**. Left shoulder on heavy volume, head on high volume to a higher level, right shoulder on **less** volume failing to reach the head. Breakout confirmed by a close **3% below the neckline**; the target decline equals the **head-to-neckline distance** projected down.
- **Inverse head and shoulders: reversal** at a **bottom**; the **RSI typically shows oversold** at the head; the right shoulder is a **higher low**.
- **Double top:** second top on **lower volume**; confirmed when the neckline breaks **downwards**. **Double bottom:** first bottom on **heavy volume**, second **dull**, volume **increases on the breakout**; confirmed when the neckline breaks **upwards**.
- **Consolidation rectangle:** sideways range, **continuation**; assume the prevailing trend continues until proven otherwise.
- **Triangles: symmetrical** (breakout either way), **ascending** (bullish), **descending** (bearish). Four points of contact.
- **Flags and pennants: continuation** after a sharp move. Flag = **parallel** lines, **5 days to 3**–**5 weeks**; pennant = **converging** lines. In a rising market the flag slopes **down**; in a falling market it slopes **up**.
- **Wedges: converging** lines both moving the same direction. **Rising wedge** appears in a **falling** trend and is followed by a sharp fall — characteristic of **bear market rallies**. **Falling wedge** interrupts a **rising** trend.
- **Gaps: breakaway** at the beginning, **runaway** in the middle, **exhaustion** at the end. Gaps are **not always** filled, so do not trade on that assumption alone.
- **Elliott Wave: R.N. Elliott, 1938**; **5-wave advance and 3-wave retreat**; **Golden Ratio 1.618**; turning points at **Fibonacci** levels (61.8, 161.8).
- **Quantitative analysis:** abstracts market reality into models to find **price anomalies and arbitrage**; uses statistics and calculus; quants look at **second and third order** derivatives, not just delta.
- **LTCM** is the cautionary example of quantitative models failing despite mathematical elegance.
- **Trap:** the Dow Theory identifies **direction, not magnitude**.
- **Trap:** a **rising wedge** is bearish and a **falling wedge** is a continuation of an uptrend — the names describe the shape, not the implication.
- **Trap:** an **ascending triangle** is bullish and a **descending triangle** is bearish, but a **symmetrical triangle** can break either way.

- **Trap:** the RSI being persistently high in a steady uptrend is **not** a sell signal.

Chapter 12 — Case Studies

The examination includes **3 case studies carrying 9 questions** — 3 questions per case study, about a tenth of the paper. The ten case studies in the official guide are the model for those questions: each sets a named client or scenario, then tests one chapter's method applied to it. This chapter works through all ten, with the arithmetic shown.

Learning Objectives

The candidate should be able to:

- Apply the investor-classification framework to a named client.
- Compute expected return, standard deviation, coefficient of variation and real future value.
- Read economic indicators and business-cycle phases to select asset classes.
- Attribute a price fall to the correct behavioural driver.
- Compute a carry trade's gain under three exchange-rate scenarios.
- Compute prospective P/E, EPS growth and PEG, and act on the result.
- Explain the DCF steps and compute enterprise value.
- Read bond prices, yields and duration to judge interest-rate sensitivity.
- Match a strategic asset allocation to a client's objectives and constraints.
- Compute warrant intrinsic value, minimum gain and read an FX quote.
- Distinguish unit trusts, business trusts and REITs, and select the vehicle that fits an income objective.

12.1 Individual versus Institutional Investor

Scenario. John Lim is moving from stockbroking to a private bank managing accredited investors. In his current job he works mostly with institutional investors, so he must adjust to the differences in client type. He asks you, as his soon-to-be colleague, to explain the key differences.

Question 1 — Which of the following is an accredited investor? (Select all that apply)

- a. An entity with S$5 million worth of net assets.
- b. A university lecturer with $500,000 in net personal assets.
- c. An accountant that earns S$400,000 a year.
- d. A university endowment trust that has S$6 billion in investible funds.

Answer: c and d.

Options (a) and (b) are incorrect because neither the individual nor the entity qualifies. Option (c) is an accredited investor because the individual earns **more than S**$300,000 ** a year. Option (d) is an accredited investor because the entity has net assets * exceeding S$**10 million** in value.

Question 2 — Which of the following is not likely to be a return objective of an institutional investor?

- a. Current income to meet recurring liquidity needs.
- b. Long-term capital gain to beat inflation.
- c. Potential for higher yields.
- d. Regular savings to fund children's education.

Answer: d.

Options (a) to (c) can be return objectives of both individual and institutional investors. Option (d) is likely to be **unique to an individual investor**.

Question 3 — Which of the following shows the key differences between an accredited investor and a retail investor? (Select all that apply)

- a. A retail investor is likely to be more risk averse than an accredited investor.
- b. The accredited investor can withstand a higher level of monetary risk than the retail investor.
- c. A retail investor has a longer investment time horizon than an accredited investor.

- d. An accredited investor is able to access a wider range of investment products than a retail investor.

Answer: a, b and d.

Option (a) is correct: retail investors usually trade in small amounts and less frequently, and have little influence over corporate decisions or the capital markets. Option (b) is correct: given their wealth, accredited investors are considered financially savvy and better able to understand financial instruments, so they can withstand more monetary risk. Option (c) is **incorrect** because the investment time horizon depends on the **needs of the investor, not the type**. Option (d) is correct: accredited investors may access more complex and riskier products such as **unrated bonds** and products requiring neither a prospectus nor a reasonable basis for recommendation by a CMI representative.

12.2 Risk, Return and Time Value Calculations

Scenario. Edna Wong has just turned **35** and wants to set aside **$200,000 for 30 years** in an investment with the following annual returns under different economic scenarios. The average long-term inflation rate is expected to be **2%**.

State of economy	Probability	Return
Boom	0.3	12%
Normal	0.5	8%
Recession	0.2	−6%

Question 1 — The expected rate of return is:

- a. 5.30%
- b. 6.40%
- c. 8.45%
- d. 9.28%

Answer: b.

The expected return is the probability-weighted average of the returns across economic states:

$$E(R) = 0.3(0.12) + 0.5(0.08) + 0.2(-0.06) = 0.0640 = 6.40\%$$

Question 2 — The coefficient of variation associated with this investment is:

- a. 0.47
- b. 0.68
- c. 1.01
- d. 1.33

Answer: c.

Variance is the probability-weighted average of the squared deviations from the mean:

$$\sigma^2 = 0.3(0.12 - 0.064)^2 + 0.5(0.08 - 0.064)^2 + 0.2(-0.06 - 0.064)^2 = 0.004144$$

$$\sigma = \sqrt{0.004144} = 0.064374 = 6.4374\%$$

$$CV = \frac{0.064374}{0.064} = 1.01$$

Question 3 — Assuming the expected return is the average return for the next thirty years, the size of Edna's nest egg when she turns 65, in today's dollars, is:

- a. $709,264.72
- b. $1,610,176.77
- c. $2,279,904.66

- d. $2,865,831.06

Answer: a.

With expected inflation of 2%, the real rate of return is:

$$\frac{1+0.064}{1+0.02} - 1 = 0.0431 = 4.31\%$$

Future value in real terms:

$$200{,}000 \times (1.0431)^{30} = \$709{,}264.72$$

12.3 Key Drivers of Market Movement and Assets

Scenario. You are part of a team producing the market outlook for the US, China and Southeast Asia over the next 12 months. Your boss asks for a standard list of items to cover, forming an outline of the key topics the research team should provide for the sales force and clients.

Question 1 — One key concern of your US clients is the timing of a future economic downturn. Which of the following suggests a potential downturn?

- a. Average duration of unemployment is increasing.
- b. The 10-year T-bond spread over the Fed Funds rate is increasing.
- c. Disposable personal incomes reported by the Bureau of Economic Analysis start to rise.
- d. New orders by suppliers start to decline.

Answer: d.

Option (a) is a **lagging** indicator, so it is incorrect. Option (c) is a **coincident** indicator and its direction is inconsistent with a downturn. Option (b) is a leading indicator, but an **increase** in the spread generally bodes **well** for economic health. Option (d) is correct because suppliers normally cut orders when they expect demand to drop.

Question 2 — Strategists postulate a slowdown in the Chinese economy occurring with rising interest rates. Which asset classes are likely to do well?

- a. Cash, real estate
- b. Defensive stocks, investment grade bonds
- c. Cyclical stocks, high-yield bonds
- d. Growth stocks, precious metals

Answer: b.

A slowdown with rising interest rates is characteristic of the **late bull to early bear** phase. Defensive stocks and investment grade bonds are likely to do well.

Option (a) is incorrect: although raising cash may be wise, holding real estate is risky because it is typically **leveraged** and rates are rising. Option (c) is possibly the worst: cyclical stocks perform worse than the market when things turn down, and high-yield issuers may default. Option (d) is incorrect: some precious metals allocation may help in uncertainty, but growth stocks are less likely to meet growth expectations and may see **P/E multiple contraction**.

Question 3 — Uninformed fund managers reduced positions out of fear because news spread that many were doing likewise. Even though the scenario was unlikely, massive selling triggered computer-driven strategies. What is the root cause of the price fall?

- a. Herd instinct
- b. Loss aversion
- c. Cognitive bias
- d. Algorithmic trading

Answer: a.

The managers simply followed the crowd without exercising independent decision-making or due diligence — **herd instinct**. Loss aversion may have influenced them and algorithmic trading may have **exacerbated** the fall, but neither provided the **original impetus**. There is insufficient information about how the original discussants formed their conclusions to attribute the fall to cognitive bias.

12.4 Foreign Exchange Carry Trade

Scenario. An investor is tempted by **IDR 100 billion** of Indonesian government bonds. IDR-denominated 1-year government bonds pay **6%**, while Singapore banks charge **1.5%** for investment-related loans. SGD/IDR stands at **10,076.68**. Assess three exchange-rate scenarios.

Question 1 — Ignoring bond price changes, what is the annual gain in SGD if the exchange rate stays flat?

- a. SGD 148,858.55
- b. SGD 446,575.66
- c. SGD 595,434.21
- d. SGD 9,923,903.51

Answer: b.

With a flat rate there is no currency gain or loss, so work in SGD. The IDR 100bn purchase is funded by an SGD-equivalent loan:

$$\frac{100{,}000{,}000{,}000}{10{,}076.68} = \text{SGD } 9{,}923{,}903.51$$

The investor earns 6% on the bonds and pays 1.5% on the loan, so the annual gain is:

$$(0.06 - 0.015) \times 9{,}923{,}903.51 = \text{SGD } 446{,}575.66$$

Question 2 — What is the total gain if the IDR weakens against the SGD by 10% over the year?

- a. −SGD 605,357.18
- b. −SGD 942,771.98
- c. SGD 605,357.18
- d. SGD 942,771.98

Answer: a.

The loan is **SGD 9,923,903.51**, so annual interest expense is:

$$0.015 \times 9{,}923{,}903.51 = \text{SGD } 148{,}858.55$$

Interest income is **IDR 6 billion** (6% of IDR 100bn). With the IDR weakening 10%, the conversion rate is:

$$\frac{10{,}076.68}{0.9} = 11{,}196.31$$

SGD-equivalent interest income:

$$\frac{6{,}000{,}000{,}000}{11{,}196.31} = \text{SGD } 535{,}890.84$$

The face value of the bonds converts back at 11,196.31:

$$\frac{100{,}000{,}000{,}000}{11{,}196.31} = \text{SGD } 8{,}931{,}514.04$$

Total gain:

$$(8{,}931{,}514.04 - 9{,}923{,}903.51) + (535{,}890.84 - 148{,}858.55) = -\text{SGD } 605{,}357.18$$

The interest income exceeds the borrowing expense, but the unfavourable FX movement has depreciated both the investment and the income. **The carry trade is not profitable.**

Question 3 — If in a year SGD/USD is 0.7 and USD/IDR is 13,086.60, how does the gain compare with Question 1?

- a. Lower gain
- b. Same gain
- c. Higher gain
- d. Insufficient information

Answer: c.

The SGD/IDR cross rate is:

$$0.7 \times 13{,}086.60 = 9{,}160.62$$

Since **9,160.62 is lower than 10,076.68**, the IDR has **appreciated** against the SGD. The IDR-denominated investment and income appreciate in value, so the gain is **higher** than the SGD 446,575.66 in Question 1.

12.5 Equities Analysis

Scenario. Mr Quek has been recommended **Mvidia**, a semiconductor company with strong earnings growth. Demand for semiconductors in the PC industry remains sluggish, but companies focused on **high-performance computing (HPC)** — cloud computing, big data, artificial intelligence — are better positioned for strong growth. He has also heard that Mvidia's P/E is too high. Mvidia's stock price is **USD 40**.

Year	EPS	EPS change %
FY18A	1.12	−5.0
FY19E	1.08	−3.6
FY20F	2.57	+138.0
FY21F	4.19	+63.0
FY22F	4.67	+11.4
FY23F	5.81	+24.4

Question 1 — What is the prospective P/E for Mvidia and why?

- a. The high ratio of 35.71 is likely because the market has overvalued Mvidia.
- b. The high ratio of 35.71 is likely because the market is factoring in Mvidia's growth potential.
- c. The high ratio of 37.04 is likely because the market has overvalued Mvidia.
- d. The high ratio of 37.04 is likely because the market is factoring in Mvidia's growth potential.

Answer: d.

$$\text{Prospective P/E} = \frac{40}{1.08} = 37.04x$$

P/E ratios can be high because of strong expected future growth, so the high prospective P/E is likely because investors are factoring in Mvidia's **high double-digit long-term growth potential**.

Question 2 — What is the 5-year compounded annualised EPS growth rate?

- a. 31.57%
- b. 38.99%
- c. 40.01%
- d. 52.30%

Answer: b.

$$\text{5-year EPS growth} = \left[\left(\frac{5.81}{1.12}\right)^{1/5} - 1\right] \times 100\% = 38.99\%$$

Question 3 — What recommendation would you give using Mvidia's prospective PEG ratio?

- a. Buy Mvidia given the PEG ratio of 0.92.
- b. Sell Mvidia given the PEG ratio of 0.92.
- c. Buy Mvidia given the PEG ratio of 0.95.
- d. Sell Mvidia given the PEG ratio of 0.95.

Answer: c.

$$\text{Prospective PEG} = \frac{37.04}{38.99} = 0.95$$

Because the PEG is **less than 1**, Mvidia is **slightly undervalued** even after accounting for its strong expected growth, so Mr Quek can consider **buying** the stock.

12.6 Estimating the Fair Value of a Stock

Scenario. A new client wants a brief explanation of the steps your firm's research team undertakes when estimating a company's fair value.

Question 1 — Free cash flows to a levered firm should be discounted using the firm's:

- a. Cost of ordinary equity only.
- b. Cost of preferred equity only.
- c. Cost of debt only.
- d. Weighted average cost of capital.

Answer: d.

The appropriate discount rate must account for capital from **all sources** for which a return is required. Option (a) would be correct only if the firm were **all-equity funded**, which it is not because it is levered. Options (b) and (c) are incorrect under any circumstance, since a firm must have ordinary equity.

Question 2 — Which of the following is NOT part of estimating free cash flows to the firm (FCFF)?

- a. Interviewing company management on its revenue growth potential.
- b. Consulting industry specialists on expected profitability margins.
- c. Estimating the amounts of future fixed asset investments.
- d. Analysing historical ratios to make interest expense projections.

Answer: d.

Revenue growth potential and profitability margins are needed to derive **EBIT**, a key FCFF input, so (a) and (b) are part of the estimate. Capital expenditure is another key input, so (c) is part of it. **Interest expense is excluded** from the FCFF computation, so (d) is not part of it.

Question 3 — Enterprise value can be calculated using _____. (Select all that apply)

- a. Sum of the present values of the FCFFs for the projection period plus the terminal value.
- b. Sum of the present values of the FCFFs for the projection period plus the present value of the terminal value.

- c. Sum of the FCFFs for the projection period plus the present value of the terminal value.
- d. Sum of the estimated market values of debt, common stock and preferred stock.

Answer: b and d.

Enterprise value can be calculated in two equivalent ways: the worth today of the cash flows the firm generates, or the value today of the claims on those cash flows. Option (a) fails to discount the terminal value; option (c) fails to discount the FCFFs in the projection period.

12.7 Fixed-Income Investments

Scenario. David is offered three USD fixed income instruments. The market expects interest rates to **rise** over the next year as the commodities, offshore oil, coal mining and energy industries come out of a slump.

Bond	Maturity	Yield to maturity	Rating
New Offshore Oil Corp: 10-year 6% **callable** bond	5 years left	4.5%	A−
Diversified Commodities Company: 10-year **4%** bond	6 years left	6.0%	BBB
Indonesian Coal Mining & Energy Company: **8%** bond	4 years left	7.0%	unrated

Question 1 — Why can some bonds be bought below face value?

- a. The Indonesian Coal Mining & Energy Company bonds are unrated.
- b. The New Offshore Oil Corp bonds are callable.
- c. The YTM of the Diversified Commodities Company bonds is higher than the coupon rate.
- d. The YTMs of the New Offshore Oil Corp and Indonesian Coal Mining & Energy Company bonds are lower than their respective coupon rates.

Answer: c.

A bond must be priced at a **discount** for its YTM to be **higher** than its coupon rate. Option (d) is wrong because those bonds will be priced at a **premium** ($4.5\% < 6\%$, and $7.0\% < 8\%$). While investors may require higher yields for unrated and for callable bonds, that does not necessarily mean the yields exceed the respective coupon rates, so (a) and (b) are incorrect.

Question 2 — If David holds his bond to maturity, what should he be most concerned with?

- a. A bond is callable.
- b. Interest rates are increasing.
- c. Interest rates are decreasing.
- d. The coupon rate of a bond.

Answer: a.

A callable bond can be redeemed before maturity, and if that happens the investor faces **reinvestment risk** from having to reinvest the proceeds at lower yields.

Holding to maturity means the **YTM**, not the coupon rate, is the relevant measure — though to realise the YTM, coupons must be reinvested at the same yield. Option (a) is a much bigger concern than (d). Changes in interest rates affect price, but that is of little concern to a hold-to-maturity investor because the price **converges to par** on maturity, so (b) and (c) are incorrect.

Question 3 — Which scenario is likely in the coming year?

- a. Diversified Commodities Company bond price is likely to decrease the most in percentage terms.
- b. New Offshore Oil Corp bonds are likely to be called.
- c. Indonesian Coal Mining & Energy Company bonds are likely to default.
- d. Indonesian Coal Mining & Energy Company bonds are likely to increase the least in percentage terms.

Answer: a.

With rates expected to rise, bond prices are likely to fall. The Diversified Commodities Company bonds have the **longest maturity (6 years) and lowest coupon (4%)**, so their prices are **most sensitive** to rate changes.

Option (d) is incorrect: although the Indonesian bonds are least sensitive because of the shortest maturity and highest coupon, a rate rise is likely to **decrease** the price, not increase it. Option (b) is incorrect because rising rates make a call **unlikely**. Option (c) is incorrect: being unrated does not necessarily mean poor credit — the company may simply have chosen not to seek ratings — and with the industries expected to recover, concluding default is a stretch.

12.8 Portfolio Management

Scenario. Mr Walter Koh is a retiree with $2 million** of investable assets, all currently passively managed. He is willing to hav
*** 500,000** managed more actively and is prepared to risk $100,000** of it in full. He plans to gift about ** **100,000** from this actively managed fund **three months from now** to an art endowment fund. He is not a big spender but hopes to fund at least part of his annual expenses, which are **highly predictable**, from his active portfolio for the next **twenty years**. He is otherwise not concerned about liquidity.

Asset class	Portfolio A	Portfolio B	Portfolio C	Portfolio D
Money market funds	20%	10%	20%	20%
Treasury bills	10%	0%	10%	10%
Long-term government bonds	10%	80%	10%	**50%**
High yield bonds	10%	0%	40%	0%
Developed-market equities	50%	10%	20%	20%

Castor & Pollux shares have a **beta of 1.2** and a **past return of 9.5%**, and have been recommended on their alpha potential after thorough fundamental analysis. The **risk-free rate is 3%** and the **market risk premium is 6%**.

Question 1 — Mr Koh does not believe one can achieve superior returns through fundamental analysis. Which form(s) of market efficiency must he necessarily believe in? (Select all that apply)

- a. Weak form
- b. Semi-weak form
- c. Semi-strong form
- d. Strong form

Answer: a and c.

Conducting fundamental analysis to identify a mispriced stock assumes all **publicly available** information is not already reflected in the price. So Mr Koh must believe the market is **semi-strong form** efficient. Because all publicly available information **includes historical price and volume information**, he must also believe it is **weak form** efficient. There is insufficient information about his views on **private** information to conclude he believes in strong-form efficiency. "Semi-weak form" is not a standard classification.

Question 2 — Jensen's alpha for Castor & Pollux is:

- a. -2.9%
- b. -0.7%
- c. 0.7%
- d. 2.9%

Answer: b.

Using CAPM, the required return is:

$$0.03 + 1.2(0.06) = 10.2\%$$

With an actual return of 9.5%:

$$\alpha = 9.5\% - 10.2\% = -0.7\%$$

Question 3 — Which portfolio best aligns with Mr Koh's objectives?

- a. Portfolio A
- b. Portfolio B
- c. Portfolio C
- d. Portfolio D

Answer: d.

Mr Koh needs **$100,000**, which is **20%** of his $500,000 portfolio, as a gift shortly, so he should allocate that to **money market investments**, the most liquid asset class. Portfolio B is incorrect because of its low allocation to money market funds.

Portfolio C is incorrect because its combined **60%** allocation to high-yield bonds and equities is likely too risky given he is only willing to risk up to $100,000. Portfolio A is incorrect for the same reason, and its allocation to long-term government bonds is likely too low to fund his annual expenses.

Portfolio D is correct: sufficient money market funds for the gift, an equity allocation that respects his risk aversion, and a sufficiently high allocation to long-term government bonds so the coupons can fund his annual expenses.

12.9 Warrants and Foreign Exchange

Scenario. Ms JL Tan has two questions. First, she is looking at the structured warrants of **Poseidon Lines**, a shipping company. Poseidon shares trade at $2.50 * *anditswarrantsat * *\mathbf{1.30}$. The warrants have a **conversion ratio of 1:1** and an **exercise price of $1.50**. Second, she has sold some USD-denominated ADRs and will receive **USD 50,000 in 2 business days**, which she wants to convert to SGD.

Question 1 — What is the intrinsic value of Poseidon's structured warrant?

- a. $0
- b. $1
- c. $1.50
- d. $2.50

Answer: b.

The warrant is in-the-money, so:

$$\text{Intrinsic value} = \frac{2.50 - 1.50}{1} = \$1$$

Question 2 — You expect Poseidon's share price to rise to $4.00. What is Ms Tan's minimum gain on the warrants in percentage terms if she buys today?

- a. 92.31%
- b. 207.69%
- c. 307.69%
- d. Impossible to determine because the future price of the structured warrant is not known with certainty.

Answer: a.

If the share price rises to $4.00, the warrant's intrinsic value is:

$$\frac{4.00 - 1.50}{1} = \$2.50$$

Because the conversion ratio is 1, the **minimum value** of the warrant is its intrinsic value of $2.50. The minimum percentage gain is:

$$\frac{2.50}{1.30} - 1 = 92.31\%$$

Option (d) is incorrect: although the future warrant price is unknown, only the future **minimum** price is needed to compute the minimum percentage gain.

Question 3 — Ms Tan has two quotes. The best quote at which she can sell her ADR proceeds for SGD is:

Quote A: USD/SGD 1.2500 – 1.2550 Quote B: USD/SGD 1.2475 – 1.2515

- a. USD/SGD 1.2500
- b. USD/SGD 1.2550
- c. USD/SGD 1.2475
- d. USD/SGD 1.2515

Answer: a.

All quotations give the number of SGD (the counter currency) per one USD (the base currency). For each quote, the **first rate is the price at which the market maker will buy USD** and the second is the price at which it will sell USD. Since Ms Tan wants to **sell** USD, the **first rate** of each quote applies, and the **highest** of those is best: **1.2500**.

12.10 Unit Trusts, REITs and Business Trusts

Scenario. Mr James Lim finds the jargon of collective investment schemes confusing. He asks you to demystify the differences between unit trusts, business trusts and REITs. He has investable funds and is deciding whether to invest in a property or one of these schemes. He is **more interested in a steady stream of income** than in large trading gains.

Question 1 — The common features among unit trusts, business trusts and REITs is/are that: (Select all that apply)

- a. They are established under a trust deed.
- b. They are exchange-traded.
- c. They allow for diversification.
- d. They are professionally managed.

Answer: a, c and d.

All three are established under a **trust deed**. By pooling and investing funds, all three allow **diversification**, albeit to different extents. The assets under all three are **professionally managed**. Option (b) is incorrect: while all REITs are listed, unit trusts are typically **not** listed and some business trusts are unlisted too.

Question 2 — The main difference(s) between business trusts and REITs is/are that: (Select all that apply)

- a. The trustee and manager are separate in a REIT but not in a business trust.
- b. Business trusts do not face investment restrictions, unlike REITs.
- c. Business trusts are unlisted whereas REITs are listed.
- d. There is a 75% gearing limit applicable on REITs but only a 45% gearing limit on business trusts.

Answer: a and b.

Unlike a REIT, a business trust is a **single responsible entity** organisation with a **trustee-manager**, so (a) is correct. Business trusts can actively engage in any business area, whereas REITs must invest only in real estate for rental income, so (b) is correct. Option (c) is incorrect because although some business trusts are unlisted, **most are listed**. Option (d) is incorrect: there is **no gearing limit** for business trusts and a **45%** limit for REITs.

Question 3 — Given Mr Lim's objectives, the scheme most likely to suit his need is:

- a. A unit trust described as an aggressive fund.
- b. A business trust comprising investments in golf course assets.
- c. A REIT that focuses on heartland retail malls.
- d. All of the above.

Answer: c.

An **aggressive** unit trust generates returns from capital gains and is unlikely to provide a steady income stream, so (a) is incorrect. Cash flows from **golf course** assets are highly sensitive to economic fluctuations, and there is **no mandatory minimum distribution** for business trusts, so (b) will not provide steady

income either. A **REIT** is often seen as an alternative to physical real estate and typically distributes at least **90% of annual income**, which is likely to be more stable for **heartland retail malls**. Option (c) is therefore the best answer, which also makes (d) incorrect.

Exam cues

- **Case studies carry 3 questions each, 9 in total** — about a tenth of the 90-question paper. Each tests one chapter's method applied to a named client.
- **Accredited investor tests:** individual — net personal assets **> S2$m * *(residencecappedat S1m), financial assets > **S1$m * *, orincome * * ≥ S**300k** in the preceding 12 months; entity — net assets > **S$10m**.
- **Time horizon depends on the investor's needs, not the investor type.** This is the standard distractor.
- **Expected return**
$$= \sum r_i p_i$$
. Worked: 0.3(12%) + 0.5(8%) + 0.2(−6%) = **6.40%**.
- **Variance and CV:** variance **0.004144**, **6.4374%**, CV **1.01**.
- **Real future value:** real rate = (1.064/1.02) − 1 = **4.31%**; $200,000 \times 1.0431^{30}$ = **709,264.72**.
- **Leading indicators** predict; **coincident** confirm the present; **lagging** confirm the past. **New orders declining** signals a downturn; **average duration of unemployment** is lagging; **disposable personal income** is coincident.
- **Slowdown with rising rates = late bull to early bear → defensive stocks and investment grade bonds**.
- **Herd instinct** is the root cause when managers follow the crowd without independent analysis; loss aversion and algorithmic trading may **exacerbate** but do not originate.
- **Carry trade worked example:** IDR 100bn at 10,076.68 = **SGD 9,923,903.51** loan. Flat rate → **+SGD 446,575.66**. IDR weakens 10% (rate 11,196.31) → **−SGD 605,357.18**. Cross rate 0.7 × 13,086.60 = **9,160.62** (IDR appreciated) → **higher gain**.
- **Prospective P/E** = price / next-year EPS. Mvidia: 40/1.08 = **37.04x**. A high P/E usually reflects **growth expectations**, not overvaluation.
- **5-year EPS growth** = (5.81/1.12)^(1/5) − 1 = **38.99%**.
- **PEG** = prospective P/E / EPS growth = 37.04/38.99 = **0.95**. **Below 1 → undervalued → buy.**
- **FCFF is discounted at WACC**, because capital comes from all sources. **Interest expense is excluded** from FCFF.
- **Enterprise value** = PV of FCFFs + **PV** of terminal value = market value of debt + common stock + preferred stock.
- **Bond at a discount** means **YTM > coupon**. New Offshore Oil: 4.5% < 6% → **premium**. Indonesian Coal: 7.0% < 8% → **premium**. Diversified Commodities: 6.0% > 4% → **discount**.
- **Hold-to-maturity investor's main concern is a call feature** (reinvestment risk), not price volatility, because price converges to par.
- **Longest maturity and lowest coupon = most rate-sensitive.** Diversified Commodities (6 years, 4%) falls most when rates rise.
- **Unrated poor credit.** A company may simply choose not to seek a rating.
- **Fundamental analysis failing implies semi-strong efficiency**, which necessarily implies **weak-form** efficiency too. "Semi-weak" is not a classification.
- **Jensen's alpha** = actual − [
$$R_f + \beta(R_m - R_f)$$
]. Castor & Pollux: 9.5% − [3% + 1.2(6%)] = **−0.7%**.
- **Portfolio selection:** match the **liquidity need** (20% for the gift), the **risk budget** ($100,000 of $500,000), and the **income need** (long-term government bond coupons).
- **Warrant intrinsic value** = (SP − EP)/n. Poseidon: (2.50 − 1.50)/1 = **$1**.
- **Minimum gain** uses the **minimum value** (intrinsic value), not the unknown future warrant price: 2.50/1.30 − 1 = **92.31%**.
- **Selling USD:** use the **bid** (first rate) and pick the **highest** — **1.2500**.
- **Common to unit trusts, business trusts and REITs:** trust deed, diversification, professional management. **Not** exchange-trading — unit trusts are typically unlisted.
- **REIT vs business trust:** separate trustee and manager vs **single trustee-manager**; investment restrictions vs **none**; **45%** gearing cap vs **no cap**; **90%** mandatory distribution vs **no minimum**.

- **Income objective → REIT**, because of the **90%** distribution requirement and stable heartland retail cash flows.
- **Trap:** a high P/E is not automatically overvaluation — check the growth rate with PEG before concluding.
- **Trap:** an unrated bond is not automatically a default risk.
- **Trap:** for a hold-to-maturity investor, interest rate risk is largely irrelevant; the call feature is the real risk.

Appendix A — Review Questions

Candidates should note that the **sole purpose of the Review Questions is to familiarise candidates with the scope and general nature of the examinations, and the format of the examination questions**. They are **not intended as preparatory study material**, and **do not cover all the material tested**. The correct answer for each question is in **bold** in the source booklet and is reproduced as **Answer** below.

All **50** review questions appear here, in the booklet's own order and chapter grouping, so the appendix can be worked as a single timed set. The booklet prints the correct option in bold; that convention is preserved.

Chapter 2 — Risk, Return and Time Value Calculations

1. As an investment manager, you are assessing a project with an immediate cash outflow followed by a series of cash inflows over the next 10 years. You calculate the project's net present value (NPV) — the sum of the present values of the cash inflows minus the initial cash outflow — and also its internal rate of return (IRR). You then discover you have **underestimated the discount rate**. What happens when you correct for the underestimation?

- **a. There will be a reduction to NPV with no change to IRR.**
- **b.** There will be no change to both NPV and IRR.
- **c.** There will be an increase to NPV and a reduction to IRR.
- **d.** Both NPV and IRR will increase.

Answer: a. NPV discounts the cash flows at the given rate, so raising the rate lowers NPV. IRR is the rate at which NPV equals zero and is a property of the cash flows alone, so it does not change.

2. A diversified portfolio has an expected return of **6%** and a standard deviation of **12%**. There is a **68%** probability of a return between **−6%** and **+18%** (plus or minus one standard deviation). Returns are normally distributed, so the probability of a loss exceeding 6% is **16%**. We can expect the portfolio's loss to exceed 6% once every _____.

- a. 5–6 years.
- **b. 6–7 years.**
- c. 7–8 years.
- d. 8–9 years.

Answer: b. A 16% probability is about **1 in 6.25**, so roughly once every 6–7 years.

3. Which statements about value at risk (VaR) are TRUE?

- a. It is not an applicable measure for loss probability on assets such as bonds, shares and currencies.
- b. It measures and quantifies the minimum possible loss of an investment over a specific time and within a confidence level.
- c. It can be calculated by applying Monte Carlo simulation or the historical method.
- d. It can be used when determinants of future returns differ from those of past returns.

Options: (a) and (d) · (c) and (d) · (a) and (c) · **(b) and (c)**

Answer: (b) and (c). VaR quantifies the loss at a stated confidence level and can be computed historically or by Monte Carlo. It applies to bonds, shares and currencies, and its historical form assumes past determinants persist.

4. Which of the following are forms of operational risk?

- a. A flash crash caused by a massive erroneous sell order.
- b. Poor system infrastructure leading to technical disruptions.

- c. Human errors resulting in erroneous trades.
- d. Inability to access funding due to a credit downgrade of the borrower.

Options: **(a), (b) and (c)** · (b) and (c) · (b), (c) and (d) · (a) and (b)

Answer: (a), (b) and (c). Option (d) is **funding liquidity risk**, not operational risk.

Chapter 3 — Key Drivers of Market Movements and Asset Values

5. Which of the following is a **leading** indicator?

- a. Average duration of unemployment.
- **b. The 10-year T-bond spread over the Fed Funds rate.**
- c. Nonfarm payroll employment.
- d. The average prime rate charged by banks.

Answer: b. The 10-year spread is one of the ten CLEI components. Average duration of unemployment and the average prime rate are **lagging**; nonfarm payrolls are **coincident**.

6. A central bank lowers the reserve requirement ratio. What is the most direct effect?

- a. The money supply contracts.
- **b. The money supply expands through credit creation.**
- c. Bond prices fall.
- d. The exchange rate appreciates immediately.

Answer: b. Lowering required reserves expands the money supply and multiplies lending through the banking system.

7. Which of the following best describes the **liquidity preference theory** of the yield curve?

- a. The curve's shape reflects investors' interest rate expectations.
- **b. The curve should always be upward sloping, because investors require a premium to hold longer maturities.**
- c. Each maturity's yield is set solely by supply and demand in that segment.
- d. The curve is flat when the market is ambivalent.

Answer: b. Liquidity preference holds that the curve is upward sloping because long-maturity securities are more price-volatile and require a higher return.

Chapter 4 — Foreign Exchange

8. Spot USD/SGD is **1.2520** and the 30-day forward points are **−10**. The 30-day forward rate is:

- **a. 1.2510**
- b. 1.2530
- c. 1.2520
- d. 1.2500

Answer: a. Negative points are subtracted: $1.2520 - 0.0010 = \mathbf{1.2510}$, a forward discount.

9. A Singapore investor holds a JPY portfolio that returned **121.88%** while SGD/JPY rose **18.89%**. The currency-adjusted return in SGD is approximately:

- a. 140.77%
- **b. 86.62%**
- c. 102.99%
- d. 121.88%

Answer: b. The reference currency (SGD) is the **first** currency of the pair, so divide: $(1 + 1.2188)/(1 + 0.1889) - 1 = \mathbf{86.62\%}$.

10. Which of the following is a **non-deliverable forward** characteristic?

- a. Physical delivery of the domestic currency at maturity.
- b. Exchange of the full notional amount at settlement.
- **c. Cash settlement of the difference between the contracted rate and the spot rate, with no exchange of notional.**

- d. A tenor of five years or more.

Answer: c. An NDF is cash-settled with no delivery of the domestic currency and no exchange of notional; the normal tenor is **1 month to 1 year**.

Chapter 5 — Company Analysis and Understanding Financial Statements

11. A company has current assets of $500,000 * *and current liabilities of * *\textbf{250,000}$. Its current ratio is:

- a. 0.5
- **b. 2.0**
- c. 1.5
- d. 2.5

Answer: b. 500,000 / 250,000 = **2.0**.

12. Which of the following would **improve** a company's reported ROE without any operational improvement?

- a. Increasing the dividend payout ratio.
- **b. Writing down goodwill against equity reserves.**
- c. Increasing inventory levels.
- d. Lengthening the depreciation period.

Answer: b. A goodwill write-down reduces equity (the denominator), raising ROE in later periods without any operational change.

13. Which of the following is a **non-cash** expense?

- a. Staff costs.
- b. Interest expense.
- **c. Depreciation.**
- d. Tax paid.

Answer: c. Depreciation is an accounting allocation, not a cash outflow, which is why it is added back in the cash flow statement.

13a. Derrick has prepared his company's financial statements for the year ended 31 December 2019, which reflect an operating profit of $280,000 * *, current assets of * *\textbf{510,000}$ and total assets of $833,000 * *. He has since discovered that the depreciation figure he used was overstated, as he double counted the depreciation cost of * *\textbf{55,000}$. Which of the following statements is TRUE?

- **a. The revised operating profit for the year will be $335,000.**
- b. The revised depreciation expense figure will be $110,000.
- c. The revised current assets figure will be $565,000.
- d. The revised total assets figure will be $778,000.

Answer: a. Overstating depreciation by $55,000 understated operating profit by the same amount, so the correction adds it back $280,000 + 55,000 = **335,000**. Depreciation is a non-current asset adjustment, so current assets are unaffected — option (c) is wrong. Total assets rise by $55,000 to $888,000, not fall to $778,000.

13b. The purposes of the statement of cash flows are to:

- a. Evaluate liquidity.
- b. Facilitate banking relationships.
- c. Analyse cash position.
- d. Compare investment performance.

Options: (a) and (b) · **(a), (b) and (c)** · (a) and (c) · (a) and (d)

Answer: (a), (b) and (c). The statement of cash flows evaluates liquidity, facilitates banking relationships and analyses the cash position. Comparing investment performance is the job of return measurement, not the cash flow statement.

Chapter 6 — Equity Securities

14. A company has **300 million** shares at $2.00 * *and announces a * *1 - for - 3 * *rights issue at * *\textbf{1.90}$. The theoretical ex-rights price is:

- a. $1.50
- **b. $1.975**
- c. $2.00
- d. $1.90

Answer: b. New shares 100m at $1.90 = $190m; total market value $600m + $190m = $790m over 400m shares = **1.975**.

15. A stock trades at $34.68 * *with a consensus next - year EPS of * *2.86**. Its prospective P/E is:

- a. 12.8x
- **b. 12.1x**
- c. 11.7x
- d. 13.5x

Answer: b. 34.68 / 2.86 = **12.1x**.

16. Which index weighting scheme automatically adjusts for bonus and rights issues?

- **a. Value-weighted**
- b. Price-weighted
- c. Equal-weighted
- d. Geometric average

Answer: a. Capital changes do not affect market capitalisation, so a value-weighted index adjusts automatically. A price-weighted index needs a divisor adjustment.

16a. Which of the following statements are TRUE about preference shares?

- a. The preference rate of dividend is usually fixed and specified.
- b. The issuer is not obliged to pay the preference dividend if there are poor earnings.
- c. Preference shares can be issued as redeemable shares.
- d. The preference dividend must be paid after the ordinary dividend has been paid.

Options: **(a), (b) and (c)** · (b), (c) and (d) · (a), (c) and (d) · (a), (b), (c) and (d)

Answer: (a), (b) and (c). The preference dividend is usually a fixed, specified rate; the issuer is not obliged to pay it when earnings are poor, and a missed preference dividend is not a default; and preference shares may be issued as redeemable. Statement (d) reverses the priority — preference dividends rank **before** ordinary dividends.

16b. A price-weighted index consists of four stocks. One of the stocks is subject to a 3-for-1 stock split. What is most likely to happen to the divisor for the index?

- **a. Decrease.**
- b. Increase.
- c. Nullify.
- d. Remain unchanged.

Answer: a. A 3-for-1 split cuts that stock's price to a third. Because the index is the sum of prices divided by the divisor and the numerator has fallen, the divisor must be reduced to keep the index level unchanged.

16c. MaMa Rich is a listed food and beverage firm. Its stock is slightly more volatile than the market, with a beta of **1.3**, and the market's expected rate of return is **9.33%**. The US 5-year Treasury bill rate is currently **5.8%**. What is the firm's cost of equity capital?

- a. 9.515%
- b. 10.123%
- **c. 10.389%**
- d. 11.761%

Answer: c. CAPM: R = Rf + (Rm − Rf) = 5.8 + 1.3(9.33 − 5.8) = 5.8 + 1.3(3.53) = 5.8 + 4.589 = **10.389%**.

16d. The price-to-sales ratio (P/Sales) is a helpful valuation technique when examining firms _____.

- a. That have negative earnings.
- b. In the technology sector only.

- c. That are highly leveraged.
- **d. In the start-up or early stage of their life cycle.**

Answer: d. P/Sales is useful where earnings are negative or unstable, which is typical of start-ups and early-stage firms. It is not confined to technology, and it does not correct for leverage — the ratio is conceptually flawed because the numerator relates to shareholders while sales are generated from funds supplied by both shareholders and debtholders.

16e. Which of the following are **not** advantages of American Depositary Receipts (ADRs)?

- a. They allow US investors to invest in foreign companies.
- b. They are denominated in US dollars and pay dividends in US dollars.
- c. They are traded on US exchanges.
- d. They eliminate currency risk.

Options: (a) and (b) · (b) and (c) · **(d) only** · (a), (b) and (c)

Answer: (d) only. ADRs give US investors access to foreign companies, are USD-denominated and pay USD dividends, and trade on US exchanges. They do **not** eliminate currency risk: the underlying shares remain denominated in the issuer's home currency, so the ADR price moves with the exchange rate.

Chapter 7 — Deposits and Fixed-Income Securities

17. A T-bill with **90 days** to maturity, face value $100,000 * *, sells for * *$**98,000**. The bank discount yield is:

- a. 2.04%
- **b. 8.0%**
- c. 8.16%
- d. 7.35%

Answer: b. (2,000/100,000) × (360/90) = **8.0%**.

18. A **10-year** bond with par **$1,000** and an **8%** coupon, with a required yield of **6%** and annual coupons, is priced at approximately:

- a. $1,000.00
- **b. $1,147.20**
- c. $1,148.77
- d. $920.00

Answer: b. Discounting eight $80 coupons and the $1,000 principal at 61,147.20**. The semi-annual equivalent is $1,148.77.

19. Which bond has the **highest** convexity?

- a. A 5-year 8% coupon bond.
- b. A 10-year 8% coupon bond.
- c. A 10-year 4% coupon bond.
- **d. A 10-year zero-coupon bond.**

Answer: d. The lower the coupon, the higher the convexity; a zero-coupon bond has the highest convexity of all.

20. A convertible bond has a conversion ratio of **40**, a market price of $1,100 ** and an underlying share price of * * $**24**. The conversion premium in percentage terms is:

- a. 4.2%
- **b. 14.6%**
- c. 27.5%
- d. 51.0%

Answer: b. Conversion value = 40 × $24 = 960. $Premium = (1{,}100 - 960)/960 =$ **14.6%**.

20a. As compared with bonds selling at par, deep discount bonds such as zero-coupon bonds will have:

- a. Greater reinvestment risk.
- **b. Greater price volatility.**
- c. Less call protection.

- d. None of the above.

Answer: b. A zero-coupon bond has **no** reinvestment risk if held to maturity, because there is no coupon to reinvest — so (a) is the opposite of the truth. Its price is the most sensitive to interest rate changes, because all of its cash flow arrives at maturity.

20b. What is the main feature of a callable bond?

- **a. It gives the issuer the option to redeem the bond before maturity.**
- b. It gives the buyer the option to redeem the bond before maturity.
- c. It gives the issuer the option to redeem the bond after maturity.
- d. It gives the buyer the option to redeem the bond after maturity.

Answer: a. The call option belongs to the **issuer**, not the buyer, and it is exercisable **before** maturity. The buyer's equivalent right is the put provision.

20c. Which of the following are potential risks that may arise when investing in foreign government bonds?

- a. Corporate sector downgrade risk.
- b. Sovereign credit risk.
- c. Currency risk.
- d. Political risk.

Options: (a), (b) and (c) · **(b), (c) and (d)** · (c) and (d) · (a) and (c)

Answer: (b), (c) and (d). A foreign government bond carries sovereign credit risk, currency risk and political risk. Corporate sector downgrade risk is not a feature of a sovereign issue.

20d. What is the value of a bond that matures in **3 years**, has an annual coupon of $110 * *, and a par value of * *$**1,000**? Assume a required rate of return of **11%**.

- a. $330
- **b. $1,000**
- c. $970
- d. $1,330

Answer: b. The coupon rate is 110/1,000 = **11%**, exactly equal to the required return, so the bond prices at **par**. No discounting is needed once the coupon rate and the yield are seen to match.

20e. What happens to a bond's yield-to-maturity (YTM) if there is an increase in the bond price?

- a. The bond's YTM rises.
- **b. The bond's YTM falls.**
- c. The bond's YTM remains unchanged.
- d. The bond's YTM could fluctuate, depending on market conditions.

Answer: b. Price and yield are inversely related. A higher price means a lower yield to maturity.

Chapter 8 — Portfolio Management

21. A portfolio has an expected return of **18%** and a standard deviation of **20%**; a second has **9%** and **10%**. With weights of **40%** and **60%** and a correlation of **0.8**, portfolio risk is:

- a. 14.00%
- b. 10.00%
- **c. 13.30%**
- d. 2.00%

Answer: c. $\sqrt{(0.4^2 \times 0.2^2 + 0.6^2 \times 0.1^2 + 2 \times 0.4 \times 0.6 \times 0.8 \times 0.2 \times 0.1)} = \sqrt{0.01768} =$ **13.30%**.

22. A portfolio returned **16%** with a standard deviation of **20%**; the risk-free rate is **6%**. Its Sharpe ratio is:

- a. 0.27
- **b. 0.50**
- c. 0.39
- d. 0.80

Answer: b. $(0.16 - 0.06)/0.20 = \mathbf{0.50}$.

23. Which measure is **most appropriate** for evaluating a portfolio manager?

- a. Dollar-weighted rate of return.
- **b. Time-weighted rate of return.**
- c. Current yield.
- d. Total return.

Answer: b. The TWR is unaffected by the size and timing of client cash flows, which the manager does not control.

24. A portfolio has a beta of **1.2**, the risk-free rate is **3%** and the market risk premium is **6%**. Its CAPM required return is:

- a. 7.2%
- b. 9.0%
- **c. 10.2%**
- d. 12.0%

Answer: c. $3\% + 1.2(6\%) = \mathbf{10.2\%}$.

24a. If you wish to eliminate the risk of your single-security portfolio, you will select a security that is _____ to your portfolio.

- a. Perfectly positively correlated.
- **b. Perfectly negatively correlated.**
- c. Not correlated.
- d. Highly correlated.

Answer: b. Combining two securities with perfect negative correlation can eliminate portfolio risk altogether. This is the principle behind hedging strategies. Zero correlation reduces risk but does not eliminate it.

24b. Which of the following will affect the expected return of a portfolio under the Capital Asset Pricing Model?

- a. Risk-free rate.
- b. Idiosyncratic risk of an individual security.
- c. Systematic risk of the portfolio.
- d. Market risk premium.

Options: (a), (b) and (c) · (b), (c) and (d) · **(a), (c) and (d)** · (a), (b), (c) and (d)

Answer: (a), (c) and (d). CAPM prices only systematic risk, so idiosyncratic risk is not compensated — it can be diversified away. The risk-free rate, the portfolio's beta and the market risk premium are the three inputs.

24c. John recently bought a financial product with an initial investment capital of $500,000 * *. At the end of the product's tenure his investment value is * *$**550,000**, and he received distributions worth **$10,000** in the interim. What was the return on John's investment?

- a. 2%
- b. 8%
- **c. 12%**
- d. 10%

Answer: c. Total return includes distributions: $(550{,}000 - 500{,}000 + 10{,}000)/500{,}000 = 60{,}000/500{,}000 = \mathbf{12\%}$. Ignoring the distribution gives 10%, which is the trap.

24d. When analysing a portfolio with frequent injections of capital, what would be the best measure for evaluating the investment manager's performance?

- a. Simple return.
- b. Expected return.
- **c. Time-weighted return.**
- d. Dollar-weighted return.

Answer: c. The TWR removes the effect of the size and timing of client cash flows, which the manager does not control. The DWR is the IRR and is dominated by those flows.

24e. The Sharpe performance measure is defined as:

- a. The difference between the actual return on the portfolio and the return that would be expected from the capital asset pricing model.
- b. The portfolio's excess return divided by the portfolio's beta.
- c. The difference between the actual return on the portfolio and the return that would be expected from the capital asset pricing model, divided by the portfolio's total risk.
- **d. The portfolio's excess return divided by the portfolio's total risk.**

Answer: d. Sharpe = (Rp − Rf)/ p. Option (a) describes Jensen's alpha, and option (b) describes the Treynor ratio.

Chapter 9 — ETFs, Unit Trusts, REITs and Insurance

25. Which of the following is **not** covered by the Singapore Deposit Insurance Scheme?

- a. A Singapore dollar savings account.
- b. A Singapore dollar fixed deposit.
- **c. A foreign currency deposit.**
- d. Monies under the CPF Investment Scheme.

Answer: c. The scheme covers **Singapore dollar-denominated** deposits only. Foreign currency deposits, structured deposits and investment products are excluded.

26. A REIT must invest at least what proportion of its fund in income-producing real estate?

- a. 45%
- b. 90%
- **c. 75%**
- d. 50%

Answer: c. The Code on CIS requires a minimum of **75%** in income-producing real estate. The **90%** figure is the distribution requirement.

27. Which of the following is a feature of a **business trust** but not a REIT?

- a. A 45% gearing limit.
- b. A mandatory 90% distribution.
- **c. No gearing limit and no investment restrictions.**
- d. A separate trustee and manager.

Answer: c. Business trusts have **no gearing limit** and **no investment restrictions**, and use a single trustee-manager.

28. An investor pays **$1,000** a year for **5 years** at a **3%** discount rate. The present value is:

- a. $5,000.00
- **b. $4,579.71**
- c. $4,212.36
- d. $5,309.14

Answer: b. 1,000 × [1 − (1.03)]/0.03 = **$4,579.71**.

28a. Which of the following are advantages of investing in ETFs?

- a. Access to a superior hedge fund manager.
- b. Small capital outlay.
- c. Low costs.
- d. Transparency.

Options: (a), (b) and (d) · (b) and (c) · (a), (c) and (d) · **(b), (c) and (d)**

Answer: (b), (c) and (d). ETFs offer a small capital outlay, low costs and transparency. They give no access to a hedge fund manager, which is a feature of a fund of hedge funds.

28b. Which of the following is **not** a feature of real estate investment?

- a. Provides security of income.
- b. Provides a hedge against inflation.

- **c. Requires a high capital outlay.**
- d. Requires annual injection of capital.

Answer: c. A high capital outlay is a genuine feature of direct real estate investment, so it cannot be the exception. The statement that does not hold is that real estate requires an annual injection of capital: the asset is held, not topped up each year.

28c. Which of the following statements are CORRECT?

- a. Apart from REITs, trusts that invest in real estate can be structured as business trusts.
- b. A business trust has a single responsible entity that serves as both the trustee and the manager.
- c. A business trust has no gearing limit.
- d. Both REITs and business trusts are required to distribute a minimum of 90% of annual income.

Options: (a), (c) and (d) · **(a), (b) and (c)** · (b), (c) and (d) · (a), (b), (c) and (d)

Answer: (a), (b) and (c). Statement (d) is wrong: only a REIT carries the 90% distribution requirement. A business trust has no mandatory minimum.

28d. Which of the following are main characteristics of hedge funds?

- a. Use of leverage to increase potential returns.
- b. Measured against absolute returns rather than relative returns.
- c. Lower fees than other financial products.
- d. Relatively unrestricted investment style.

Options: **(a), (b) and (d)** · (b), (c) and (d) · (a), (b) and (c) · (a), (c) and (d)

Answer: (a), (b) and (d). Hedge funds use leverage, target absolute returns and are relatively unrestricted. Their fees are **higher**, not lower, than those of conventional funds — typically a management fee plus a performance fee.

28e. Which statement best describes private equity investments?

- a. Investments that are exclusive to private banking clients.
- **b. Investments in companies that are not publicly listed.**
- c. Investments in companies that operate in the private sector.
- d. Investments that are not publicly disclosed to outsiders and non-clients.

Answer: b. Private equity is investment in companies that are not publicly listed. The private sector is not the defining test — a listed company is in the private sector too — and private equity is not restricted to private banking clients.

Chapter 10 — Warrants

29. A warrant has a share price of $2.50 **, an exercise price of ** **1.50** and a conversion ratio of **1**. Its intrinsic value is:

- a. $0
- **b. $1.00**
- c. $1.50
- d. $2.50

Answer: b. $(2.50 - 1.50)/1 =$ **$1.00**.

30. A warrant trades at $1.30 ** with a share price of ** **2.50** and an exercise price of **$1.50**, conversion ratio 1. Its gearing ratio is approximately:

- a. 1.9x
- **b. 1.9x to 2.0x depending on the share price used**
- c. 8.8x
- d. 0.5x

Answer: b. Gearing = SP/(n × WP) = 2.50/1.30 = **1.92x**.

31. Which factor **reduces** a company warrant's price?

- a. An increase in the underlying share price.
- b. An increase in volatility.

- **c. An increase in dividend expectations.**
- d. An increase in the remaining life of the warrant.

Answer: c. Warrant holders do not receive dividends, so higher dividend expectations reduce the call warrant's price.

31a. Company A's warrant is trading at $0.35 * *; itsexercisepriceis * *\mathbf{0.50}$ while the share price is **$0.65**. What is its intrinsic value?

- a. $0.85
- **b. $0.15**
- c. $0.30
- d. $1.00

Answer: b. Intrinsic value = (SP − EP)/n = (0.65 − 0.50)/1 = **$0.15**.

31b. A warrant with an exercise price of $1.00 * *istradingat * *\mathbf{1.20}$. One warrant converts to one share. What is its conversion price?

- a. $1.00
- b. $1.20
- **c. $2.20**
- d. $0.20

Answer: c. Conversion price = n × WP + EP = 1 × 1.20 + 1.00 = **$2.20**. This is the total the investor pays to convert warrants into one share.

31c. The market price of Company B's common share is $5 * *andtheexercisepriceofthewarrantis * *\mathbf{3.50}$. Two warrants convert into one common share. The warrant price is **$2**. What is the warrant premium?

- a. 10%
- **b. 50%**
- c. 57%
- d. 100%

Answer: b. Premium () $= nWP + EP - SP = 2(2.00) + 3.50 - 5.00 = **2.50**$. Premium (%) = 2.50/5.00 × 100 = **50%**. The conversion ratio of 2 is the trap: using n = 1 gives (2.00 + 3.50 − 5.00)/5.00 = 10%, which is option (a).

31d. A warrant is trading at $0.40 * *.Theexercisepriceis * *\mathbf{0.80}$. The share price is **$2.40**. What is its gearing? Assume the warrant is company-issued, so the conversion ratio is 1.

- a. 2x
- **b. 6x**
- c. 3x
- d. 5x

Answer: b. Gearing = SP/(nWP) = 2.40/0.40 = **6x**.

31e. Which of the following will directly affect the price of a company warrant?

- a. The length of time before the expiration date.
- b. The coupon rate of bonds issued by the company.
- c. The underlying share price.
- d. The number of warrants issued by the company.

Options: (a), (b) and (c) · **(a), (c) and (d)** · (b) and (d) · (a) and (c)

Answer: (a), (c) and (d). Time to expiry, the underlying share price and the size of the warrant issue all affect the warrant price directly. The coupon rate on the company's bonds is not a direct determinant.

Chapter 11 — Technical Analysis and Quantitative Analysis

32. An RSI reading of **75** most likely indicates:

- a. An oversold condition.
- **b. An overbought condition.**
- c. A neutral market.
- d. A trend reversal has already occurred.

Answer: b. An RSI above **70** is considered overbought; below **30** is oversold.

33. Which chart pattern is a **reversal** pattern?

- a. Ascending triangle.
- b. Flag.
- **c. Head and shoulders.**
- d. Consolidation rectangle.

Answer: c. Head and shoulders is a reversal pattern. Triangles, flags and rectangles are continuation patterns.

34. Under the Dow Theory, the secondary movement typically retraces what proportion of the previous advance?

- a. One-tenth to one-quarter.
- **b. One-third to two-thirds.**
- c. Three-quarters to nine-tenths.
- d. The full amount.

Answer: b. The secondary movement usually retraces **one-third to two-thirds** of the previous advance or decline.

34a. The technical analyst is one who:

- a. Buys shares whose intrinsic value is higher than the market price.
- b. Tends to be long-term oriented.
- c. Studies the prospects of earnings growth in forecasting future price trends.
- **d. Studies price and volume patterns of stocks.**

Answer: d. Technical analysis studies historical market data — price changes and trading volume. Options (a) and (c) describe fundamental analysis, and technical analysts are typically shorter-term oriented than fundamental analysts.

34b. Which of the following indicates a buy signal to technical analysts?

- **a. The stock breaks through the moving average line from below.**
- b. A head and shoulders pattern is formed with a breakout below the neckline by 3% of the share price.
- c. Both the Dow Jones Industrial Average and the Dow Jones Transportation Average are moving down.
- d. The advance-decline line is falling in a rising market.

Answer: a. A break upward through the moving average is a bullish crossover. A head and shoulders breakout below the neckline is a sell signal, both averages moving down confirms a downtrend, and a falling advance-decline line in a rising market signals a market peak.

34c. Which of the following are bullish indicators?

- a. Ascending triangles.
- b. Descending triangles.
- c. Double bottoms.
- d. Reverse head and shoulders.

Options: **(a), (c) and (d)** · (b), (c) and (d) · (b) and (c) · (c) and (d)

Answer: (a), (c) and (d). Ascending triangles, double bottoms and inverse (reverse) head and shoulders are all bullish. A descending triangle is a bearish indicator.

34d. According to the Dow Theory, daily fluctuations are considered:

- a. The secondary trend.
- b. A moving average.
- c. The primary trend.
- **d. Random and have no long-term forecasting value.**

Answer: d. The Dow Theory holds that daily fluctuations are random noise with no forecasting value. Only the primary and secondary movements carry information.

34e. A low relative strength indicator (RSI) reading, below 20, could indicate that the stock is:

- a. At a market peak and likely to head down.
- b. Overbought.
- **c. Oversold.**
- d. Performing well compared with other stocks in the same risk category.

Answer: c. An RSI below 30 signals an oversold condition; below 20 is deeply oversold. Above 70 is overbought.

Chapter 12 — Case Studies

35. An individual earns **S$400,000** a year. Under the SFA, is the individual an accredited investor?

- **a. Yes, because income in the preceding 12 months is at least S$300,000.**
- b. No, because net personal assets are not stated.
- c. No, because only net personal assets qualify.
- d. Yes, but only if the individual opts in.

Answer: a. Any one of the three individual tests suffices; income of at least **S$300,000** in the preceding 12 months qualifies.

36. A carry trade borrows in a **1.5%** currency and invests in a **6%** currency, with the exchange rate unchanged. On an SGD-equivalent exposure of **SGD 9,923,903.51**, the annual gain is:

- a. SGD 148,858.55
- **b. SGD 446,575.66**
- c. SGD 595,434.21
- d. SGD 9,923,903.51

Answer: b. $(0.06 - 0.015) \times 9{,}923{,}903.51 =$ **SGD 446,575.66**.

37. A stock trades at $40 * *with next - year EPS of * *$**1.08** and a 5-year EPS growth rate of **38.99%**. Its PEG ratio is approximately:

- a. 0.92
- **b. 0.95**
- c. 1.05
- d. 1.37

Answer: b. Prospective P/E = 40/1.08 = 37.04; PEG = 37.04/38.99 = **0.95**.

Answer key at a glance

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	a	11	b	21	c	31	c
2	b	12	b	22	b	31a	b
3	(b) and (c)	13	c	23	b	31b	c
4	(a), (b) and (c)	13a	a	24	c	31c	b
5	b	13b	(a), (b) and (c)	24a	b	31d	b
6	b	14	b	24b	(a), (c) and (d)	31e	(a), (c) and (d)
7	b	15	b	24c	c	32	b
8	a	15a	(a), (b) and (c)	24d	c	33	c
9	b	15b	a	24e	d	34	b
10	c	15c	c	25	c	34a	d
		15d	d	26	c	34b	a
		15e	(d) only	27	c	34c	(a), (c) and (d)
		16	a	28	b	34d	d
		16a	(a), (b) and (c)	28a	(b), (c) and (d)	34e	c
		16b	a	28b	c	35	a
		16c	c	28c	(a), (b) and (c)	36	b
		16d	d	28d	(a), (b) and (d)	37	b
		16e	(d) only	28e	b		
		17	b	29	b		
		18	b	30	b		

Q	Ans	Q	Ans	Q	Ans	Q	Ans
		19	d	20a	b		
		20	b	20b	a		
				20c	(b), (c) and (d)		
				20d	b		
				20e	b		

What these questions test

- **Q1–Q4** test the distinction between NPV and IRR, the normal distribution landmarks, VaR's construction, and the boundary between operational and funding-liquidity risk.
- **Q5–Q7** test the three indicator classes, credit creation, and the three yield-curve theories.
- **Q8–Q10** test forward points, the currency-adjustment divide-or-multiply rule, and NDF mechanics.
- **Q11–Q13** test the current ratio, the ROE manipulation route through goodwill, and the non-cash nature of depreciation.
- **Q13a–Q13b** test the effect of a depreciation error on operating profit and total assets, and the stated purposes of the cash flow statement.
- **Q14–Q16** test rights-issue arithmetic, prospective P/E, and index weighting.
- **Q15a–Q15e** test preference share characteristics, the price-weighted divisor after a split, CAPM with a beta of 1.3, the P/Sales use case, and the limits of ADRs.
- **Q16a–Q16e** repeat the preference share, divisor, CAPM, P/Sales and ADR tests under the booklet's own numbering, so the appendix can be worked against it.
- **Q17–Q20** test the bank discount yield, bond pricing, convexity ordering, and the conversion premium.
- **Q20a–Q20e** test zero-coupon price volatility, the direction of the call option, foreign government bond risks, par pricing when coupon equals yield, and the price-yield inverse relation.
- **Q21–Q24** test the two-asset risk formula, the Sharpe ratio, the TWR-versus-DWR choice, and CAPM.
- **Q24a–Q24e** test perfect negative correlation, the CAPM inputs, total return including distributions, the TWR for frequent injections, and the Sharpe definition.
- **Q25–Q28** test deposit insurance exclusions, the REIT 75% investment test, business trust features, and annuity present value.
- **Q28a–Q28e** test ETF advantages, real estate features, business trust versus REIT characteristics, hedge fund characteristics, and the definition of private equity.
- **Q29–Q31** test warrant intrinsic value, gearing, and the dividend effect.
- **Q31a–Q31e** test warrant intrinsic value, conversion price, the premium with a conversion ratio of 2, gearing, and the determinants of a warrant's price.
- **Q32–Q34** test RSI thresholds, reversal versus continuation patterns, and Dow Theory retracement.
- **Q34a–Q34e** test the technical analyst's remit, buy signals, bullish patterns, Dow Theory on daily fluctuations, and a deeply oversold RSI.
- **Q35–Q37** test the accredited investor income test, carry trade arithmetic, and PEG.

Exam cues

- **NPV falls when the discount rate rises; IRR does not change** — IRR is a property of the cash flows alone.
- **±1 standard deviation = 68%; ±2 = 95.4%.** A 16% tail probability is about **1 in 6.25**.
- **VaR** can be computed historically or by Monte Carlo; it applies to bonds, shares and currencies.
- **Funding liquidity risk is not operational risk.**
- **Leading:** 10-year T-bond spread over Fed Funds. **Coincident:** nonfarm payrolls. **Lagging:** average duration of unemployment, average prime rate.
- **Lower reserve requirements expand the money supply** through credit creation.
- **Liquidity preference:** the curve is always upward sloping because long maturities need a premium.
- **Forward points:** negative → subtract; positive → add. 1.2520 with −10 points = **1.2510**.
- **Currency adjustment:** reference currency first in the pair → **divide**; second → **multiply**.
- **NDF:** cash-settled, no delivery, no notional exchange, tenor **1 month to 1 year**.
- **Current ratio** = current assets / current liabilities.
- **Goodwill write-downs raise ROE** by shrinking the equity denominator.
- **Depreciation is non-cash** and is added back in the cash flow statement.
- **Rights issue:** theoretical ex-rights price = (old market value + new capital) / new share count.
- **Prospective P/E** = price / next-year EPS.

- **Value-weighted indices adjust automatically** for capital changes.
- **Bank discount yield** = (D/F) × (360/t).
- **Bond price** falls as the required yield rises above the coupon.
- **Zero-coupon bonds have the highest convexity.**
- **Conversion premium (%)** = (market value of CB − conversion value) / conversion value.
- **Portfolio risk is not a weighted average** of constituent risks.
- **Sharpe** = excess return / standard deviation.
- **TWR is the correct measure for judging a manager.**
- **CAPM** =
$$R_f + \beta(R_m - R_f)$$
.
- **Deposit insurance covers S$ deposits only.**
- **REIT: 75% investment test, 90% distribution test, 45% gearing cap.**
- **Business trust: no gearing cap, no investment restrictions, single trustee-manager.**
- **Annuity PV** =
$$C[1-(1+i)^{-n}]/i$$
.
- **Warrant intrinsic value** = (SP − EP)/n; **gearing** = SP/(n × WP).
- **Dividends reduce a call warrant's price.**
- **RSI above 70 overbought, below 30 oversold.**
- **Head and shoulders is a reversal pattern.**
- **Dow Theory secondary movement retraces one-third to two-thirds.**
- **Accredited investor: any one of the three individual tests suffices.**
- **Carry trade gain** = interest differential × SGD-equivalent exposure when the rate is flat.
- **PEG below 1 suggests undervaluation.**

Appendix B — Formulae Sheet

This formulae sheet is provided during the examination. The examination might not cover all formulas provided. Every symbol is defined where it first appears.

Risk, Return and Time Value Calculations

Measures of return

$$\text{Total Return on a Security} = \frac{\text{Income} + \text{Capital Gain or Loss}}{\text{Purchase Price}}$$

$$\text{Expected Rate of Return} = E(R) = \sum_{i=1}^{n} r_i p_i$$

where

$$E(R)$$

is the expected return on a security,

$$r_i$$

the

$$i$$

th possible return,

$$p_i$$

the probability of that return, and

$$n$$

the number of possible returns.

Measures of risk

$$\text{Variance of Returns} = \sigma^2 = \sum p_i \left[r_i - E(R)\right]^2$$

$$\text{Standard Deviation of Returns} = \sigma = \sqrt{\sigma^2}$$

$$\text{Coefficient of Variation} = CV = \frac{\text{Standard Deviation of Returns}}{\text{Expected Rate of Return}}$$

Time value of money

$$\text{Simple Interest} = \frac{\text{Principal} \times \text{Interest Rate (in \%)} \times \text{Interest Period (in days)}}{36{,}500}$$

$$FV_n = PV(1+k)^n$$

where

FV_n

is the future value at the end of

n

years,

PV

the present value or original amount invested at the beginning of the first year,

n

the number of years of compounding, and

k

the annual interest rate.

$$CAGR = \left[\frac{FV}{PV}\right]^{1/n} - 1$$

where

$CAGR$

is the compounded annual growth rate,

FV

the future value,

PV

the present value, and

n

the number of years.

$$FV_n = PV\left(1 + \frac{k}{m}\right)^{mn}$$

where

m

is the number of interest periods in the year.

Present value of a single cash flow

$$PV = \frac{FV_n}{(1+k)^n}$$

where

$$PV$$

is the present value of the future sum,

$$FV_n$$

the future value at the end of

$$n$$

years,

$$n$$

the number of years to receipt, and

$$k$$

the annual discount or interest rate.

Key Drivers of Market Movements and Asset Values

Quantity theory of money

$$M \times V = P \times Q$$

where

$$M$$

is the stock of money (currency outside banks plus demand deposits),

$$V$$

the velocity of money circulation (the number of times per period an average unit of money changes hands),

$$P$$

the general price level or price index, and

$$Q$$

the number of transactions in an economy during a year.

Foreign Exchange

Forward foreign exchange contract

$$\text{Forward Points} = \frac{S \times (R_c - R_b) \times n/360}{1 + (R_b \times n/360)}$$

where

$$S$$

is the spot rate,

$$R_c$$

the annualised interest rate of the counter currency,

$$R_b$$

the annualised interest rate of the base currency, and

$$n$$

the number of days from the spot date.

Interest rate parity

$$F = S\left[\frac{1 + R_c\left(\frac{n}{360}\right)}{1 + R_b\left(\frac{n}{360}\right)}\right]$$

where

$$F$$

is the forward rate,

$$S$$

the spot rate,

$$R_c$$

the annualised counter-currency rate,

$$R_b$$

the annualised base-currency rate, and

$$n$$

the number of days.

Company Analysis and Understanding Financial Statements

Per share ratios

$$\text{Book Value Per Share} = \frac{\text{Shareholders' Equity}}{\text{Total Number of Shares Outstanding}}$$

$$\text{Fully Diluted Earnings Per Share} = \frac{\text{Net Earnings}}{\text{Fully-Diluted Number of Shares}}$$

$$\text{Dividend Per Share} = \frac{\text{Dividends Paid}}{\text{Total Number of Shares Outstanding}}$$

Liquidity ratios

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Quick Ratio} = \frac{\text{Cash} + \text{Marketable Securities} + \text{Accounts Receivable}}{\text{Current Liabilities}}$$

$$\text{Cash Ratio} = \frac{\text{Cash} + \text{Near-Cash}}{\text{Current Liabilities}}$$

$$\text{Cash Flow Ratio} = \frac{\text{Cash Flow From Operating Activities}}{\text{Current Liabilities}}$$

Leverage ratios

$$\text{Debt/Equity Ratio} = \frac{\text{Total Debt}}{\text{Total Shareholders' Equity}}$$

$$\text{Debt-to-Total Assets Ratio} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

$$\text{Total Liabilities-to-Total Assets Ratio} = \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

$$\text{Interest Coverage} = \frac{\text{Earnings Before Interest and Taxes}}{\text{Interest Expense}}$$

$$\text{EBITDA Interest Coverage} = \frac{\text{EBITDA}}{\text{Interest Expense}}$$

$$\text{Total Fixed Charge Coverage} = \frac{\text{Earnings Before Interest, Taxes and Lease Payments}}{\text{Interest Expense} + \text{Lease Payments} + \dfrac{\text{Preferred Dividend}}{1 - \text{Tax Rate}}}$$

Operating efficiency / activity ratios

$$\text{Total Asset Turnover} = \frac{\text{Sales}}{\text{Average Total Assets}}$$

$$\text{Inventory Turnover} = \frac{\text{Sales}}{\text{Average Inventory}}$$

$$\text{Days to Sell Inventory} = \frac{365 \text{ days}}{\text{Inventory Turnover}} = \frac{\text{Average Inventory}}{\text{Sales}/365}$$

$$\text{Receivables Turnover} = \frac{\text{Sales}}{\text{Average Accounts Receivables}}$$

$$\text{Collection Period} = \frac{365}{\text{Receivables Turnover}} = \frac{\text{Average Accounts Receivables}}{\text{Sales}/365}$$

Profitability ratios

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Sales}}$$

$$\text{Operating Profit Margin} = \frac{\text{Earnings Before Interest and Tax}}{\text{Sales}}$$

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Sales}}$$

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Average Common Equity}}$$

$$\text{Pre-tax Return on Assets} = \frac{\text{Pre-Tax Income} + \text{Interest Expense}}{\text{Average Total Assets}}$$

$$\text{After-tax Return on Assets} = \frac{\text{Net Income} + \text{Interest Expense}(1 - \text{Tax Rate})}{\text{Average Total Assets}}$$

$$\text{Earnings Per Share} = \frac{\text{Net Income Available to Common Shareholders}}{\text{Average Number of Common Shares Outstanding}}$$

Equity Securities

Revised net asset value per share

$$RNAVPS = \frac{\text{Revised Asset Value} - \text{Market Value of Liabilities}}{\text{Total Outstanding Number of Shares}}$$

Valuation ratios

$$\text{Price-Earnings Ratio (P/E)} = \frac{\text{Market Price Per Share}}{\text{Earnings Per Share}}$$

$$\text{Price-Earnings Growth Ratio (PEG)} = \frac{P/E}{\%\text{ Growth in Earnings}}$$

$$\text{Price-to-Book Value Ratio (P/BV)} = \frac{\text{Market Price Per Share}}{\text{Book Value Per Share}}$$

$$\text{Price-to-Revised Net Asset Value Ratio (P/RNAV)} = \frac{\text{Market Price Per Share}}{(\text{Market Value of Assets} - \text{Liabilities})/\text{Shares Outstanding}}$$

$$\text{Price-to-Sales Ratio} = \frac{\text{Current Market Price Per Share}}{\text{Sales Per Share}}$$

$$\text{Enterprise Value-to-Cash Flow Ratio} = \frac{\text{Market Cap} + \text{Debt} + \text{Preferred Shares} - \text{Cash}}{\text{Cash Flow from Operations}}$$

$$\text{Enterprise Value-to-EBITDA Ratio} = \frac{\text{Market Cap} + \text{Debt} + \text{Preferred} + \text{Minority Interest} + \text{Unfunded Pension} - \text{Associat}}{\text{Earnings Before Interest, Taxes, Depreciation and Amortisati}}$$

Dividend yield

$$\text{Dividend Yield} = \frac{\text{Dividend Per Share}}{\text{Market Price Per Share}} \times 100$$

Dividend discount model

$$P_0 = \frac{D_1}{(1+k)} + \frac{D_2}{(1+k)^2} + \frac{D_3}{(1+k)^3} + \cdots + \frac{D_\infty}{(1+k)^\infty}$$

where

$$P_0$$

is the intrinsic or theoretical value of the share today,

$$D_t$$

the dividends expected at the end of period

$$t$$

, and

$$k$$

the required rate of return.

With a subsequent sale of the share:

$$P_0 = \frac{D_1}{(1+k)} + \frac{D_2}{(1+k)^2} + \cdots + \frac{D_n}{(1+k)^n} + \frac{P_n}{(1+k)^n}$$

where

$$n$$

is the holding period and

$$P_n$$

the expected selling price at the end of period

$$n$$

.

Constant growth dividend discount model

$$P_0 = \frac{D_0(1+g)}{(1+k)} + \frac{D_0(1+g)^2}{(1+k)^2} + \frac{D_0(1+g)^3}{(1+k)^3} + \cdots + \frac{D_0(1+g)^\infty}{(1+k)^\infty}$$

where

$$D_0$$

is the current dividend and

$$g$$

the constant growth rate. This simplifies to:

$$P_0 = \frac{D_1}{k-g}$$

where

$$D_1$$

is the dividend expected at the end of period 1.

$$\text{Required Rate of Return } (k) = \frac{D_1}{P_0} + g$$

$$g = \text{Retention Rate} \times \text{Return on Equity} = RR \times ROE$$

Zero-growth model

$$P_0 = \frac{\text{Dividend}}{\text{Discount Rate}}$$

Multi-stage growth dividend discount model

For two-stage dividend growth:

$$P_0 = \sum_{t=1}^{n} \frac{D_0(1+g_1)^t}{(1+k)^t} + \frac{\frac{D_n(1+g_c)}{k-g_c}}{(1+k)^n}$$

where

$$P_0$$

is the intrinsic value today,

$$D_0$$

the current dividend,

$$g_1$$

the growth rate when dividends are more predictable,

$$g_c$$

the constant growth rate when dividends are less predictable,

$$k$$

the required rate of return,

$$n$$

the number of periods of supernormal or subnormal growth, and

$$D_n$$

the dividend at the end of the abnormal growth period.

Deposits and Fixed-Income Securities

T-bill yield

$$Y_d = \frac{D}{F} \times \frac{360}{t}$$

where

Y_d

is the annualised yield on a bank discount basis (as a decimal),

D

the dollar discount (face value minus price),

F

the face value, and

t

the number of days remaining to maturity.

Bond valuation

With annual interest:

$$P = \sum_{t=1}^{n} \frac{C_t}{(1+k)^t} + \frac{M}{(1+k)^n}$$

where

P

is the present value of the bond today,

C_t

the coupon payments for period

t

,

M

the maturity value,

n

the number of years to maturity, and

k

the required rate of return.

With semi-annual interest:

$$P = \sum_{t=1}^{2n} \frac{C_t}{(1+k/2)^t} + \frac{M}{(1+k/2)^{2n}}$$

Zero-coupon bond:

$$P = \frac{M}{(1+k/2)^{2n}}$$

Yield measures

$$\text{Current Yield} = \frac{\text{Annual Interest Payment}}{\text{Bond Price}}$$

Yield-to-maturity

$$k$$

solves:

$$P = \frac{C_1}{1+k/2} + \frac{C_2}{(1+k/2)^2} + \frac{C_3}{(1+k/2)^3} + \cdots + \frac{C_{2n}}{(1+k/2)^{2n}} + \frac{M}{(1+k/2)^{2n}}$$

$$\text{Effective Annual Yield} = (1 + \text{Semi-Annual Interest Rate})^2 - 1$$

$$\text{Approximate Yield-to-Maturity} = \frac{\text{Annual Interest Payment} \pm \text{Amortised Gain or Loss}}{(\text{Current Market Price} + \text{Par Value})/2}$$

Yield-to-maturity for a zero-coupon bond:

$$\frac{k}{2} = \left[\frac{\text{Maturity Value}}{\text{Price}}\right]^{1/2n} - 1$$

$$\text{Taxable Equivalent Yield} = \frac{\text{Tax Exempt Yield}}{1 - \text{Marginal Tax Rate}}$$

Analysis of convertible bonds

$$\text{Conversion Value} = \text{Conversion Ratio} \times \text{Share Price}$$

$$\text{Conversion Ratio} = \frac{\text{Par Value of Convertible Bond}}{\text{Conversion Price}}$$

$$\text{Conversion Parity Price} = \frac{\text{Market Price of Convertible Bond}}{\text{Conversion Ratio}}$$

$$\text{Conversion Premium (\$)} = \text{Market Value of Convertible Bond} - \text{Conversion Value}$$

$$\text{Conversion Premium (\%)} = \frac{\text{Market Value of Convertible Bond} - \text{Conversion Value}}{\text{Conversion Value}}$$

$$\text{Breakeven Time/Payback} = \frac{\text{Conversion Premium}}{\text{Bond Income} - \text{Dividend Income}}$$

$$\text{Premium Over Straight Value (\%)} = \frac{\text{Market Price of Convertible Bond}}{\text{Investment Value}} - 1$$

Portfolio Management

Calculating portfolio return and risk

$$\text{Expected Return of a Portfolio} = R_p = \sum_{i=1}^{n} w_i r_i$$

where

R_p

is the expected portfolio return,

w_i

the portfolio weight for security

i

,

r_i

the expected return on security

i

,

n

the number of securities, and

$\sum w_i = 1.0$

.

$$\text{Portfolio Risk (two assets)} = \sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2}$$

where

w_1

and

w_2

are the portfolio weights,

σ_1

and

σ_2

the standard deviations, and

$\rho_{1,2}$

the correlation coefficient.

Capital market line

$$E(R_p) = R_f + \frac{E(R_m) - R_f}{\sigma_m}\sigma_p$$

where

$E(R_p)$

is the expected return on any efficient portfolio P on the CML,

R_f

the risk-free rate,

$E(R_m)$

the expected return on the market portfolio,

σ_m

the standard deviation of market returns, and

σ_p

the standard deviation of portfolio P's returns.

Capital asset pricing model

$$E(R_i) = R_f + \beta_i(R_m - R_f)$$

where

$$E(R_i)$$

is the expected return on security or portfolio

$$i$$

,

$$R_f$$

the risk-free rate,

$$R_m$$

the expected market return,

$$\beta_i$$

the systematic risk relative to the market, and

$$R_m - R_f$$

the market risk premium.

Arbitrage pricing theory

$$R_i = E(R_i) + b_{i1}f_1 + b_{i2}f_2 + \cdots + b_{in}f_n + e_i$$

where

$$R_i$$

is the actual return on security

$$i$$

in a period,

$$E(R_i)$$

the expected return,

$$f$$

the deviation of a systematic factor from its expected value,

$$b_i$$

the sensitivity of security

$$i$$

's returns to a factor, and

$$e_i$$

a random error term unique to security

$$i$$

.

Measuring investment returns

No deposits or withdrawals:

$$R_p = \frac{V_E - V_B}{V_B}$$

where

$$V_E$$

is the end value and

$$V_B$$

the beginning value.

Deposits or withdrawals at the **end** of the measurement period:

$$R_p = \frac{(V_E + \text{Withdrawals} - \text{Deposits}) - V_B}{V_B}$$

Deposits or withdrawals at the **beginning** of the measurement period:

$$R_p = \frac{V_E - (V_B + \text{Deposits} - \text{Withdrawals})}{V_B + \text{Deposits} - \text{Withdrawals}}$$

Annualising returns:

$$\text{Annual Return} = [(1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)] - 1$$

where

$$r_1, r_2, r_3, r_4$$

are the quarterly returns.

Modified Dietz method:

$$R_p = \frac{V_E - V_B - CF}{V_B + WCF}$$

where

$$CF$$

is the net cash flow over the period and

$$WCF$$

the weighted cash flow:

$$WCF = \sum_i CF_i \frac{n - t_i}{n}$$

where

$$CF_i$$

is the cash flow on day

$$t_i$$

,

$$t_i$$

the number of days from the beginning of the period until

$$CF_i$$

, and

$$n$$

the number of days in the period.

Risk-adjusted measures of performance

$$\text{Sharpe ratio} = \frac{R_p - R_f}{\sigma_p}$$

where

$$R_p$$

is the average portfolio return,

$$R_f$$

the risk-free rate, and

$$\sigma_p$$

the standard deviation of portfolio returns.

$$\text{Treynor ratio} = \frac{R_p - R_f}{\beta_p}$$

where

$$\beta_p$$

is the portfolio beta.

Jensen's alpha. The CAPM expected return is:

$$R_e = R_f + \beta_p(R_m - R_f)$$

The portfolio's alpha is the difference between the actual return

$$R_a$$

and the CAPM-expected return:

$$\alpha_p = R_a - R_e = R_a - [R_f + \beta_p(R_m - R_f)]$$

$$\text{Information ratio} = \frac{\alpha_p}{\sigma_p}$$

where

$$\alpha_p$$

is Jensen's alpha and

$$\sigma_p$$

the excess risk.

$$\text{Sortino ratio} = \frac{R_p - R_f}{\sigma_{\text{negative}}}$$

where

$$\sigma_{\text{negative}}$$

is the average downside volatility.

ETFs, Unit Trusts, REITs and Insurance

Net asset value

$$\text{NAV} = \frac{\text{Fund Securities} + \text{Other Fund Assets} - \text{Liabilities (including management fees)} - \text{Debt used for leveraging}}{\text{Total fund units outstanding}}$$

Annuity calculations

$$\text{Present value, } PV_{\text{annuity}} = C \times \left[\frac{1-(1+i)^{-n}}{i}\right]$$

$$\text{Future value, } FV_{\text{annuity}} = C \times \left[\frac{(1+i)^{n}-1}{i}\right]$$

where

$$C$$

is the cash flow per period,

$$i$$

the interest rate, and

$$n$$

the number of payments.

Warrants

Warrant valuation

$$\text{Intrinsic Value} = \frac{SP-EP}{n}$$

where

$$SP$$

is the share price,

$$EP$$

the exercise price, and

$$n$$

the conversion ratio.

$$\text{Conversion Price} = n \times WP + EP$$

where

$$WP$$

is the warrant price.

$$\text{Warrant Premium (\$)} = n \times WP + EP - SP$$

$$\text{Warrant Premium (\%)} = \frac{n \times WP + EP - SP}{SP} \times 100$$

$$\text{Gearing Ratio} = \frac{SP}{n \times WP}$$

$$\text{Minimum Value of Warrant} = SP - EP \text{ if } SP > EP; \qquad = 0 \text{ if } SP \leq EP$$

Exam cues

- **Six return measures:** total, expected, annualised, real, currency-adjusted, IRR. Only total and expected appear on this sheet.
- **Risk measures:** variance, standard deviation, coefficient of variation. CV is the only one expressed per unit of return.
- **TVM:**
$$FV_n = PV(1+k)^n$$
;
$$FV_n = PV(1+k/m)^{mn}$$
;
$$PV = FV_n/(1+k)^n$$
;
$$CAGR = (FV/PV)^{1/n} - 1$$
. Simple interest uses **36,500** as the denominator.
- **Fisher's equation of exchange:**
$$M \times V = P \times Q$$
.
- **Forward points:**
$$S(R_c - R_b)(n/360)/[1 + R_b(n/360)]$$
. **Interest rate parity:**
$$F = S[(1 + R_c n/360)/(1 + R_b n/360)]$$
.
- **Per-share denominators differ:** BVPS uses **total shares outstanding**; diluted EPS uses **fully diluted shares**; DPS uses **total shares outstanding**.
- **Liquidity ratios:** current, quick (excludes inventory), cash, cash flow.
- **Leverage ratios:** debt/equity, debt/total assets, total liabilities/total assets, interest coverage, EBITDA interest coverage, total fixed charge coverage (preferred dividend grossed up by dividing by 1 − tax rate).
- **Efficiency ratios use AVERAGE balances** in the denominator.
- **Profitability:** gross, operating, net margins; ROE; pre-tax and after-tax ROA; EPS.
- **Valuation ratios:** P/E, PEG, P/BV, P/RNAV, P/Sales, EV/CF, EV/EBITDA. **EV/EBITDA adds minorities and unfunded pension and subtracts associates and cash.**
- **DDM family:** general DDM, DDM with a subsequent sale, constant growth (Gordon), zero growth, two-stage.
$$g = RR \times ROE$$
.
- **T-bill yield:**
$$(D/F)(360/t)$$
.
- **Bond pricing:** annual, semi-annual (double the periods, halve the rate), zero-coupon.
- **Yield measures:** current yield, YTM, effective annual yield, approximate YTM, zero-coupon YTM, taxable equivalent yield.
- **Convertible measures:** conversion value, conversion ratio, conversion parity price, conversion premium in dollars and per cent, breakeven time, premium over straight value.
- **Portfolio risk** uses the two-asset formula with the correlation term; **portfolio return** is a simple weighted average.
- **CML uses total risk ( ); CAPM/SML uses beta.**
- **APT** is the multifactor model.
- **Return measurement:** no cash flow, cash flow at end, cash flow at start, annualising, modified Dietz with weighted cash flow.
- **Risk-adjusted measures:** Sharpe ( ), Treynor ( ), Jensen's alpha, information ratio, Sortino (downside).
- **NAV** subtracts liabilities including management fees and any debt used for leveraging.
- **Annuity PV and FV** use the same bracket structure with the sign of the exponent reversed.

- **Warrant formulas:** intrinsic value, conversion price, premium in dollars and per cent, gearing, minimum value.
- **Trap:** the **approximate YTM** formula uses the **average** of current price and par in the denominator, not the current price alone.
- **Trap: EV/CF** subtracts cash; **EV/EBITDA** subtracts cash **and** associates but adds minorities and unfunded pension.
- **Trap:** the **Sortino** denominator is **downside** volatility, not total standard deviation.

Appendix C — Essential Readings

The source list runs to **seven items**. Candidates should complete the essential readings before attempting the examination. Use the current edition of each; page references in older editions will not match.

Core investment texts

#	Instrument
1	*Analysis of Investments and Management of Portfolios* — Frank K. Reilly, Keith C. Brown
2	*Fundamentals of Investment Management* — Geoffrey A. Hirt, Stanley B. Block
3	*Essentials of Investments* — Zvi Bodie, Alex Kane, Alan J. Marcus
4	*Fundamentals of Investments: Valuation and Management* — Bradford D. Jordan, Thomas W. Miller, Steven D. Dol
5	*Investments: Principles of Portfolio and Equity Analysis* — Michael G. McMillan et al.

Technical analysis

#	Instrument
6	*Technical Analysis Plain and Simple: Charting the Markets in Your Language* — Michael N. Kahn

Portfolio management

#	Instrument
7	*Managing Investment Portfolios: A Dynamic Process* — John L. Maginn, Donald L. Tuttle, Jerald E. Pinto, Dennis

How to read the list efficiently

- **Items 1–5 overlap heavily.** They are general investments texts covering risk and return, asset classes, valuation and portfolio theory. Read **one** thoroughly — Bodie, Kane and Marcus (item 3) is the most compact for the risk-return and CAPM material, and Reilly and Brown (item 1) is the most comprehensive on asset classes. Use the others as reference for a topic the first does not explain well.
- **Item 5** is the CFA Institute's portfolio and equity analysis text and maps most closely onto Chapters 6 and 8 of this syllabus — equity valuation, relative valuation ratios and portfolio management.
- **Item 6** is the only technical analysis text and covers Chapter 11. Read it for the chart patterns and indicators; the examination tests recognition and interpretation, not construction.
- **Item 7** is the reference work on the portfolio management process, investment policy statements and performance measurement — Chapters 8 and 9.
- **Nothing on the list covers Chapters 1, 4, 7, 10 or 12 in depth.** For those, the official study guide and this teaching guide are the primary sources: Chapter 1 (investor classification and market structure), Chapter 4 (FX quotations and forwards), Chapter 7 (fixed income, duration and convertibles), Chapter 10 (warrants) and Chapter 12 (case studies) are syllabus-specific and not covered by general investment texts at this level of detail.
- **Singapore-specific content** — the SFA investor definitions, the Deposit Insurance Scheme, SGS, the CPF Investment Scheme, the Code on CIS, REIT rules and the Business Trusts Act — is not in any of these books. Use the MAS and SGX publications referenced in the chapters.

Exam cues

- **Seven items** on the list, across **five publishers**.
- **Items 1–5** are general investments texts; **item 6** is technical analysis; **item 7** is portfolio management.
- **Bodie, Kane and Marcus** is the most compact for risk-return and CAPM; **Reilly and Brown** is the most comprehensive on asset classes.
- **McMillan et al.** maps most closely to the equity valuation and portfolio chapters.
- **Maginn et al.** is the reference for the IPS and performance measurement.
- **Kahn** is the only technical analysis text.
- **No listed text covers** investor classification, FX mechanics, fixed income duration and convertibles, warrants, or the case studies at syllabus depth — use the official guide and this teaching guide.
- **Singapore regulatory content is not in any listed text.** Use MAS and SGX sources.
- **Trap:** the list is a reading list, not a syllabus. Reading all seven does not substitute for working the calculations in Chapters 2, 6, 7, 8 and 10.

Appendix D — Glossary

Terms are grouped by the chapter that uses them, so a term can be found from the topic you are revising rather than from its first letter. Each group ends with the chapter reference.

Investor classification and market structure

- **Accredited investor (AI)** — an individual or entity meeting the SFA s 4A wealth or income tests, presumed financially savvy and able to bear more risk, with access to more complex products. **Chapter 1.**
- **Expert investor (EI)** — a person whose business involves acquiring, disposing of or holding capital markets products; protected less than a retail investor because of occupational knowledge, not wealth. **Chapter 1.**
- **Institutional investor (II)** — a government, statutory board, bank, insurer, CMS licensee, approved exchange, pension fund or CIS; the widest product access. **Chapter 1.**
- **Retail investor** — anyone outside the AI, EI and II definitions; the highest level of protective regulation. **Chapter 1.**
- **Primary market** — where new securities are issued and the **issuer** receives the cash. **Chapter 1.**
- **Secondary market** — where already-issued securities trade and the **selling investor** receives the cash; supplies liquidity. **Chapter 1.**
- **Marketability** — the likelihood of selling a security quickly; necessary but not sufficient for liquidity. **Chapter 1.**
- **Price continuity** — prices do not change much between transactions absent new information. **Chapter 1.**
- **Internal efficiency** — low transaction costs across reaching the market, brokerage and transfer. **Chapter 1.**
- **Price discovery** — the process by which trading establishes a transparent price; the most important function of a market. **Chapter 1.**
- **Dealer** — trades as principal, has a vested interest in the transaction and earns the bid-offer spread. **Chapter 1.**
- **Broker** — acts as agent for a commission and does not take the other side. **Chapter 1.**

Risk, return and time value

- **Total return** — current income plus capital gain or loss, divided by the purchase price. **Chapter 2.**
- **Expected rate of return** — the probability-weighted average of possible returns. **Chapter 2.**
- **Real rate of return** — the nominal return adjusted for inflation:

$$(1 + r)/(1 + \text{inflation}) - 1$$

  . **Chapter 2.**
- **Currency-adjusted return** — the return restated in the investor's reference currency; divide when the reference currency is first in the pair, multiply when it is second. **Chapter 2.**
- **Internal rate of return (IRR)** — the rate equating all cash outflows and inflows at one point in time; the **money-weighted** return. **Chapter 2.**

- **Compounded annual growth rate (CAGR)** — the effective annual rate on the initial cash flow; the **time-weighted** return. **Chapter 2.**
- **Variance and standard deviation** — measures of the dispersion of returns from the expected return. **Chapter 2.**
- **Volatility** — the standard deviation of the security's returns; historical volatility differs from implied volatility. **Chapter 2.**
- **Coefficient of variation (CV)** — risk per unit of expected return. **Chapter 2.**
- **Z-score** — the number of standard deviations a return deviates from the expected return. **Chapter 2.**
- **Value at risk (VaR)** — the loss at a stated tail probability, for example 2.5% or 0.5%. **Chapter 2.**
- **Systematic risk** — market-wide risk that cannot be diversified away but can be hedged. **Chapters 2, 8.**
- **Non-systematic (firm-specific) risk** — risk peculiar to a firm or industry, reducible to near zero by diversification. **Chapters 2, 8.**
- **Basis point** — 0.01%, or 0.0001. **Chapters 2, 7.**

Market drivers

- **Fiscal policy** — government expenditure and taxation used to influence aggregate demand and supply. **Chapter 3.**
- **Monetary policy** — management of money supply and the cost of funding. **Chapter 3.**
- **Multiplier effect** — fiscal spending rippling through industries, raising incomes and further consumption. **Chapter 3.**
- **Leading indicator** — peaks and troughs **precede** the economy; a short-term predictor. **Chapter 3.**
- **Coincident indicator** — peaks and troughs **align with** the economy; reflects the present. **Chapter 3.**
- **Lagging indicator** — peaks and troughs **follow** the economy; confirms turning points. **Chapter 3.**
- **Purchasing Managers' Index (PMI)** — a leading indicator; **50** divides expansion from contraction. **Chapter 3.**
- **Industry life cycle** — early development, rapid expansion, mature growth, stabilisation, decline. **Chapter 3.**
- **Porter's five forces** — rivalry, new entrants, substitutes, buyer power, supplier power. **Chapter 3.**
- **Herd instinct** — following the crowd without independent decision-making; strongest when information is scarce. **Chapter 3.**
- **Loss aversion** — from prospect theory; preferring less uncertainty for gains and more for losses, so investors sell winners and hold losers. **Chapter 3.**
- **Cognitive bias** — concluding from past experience, preferences or predetermined beliefs; confirmation bias is the common form. **Chapter 3.**

Foreign exchange

- **Base currency** — the first currency of a pair; the investor's reference currency. **Chapter 4.**
- **Counter (term) currency** — the second currency of a pair. **Chapter 4.**
- **Pip** — 0.0001, the fourth decimal place; the smallest FX price movement. **Chapter 4.**
- **Bid** — the price at which the dealer **buys**. **Chapter 4.**
- **Offer (ask)** — the price at which the dealer **sells**. **Chapter 4.**
- **Direct quote** — foreign currency per unit of domestic currency. **Chapter 4.**
- **Indirect quote** — domestic currency per unit of foreign currency; the reciprocal of the direct quote. **Chapter 4.**
- **Cross rate** — a rate between two non-USD currencies derived from two USD rates. **Chapter 4.**
- **Forward points (swap points)** — the interest differential expressed in FX rate terms; added for a premium, subtracted for a discount. **Chapter 4.**
- **Interest rate parity** — the forward premium or discount equals the interest rate differential for the same maturity. **Chapter 4.**
- **Covered interest arbitrage** — borrowing in one currency, investing in another and covering forward to lock a risk-free profit when parity is violated. **Chapter 4.**
- **Non-deliverable forward (NDF)** — a cash-settled forward with no delivery of the domestic currency and no exchange of notional. **Chapter 4.**
- **Carry trade** — borrowing in a low-yield currency to invest in a higher-yield one, capturing the differential. **Chapters 4, 12.**

- **Continuous Linked Settlement (CLS)** — a settlement mechanism that reduces FX delivery risk. **Chapter 4.**

Company analysis

- **Balance sheet** — a statement of financial position at a point in time: assets, liabilities and shareholders' equity. **Chapter 5.**
- **Common size statement** — every item expressed as a percentage of total assets or of sales. **Chapter 5.**
- **Goodwill** — the premium paid above the book value of an acquired company; written down against equity reserves. **Chapter 5.**
- **Treasury stock** — shares bought back and retired, benefiting remaining shareholders. **Chapter 5.**
- **Hidden reserves** — value not reflected in book value, from rezoned land or provisions not written back. **Chapters 5, 6.**
- **Current ratio** — current assets / current liabilities. **Chapter 5.**
- **Quick (acid test) ratio** — current assets excluding inventory, over current liabilities. **Chapter 5.**
- **Gearing (debt-to-equity)** — total debt / total shareholders' equity. **Chapter 5.**
- **Interest coverage** — EBIT / interest expense. **Chapter 5.**
- **Total fixed charge coverage** — earnings before interest, taxes and lease payments over interest plus lease payments plus the grossed-up preferred dividend. **Chapter 5.**
- **Cash conversion cycle** — the period between paying suppliers and collecting from customers. **Chapter 5.**
- **Window dressing** — manipulating reporting-date ratios by collecting receivables early, cutting inventory and delaying purchases. **Chapter 5.**
- **Quality of earnings** — realistic, repeatable earnings supported by conservative accounting and adequate provisioning. **Chapter 5.**

Equity securities

- **Residual claim** — ordinary shareholders are paid last on liquidation. **Chapter 6.**
- **Limited liability** — shareholders can lose only their original investment. **Chapter 6.**
- **Preference share** — a hybrid with a fixed dividend, no vote except in distress, ranked above ordinary and below bonds. **Chapter 6.**
- **Rights issue** — new capital raised from existing shareholders pro rata; prevents dilution. **Chapter 6.**
- **Bonus issue** — additional shares from reserves; raises no capital. **Chapter 6.**
- **Stock split** — divides existing shares to lower the unit price; raises no capital. **Chapter 6.**
- **Reverse stock split** — consolidates shares, often to escape penny-stock restrictions. **Chapter 6.**
- **Value-weighted index** — weights by market capitalisation; adjusts automatically for capital changes. **Chapter 6.**
- **Price-weighted index** — arithmetic average of prices; high-priced shares dominate. **Chapter 6.**
- **Equal-weighted index** — all constituents carry equal weight. **Chapter 6.**
- **Efficient frontier** — the set of portfolios offering the highest return for each level of risk. **Chapters 6, 8.**
- **Book value** — assets net of liabilities as stated on the balance sheet. **Chapter 6.**
- **Liquidation value** — the value realised on a forced sale of assets. **Chapter 6.**
- **Revised net asset value (RNAV)** — book value restated to market value. **Chapter 6.**
- **Free cash flow to firm (FCFF)** — after-tax operating cash flow less capital expenditure, plus non-cash charges. **Chapter 6.**
- **WACC** — the weighted average of the cost of equity and after-tax cost of debt. **Chapter 6.**
- **Terminal value** — the present value at the end of the projection period of all subsequent cash flows. **Chapter 6.**
- **P/E ratio** — price per share / earnings per share; trailing or leading. **Chapter 6.**
- **PEG ratio** — P/E divided by expected EPS growth; below 1 suggests undervaluation. **Chapters 6, 12.**
- **P/BV ratio** — price per share / book value per share; suited to banks, insurers and property companies. **Chapter 6.**
- **Enterprise value (EV)** — market capitalisation plus debt and preferred shares, less cash. **Chapter 6.**
- **EV/EBITDA** — the enterprise multiple; unaffected by capital structure. **Chapter 6.**

- **Earnings yield** — the inverse of P/E; compared with bond yields to give the **yield gap**. **Chapter 6.**
- **Depositary receipt (DR)** — a security trading in an investor's home market representing a foreign underlying share. **Chapter 6.**
- **ADR levels** — Level 1 OTC, Level 2 exchange-listed, Level 3 exchange-listed with capital raising. **Chapter 6.**
- **Greenshoe (over-allotment) option** — lets underwriters stabilise an IPO price by buying back shares. **Chapter 6.**
- **Claw-back** — reallocation of IPO shares between institutional and retail tranches. **Chapter 6.**

Fixed income

- **Par (face) value** — the redemption value repaid at maturity; usually $1,000. **Chapter 7.**
- **Coupon rate** — the periodic interest rate applied to par value. **Chapter 7.**
- **Term bond** — a single maturity date. **Chapter 7.**
- **Serial bond** — several maturity dates. **Chapter 7.**
- **Bank discount yield** —
$$(D/F)(360/t)$$
; based on face value, so not a true return. **Chapter 7.**
- **Accrued interest** — interest earned since the last coupon, paid by the buyer to the seller. **Chapter 7.**
- **Eurobond** — underwritten internationally, offered simultaneously in several countries, issued outside any single jurisdiction. **Chapter 7.**
- **Zero-coupon bond** — no coupons, sold at a deep discount; no reinvestment risk, highest convexity. **Chapter 7.**
- **Floating rate bond** — coupon reset periodically against a benchmark such as SIBOR. **Chapter 7.**
- **Covered bond** — highly rated paper backed by bank assets; no default since 1769. **Chapter 7.**
- **High-yield (junk) bond** — rated below BBB; trades on creditworthiness rather than rates. **Chapter 7.**
- **Payment-in-kind (PIK) bond** — the issuer may pay cash or issue a similar bond. **Chapter 7.**
- **Perpetual bond** — no maturity; ranks before equities on default. **Chapter 7.**
- **Call provision** — the issuer's right to redeem early; raises the yield. **Chapter 7.**
- **Put provision** — the holder's right to sell back at par; lowers the yield. **Chapter 7.**
- **Sinking fund** — annual retirement of part of the issue; reduces credit risk. **Chapter 7.**
- **Credit spread** — the yield difference between bonds of the same maturity and different credit quality. **Chapter 7.**
- **Credit default swap (CDS)** — a contract transferring default risk for a premium. **Chapter 7.**
- **Collateralised debt obligation (CDO)** — pooled bonds (CBO) or loans (CLO) repackaged and sold in tranches. **Chapter 7.**
- **Duration** — the linear price sensitivity to small yield changes; higher for lower coupons, lower yields and longer maturities. **Chapter 7.**
- **Convexity** — the curvature of the price-yield relationship; higher for lower coupons, highest for zero-coupon bonds. **Chapter 7.**
- **PVBP (PV01)** — the change in portfolio value for a one basis point rate move. **Chapter 7.**
- **Current yield** — annual interest / price; ignores capital gain or loss. **Chapter 7.**
- **Yield-to-maturity (YTM)** — the discount rate equating present value to price. **Chapter 7.**
- **Bond-equivalent yield** — the semi-annual YTM doubled. **Chapter 7.**
- **Taxable equivalent yield (TEY)** — tax-exempt yield / (1 − marginal tax rate). **Chapter 7.**
- **Reinvestment risk** — the risk that coupons cannot be reinvested at the same rate; rises with coupon and maturity. **Chapter 7.**
- **Term structure (yield curve)** — the relationship between YTM and maturity. **Chapter 7.**
- **Expectations theory** — the curve's shape reflects interest rate expectations. **Chapter 7.**
- **Liquidity preference theory** — the curve is always upward sloping because long maturities need a premium. **Chapter 7.**
- **Market segmentation theory** — each maturity's yield is set by supply and demand in that segment. **Chapter 7.**
- **Conversion parity price** — market price of the convertible / conversion ratio. **Chapter 7.**
- **Conversion premium** — the excess of the convertible's market price over its conversion value. **Chapter 7.**
- **Breakeven time** — conversion premium / (bond income − dividend income). **Chapter 7.**

Portfolio management

- **Efficient Market Hypothesis (EMH)** — prices fully reflect available information; weak, semi-strong and strong forms. **Chapter 8.**
- **Random walk** — price movements are independent over time. **Chapters 8, 11.**
- **Passive strategy** — buy-and-hold or index tracking, minimising cost. **Chapter 8.**
- **Active strategy** — security selection, sector rotation and market timing to beat a benchmark. **Chapter 8.**
- **Correlation coefficient ( )** — the extent to which two securities' returns move together; +1 to −1. **Chapter 8.**
- **Market portfolio (M)** — the fully diversified portfolio of all risky assets, containing only systematic risk. **Chapter 8.**
- **Capital market line (CML)** — the risk-return trade-off for **efficient** portfolios, using total risk. **Chapter 8.**
- **Security market line (SML)** — the risk-return trade-off for **all** assets, using beta. **Chapter 8.**
- **Beta** — the sensitivity of a security's return to the market; the market's beta is 1. **Chapter 8.**
- **Alpha** — the excess of actual return over the model-predicted return. **Chapter 8.**
- **Arbitrage pricing theory (APT)** — a multifactor equilibrium model; factors must be pervasive, influence expected return and be unpredictable. **Chapter 8.**
- **Investment policy statement (IPS)** — the client's mandate, recording objectives and constraints. **Chapter 8.**
- **Strategic asset allocation (SAA)** — the long-run policy mix, a constant-mix strategy. **Chapter 8.**
- **Tactical asset allocation (TAA)** — short-run overweighting and underweighting within SAA ranges. **Chapter 8.**
- **Dollar-weighted return (DWR)** — the IRR; sensitive to the size and timing of cash flows. **Chapter 8.**
- **Time-weighted return (TWR)** — links sub-period returns; unaffected by client cash flows, so preferred for judging managers. **Chapter 8.**
- **Modified Dietz method** — approximates TWR by weighting each cash flow by the time held. **Chapter 8.**
- **GIPS** — the Global Investment Performance Standards; require total return, TWR, monthly valuation and disclosure. **Chapter 8.**
- **Sharpe ratio** — excess return per unit of **total** risk. **Chapter 8.**
- **Treynor ratio** — excess return per unit of **systematic** risk. **Chapter 8.**
- **Jensen's alpha** — actual return less the CAPM-expected return. **Chapters 8, 12.**
- **Information ratio** — alpha per unit of excess risk. **Chapter 8.**
- **Sortino ratio** — excess return per unit of **downside** volatility. **Chapter 8.**
- **Bogey** — the passive benchmark against which attribution is measured. **Chapter 8.**
- **Performance attribution** — decomposing excess return into asset allocation and selection components. **Chapter 8.**

Funds, REITs and insurance

- **Collective investment scheme (CIS)** — a pooled investment vehicle governed by the Code on CIS. **Chapter 9.**
- **Variable Capital Company (VCC)** — a fund structure allowing free issue and redemption of shares, dividends from capital, and umbrella sub-funds. **Chapter 9.**
- **Exchange-traded fund (ETF)** — a passively managed fund tracking an index, traded during exchange hours. **Chapter 9.**
- **Net asset value (NAV)** — fund assets less liabilities, divided by units outstanding. **Chapter 9.**
- **Physical (cash) replication** — holding the actual index securities, by full replication or representative sampling. **Chapter 9.**
- **Synthetic replication** — using swaps and derivatives to mimic the index; lower tracking error, higher counterparty risk. **Chapter 9.**
- **Tracking error** — the divergence between an ETF's return and its index. **Chapter 9.**
- **Smart beta** — weighting securities on performance criteria rather than market capitalisation. **Chapter 9.**
- **Securities lending** — lending portfolio shares for collateral and interest; creates counterparty risk. **Chapter 9.**
- **Cash equitisation** — converting cash to equities via ETFs to reduce cash drag. **Chapter 9.**

- **Unit trust** — a CIS pooling funds under an independent trustee. **Chapter 9.**
- **Open-end fund** — issues units continuously at NAV. **Chapter 9.**
- **Closed-end fund** — a fixed number of units priced by market demand and supply. **Chapter 9.**
- **Offer price** — the price at which an investor buys units; NAV plus purchase expenses, cash, accrued income and the initial charge. **Chapter 9.**
- **Bid price** — the price at which an investor sells units; NAV less sale expenses, plus cash and accrued income. **Chapter 9.**
- **Front-end load** — a sales charge of 1.5–5% of NAV. **Chapter 9.**
- **Back-end load** — a redemption fee, often waived after a holding period. **Chapter 9.**
- **High watermark** — the profit level a manager must exceed before a performance fee is paid. **Chapter 9.**
- **Total expense ratio** — total expenses including management fees, over NAV. **Chapter 9.**
- **Retrocession** — a distribution fee paid by the fund house to the distributor; a conflict of interest that reduces NAV. **Chapter 9.**
- **REIT** — a CIS investing at least 75% in income-producing real estate and distributing at least 90% of taxable income. **Chapter 9.**
- **Business trust** — a hybrid structure under the Business Trusts Act 2004, not a separate legal entity, with a single trustee-manager. **Chapter 9.**
- **Stapled security** — two or more securities traded as one instrument. **Chapter 9.**
- **UCITS** — the EU directive providing a passport for cross-border fund distribution. **Chapter 9.**
- **Hedge fund** — a fund using short selling, derivatives and leverage with absolute return targets. **Chapter 9.**
- **Lock-up period** — a period during which hedge fund investors cannot redeem. **Chapter 9.**
- **Gate provision** — a limit on withdrawals at a given redemption date. **Chapter 9.**
- **Private equity J-curve** — negative early cash flows during the 5–6 year investment period, with harvest from year 6. **Chapter 9.**
- **Whole life policy** — life cover plus cash value, payable on death whenever it occurs. **Chapter 9.**
- **Endowment policy** — payable on maturity or death, whichever is earlier. **Chapter 9.**
- **With-profits policy** — benefits linked **indirectly** through annual valuation and a reversionary bonus. **Chapter 9.**
- **Investment-linked policy** — policy values linked **directly** to underlying unit values. **Chapter 9.**
- **Immediate annuity** — payments begin at purchase, funded by a single premium. **Chapter 9.**
- **Deferred annuity** — payments begin at a future date. **Chapter 9.**
- **Dollar cost averaging** — investing equal amounts over time, accumulating more units when prices are low. **Chapter 9.**

Warrants

- **Warrant** — a right, not an obligation, to buy or sell a specified number of securities at a fixed price within a specified period. **Chapter 10.**
- **Company warrant** — issued by the company on its own shares; call only; new shares issued on exercise. **Chapter 10.**
- **Structured warrant** — issued by a third party such as a bank; call or put; cash-settled; no new shares. **Chapter 10.**
- **Conversion ratio** — the number of warrants needed to convert into one share. **Chapter 10.**
- **Intrinsic value** — $(SP - EP)/n$ when in-the-money; zero otherwise. **Chapter 10.**
- **Conversion price** — the total cost of one share via the warrant: $n \times WP + EP$. **Chapter 10.**
- **Warrant premium** — the extra cost of the warrant route over buying the share outright. **Chapter 10.**
- **Gearing ratio** — $SP / (n \times WP)$; the leverage the warrant provides. **Chapter 10.**

Technical and quantitative analysis

- **Technical analysis** — forecasting price movements from historical market data; not concerned with fair value. **Chapter 11.**
- **Advance-decline (A/D) line** — cumulative advances minus declines across all securities; measures market breadth. **Chapter 11.**
- **Moving average** — the average closing price over a period; smooths erratic price changes. **Chapter 11.**
- **MACD** — the difference between a fast and a slow EMA, with a signal line; commonly (12,26,9).

Chapter 11.

- **Bollinger Bands** — a 20-day SMA plus and minus two standard deviations. **Chapter 11.**
- **Support level** — a price range where a substantial increase in demand is expected. **Chapter 11.**
- **Resistance level** — a price range where buyers refuse to pay more and sellers take profits; the overhang. **Chapter 11.**
- **Breakout** — a move through support or resistance, confirming the trend. **Chapter 11.**
- **Dow Theory** — primary, secondary and daily movements; confirmed when the DJIA and DJTA agree. **Chapter 11.**
- **Relative Strength Index (RSI)** — a momentum indicator from 0 to 100; above 70 overbought, below 30 oversold. **Chapter 11.**
- **Head and shoulders** — a reversal pattern at a peak; the inverse form marks a bottom. **Chapter 11.**
- **Double top / bottom** — a reversal pattern confirmed when the neckline breaks. **Chapter 11.**
- **Consolidation rectangle** — a sideways continuation pattern. **Chapter 11.**
- **Triangle** — symmetrical (either direction), ascending (bullish) or descending (bearish). **Chapter 11.**
- **Flag and pennant** — short continuation patterns after a sharp move; parallel lines versus converging lines. **Chapter 11.**
- **Wedge** — converging lines both moving the same direction; a rising wedge is bearish. **Chapter 11.**
- **Gap** — a price range with no transactions; breakaway, runaway or exhaustion. **Chapter 11.**
- **Elliott Wave Theory** — 5-wave advances and 3-wave retreats exhibiting the Golden Ratio 1.618. **Chapter 11.**
- **Quantitative analysis** — abstracting market reality into models to find price anomalies and arbitrage. **Chapter 11.**

Exam cues

- **Grouped by chapter, not alphabetically** — find a term from the topic you are revising.
- **Four investor categories:** II, AI, EI, retail. **AI thresholds:** S2$mnetpersonalassets(residencecappedatS$1m), S1$mfinancialassets, orS$300k income; entity S$10m.
- **Primary market: issuer gets the cash. Secondary market: selling investor gets the cash.**
- **Dealer = principal earning the spread; broker = agent earning commission.**
- **Six return measures; three risk measures (variance, standard deviation, CV).**
- **IRR is money-weighted; CAGR is time-weighted.**
- **Systematic risk cannot be diversified but can be hedged; non-systematic can be diversified away.**
- **Leading precedes, coincident aligns, lagging follows. PMI 50 is the dividing line.**
- **Base currency first; pip = 0.0001; bid is where the dealer buys.**
- **Forward points: add for a premium, subtract for a discount; the higher-rate currency is at a discount.**
- **NDF: cash-settled, no delivery, no notional exchange.**
- **Balance sheet is a snapshot; common size uses total assets or sales as 100%.**
- **Goodwill write-downs shrink equity and raise ROE.**
- **Efficiency ratios use average balances.**
- **Residual claim and limited liability define an ordinary share.**
- **Rights issue raises capital; bonus issue and stock split do not.**
- **Value-weighted adjusts automatically; price-weighted needs a divisor.**
- **PEG below 1 suggests undervaluation.**
- **EV subtracts cash; EV/EBITDA also subtracts associates and adds minorities and unfunded pension.**
- **Bank discount yield uses face value, so it overstates the true return.**
- **Zero-coupon bonds have no reinvestment risk and the highest convexity.**
- **Duration rises with lower coupon, lower yield and longer maturity.**
- **CML uses total risk; SML uses beta.**
- **TWR is the correct measure for judging a manager.**
- **Sharpe uses ; Treynor uses ; Sortino uses downside .**
- **REIT: 75% investment, 90% distribution, 45% gearing. Business trust: no gearing cap, no investment restrictions.**
- **Company warrant: call only, new shares issued. Structured warrant: call or put, cash-settled.**
- **RSI above 70 overbought, below 30 oversold.**
- **Head and shoulders is a reversal; triangles, flags and rectangles are continuations.**

- **Trap:** a **dealer** and a **broker** are not interchangeable.
- **Trap: marketability** is necessary but not sufficient for **liquidity**.
- **Trap:** a **beta of zero** does not make an asset risk-free.
- **Trap:** an **unrated** bond is not automatically a default risk.

Exam method and final revision checklist

The CM-EIP paper is **90 questions in 2 hours 30 minutes** — about **100 seconds per question**, including the three case studies that carry nine questions between them. The pass mark is **70%**, so **63 correct answers**. That leaves room for 27 errors and no more, which means the method matters as much as the knowledge.

How to work the paper

1. **Answer every question.** There is no penalty for a wrong answer, so never leave a blank. If time is short, fill the remaining answers with your best judgement and return only if time allows.
2. **Read the stem for the qualifier first.** Words such as *least likely*, *most appropriate*, *not*, *except* and *which of the following is TRUE* decide the answer more often than the technical content does.
3. **Classify the question before computing.** Most calculation questions name the measure they want. Identify whether it is a return measure, a risk measure, a valuation ratio, a bond measure or a warrant measure, then recall the formula from Appendix B.
4. **Check the convention. 365-day** for Singapore Government Securities, **360-day** for everything else. Semi-annual coupons double the periods and halve the rate. Currency adjustment divides when the reference currency is first in the pair and multiplies when it is second.
5. **Sanity-check the magnitude.** A bond priced at $1,147 against a $1,000 par with an 8% coupon and a 6% yield is plausible; $1,014 is not. A P/E of 12x on a mature industrial and 37x on a high-growth semiconductor are both plausible; 370x is not.
6. **For multiple-response questions, evaluate each option independently.** Do not look for a pattern in the letters. A statement is true or false on its own merits.
7. **Do the case studies last if they are grouped at the end**, or first if you prefer to bank the applied marks while fresh. Either way, read the scenario once and note the client's objective, horizon, risk tolerance and any constraint — the questions usually turn on one of those.
8. **Leave the hardest calculation for a second pass.** Mark it, move on, and return. Two minutes lost on one question costs two easier questions.

The ten distinctions that decide the most marks

Distinction	The rule
Primary vs secondary market	Primary: the **issuer** receives the cash. Secondary: the **selling investor** does
Dealer vs broker	Dealer trades as principal and earns the **spread**; broker acts as agent for a **comm**
Money market vs capital market	Money market: **1 year**. Capital market: **> 1 year**
Direct vs indirect quote	Direct: foreign/domestic. Indirect: domestic/foreign — the **reciprocal**
Discount vs premium bond	Coupon **below** yield → discount. Coupon **above** yield → premium. Equal → par
Duration vs convexity	Duration is the **linear** first-order sensitivity; convexity is the **curvature**, the seco
CML vs SML	CML: **efficient portfolios**, **total risk**. SML: **all assets**, **beta**
DWR vs TWR	DWR = IRR, driven by client cash flows. TWR = manager skill, unaffected by th
REIT vs business trust	REIT: **90%** distribution, **45%** gearing cap, real estate only. Business trust: **no m**
Company vs structured warrant	Company: **call only**, **new shares issued**, years. Structured: **call or put**, **cash-**

Before the examination, be able to explain the difference between

- An **institutional**, **accredited**, **expert** and **retail** investor, and the statutory thresholds for each.
- A **financial asset** and a **real asset**, and why the distinction matters to an adviser.
- **Money market** and **capital market** instruments, and which are quoted on a discount basis.
- **Total return**, **expected return**, **annualised return**, **real return**, **currency-adjusted return** and **IRR** — and which question each answers.
- **Variance**, **standard deviation**, **coefficient of variation** and **value at risk**.
- **Systematic** and **non-systematic** risk, and how each is managed.

- **Fiscal** and **monetary** policy, and how each transmits to asset prices.
- **Leading**, **coincident** and **lagging** indicators, with one example of each.
- **Direct**, **indirect** and **cross** FX quotations, and the bid-offer spread.
- **Forward points** and **interest rate parity**, and why the higher-rate currency trades at a forward discount.
- A **non-deliverable forward** and why it exists.
- **Qualitative** and **quantitative** company analysis.
- The **balance sheet**, **income statement** and **cash flow statement**, and what each reveals that the others do not.
- **Liquidity**, **leverage**, **efficiency** and **profitability** ratios, and which use average balances.
- **Ordinary** and **preference** shares, and the claim order on liquidation.
- A **rights issue**, **bonus issue**, **stock split** and **reverse split**.
- **Value-weighted**, **price-weighted** and **equal-weighted** indices.
- **Net asset**, **absolute (DCF)** and **relative** valuation.
- **P/E**, **PEG**, **P/BV**, **P/Sales**, **EV/CF** and **EV/EBITDA**, and when each is appropriate.
- **Deposits**, **money market instruments** and **bonds**, and the deposit insurance limits.
- **Duration** and **convexity**, and what each implies for price risk.
- **Current yield**, **YTM**, **bond-equivalent yield** and **taxable equivalent yield**.
- The **expectations**, **liquidity preference** and **market segmentation** theories of the yield curve.
- **Conversion value**, **conversion parity price**, **conversion premium** and **breakeven time** for a convertible bond.
- The **weak**, **semi-strong** and **strong** forms of the efficient market hypothesis.
- **Passive** and **active** management, and the three active strategies.
- **Strategic** and **tactical** asset allocation.
- The **Sharpe**, **Treynor**, **Jensen**, **information** and **Sortino** measures, and which risk each uses.
- **Performance attribution** and the difference between allocation and selection effects.
- An **ETF**, a **unit trust**, a **REIT**, a **business trust** and a **stapled security**.
- **Whole life**, **endowment**, **with-profits** and **investment-linked** insurance.
- An **immediate** and a **deferred** annuity, and why annuities hedge longevity risk.
- **Dollar cost averaging** and **market timing**.
- A **company warrant** and a **structured warrant**, and the four warrant valuation measures.
- **Technical analysis** and **fundamental analysis**, and the assumptions that separate them.
- **Support**, **resistance**, **breakout** and the main reversal and continuation patterns.

Final week

- **Re-read every chapter's Exam cues block.** They are written to be the last thing you read.
- **Work the Appendix A review questions** without looking at the answers, then check.
- **Rehearse the formulae sheet** in Appendix B until you can locate any formula within seconds. It is provided in the examination, so the skill is retrieval, not memorisation.
- **Practise the case studies in Chapter 12** as full sets of three questions, timed.
- **Take at least two full mock exams** on this site at the real length and pass mark.
- **Confirm the logistics:** original NRIC or passport, a **non-programmable** calculator, arrival about 30 minutes early, and no notes or smart devices.

Exam cues

- **90 questions, 150 minutes, 70% pass — 63 correct.** About **100 seconds per question**.
- **Never leave a blank.** There is no penalty for a wrong answer.
- **Read the qualifier first:** *least likely*, *most appropriate*, *not*, *except*, *TRUE*.
- **Classify before computing** — return, risk, valuation, bond or warrant measure.
- **Conventions: 365** for SGS, **360** for everything else; semi-annual doubles periods and halves the rate; currency adjustment divides when the reference currency is first in the pair.
- **Sanity-check magnitudes** before committing to an answer.
- **Multiple response:** evaluate each option independently; do not look for letter patterns.
- **Case studies:** note the client's objective, horizon, risk tolerance and constraints first.
- **Mark and return** on the hardest calculation rather than losing two easier marks.
- **The ten highest-value distinctions** are listed in the table above — learn them as pairs.
- **The formulae sheet is provided**, so practise **retrieval**, not memorisation.
- **Trap:** the pass mark is **70%**, not 75% — do not carry over the FMRP or RES 12B figure.

- **Trap:** the paper includes **3 case studies of 9 questions**, not a separate case-study section worth a fifth of the marks.